

The Royal Mint

by

G. F. Ansell

The Royal Mint
by
G. F. Ansell

The Project Gutenberg eBook of The Royal Mint

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The Royal Mint

its working, conduct, and operations, fully and practically explained

Author: George Frederick Ansell

Release date: May 24, 2026 [eBook #78741]

Language: English

Original publication: London: Effingham Wilson, 1871

Other information and formats: www.gutenberg.org/ebooks/78741

Credits: deaurider and the Online Distributed Proofreading Team at <https://www.pgdp.net> (This file was produced from images generously made available by The Internet Archive)

*** START OF THE PROJECT GUTENBERG EBOOK THE ROYAL
MINT ***

THE ROYAL MINT:

***Its Working, Conduct, and Operations,
Fully and Practically Explained;***

WITH
SUGGESTIONS FOR ITS BETTER SCIENTIFIC
AND OFFICIAL MANAGEMENT.
RE-WRITTEN, ENLARGED, AND REVISED TO THE
PRESENT TIME. TO WHICH IS ADDED,
AS AN APPENDIX,
A LETTER ADDRESSED TO THE RIGHT HON. LORD KINNAIRD, K.T.

BY GEORGE FREDERICK ANSELL,
ANALYTICAL CHEMIST.
LATE OF THE ROYAL MINT,
FORMERLY OF THE ROYAL COLLEGE OF CHEMISTRY.

ILLUSTRATED WITH ENGRAVINGS.

“To have a thing is little if you’re not allowed to show it;
And to know a thing is nothing unless others know you know it.”

LORD NEAVES.

THIRD EDITION.



LONDON:
EFFINGHAM WILSON, ROYAL EXCHANGE.
1871.

Entered at Stationers' Hall.]

[All rights reserved.

Ipswich, *March 1st, 1871.*

DEAR MR. ANSELL,

I have no objection to your appending to the third edition of your book your letter to me in regard to your case, and my correspondence with Mr. Gladstone and the Chancellor of the Exchequer, but if you add in addition to this correspondence the report of the debate on my motion for a committee to inquire into the management of the Mint, I should wish to make some few remarks on the subject in question; but before doing so I must take the opportunity of thanking you for letting me see Mr. Fremantle's letter in answer to your offer, made at my suggestion, to meet Mr. Fremantle with a view to rendering him some assistance (which from your great experience you are well able to do) in the management of the Mint, which I have reason to know both he and Mr. Lowe find *somewhat difficult*.

The answer is evidently dictated by Mr. Lowe. After his letter^[1] to me of the 29th April, 1869, I might have known that he was not likely to do anything which might appear as if he owned himself in the wrong. This letter, indeed, showed how annoyed Mr. Lowe was at my having seen that most malicious report to the Treasury by Dr. Graham, which has been the cause of your having been so unjustly dealt with, to the ruin, in a great measure, of your prospects in life. Dr. Lyon Playfair at the time threatened that if I moved to inquire into the management of the Mint, he would move that that report be laid on the table of the House. This would not have deterred me, as the report *itself* would have been the best refutation of the *insinuation* it contained that you were dismissed because you were guilty of the Mint practice of purloining gold (whereas the note^[2] relieving you from further attendance stated that the short work at the Mint was the cause). Dr. Graham did not, he said, believe the insinuation, and to show that he did not believe it, although he had sent in the report to the Treasury, he authorised me to offer you the situation (at an increased salary) filled by Mr. Goodwin, who was to be retired, which of course you refused as being far inferior to the one you held; and in further proof he wrote you that handsome testimonial^[3] which I read to the House of Lords.

The report, however, *had served its purpose* in preventing you from being appointed to an office in the Mint.

I am sure Dr. Graham regretted that he had been induced by his brother, whose animus against you was known throughout the Mint, to send in such a report to the Treasury. Dr. Lyon Playfair's threat, as I have said, would not have deterred me from moving for a committee to inquire into the malpractices of the Mint, but I confess that I had great scruples in regard to Dr. Graham himself. I shall never forget the expression of his face, and the fearful agitation the mention of my intention always caused.

I confess I did hesitate to place a man of Dr. Graham's great scientific attainments in such a position as an inquiry would have done, seeing also that his brother was just dead, and that I knew individually Dr. Graham was in no way implicated in the malpractices which formerly prevailed in the Mint; for, as you know, Dr. Graham had soon after his appointment set to work to clear out that "Augean stable," which, with your assistance, he had almost accomplished till in an evil moment he appointed his brother, to whose intemperate habits I referred in the House of Lords,^[4] to a responsible office in the Mint.

Dr. Graham's death following so soon after that of his brother, I was greatly rejoiced at having delayed bringing the subject of the Mint before Parliament, because I should have felt that I had been to some extent instrumental in hastening that event by drawing attention to the state of that Department.

On the occasion of the reading to me, by Dr. Lyon Playfair, of that wicked report to the Treasury (to which I have previously alluded), in the presence of Dr. Graham and of Mr. C. W. Fremantle, I urged, as Dr. Graham exculpated you from all blame *insinuated* by the report, that in common justice you should be reinstated in some office in the Mint, you having, with my advice, refused the position offered by Dr. Graham through me, when Mr. Fremantle remarked as a reason against your being appointed that "you were so unpopular with the officials;" such unpopularity being in my opinion the natural result of the work you had done at the request of Dr. Graham, and in the service of the Mint. I say the work you had done, and use the expression on the authority of Dr. Graham, copies of whose letters to the Treasury are now in my possession, and in which he distinctly states

that to you mostly were due the reforms. There is no doubt, therefore, but that you were exceedingly unpopular just in the sense that a detective would be amongst the “light-fingered gentry,” and had I obtained the committee for which I moved I could have shown the true grounds of your unpopularity. I have collected some further valuable information showing the necessity for a rigid and full inquiry into the affairs of the Mint, which will show that the officials of a public department are not justified in putting into the hands of a young and rising member of the House of Lords statements which are not only erroneous, but absolutely false, so that Lord Lansdowne was obliged to declare that “he had on a previous occasion overstated the amount in regard to the profit on silver;”^[5] but again was he grossly misled by the Mint officials when he stated “that there was a gain of £5,000 per annum on the silver coinage”^[6] (a considerable reduction on the former statement) while the *return* showed a positive and very great loss. It would appear that the Mint officials mystify him, and that *he*, not *I*, confuse two things.

I alluded in the House to a job by Mr. Lowe in appointing Mr. Roberts, and Lord Lansdowne in reply stated that Mr. Roberts was a *very eminent chemist* who had proved his capability by “*devising* a system for depriving brittle gold of the obnoxious properties on which I recently commented.” Have you heard that the plan has been tried and found to be a complete failure!!! I am also informed that Mr. Roberts *did not devise* the process in question, but that it was invented by a really eminent chemist, who did not make the inquiry with a view to its application under the circumstances which arise in the Mint, where operations are conducted on so large a scale. The process is described in the British Association Reports for 1848, and more fully in the *Philosophical Magazine* for 1850.

I have just received the Report of the Deputy-Master of the Mint, on European Mints, which I have only had time to glance over. The report is evidently framed with a view of meeting the allegations I made on the appointment of the commission. It would appear, moreover, that Mr. Fremantle and his colleagues have made a *very careful study of your book!!!* which might have been done without the pleasant trip. For the general public the reports are interesting as showing how foreign mints are worked, but as far as the management of the English Mint is concerned, and how a department which is a serious expense to the country, and from

which no one but the Bank of England and the employés derive any profit, can be made, as it ought to be, a source of revenue, the report does not show. I see it is proposed to build new offices, “so constructed as to diminish facilities for speculation!!” I think the real question is, would it not be better TO COIN BY CONTRACT—this might be tried, at all events for a time, before the new offices are commenced.

I almost fear this Mint question will not be taken up in the House of Commons, as members are afraid to “tackle” Mr. Lowe, and in these fearfully excited times the subject is *too dry* for members to interest themselves in; but I cannot understand how a Government bent on economy and reforming abuses in the several Departments can allow such wasteful extravagance and useless losses to go on, unless indeed Mr. Lowe is dreaded as much at the Council Board as in the House of Commons.

Believe me,

Dear Mr. Ansell,

Yours faithfully,

(Signed) KINNAIRD.

To G. F. Ansell, Esq.,
27, Bernard Street, Russell Square, London, W.C.

PREFACE TO THIRD EDITION.

Influential friends have advised me to issue the present as the *third* edition of “The Royal Mint” on the ground that a Treatise on Coining which I wrote, at the request of Mr. Graham, for “Tomlinson’s Cyclopædia,” of which Mr. Tomlinson allowed me to have a hundred copies printed and bound separately for private circulation, was really the first edition of the more extended work I have recently published. The copies of that first edition were soon absorbed: indeed, owing to the fact that there is little literature in relation to the Royal Mint, I was flattered by foreign governments and foreign mints seeking from me copies of that book. Such applications gave me pleasure, and as far as I was able I complied with the requests preferred, except in the case of the Spanish mint, for which institution Mr. Graham asked of me a copy when, unfortunately, I had parted with the last at my disposal. I was further gratified by the fact that the Imperial Government of France sought permission to translate this book into French, with a view to give a copy to each workman in the mints of that nation.

Under considerable pressure I re-wrote the book above spoken of for public use, but omitted to style it, as I should have done, the *second* edition. This, in its turn, was most kindly received by those whose opinions I value in both Houses of Parliament, as well as by those who understand in an especial degree the subject of which I treated; and I have reason to know that it has found its way to all, even the most distant, parts of the world. It was produced, but not replied to, by Ministers in both Houses of Parliament, and remains yet uncontradicted ([see page 198](#)); indeed, I may assume that it has been officially confirmed; for on the motion of the Right Honourable Lord Kinnaird, returns were made to the House of Lords, which are verbatim copies of some of the most damaging pages in my book up to and inclusive of the dates then reached (see pages [91](#), [97](#), [102](#), [133](#), [135](#), [197-8](#)). I say thus much in gratitude, for the subject I have chosen is one which could hardly have induced me to expect for it so generous a reception. I wrote it hoping only that the public would acquaint themselves with the manner in which one of the principal Departments of the

Government was and is conducted. Lord Kinnaid, in the letter which he has done me the honour to write, and permits me to print in these pages, expresses an opinion—one that is largely entertained—that Mr. Fremantle and his fellow-travellers to European mints are greatly indebted to my books for the information on which they found their opinions, as expressed in the reports on foreign mints. For their own sakes as well as for the benefit of the nation, I would that those gentlemen had followed in their entirety my suggestions, because such a course would have enabled them to avoid those difficulties which beset them, and pitfalls into which they will stumble in the carrying out the designs expressed in these reports on European mints; these I will discuss in the fitting place, but here I desire to thank those gentlemen for such attention as they have given to my opinions.

The leading newspapers and periodicals have reviewed or noticed the books in a kindly spirit, with but one exception, in which the writer without refuting one statement indulged in personalities. To those gentlemen of the Press who have spoken candidly and generously, I offer my best thanks. I have endeavoured to correct the faults they pointed out—unless, indeed, where principle was concerned; and, in deference to the expressed opinions of some of my reviewers, I, now that a new edition is called for, publish, by his Lordship's authority, a letter which I wrote to Lord Kinnaid at his Lordship's request, giving a history of my dissociation from the Mint, in which, I hope, will be found evidence confirming the statements I make throughout the book.

I do not attempt to disguise the fact that I consider myself to have been seriously wronged both by Mr. Thomas Graham and by the Treasury, still I consider, wisely or unwisely, that my personal grievances are of less interest to the public than is the principle involved. I, however, deem it right to invite attention now to the subject of my complaint,—fully set forth in the Appendix,—because it is illustrative of the treatment a public servant will surely receive if he should follow the intimation conveyed by the present Government, and, neglecting the fact of his own relative position, fulfil the duties of his office as he would do if the matter in hand were one of personal concern. The Government of the Right Honourable W. E. Gladstone, from their bench in the House of Commons, promised that “if each in his especial sphere would faithfully fulfil the duties required in his department irrespective of his nominal position,” “such public servant

should be duly rewarded:”—the Ministers might have added, and would have so added had they been candid, “with immediate discomfort and by ultimate dismissal, and that justice would be sought at *their* hands by such public servant in vain!”

The effect of a rule which obliges Ministers to support the “head of a Department” “*under any and all circumstances,*” will be apparent to all evenly balanced minds; and I from my own knowledge, without fear of contradiction—except, indeed, official denial, which is now so perfectly understood—state that the man who best succeeds in a Government office is he who does unreasoningly and undoubtingly just what he is told to do, and avoids giving offence to his superior officers by pointing out any irregularity or improvement.

Upon the publication by Mr. Thomas Graham of the malevolent report he had written to the Treasury, I, when that fact was made known to me, obtained a high opinion, and in pursuance of advice then given, it was my intention to proceed by action at law against Mr. Graham for libel, but his early death made that step impossible.

With regard to the publication of the correspondence, it is right that I should say that Lord Kinnaird had not intended to draw attention to my case; but, when his Lordship found that the Marquess of Lansdowne and His Grace the Duke of Argyll led off against me in the House of Lords, as the Right Honourable Robert Lowe had done most bitterly in the House of Commons on the previous Friday, he with great generosity threw off the natural reluctance he had entertained to hurting the feelings of Mr. Graham’s friends, and defended me with words and documents so nobly that I can never sufficiently thank his Lordship, or fittingly acknowledge the indebtedness I thereby incurred.

To Mr. Lowe, Lord Lansdowne, and the Duke of Argyll must be attributed any pain which may arise to the friends of Mr. Graham, for Lord Kinnaird desired to spare them; for myself, however, I would say that if either of the Ministers named had obligingly pointed out any presumed false statement contained in the book which has so far excited their anger, and proved its untruthfulness, instead of saying hard words of the Author, I should have been less mindful of the sage instructions given on a celebrated occasion to an eminent Advocate, “It is a weak case, bully the counsel.”

Some of my readers will perhaps consider that the following pages are overlaid with a strong personal feeling, and may therefore feel disposed to lay the book aside. To such I would remark that it is impossible to follow a pursuit of any kind earnestly, unless it be made a matter of undivided interest. Besides, it must not be forgotten that I was invited to go to the Mint solely on the ground of my known determination to resist abuses. Having devoted fifteen years to the subject of the Mint, it is not to be supposed that I can view its mismanagement with less concern now than I did when selected for appointment to that Department, that I might repress irregularities which were then known to exist, and which appear to have been as yet uneradicated.

With such remarks, I invite a perusal of my new edition by such members of the legislature as are really desirous of reforming the abuses which exist in the Royal Mint.

PREFATORY LETTER TO THE FIRST EDITION.

TO

THOMAS GRAHAM, ESQ., F.R.S.,

MASTER OF THE MINT, ETC. ETC.

SIR,

I have written a short article for “Tomlinson’s Cyclopædia,” for the purpose of explaining (as far as the limited space at my disposal would allow) some of the details of the process of Coining, and by permission of Mr. Tomlinson I am allowed to have a few copies of that article printed separately for private circulation.

I take advantage of this opportunity to acknowledge most gratefully that the information I have been so fortunate as to obtain on the subject of Coining has been gained in consequence of the opportunities you have afforded me in my position in the Royal Mint. With heart-felt thanks for the privileges you have accorded me, and for the support you have given me,

I beg to remain, Sir,

Your most faithful servant,

GEO. F. ANSELL.

ROYAL MINT, *August, 1862.*

PREFATORY LETTER TO THE SECOND EDITION.

27, Bernard Street, Russell Square, London, W.C.,
November 8th, 1869.

MY LORD,

I have ventured to ask permission to dedicate this book to your Lordship, as a mark of my sense of the great kindnesses you have been pleased to show to me through a series of years. As you, my Lord, will perhaps permit me to remind you, generous efforts have been made by many with a view to reinstate me in a position of which they and you consider me to have been unjustly deprived—an opinion, I believe, your Lordship has formed from a knowledge of all the facts.

The recent death of Mr. Thomas Graham puts an end to any personal feeling between him and myself, whoever was right or whoever was wrong, or even if there were faults on both sides, the matters which caused a severance are better forgotten—our differences are beyond human adjustment.

The information I have attempted to give in the following pages is given solely with a view to throw as much light as is possible upon a subject which is now attracting very considerable attention, and which has become of national importance.

It will be to me a great pleasure if I may thus express the gratitude I feel for the valuable assistance I have received from your Lordship during a time of great trial and anxiety.

I have the honour to be,

My Lord,

Your Lordship's most faithfully obliged and humble servant,

GEORGE F. ANSELL.

To the Right Hon. Lord Kinnaird, K.T., F.R.G.S.,
Lord-Lieutenant of Perthshire, &c. &c. &c.

PREFACE TO SECOND EDITION.

Perhaps it will be well that I should give a brief outline of the circumstances antecedent to and attending my association with the Royal Mint, because the facts and opinions which will be given in the following pages appear to require such a notice, that their worth or worthlessness may be estimated at its true value, as well as that the sources of my information, and the modes of obtaining it, may be in the hands of my readers.

In the early part of the year 1856 the Master of the Mint found himself so beset with difficulties arising from irregularities committed by those who should have supported him, that he formed a determination to engage a person who should be indisputably a faithful and intelligent officer in the Coining Department, and who at the same time had sufficient personal influence to check those irregularities which he knew were taking place, but which were beyond his personal supervision. These irregularities were known to the Government of that period, and had caused so vast an expense that that Government, which was presided over by Lord Palmerston, had intimated to the Master, that unless the Mint could be conducted more satisfactorily and economically, it would be broken up as an Imperial establishment, and thus necessitate the placing of the coinage in the hands of contractors. With this view papers were printed and issued to various firms; but Mr. Graham, being very sincere and energetic in his desire to so conduct the Mint as that this necessity should not arise, sought the advice of engineers and others, with a view to finding such a man as combined within himself the qualities which he saw to be necessary. Finally, by the advice of Dr. A. W. Hofmann, he called upon me at my residence in October, 1856, and related to me in detail the facts above alluded to, and explained precisely the position of affairs at the Mint, giving me the names of the troublesome persons, and showing his own position to be so intolerable, that without some one on whom he could rely, it would be impossible for him to continue his Mastership. Mr. Graham then proceeded to tell me what had passed between Dr. Hofmann and himself as to my fitness for the work required, and explained clearly the impossibility of placing me in a secure position or in high office at first, but that if I would accept a supernumerary

clerkship temporarily, he would, when I had effected the object of his desires, advise my promotion to the office then held by Mr. W. T. Brande, when that office should become vacant. With this understanding I agreed to accept the position proposed if duly appointed; but that I might be able to carry out such reforms as were necessary, I insisted on sufficient power being given to me by himself or by the Government. In pursuance of this agreement, Mr. Graham recommended my appointment in a letter to the Treasury, dated 29th October, 1856, in the following terms:—

“I desired to introduce a young man, recommended by scientific and technical information available in coining, by energy of character, and by tried ability in the supervision of workmen—a faculty by no means common. After applying to Mr. William Fairbairn and to Mr. George Rennie, who both recommended candidates, and making inquiry in various other quarters, I have been led to propose the name of Mr. George Frederick Ansell as temporary clerk. Mr. Ansell was educated under Dr. Hofmann, and acted for some years as his laboratory assistant. He has since been Scientific Director in the Panopticon, Leicester Square, which was lately broken up. He appears to be a person of superior education and ability, and great activity and vigour of character, and, as I have been assured, has shown much discretion in the management of both pupils and workmen.”

In accordance with this recommendation I was appointed to a supernumerary clerkship, and took office in the Rolling Room of the Royal Mint; but before doing so I particularly inquired of Mr. Graham, in the presence of Mr. W. H. Barton, the then Deputy-Master, what authority I should have, and how far I should be supported if I attempted to enforce obedience, and whether I was at liberty to call for statements and examine original documents and books. He verbally authorised me to examine such books as I wished to examine, and desired Mr. Barton to give me such statements or information as I might ask for; and as to my authority, he said, “If you order the men to dance a hornpipe on the table, they shall do it, and all orders shall pass through your hands.” With such power, and a salary of £120 a year, or rather £10 a month, I took office on the 12th November, 1856.

My personal experience of men and manners in the Mint is recorded in another book, but in this I feel it right to adhere rigidly to explanations of the processes of coining, and to treat of the expense and loss attending the

manipulation of the precious metals; and in the latter pages I will demonstrate what formerly was, and what should be, the cost of producing a coined sovereign, each statement being the result of actual experience, and capable of proof.

THE ROYAL MINT.

Coining is the term applied to the processes employed in the manufacture of money. It is proposed to describe the present mode of manufacture; but it may be as well to state, that in remote periods money was made by cutting out a piece of metal somewhat of the form of the intended coin, and imparting the device to it by the blow of a sledge-hammer. For this purpose the blank piece of metal was laid upon a die, say the obverse, fixed into a block of wood or stone, supposed to have been so large as to absorb the vibration caused by the blow, and to a great extent prevent the quivering which would naturally arise, and cause unsteadiness. The workman then took the other die, say the reverse, and passing it through a folded sheet of lead, in order to avoid the shock to the hand, he placed the engraved part on the blank, which was resting on the lower die, and held it firmly while another workman struck it with a sledge-hammer. It is worthy of remark that a piece of lead such as that described is in the Museum of Dies at the Royal Mint; and although its surface, by the action of the atmosphere and other causes, is now converted into carbonate of lead, indentations caused by the tips of the workman's fingers are evident. This piece is believed to have been used with a die of Edward IV. At a later period the upper die was held in a twisted hazel stick. After each blow the dies were what is technically termed locked together; that is, the lower and upper dies were made to fit into the partly-formed coin, so that neither die could turn without turning the whole mass, and then a second or third blow was given, till the coin was completed. An improvement on this method was effected by fitting the two dies into rods of iron, which may be represented as a pair of tongs; the flat parts which are used to take a coal would then be the position occupied by the dies. This plan saved the operator some risk of bruised fingers, but the process was essentially the same as the original one; and to money produced by either means was applied the term *hammered* money, in contradistinction to *milled* money; that is, money which was made from blanks obtained from fillets which had been rolled in a very rough kind of rolling mill, driven by horse power—the germ from which sprang the present machinery. On this point it is perhaps

of interest to quote a passage from the Report made to the Lords of the Treasury in 1695 by Mr. William Lowndes, who says:—"All the moneys we have now in England, both gold and silver, are reducible to two sorts, one stamped with the hammer, and the other pressed with an engine called the mill. The gold or silver of the hammered money is first cast from the melting-pot into long bars, those bars are cut with shears into several square pieces of exact weight for sovereigns, angels, crowns, half-crowns, shillings, &c. Then with tongs and hammer they are forged into a round shape, after which they are blanced (that is, made white or refulgent by nealing or boiling), and afterwards stampd or impressed with an hammer to make them perfect money. This method of making money with the hammer (as appears in the said red book) was practised in the reign of King Edward the first" [the book referred to is in the Exchequer] "and this kind of hammered money continued through all the reigns of succeeding Kings and Queens till about the year of our Lord 1663, when by several warrants and command of the King, Charles the Second: to wit, by one warrant, dated the fifth of November, 1662; one warrant, dated the eighth of April, 1663; and a third warrant, dated the twenty-fourth of December, 1663; the other sort, called *milled money*, was first fabricated to be current in England in this manner: first, the gold or silver is cast out of the melting-pot into long flat bars, which bars are drawn through a mill (wrought by a horse), to produce the just thickness of guineas, half-guineas, crowns, half-crowns, shillings, &c. Then with forcible engines called *cutters*, which answer exactly to the respective sizes or dimensions of the money to be made, the round pieces are cut out from the flat bar, shaped as aforesaid (the residue whereof, called *sizel*, is melted again), and then every piece is weighed and made to agree exactly with the intended weight, and afterwards carried to other engines (wrought secretly), which put the letters upon the edges of the larger silver pieces, and mark the edges of the rest with a graining. The next thing is the blanching, performed as above; and at last every piece is brought to the press, which is called the *mill* (wrought of the strength of men), and there receives the impression, which makes it perfect *milled money*."

The processes now used are as elaborate as the old methods were simple; but considering the requirements of the present day, and the enormous quantity of money produced, it must not be expected that our coins will, for depth of engraving, bear comparison with those of the

Romans, who, though succeeding in producing finished works of art, seem to have forgotten the wear and tear to which they would be subjected, and so left them, as a rule, free from a protecting edge; hence they would lose their image and superscription at a far earlier date from their birth or manufacture than would well-made coins of the present period.

It was formerly believed that gold could be sent to the Mint to be coined free of charge to the importer; such, however, was not the case. By the Act 33 Victoria, cap. 10, several Acts relating to the coinage are repealed, and all persons^[7] obtain the right to send gold to the Mint to be coined on the following terms:—

“8. Where any person brings to the Mint any gold bullion, such bullion shall be assayed and coined, and delivered out to such person, without any charges for such assay or coining, or for waste in coinage. Provided that (1) —If the fineness of the whole of the bullion so brought to the Mint is such that it cannot be brought to the standard fineness under this Act of the coin to be coined thereout without refining some portion of it, the Master of the Mint may refuse to receive, assay, or coin such bullion. (2) Where the bullion so brought to the Mint is finer than the standard fineness under this Act of the coin to be coined thereout, there shall be delivered to the person bringing the same such additional amount of coin as is proportionate to such superior fineness. No undue preference shall be shown to any person under this section, and every person shall have priority according to the time at which he brought such bullion to the Mint.”

The standard for gold is fixed by the new Act at “Twenty-two carats fine and two carats of alloy in the pound weight troy”—the same that was specified in the 3rd Edward VI., 35th Elizabeth, 2nd Charles I., and 18th Charles II. Silver is maintained at the former standard of “Eleven ounces two pennyweights fine silver and eighteen pennyweights of alloy in every pound weight troy.” Until the passing of this Act silver was coined under that of George III., which repealed the Act 18 Charles II., as regards the coining of silver free of charge. The Act 33 Victoria, cap. 10, repeals that of George III., but fails to make provision for the coining of silver, which is therefore now coined under the Common Law; yet the new Act specifies the weight and fineness of the silver coinage. The omission was made because the Chancellor of the Exchequer thought it “unfair to introduce silver to this Act as it is a source of revenue;” a little later I will examine this statement.^[8]

The Bank of England is practically the only “importer” of bullion to the Mint, and by coinage makes a considerable profit merely from its exceptional circumstances. By its charter and by Act of Parliament it is compelled to keep a stock of bullion equivalent to the excess of its authorised issue of notes—such authorised issue being secured by “Government debt” and “other securities”—which at the present time is fifteen millions; and gold, when at the Mint for coining, is esteemed as being still at the Bank. By the Act 7 and 8 Victoria, cap. 32, section 4, the Bank of England is compelled to purchase bullion, previously melted and assayed at the cost of the seller, at the rate of £3 17s. 9d. per ounce of standard gold, paying for it in Bank of England notes. Assuming that all the gold thus purchased is coined, the Bank makes a profit^[9] of about £2,000 on each million. When its authorities desire to coin gold they give notice at the Mint, and subsequently send at stated intervals “importations” of say 200 ingots of gold, each weighing about 180 ounces, and previously assayed. Upon the arrival of these ingots at the Mint, they are taken to an office, where, in the presence of an officer from the Bank of England, a small piece is cut from each ingot, and hammered out into a kind of strip. This piece is then put into a paper marked with a letter which corresponds with the mark or letter on the individual ingot from which it was taken. This having been effected for each ingot, the pieces are forwarded with their respective ingots to the Mint Office, where their weight is determined and compared with that charged by the Bank of England. The pieces which have been hammered out and placed in a marked packet are called the “assay pieces.” The assay pieces are sent to the resident assayer in the Mint, who estimates and reports to the Master the amount of pure gold in each ingot. The report of the resident assayer is then submitted to the Bank authorities, and if they find it to agree with their own assayer’s report, the ingots are retained for operations in the Mint. When it is desired to get the ingots of gold into a form fit for coining, they have to be alloyed with copper, or fine gold must be added, so as to bring them to standard or crown gold, which consists of 22 parts of pure gold and 2 parts of alloy, or 91·66 per cent. of gold and 8·33 per cent. of copper in 100 parts. Copper is usually employed because it is cheaper, and forms a harder alloy than silver.

Supposing there are 100 ingots, having a total weight of 18,000 oz., and by assay it has been found that they contain 17,900 oz. of pure gold, it is required to bring them to the standard fixed by law. A rule-of-three sum

tells how much alloy or how much gold is to be added. So, if 22 parts of pure gold require 2 parts of alloy, how much alloy will 17,900 parts require? The fourth term of the following proportion, viz., as $22:2::17,900:1,627\cdot027$, will show how much is to be added; but since there are already 100 oz. of alloy present, as indicated by the fact that 18,000 oz. contain within themselves only 17,900 oz. of pure gold, it is only necessary to add the excess required, viz., $1,527\cdot027$, so that these 100 ingots will produce $19,527\cdot027$ oz. of standard gold. Had the ingots been found to contain more alloy than is required, it would have been necessary to add so much pure gold as would have rendered the whole standard. The proportions of the metals having been determined, the whole mass is divided so as to make about 1,200 oz. of gold and alloy for each melting-pot. In practice it is customary to take the estimated amount of gold in a given number of ingots, and add so much alloy, or so much pure gold, as shall bring those ingots to standard, the whole weight being generally about 1,200 oz.; this is called *potting* the ingots. Ingots are almost invariably melted, and cast into bars for sovereigns.

I must here mention that the various illustrations I have introduced were originally used for the article "COINING" in TOMLINSON'S "CYCLOPÆDIA," and the permission to use them here has been generously granted to me by *Messrs. Virtue and Co.*, whose property they are.

In the illustrative engravings, the initial letters used to indicate the different parts are made to take their places in accordance with the order in which the various parts of the machine under description come into play, so that by tracing out the alphabet in any figure, the reader will see the consecutive action of the parts of the machine he is studying.

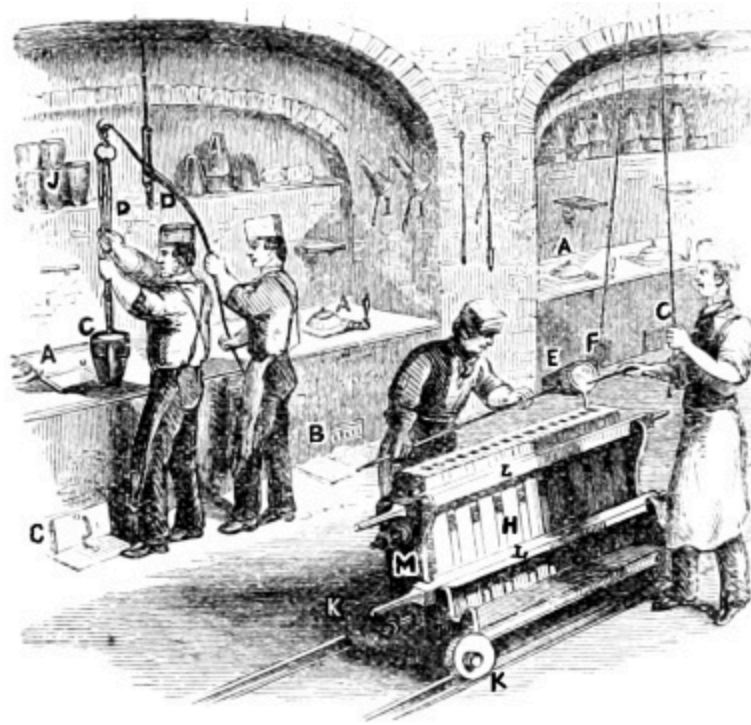


Fig. 1.—Gold Melting.

[Fig. 1](#) represents the arrangements for melting the ingots with their alloy, for producing standard gold. In the Mint there are seven furnaces: each furnace, A, is 12 inches square and 24 inches deep to the top of the bars, which are seen at the ash-pit B. The pot C is made of a mixture of Stourbridge clay and plumbago, hence called a *plumbago pot*; it is $9\frac{1}{4}$ inches deep, and 7 inches across at the top inside. The pot, previously annealed, is placed in the furnace on a *bottom* which stands upon the two centre bars. The bottom is intended to shelter the base of the pot C from the stream of air which is necessary to the combustion of the fuel, but which would damage the pot. The pot is then covered by its muffle and lid, and surrounded by fuel, which, as it burns up, warms and then heats the pot to redness, but so gradually that there is no great risk of breaking the pot. When the pot has become of a full red heat, the ingots are carefully placed in it, and the alloy is added by means of the funnel I: the pot is then covered up, and allowed to remain till the whole mass of metal has liquefied. The foreman then stirs it with a rod made of the same substance as the pot, and allows it to remain till the fluid mass has reached a peculiar temperature,

known by experience as that at which the metal, when poured into a mould, forms the most solid and workable bar. This temperature having been gained, the two outside bars at B are removed, the ash-pit is then covered to protect the men's feet, and the firing poked out. The lid and muffle are next removed from the pot, and the pot lifted by an assistant from the furnace by the crane D. The foreman then conveys the pot C, by means of a pair of tongs which clasp it, to the frame of moulds, when an assistant brings forward a loop of iron, suspended by a chain and cord from the roof of the building, and passing the loop of iron over a button on the end of the lower clamp of the tongs, as shown at E, F, supports the weight of the pot by the cord, as indicated at G. Previously to the melting, pieces of charcoal are placed at the bottom of the pot for the purpose of reducing any oxide that may be present in the alloy—and Dr. Percy has found that copper can dissolve 13·50 per cent. of oxide of copper—because oxide of copper, when dissolved in standard gold, frequently renders the mass so brittle that when the bars produced are passed between the rollers they crack and break, just as does gold containing minute quantities of lead or zinc. The influence of oxide of copper on standard gold was thus very marked in 1859, when a large proportion of the gold was so brittle, that if a bar were dropped on the floor, or tapped with a hammer, it broke, just as heated brass breaks under similar circumstances. Mr. Richard Smith had some years since pointed out that if copper in which oxide of copper is dissolved be poured through an atmosphere of coal gas, all the oxide is reduced. I therefore submitted this gold to the process thus suggested to reduce all the oxide of copper, and produced bars so tough that they could not be broken by a sledge hammer unless after repeated bending by that means. Yet this gold was not thus cured of its brittleness, but was treated peculiarly, as will be seen in a future place. ^[10]

When the metal is fused, the charcoal which was placed at the bottom is brought to the surface by stirring, and as it rises through the fluid alloy, reduces the oxide of copper, and remains on the surface to protect the alloy from the action of atmospheric oxygen. In order to prevent it from falling into the moulds, the assistant holds a piece of stick at the mouth of the pot, thus allowing the gold to pass, but stopping the charcoal. The foreman, in the case of gold, judges almost wholly by the eye, but in the case of silver by the ear, as to when the moulds are filled. The metal poured from each pot forms four bars. The moulds H are made of iron, cast into such a form that

each piece viewed from the top has the form of the letter **T**, so that three pieces placed together form two moulds; the whole of the set is held in its position by cross-bars, **L**, which fit into notches cut in the main frame **M**, where will be seen two powerful screws by which the moulds are forced tightly together after the cross-bars **L** are fixed. **J** represents the pots placed so as to be kept dry and ready for use; **K**, the wheels and tram on which the frame of moulds runs. The whole of the set of moulds having been filled, the screws at **M** are loosened, the bars, **L**, removed, and the parts of the several moulds removed, so that the bar in each mould is exposed to view. The workman takes each bar as it is exposed by a pair of tongs, and plunges it into a cistern of cold water to insure rapid cooling, and then places it on a bench, where the bars produced from each pot are separated and marked with a number, to indicate the pot from which they were poured, and the entire set with letters, to indicate the day on which the melting took place. After these distinctive marks have been placed on the bars, two assay pieces are taken from the bars from each pot.

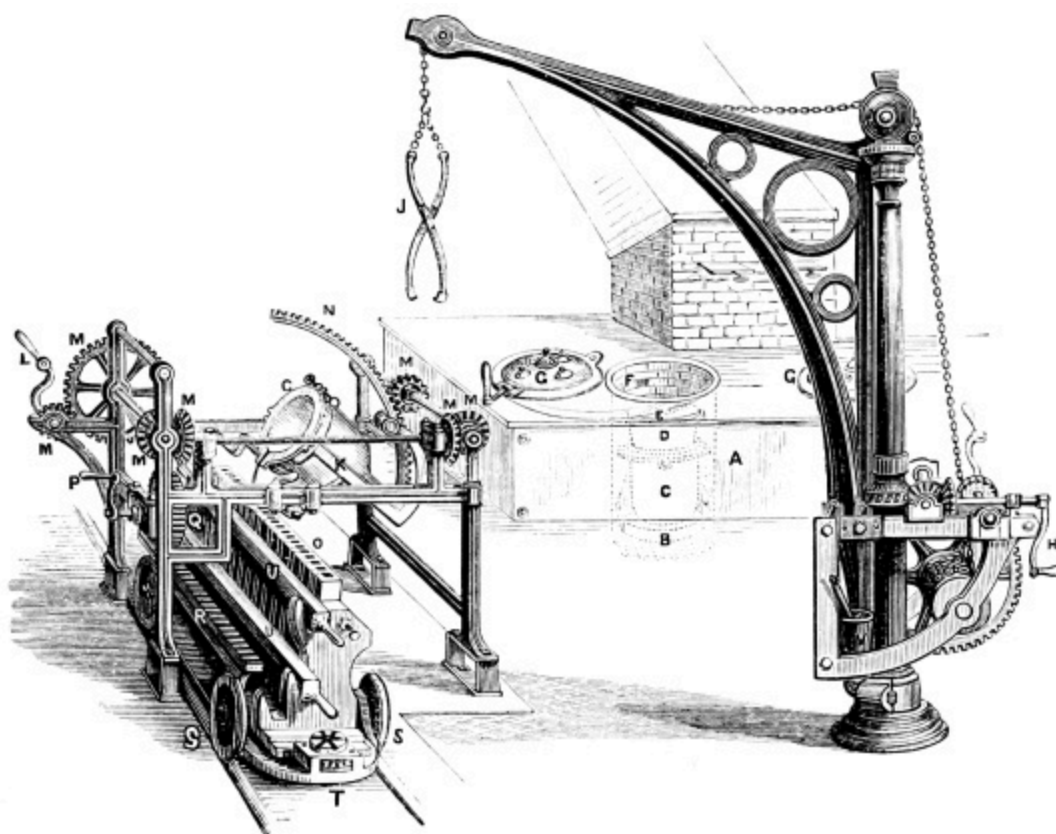


Fig. 2.—Silver Melting.

As silver melting,^[11] when discreetly conducted, is slightly different in some of its details, it will be well to describe it here. In [Fig. 2](#), A indicates the furnace with the front removed, to exhibit the internal arrangements. The furnace is circular, and is 21·5 inches across, and 31 inches deep. The bars, which are represented by dotted lines, are removable at will. The *bottom* B stands on the centre bars, just as in the case of gold melting: it is filled with coke dust, which retains any silver issuing from a small accidental crack or pin-hole in the pot, and also offers a non-conducting medium for the pot C to rest upon, it being necessary to avoid the abstraction of heat by the current of air against the base of the pot. A muffle, D, is placed on the pot, and upon this muffle the lid E rests. The pot is circular, and provided with a lip, to facilitate the pouring of the metal; it is made of wrought-iron, 12 inches across at the top, and 15 inches deep, and when melting for florins is charged with 4,800 ounces of silver. As in the case of gold, so with silver, the metal is alloyed with copper; standard silver being composed of 222 parts of pure silver and 18 parts of copper in 240 parts, or 92·50 parts of silver and 7·50 of copper in 100 parts.^[12] The calculations based on the assays from ingots are just the same as in the case of gold. The ingots are put into the pot, and the lid is placed on the muffle (which is intended to prevent metal from falling over the top of the pot during the process of fusion, as sometimes might happen when *scissel*^[13] is melted), and the furnace is then closed, so that the air enters through the bars, and passes into the chimney through the flue F. The lid of the furnace G is provided with peep-holes, by which the temperature can be regulated at will, while they also admit of a survey of the state of the furnace. When the fusion is complete, and carried so far as to raise the fluid metal to the necessary temperature, the bars of the furnace, with the exception of those which support the pot, are removed, and the fuel is poked out, because the pot at this temperature is so soft that it would be *torn* if it were pulled through the fuel. The lid and muffle are next taken off by tongs, and the crane is swung round by the handle, H, till its tongs are brought over the furnace, when, by working the handle I, the tongs J are lowered till they come to the pot; the foreman then makes them seize the pot, and by a signal gives his order for the raising of the pot from the furnace. Pieces of fuel and iron scaling are knocked off from the pot by the foreman with a brush kept in a crucible of water, indicated by V at the foot of the crane. The crane is then swung round by H, till the pot is over the cradle K, when it is lowered

and secured in the cradle by the screw, shown at C. The frame of moulds is now run under the lip of the pot, and the foreman, by means of the handle L, which will be seen to communicate with the wheels M, and the rack N, tilts the pot so that the fluid metal may pour in a good stream into one of the moulds O, until, by the sound, he judges it to be full; he then lowers the pot, while his assistant, who is also watching the filling of the moulds, turns the handle P, which, by the wheel Q fitting into the rack R, moves forward the frame of moulds so far as to allow another mould to come beneath the lip of the pot, to be filled as before. The fluid silver in the melting-pot is covered with charcoal, for the same reason as explained in gold melting, and this is kept back by means of a large piece of charcoal laid at the mouth of the melting-pot. The frame of moulds runs on wheels S, and at T is provided with a rack and pinion, by which the moulds may be moved farther from or nearer to the lip of the pot, which, as it is tilted, is continually altering its relative position to the mouth of the moulds, and would, were it not for this arrangement, pour its metal outside of, or upon the moulds, instead of into them. The moulds are formed of **T** pieces, in the same manner as the moulds for gold, and are secured in their position by the same method, the cross-bars being shown at U. The bars, having been taken from the moulds, are cooled in water, and distinctively marked, as in the case of gold. Assay pieces are cut from them, but in the case of silver three assays are taken from the bars produced from each pot.

The bars for different denominations of coins are proportioned in their width, so as to admit of two rows of blanks being cut from the fillets produced from them. The following statement gives the whole facts concerning bars for each coin, as used in the Royal Mint. It would be well to produce bars of a uniform thickness of 0·50 inch, as such bars may be cast solid, and when cast produce better fillets with half the rolling, without the cost of annealing. There would be also another advantage, for the blanks obtained from the fillets produced from bars 0·50 inch thick would not require annealing, and would thus save the cost of that process, as well as the loss occasioned by it, while the coins produced would wear immeasurably longer in circulation. There are other considerations why this thickness should be at once adopted, as will be shown when the subject of loss by coining is explained.

I am glad to notice that Mr. Fremantle, in his Report on European Mints, proposes to adopt my recommendation, as shown by the following passage, page 9:—

“... It would seem to be a question whether in a newly organised Mint considerable time and labour might not be saved by reducing the size of the moulds in which both gold and silver bars are cast. The result would be that, as bars would be thinner when first subjected to the process of rolling, the time now occupied by that process would be sensibly diminished. It should also be mentioned that although there is great difficulty in producing sound castings if the thickness of the bars should be reduced, it might nevertheless, for the metallurgical reasons which Mr. Roberts points out, be advisable to make this alteration.”

There is really no “*difficulty* in producing sound castings if the thickness should be reduced;” but I fear there are many DIFFICULTIES in the Royal Mint itself. Indeed the casting of the bronze bars—whose thickness is less than that I propose for those of gold—gives evidence of the ease with which thinner bars may be produced; yet against *these, insuperable difficulties* were urged until I experimentally demonstrated how easily they could be cast.

STATEMENT OF PARTICULARS AS REGARDS
BARS USED FOR COINING.

Metal.	Denomination of intended coin.	Length.	Breadth.	Thickness.	Average Weight.
		Inches.	Inches.	Inches.	Troy oz.
Gold	Sovereign	24	1·375	1·000	320
	Half-Sovereign	24	1·125	1·000	250
Silver	Crown	22	2·750	1·000	300
	Half-Crown	22	2·500	1·000	240
	Florin	21	2·125	1·000	220
	Shilling	21	1·437	1·000	150
	Sixpence	21	1·125	1·000	120
	Fourpence	21	0·875	1·000	96
		21	0·875	1·000	96
Bronze	Penny	24	2·500	0·375	100
	Halfpenny	24	3·000	0·375	117
	Farthing	24	3·000	0·375	117

It is to be regretted that crowns, half-crowns, and fourpences are no longer coined. It has been represented to me by many manufacturers and others who employ vast numbers of men, that the disuse of the fourpenny piece has caused considerable inconvenience in the weekly payments to their workpeople. The objection to this coin would appear to be that it is so nearly the diameter of the threepenny piece; but this is obviated by the fact that the edge of the latter is plain, while that of the fourpenny is crenated or “milled.”

The assay pieces,^[14] when cut from the bars, are placed in the divisions of a tray going from left to right, so that there can be no mistake as to the number of the pot from which the metal comes; each assay piece is then placed in a small envelope, marked with a distinctive mark, to characterise the pot from which it came. The assay pieces are sent in equal numbers to the two non-resident assayers, who determine, irrespectively of each other, the amount of gold found in each piece. Suppose we trace one assay piece, and imagine that 120 are undergoing the same process at the same time. The assay piece is flattened out into a kind of ribbon, and from it three pieces are cut, each weighing half a gramme. Each piece of this weight takes the name of a *thousand*, and is represented by the figures 1,000; but of course it may be called a pound, a ton, or by any other denomination. Each piece is wrapped in paper, with three times its weight of silver in two pieces, both metals being of the same thickness. The parcels, as they may now be called, are ready for the next process, and are arranged in their proper places from left to right in the divisions of a tray, and taken to the assaying furnace. In the assaying furnace is placed a Payen^[15] muffle, or kind of oven, perforated at places so as to allow a limited amount of atmospheric air to pass from the interior through its sides into the furnace. The muffle is surrounded with fuel, so that it is kept at a good—almost white—heat. On the floor of the muffle is sprinkled some bone ash—obtained by burning bones to whiteness—and on this bone ash is placed a set of 40 cupels, or little cups, made of compressed bone ash, and about as big as a florin, and so deep as to hold about half a teaspoonful of water. When the cupels have been in the muffle long enough to become red hot, a piece of lead of about nine times the weight of the gold to be assayed is put into each cupel, taking care not to let any lead fall over, or it would destroy the muffle. So soon as the lead has melted, the paper parcels containing the gold and the silver are

placed one in each cupel of melted lead. In a short time the gold and silver melt together, and, as it were, dissolve in the lead; then a kind of circulation of the fluid metallic mixture is observed to take place; and during this circulation, the lead, as it is presented to the surface, meets with the oxygen of the atmosphere, and combining with it, forms oxide of lead, which at that temperature is fluid, and drains into the substance of the cupel, carrying with it the copper and other impurities contained originally in the gold, leaving in the cupel a button which contains all the pure gold and the silver. The cupellation occupies about ten minutes, and at the end of that time the little mass of fused metal is observed to brighten up, the signal by which the assayer knows that the process is finished, and he withdraws the cupels one by one, tilting the fluid globule on one side, that it may incorporate with it any small globule which may chance to be upon the side of the cupel. Some assayers close the doorway of their muffles as the time of the brightening up of the assay approaches, to avoid the access of the atmosphere till the buttons have become solid, because the button of metal—by its silver—absorbs oxygen; and giving off this oxygen at the moment of solidification, spurts or opens, leaving the button hollow or ragged. This precaution is not thought necessary by all assayers, nor, indeed, is this system invariably followed, for some prefer to wrap the gold and silver in the lead foil which is to be used, and do not care to have the precious metals of identical thickness; these assayers use a smaller proportion of silver, viz., 15 parts to 6 of standard gold, or 17·50 to that which they judge to be *fine*, with a view to avoid the danger of “spurting;” they also save time by withdrawing the whole batch of cupels on a tray by means of a peel. The button is taken from the cupel as soon as it has been detached, either by dropping a little water on to it, or by allowing it to cool spontaneously, and it is then hammered out into a strip and annealed. It is next rolled into a ribbon or fillet, and again annealed, after which it is curled up into the form of a letter S. Some prefer to roll the piece into a kind of cushion, but in such a case it is more difficult to remove all the silver; the proper plan is to expose as large a surface as possible to the action of the acid in the next operation.

The S-shaped fillet is put into a flask of nitric acid of specific gravity 1·23, and the flask is placed in a little cup of brass, which stands over a small gas-burner, while the neck of the flask enters a kind of flue, through which a current of air is continually passing into the chimney of the furnace. The flask being placed in its position, the gas is lighted, and a

gentle heat applied, when the nitric acid dissolves out the silver from the S-shaped fillet, and is known to have done its work when red fumes cease to be evolved. The acid is then poured off, and the remaining sponge of metal is washed with distilled water, and boiled with concentrated nitric acid, which removes the remainder of the silver. The sponge of gold is now washed with water, to remove the nitrate of silver, and is then heated to redness in a capsule to render it tough, for in its spongy state it is so rotten that it will not bear to be touched; and although it remains spongy, it is toughened by being heated to redness. It is now called a *cornet*. This spongy state is a consequence of the addition of silver, the presence of which, and its subsequent removal, produce a separation of the particles of the gold. The gold is thus formed into a kind of network or sponge, so that the acid can get at every part of it, and remove any metal which is soluble in the acid. Diluted acid is used first, because if there be any lead left it is dissolved out, and because the action of the strong acid is so violent, that part of it would probably be carried out from the flask. Nitric acid, when boiling, is liable to form bubbles of gas, which expanding, give rise to what is called *bumping*; hence it is usual to put a charred pea into the flask, and this, floating on the surface, causes a more even flow of vapour and gases, and so prevents bumping. It is probable that the charred pea determines the boiling at one particular temperature, for it is found that water, if it be floated in a fluid of a higher boiling point than itself, may be raised considerably above its ordinary boiling point, although remaining quiescent till disturbed by a rod or point. The cornet is next weighed, and as it has been begun under the idea that it was a *thousand*—1,000—all that it weighs short of 1,000 is the alloy which has been removed, the object having been to determine how much pure gold was present in the alloy. There is, however, a source of error in the process which requires to be explained.

It is found to be practically impossible to remove every trace of silver from the cornet; it is therefore necessary to make an allowance, the amount of which is determined by a *proof*. The proof consists of a mixture of gold and silver of known proportions, so that if all the silver be removed from a thousand—1,000—the remaining cornet should weigh exactly 916·6. Four of these proofs are worked with each batch of assays—a batch being 120 assays—under precisely similar circumstances, every precaution being taken that the four shall be equally distributed over different parts of the furnace, &c. Now suppose that the proofs, instead of weighing 916·6, as

they would weigh if the whole of the silver were removed, were found each to weigh 916·9; it would be known that 000·3 of silver had been retained, and must be deducted from each of the whole batch of assays. If the assays of coins exceed the limits of from 917·6 to 915·6, they are repeated, as it is assumed that an error has arisen. The convenience of considering the $\frac{1}{2}$ gramme as 1,000 consists in the fact that 1,000 parts of standard gold contain 916·67 of gold and 83·33 of alloy; so that the result having been arrived at without any calculation, one source of error is avoided and time is saved.

Messrs. Johnson, Matthey, and Co. have invented a tray of platinum capsules or thimbles, into which the assay pieces are placed for treatment with acid, instead of into the ordinary glass vessels over gas-burners. The advantages of this invention are too manifest to require elaboration; but saving of acid, gas, labour, and risk of error are amongst the chief of them; and besides, the plan has been in successful operation in their own assay offices in Hatton Garden, as well as in other important assay offices, for several years past. The first cost would seem to be the principal objection to this plan of Messrs. Johnson, Matthey, and Co.; but this is really a small matter, for the apparatus becomes stock in trade, and its cost should be viewed simply as so much capital, whose interest is paid by the saving effected in glass, but more especially by the smaller amount of acid actually employed, while the platinum can at all times be sold for nearly its original cost as old metal.

The mode of operation is as follows:—A stand of slate is so arranged that a means of heating is made to rest on its base. For this purpose a jet of gas is preferred; where, however, gas cannot be obtained, an ordinary oil lamp or a charcoal fire may be used. On a shelf over the source of heat are placed two or three receptacles of platinum, each communicating with a vessel made of porcelain, provided with three necks and an overflow pipe. There is a kind of sieve or tray of platinum, so arranged as to carry from 16 to 100 thimbles of platinum, and provided with a handle, so that this tray, with its charge, can be manipulated at pleasure. The thimbles vary in size according to taste, but each one is cut or slit at the bottom, so that the solution of silver as it is formed may by its density fall out, and allow the clean acid to take its place.

When an operation is to be performed, the tray filled with the charged thimbles—that is, containing the assay pieces—is placed in one of the platinum receptacles or boilers, and heated to a fitting temperature; when the desired effect is produced, the tray is lifted into another receptacle, and again heated; this may be carried to three times if necessary. The products of decomposition of the acid go, with the acid which evaporates, into the porcelain vessel, where the free acid falls through the overflow pipe into a proper chamber, while the acid fumes pass into the flue through the third neck before spoken of. The parted assays having been washed by several immersions in boiling distilled water, without removal from their thimbles, have now to be dried and annealed in a platinum muffle, so formed as to fit into an ordinary muffle, and, after annealing, to be weighed in the usual manner, having saved at least 75 per cent. of the usual trouble.

It is almost needless to add that the system of *proofs* above described must be also used with this process.

The assay for silver is not so tedious, as it is finished at the point where it leaves the muffle on the cupel; but up to this point it passes through precisely the same process as the gold.

It will have been observed that the event of the process of assaying is to exhibit the proportion of *bullion* which may be present in a given weight of mixed metals, but that it does not demonstrate the character of the substance which forms the alloy, because this is assumed to be copper or silver. In practice it is found that the alloy may be a mixture of copper or silver with lead, mercury, antimony, arsenic, tin, or zinc, in varying proportions, but whose sum does not exceed the rate per centum in relation to the bullion which is allowed by law. The existence of minute proportions of lead, mercury, antimony, arsenic, tin, or zinc is a matter of great concern, for these, when present, add indefinitely to the difficulties of the coining departments.

Under the present system, complete analyses of bullion imported for coining are a practical impossibility. I would therefore propose to abolish the office of non-resident assayer, and then to appoint two resident assayers, who should conduct their operations in independent laboratories—already existent—and a condition of whose appointment should be an agreement to conduct, when so directed by the Master of the Mint, complete analyses of

the bullion submitted for assay. The information thus obtained would enable the melter, by processes subsequently explained,^[16] to remove all those substances which are *now* found to be fatal to the coining of certain bullion. The appointment of officers possessing such ability would be an equal guarantee with the present for the faithful preservation of the integrity of the coinage. This suggestion has been adopted and recommended—so far as regards the resident assayers—by Messrs. C. W. Fremantle and C. Rivers Wilson, in their “Reports on the Mint,” 1870. Indeed two assistant assayers have been recently appointed.

When the assay reports arrive, the Master determines whether the metal has been found within the limits, and if he be satisfied he writes on the assay reports “Passed,” and signs the reports. The Deputy-Master retains the reports as his warrant, and then issues his order for the delivery of the bars by the melter to the rolling room, as recommended in my report dated 29th January, 1859; but, as the accounts are kept by weight, every set of bars is weighed by the officer who receives them into that room before he gives them to his men to work into fillets.

When, in 1856, I took charge of some departments in the Royal Mint, I found that the system of weighing was extremely loose. Officers were plainly told that if they ventured to satisfy themselves as to the weight of bullion they had received—in fact, to determine whether they received what was charged to them—“they would be paid out.” This state of things led men to accept *any* weight. But I objected and resisted; for I found that the average on each day’s work as received amounted to an habitual minus of five ounces on the weight charged on silver, while on gold it was seldom so little as one ounce. In illustration I will state one case which occurred. I delivered 7920·00 ounces of gold to the Mint Office. When that gold was received, the official weigher gave me credit for 7918·15 ounces, which was a deduction of no less than 1·85 ounces. I appealed to the Master, who by written order directed the gold, the weight of which was thus disputed, to be weighed by Mr. Pilcher, the officer of the weighing room, in the presence of witnesses whom the Master nominated. Mr. Pilcher complied with this order, and gave a certificate, signed by the deputed witnesses, in which he stated that the gold under dispute weighed 7919·98 ounces—that is, that it differed in weight as charged by me to the Mint Office 9·60 grains instead of 888·00 grains, the difference which the official weigher had

deducted from the bullion I had delivered to him. The custom had been to weigh silver to 0·50 ounce, and gold to a pennyweight; but I introduced the system of weighing silver to 0·10 ounce, and gold to 0·01 ounce, and at the same time induced the Master to order new balances of superior construction for the coining department, and one specially devised by Mr. James M. Napier for use in the Mint Office. Of this balance, received after having been ordered for some years, it will be sufficient to say that it appears to be extremely accurate when properly used, and is then capable of great results; but, unfortunately, the system is such that those who are admittedly unfit may be promoted to important posts, just because they are senior, and not because of superior fitness for the work. This fine balance, therefore, becomes equal to a good rifle in the hands of a bad marksman. Great accuracy having been enforced, the weighing is now improved, but is still far from perfect, because the officers are compelled to abide by the decision of the weigher at the Mint Office, who, as before said, may be unfit for his office, while the officers themselves are debarred from all checks, by the removal from the coining department of the set of standard weights, which were bought on purpose that those gentlemen might check their own weights as to accuracy. The Master's order for the removal of these weights was in the following terms:—

“The standard weights hitherto kept in the weighing room are permanently transferred to the Mint Office. A new set of standard weights to be made for the Mint Office. The coining department weights to be examined by Mr. ———, and compared with those in the Mint Office (after these have been corrected), and reported on by him. Such examinations to be repeated every six weeks.”

It should be observed that this order was preceded by one written on the 2nd February, 1865, which directed that the Mint Office weigher should be the final judge, but that any officer might demand the re-weighing of any bullion, while the second weighing was to be final. All appeals to pass through Mr. John Graham, the order proceeding to direct that—

“The final decision is not to be called in question by any other officer of the coining department.”

The latter determination, I was informed, was intended to apply to me personally. Its immediate effect was a deduction of 0·25 ounce from the first gold I delivered, and a consequent alteration of my book to that extent by Mr. John Graham, who made a note in writing as follows:—

“This difference (error) may be accounted for by the acknowledged error of three grains heavy of the Mint Office 500 oz. weight. See Cœrtling’s Report.

(Signed) “J. G.”

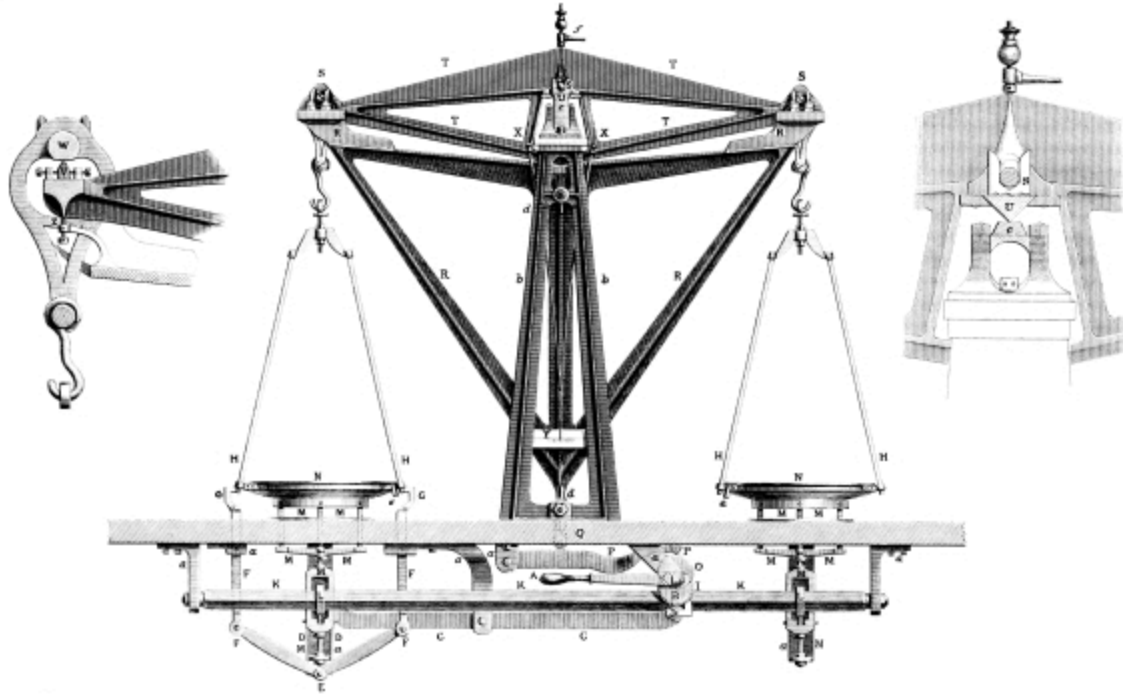
Such being the circumstances, I submitted; but the result could be in one direction only, for it had long been maintained that the reported losses were but “an hallucination,” and were, in fact, a “mere difference of weighing.”

It is true that the weights are periodically adjusted, but there is adjusting and adjusting. These facts have been dwelt upon because they have a most important bearing on the subject of loss and gain by coining, to be afterwards treated of.

Between 1856 and 1866 the old scales were removed, to make room for the superior balances of Messrs. De Grave, Short, and Fanner, the eminent scale-makers of St. Martin’s-le-Grand, of whose balances it is impossible to speak too highly, and of whom it is but fair to state that the Prize Medal was awarded to them at the International Exhibition, 1862, for their superior workmanship in balances.

Since it is necessary for the officer in charge of any department to ascertain the exact amount he gives to his workmen, and to satisfy the Master that they have returned—to within the limit of the weight of one blank of whatever denomination of coin he may be working—the bullion which they received, it became imperative to select the best balance; and in practice that made by Mr. S. R. Short, of the firm above named, was found to be the most serviceable. Mr. Short has introduced minor improvements from time to time, as experience has dictated, but the balance about to be described, after years of wear, determines to within one grain when charged with 1,200 ounces troy. The mode of proving the accuracy of a balance is to weigh as usual, and having arrived at a just determination of the weight of the matter to be weighed, to change the weights to the pan in which the matter has been weighed, and to place the matter in the pan previously occupied by the weights, so that both are now made dependent from the reverse ends of the beam; if the results be identical with the former results there can be no doubt as to the accuracy of the beam; should there be any variation, the balance must be adjusted by minute alterations of the knife-edges at its ends.

Mr. Napier has made his balance after the principle of Mr. Cotton's balance, so that it requires no adjusting screws, but I have no personal experience of the benefits of this omission.



IMPROVED BALANCE (TO WEIGH 2000 OUNCES.)

BY MR. S. R. SHORT, JUN^r.

The balance used in the rolling room of the Royal Mint is specially adapted for the purpose. Such a balance, as already noticed, must be accurate, as must also all the balances used in the Mint; but this balance requires an arrangement by which the pans can be released from the beam and held firmly while being loaded. Mr. S. R. Short has been peculiarly happy in effecting this object, as will be seen by reference to the steel engraving, which exhibits a fine illustration of Mr. Short's balance. The raising of the handle A causes the partial revolution of the cam B—represented by dotted lines—and this, as it diminishes, releases the lever C, which, by the rods D acting on the point E, permits the approach of the joints on the rods F towards the centre. The rods F work on a centre fixed beneath the table on which the balance stands, so that the closing of their one end causes the opening of the other. Thus the claws G recede from the

wedges H, upon which they had been fastened, and so release the pan, for the claws act simply as a man's hand would act in fixing the pan while it is being loaded. Immediately upon the opening of the claws another action takes place by the continued movement of the handle A, which causes the cam I to rotate upon the friction roller J, affixed to a lever proceeding from the shaft K, which at L is cut with eccentric bearings, so that by the partial revolution of K, caused by the pressure of I upon J, the series of supports M sink downwards from the pans N, thus obliging the pans, with the rods which support them, to be suspended from the supports S, which rest on the frame R. So soon as the pans are thus suspended, the continued motion of the handle A causes the cam O to permit the falling of the lever P, which at Q supports upon a friction roller the iron framework R. This framework terminates upwards at the points near S in sugar-loaf cones of gun-metal, which are intended to relieve the knife-edges of the beam from the pressure of their planes while the beam T is out of use, or while it is being loaded, for at these times the knife-edges would become seriously damaged. By the lowering of the frame R the centre knife-edge U is permitted to rest on its plane of steel *c*, which is beautifully bedded on an arch of gun-metal, as will be seen if the enlarged representation of this part, and which is shown to the right, be examined. In fact, the steel plane *c* is secured by wedges exactly shaped to it, but for the sake of illustration these wedges have been replaced in the enlarged portion of the engraving by upright pillars shown against *c*, with screws travelling through them, and intended to fix *c* rigidly in a given position. At the same instant that U comes into rest upon *c*, the knife-edges V at the ends of the beam T receive the planes W, so that the beam has now to support the weight of the matter placed in the scale-pans, and at this time is determined the actual weight of whatever that matter may be. The weigher, by an indicator suspended from the beam at X, reads the oscillations of the beam on an ivory plate at Y. At the time of weighing, the beam with its dependents is carried by the four-legged frame of iron *b*. The frame R passes through friction rollers of brass at *d*, which rollers maintain it in its relative position to the beam. The weighing having been determined, the handle A is reversed, and the beam is placed at rest; but the vibration of the machinery in the rolling room causes a continuous chatter of the knife-edges V against the planes W, therefore at each end of the beam a small screw Z is made just to touch the lower edge of the beam when it is at rest. Final adjustments which are very minute may be arranged by *f*; while the

box *e* on the pan against *G* is intended to receive small pieces of lead, which are used to compensate for the loss by the wearing of the pans. The shaft *K* is supported from the table by *a*. The beam of the balance represented in the steel engraving measures 48·00 inches from knife-edge to knife-edge; that is to say, it measures 24·00 inches from the centre knife-edge to the knife-edge at either end. Mr. S. R. Short has made for the rolling room another balance for weighing silver, and as it differs in some important details, it is deemed wise to demonstrate those alterations in [Fig. 3](#), where the supporting frame *R* will be seen to carry three upright pillars *S*¹, *S*², *S*³. The pillars *S*¹ and *S*³ are represented as being placed on the farther side of the beam, which at those parts is broken away, that the pieces of steel against which the pillars are made to touch may be seen. *S*² touches against one of these pieces of steel near the centre knife-edge *U*, and in front of the beam. The pillar *S*² terminates in a point, which passes into an inverted cone, while the point of *S*³ is made to pass into an inverted **V**, so that when the beam is raised by these pillars from its centre knife-edge, *S*² obliges it to take a specific position with regard to its distance from the centre of the plane on which its centre knife-edge rests, and *S*³ causes it to take a perfectly parallel line. *S*¹ is simply a plane which rises against a plane and steadies the other end of the beam, which by this arrangement stands as on a tripod, and is, therefore, prevented from moving, and in addition has its knife-edge invariably placed upon the same part of the plane at each time of weighing, while the frame *R*, by its continued lowering, drops the planes with equal accuracy on the end knife-edges *V*.

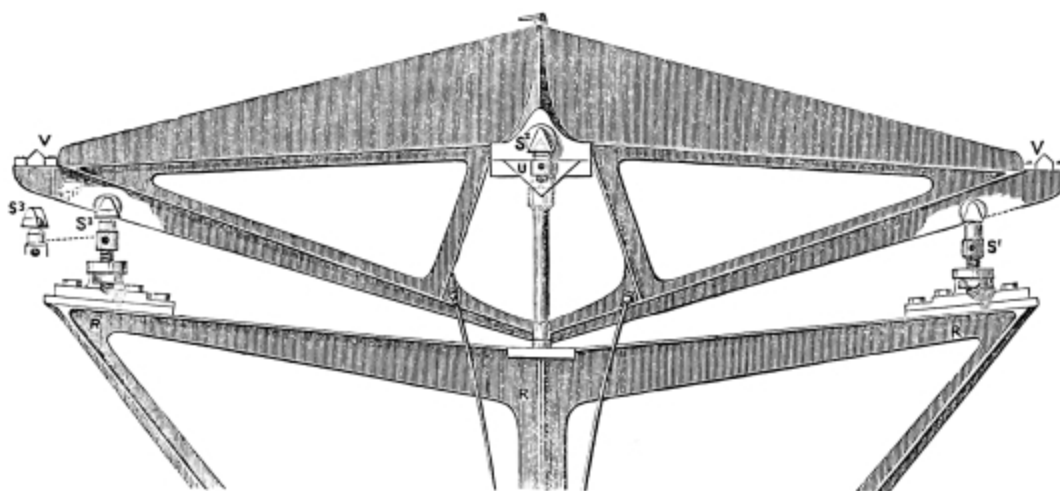


Fig 3.—Short's Plan for Levelling a Beam.

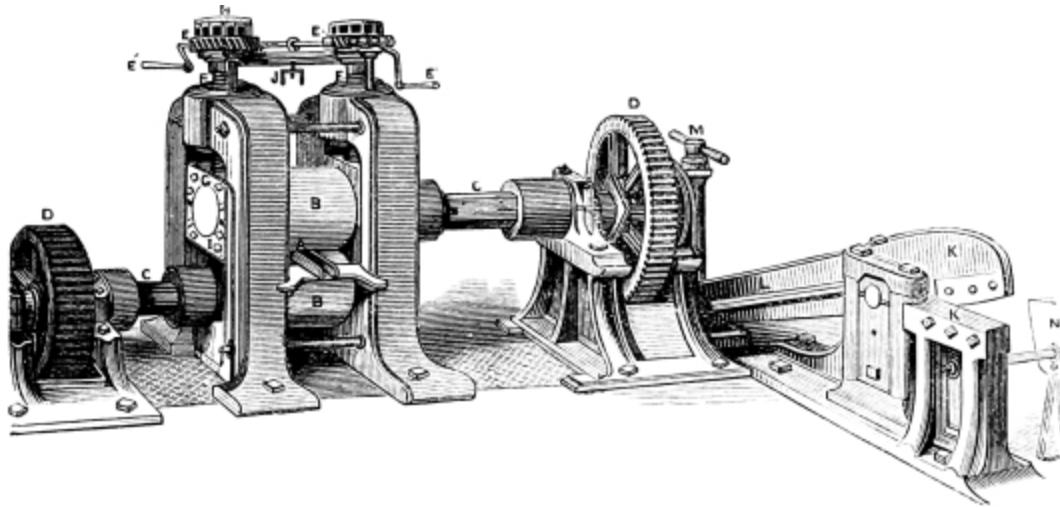


Fig. 4.—Breaking-down Mill.

The sovereign bars having been weighed by the officer, and given by him to his men, are wrought in sets of twenty; each set is called a *batch*, and each bar in the batch undergoes precisely the same process. The bar is passed into the opening A of the breaking-down mill [Fig. 4](#), where it receives a considerable compression, for the rollers B, seizing its end, drag it forward, while they roll back and retard the progress of that part of the bar which is not between them. The result is that the bar is lengthened, but not widened materially, so that length is gained at the expense of the thickness, which is regulated by the distance between the rollers. The rollers are driven by shafts and adjusting couplings C, which are themselves driven by the wheels D. The distance between the rollers is determined by the action of the lever E', which, by the endless wheels on its axle at E fitting into geared wheels, gives motion to powerful screws shown at F, which terminate in cups on the upper part of the upper brasses of the rollers B, as may be seen at G. The upper brasses are kept always against the ends of the screws by weights which are beneath the mill, but from which levers and rods terminate at the lower part of the upper brasses, at about the position indicated by I, so that the upper roller has motion either upwards or downwards at pleasure, but the motion upwards is arrested by the powerful screws F, and this point once determined by the reading of the scale H, is fixed by the clamp J. The thickness, therefore, of each bar in a batch is determined within certain limits; and when each bar in the batch has been

rolled, the mill is altered, refixed, and again the rolling goes on till each bar has passed seven times at varying pinches through these rollers. Owing to the wear of the moulds in which the bars are cast—and which is largely due to the presence of minute portions of antimony in the gold—the bars are never of uniform thickness; hence bars of every denomination are passed through the rollers on their edge, so as to reduce them to one uniform thickness, otherwise the fillets resulting would be ragged, and of unequal widths, which defect would cause them to produce blanks out of *remedy*^[17] as to weight. The rollers are set face to face; the graduated scale H is then fixed at zero; if now the rollers are separated until there be a space of 1·00 inch between them, the scale H will indicate 50; if, however, it be desired to read a higher figure, part of another revolution must be performed, causing the scale to read, say 31·50 of that revolution, when the rollers will be so far apart that sovereign bars which pass between them will, allowing for the expansion after compression, be found to be exactly 1·375 inches wide.

If now these bars be rolled on their sides, the mill must be turned down till the scale reads 45, when, if the bars be passed through, they will measure as under for width and thickness at each successive pinch. A *spring-pinch* means the passing the bar once more through the mill without altering the distance between the rollers.

**A sovereign bar passed Becomes a fillet or ribbon—
through the rolling mill— Inches wide. Inches thick.**

At	45·00	1·450	0·775
”	35·00	1·540	0·610
”	26·00	1·610	0·460
”	19·00	1·665	0·335
”	14·00	1·696	0·250
”	10·50	1·712	0·194
”	8·50	1·765	0·148
”	7·00	1·778	0·129
”	1st spring-pinch.	1·778	0·127
”	2nd ” ”	1·778	0·120
”	3rd ” ”	1·779	0·118
”	4th ” ”	1·781	0·117

When the bars have passed through the rollers at 10·50 they have become of great hardness, and of considerable length, say 6 or 8 feet. They are then taken to the shears K, where their hollow ends are cut off, and the bars cut into lengths of 18 inches; or rather, such was the practice when it was considered wise to obtain as much coined money as was possible from the bars; but recently, under other considerations, the old plan of the moneyers has been re-introduced, that of shearing the hollow ends from the bars before rolling. In this process there is needless waste, and it is to be hoped that the proper practice will be reverted to, for under that system the ends averaged 4·26 per cent., while under the re-introduced one it amounts to 7·03^[18] per cent. on the bars, and for the following reason. When the metal is poured into the moulds, it almost immediately solidifies, and while solidification is going on contracts in volume, leaving a kind of cup of metal or hollow part on the top of the bar. In the act of rolling, the bar maintains an equable width until this hollow part is reached, when it suddenly expands, and at this point the workman shears off the defective part of the bar, whereas in the other case he shears off the hollow end until his eye fails to discover the effects of the shrinkage; hence the loss of 2·77 per cent. on the produce of fillets, as well as the loss in wages for melting the bars. The shears may be regarded as large scissors driven by a drum on the same shaft that carries the driving wheel for D. The drum is eccentric, so that at each revolution the shears are caused to open and shut. The bar being placed between the jaws K, the long end L of the shears is raised by the drum, and the piece is cut off. The shears may be closed and the end of L suspended, when out of use, by a hook on the end of the screw worked by the lever M. The length at which a bar is to be cut off is regulated by the gauge shown at N.

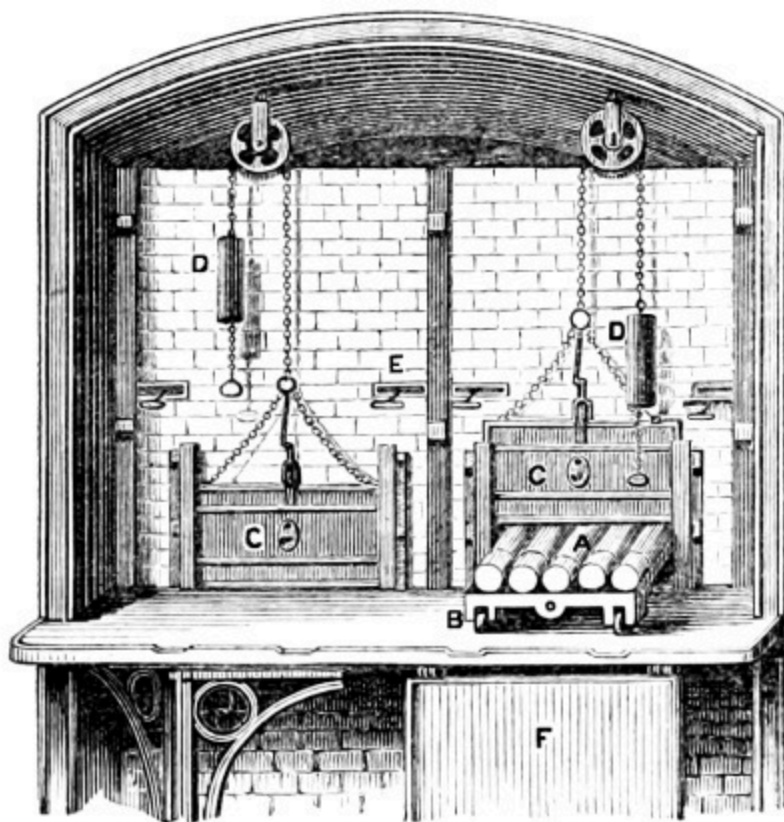


Fig. 5.—Annealing Furnace.

The sheared bars are placed in copper tubes, the tops of which are luted on with clay. It is imperative that the copper tubes should be made without solder, because this fuses at a temperature below that which is required to anneal the gold; if present it would run down upon the hot gold, and cause it to fuse and alloy with the solder, thus spoiling the work and entailing expense upon the coiner. The tubes which are used in the Royal Mint are made by Messrs. Benhams and Froud, of Chandos Street, Strand, who, after considerable pains, arrived at a method of making the tubes in such a manner as to entirely satisfy the requirements of the Mint. The tubes A are placed on an iron carriage B, which is then run into the furnace, as shown in [Fig. 5](#). The door of the furnace C is closed by raising the counterpoise D; the heat of the furnace is regulated by the damper E. The apron F is sometimes of use in annealing very long silver bars. After remaining in this furnace for twenty minutes, the carriage is withdrawn, and the tubes, taken with tongs, are plunged into cold water, to cool the gold as rapidly as

possible. The rapid cooling of gold and silver gives to each metal a peculiar character, which is of value in the after processes, and prevents the access of the atmosphere, which, in prolonged cooling, would cause the oxidation and consequent removal of so much copper that the alloyed metal would become too rich in gold for circulation as coin. The annealing of some metals is effected not so much by the continued heat as by the slow cooling; it is therefore wise to raise the metal to its full heat as rapidly as possible, and then so to arrange matters that it may cool very slowly. This method does not hold good in the case of the precious metals and of copper, for they become, under such treatment, so soft, malleable, and pasty as to stick to the machinery, and thus to cause considerable trouble and loss. After annealing, the bars, which are now called *fillets*, go again to the breaking-down mill, through which they are passed with the scale indicating 8·50, then at 7·00, and after this are submitted to another pinch without altering the scale at all; so that what is called a *spring-pinch* is given, with the intention of effecting the reduction of the fillet to one uniform thickness, for the breaking down and subsequent rolling cause the fillets to become much thicker in their middle than at their sides. The spring-pinches reduce this, while at the same time they diminish the elasticity of the metal, and fit it for the other mills. It will be seen, by reference to the above table, that the widening of the fillet is very trifling; but width may be gained to any desired extent, at the will of the workman, if the bars be submitted to a heavy pinch instead of a series of light ones. The fillet having been submitted to the fourth spring-pinch, is gauged on its side by a steel instrument, of which [Fig. 6](#) is a representation. It is a hollow wedge, which is graduated to the thousandth of an inch. Supposing that the opening from A to B were extended until it were one inch wide at A, the space would be divided between it and B into 1,000 parts, and then every fillet passed into this opening would stop at a given point, say, for instance, at 140; such being the case, every part of the same fillet should be arrested precisely at the same point. The fillets are reduced till they measure 117 on this gauge, and are consequently 0·117 inches thick. They are then passed to the next mill, where they receive four light pinches, and then to a third mill, where they receive two more very light pinches, and by means of these six pinches are reduced in thickness to 0·075 inches. They then pass to another mill, still finer than any of the preceding, and here are submitted to four very light pinches, by which they are reduced to 0·058 inches, and are finally

finished at the sixth or gauging mill, where they receive three pinches, and are then 0·053 inches thick by 1·829 wide.



Fig. 6.—Small Gauge.

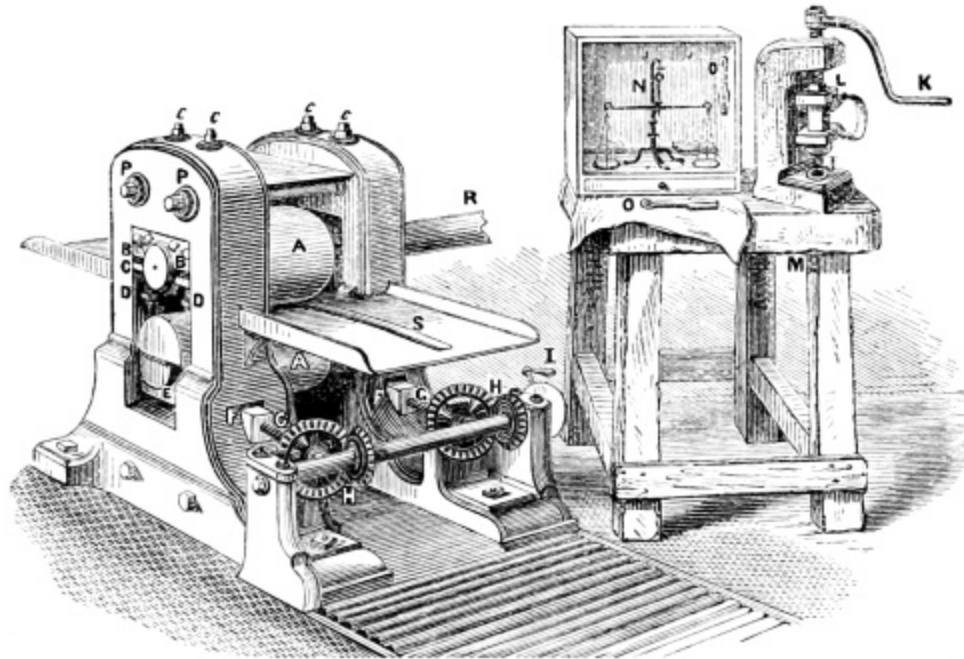


Fig. 7.—Gauging Mill.

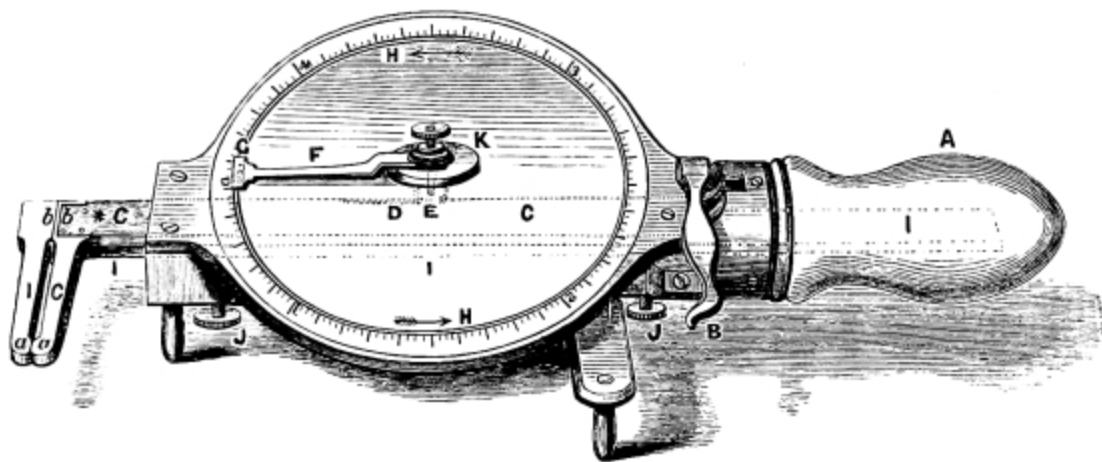


Fig. 8.—Ansell's Standard Gauge.

The gauging mill is of different construction from the other mills, as may be seen by reference to [Fig. 7](#), where the rollers A are seen in the act of reducing a fillet. The upper roller is fixed in brasses loosely clamped together; the upper brass, B, being firmly bolted to the main frame of the

mill by the screws *c*, while the lower one *C*, which carries the weight of the roller when it is running empty, is supported by spiral springs shown at *D*. The lower roller works on a brass, *E*, which rests on a wedge shown at *F*; the brass being cut to fit the wedge, so that it may become similar to a solid mass, irrespective of any motion given to the wedge. By this mode of adjusting, a difference of the 0·001 of an inch may be made with ease between the distance of the rollers, and, consequently, in the thickness of fillets which may pass between them. The wedge *F* is moved forward and backward by the screw *G*, which itself has motion from the gear work *H*, by the handle *I*. Directly a fillet is passed between the rollers, the topmost one is forced against its upper brasses, and further upward motion becomes impossible. The weight of metal in this roller gives rise to irregularities in the thickness of the fillets which pass from this mill, causing much trouble and some expense; therefore it is proposed to support the upper roller by a similar arrangement to that which affects the lower roller, and to keep it rigidly against the upper brass, but with just so much pressure as its own weight would induce were circumstances reversed, and by these means to relieve the fillet from the weight of the roller, because that has an undue influence on each end of it. The gauger of fillets requires other tests besides that of the thickness of the edge of the fillet, so he punches out a blank from an occasional fillet by a hand-press, the cutter of which is shown [at J, Fig. 7](#), worked by the handle *K*, through the screw *L*. The blank falling through the bolster of the cutter is caught at *M*, and is then weighed in the hand scales *N*, against a standard weight, from which it must not vary more than 0·50 grain. [O, Fig. 7](#), is the gauge actually used by the workman; [Fig. 6](#) represents the standard gauge used only by the officer in charge to check the work at its various stages. He has in addition a gauge of great accuracy, by which to measure the fillets at any point, as to width and thickness. This gauge will be more intelligible by reference to [Fig. 8](#). *A* is the handle, which is hollow; *B* is a lever attached to the flat rod of copper *C*, which at *D* is cut with a rack, into which a pinion *E* is made to work. The pinion *E* works on a shaft, the upper end of which carries a hand *F*, provided with a vernier *G*. If now the handle *A* be firmly held by the hand, while the thumb be made to press the lever *B* towards the end of the handle, the rod *C* is set in motion, and causes the hand *G* to travel in the direction of *H*. The rod *C* rests on another rod *I*, made of steel, and so long as to pass into the handle of the instrument. The ends of the rods *C* and *I* are fitted at *b* with steel shoulders,

and are then continued, as represented, to *a*. If it be desired to measure the thickness of a fillet, the points *a a* and *b b* are caused to open by pressure applied to B, and the fillet is placed between the points *a*, when a spring fixed in the box K brings back the rod C as soon as B is gently released, and encloses the fillet. The separation of the points *a* by the fillet causes the hand or indicator G to stand at a point from zero, which is then read. The scale is divided into 500 parts; and if the points be opened 0·50 inch, the hand makes one revolution; so that the ·001 of an inch is gained by one reading. But each 0·001 is subdivided by the vernier into ten, so that a ten-thousandth part of an inch is read without trouble. To measure the diameter of a blank coin, or the width of a fillet, it must be placed between the points *b*; but since the extreme graduation of this gauge is 0·50 inch, it is necessary, if it be desired to measure a larger diameter, to press back the lever B till the zero of the vernier G reaches 0·500 on the scale H, and hold it there while a clamp is made fast at the spot indicated by the star (*), to prevent the motion of C without I. When the clamp is fixed the rod I must be drawn out till the zero of the vernier reaches that of the scale H, when the screws J must be tightened to retain I in its new position, with half an inch permanent opening between the points *b* and between the points *a*. In a new measurement, that permanent 0·500 must be added to the reading. This arrangement admits of measuring up to 3·5000 inches, to which limit the gauge is extremely accurate. The instrument was invented by myself, because I found it difficult to convince the men that the fillet was thickest in its middle, and consequently heavier there than it should be; and, unfortunately, that the workmen habituated themselves to attributing this fault to each other, when its existence was proved. The fact was, that under the system which had prevailed, the men—with a view to make bad work for a specially-designed reason—would set the upper rollers at varying angles, so that a fillet at one mill would have one thin and one thick edge; and when that fillet passed through the next mill, the angle of the roller being altered, would make both edges alike, but the effect of the manœuvre was to push the metal into the middle of the fillet, and thus to unfit it for the draw-bench. Hence the necessity to fix where the blame should rest, and the production of this instrument, under my direction, by Mr. C. Becker, of 30, Strand, at once overcame all those difficulties.

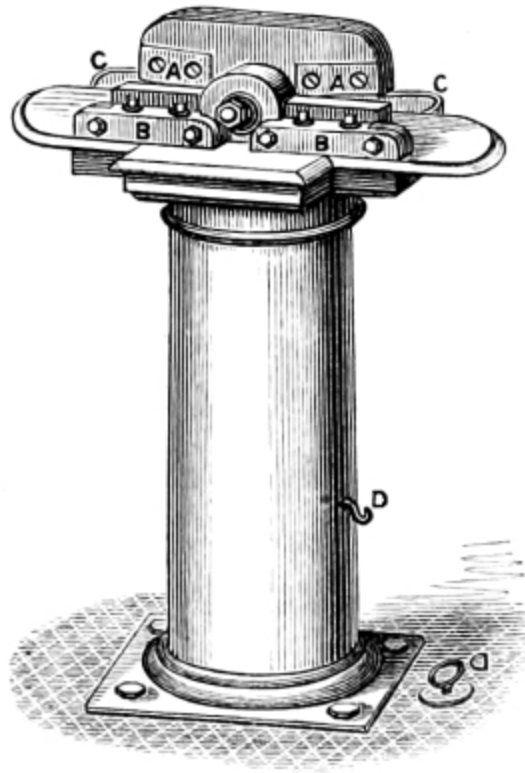


Fig. 9.—Small Shears.

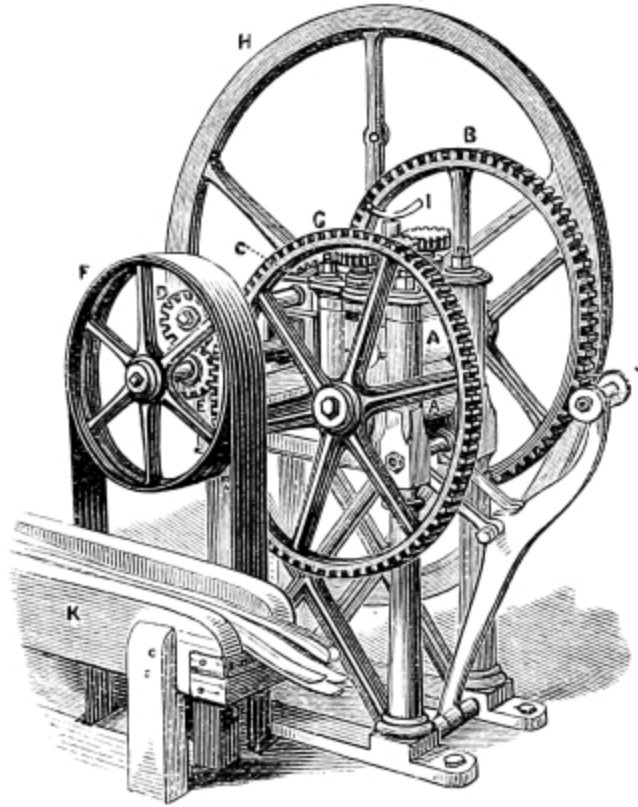


Fig. 10.—Flatting Mill.

The fillets are weighed from the rolling room to the drag room, where they are finally adjusted; for with every energy, discretion, and skill, FILLETS CANNOT BE OBTAINED OF UNIFORM THICKNESS BY SIMPLE ROLLING. In the drag room the fillets are taken to the small shears, [Fig. 9](#), by which one end of each fillet is trimmed so as to render it square. The plates A are fixed to the head of a T-shaped lever, which is caused to oscillate by a cam beneath the floor. The plates A shut against a face of steel fixed to a block, and held by the screws shown at B; if therefore the end of the fillet be passed between the plate A and the face of B, each oscillation causes the cutting off of so much as protrudes, the pieces cut off falling into the box C, which has now been enlarged so far as to form a pan all round the top with a view to catch all of them. D forms part of a chain, by which the shears are thrown out of motion. The fillets, having been trimmed so as to render their ends square, are next passed to the extent of about two inches between the rollers of a flatting mill, shown in [Fig. 10](#), which reduce that part of the fillet to about two-thirds its thickness. A A represent a pair of small rollers, the upper one of which is cut with three flat faces, so that it has three rounding and three

flat surfaces; hence, when both rollers are revolving, there are spaces with openings between them; but when the rounding faces come down, those openings are much narrowed, so that any fillet placed between them becomes thinned to just such an extent as may be deemed necessary. The rollers travel in opposite directions, so as to cause the expulsion of a fillet placed between them. The reverse motion is gained as follows:—The upper roller is driven by B, which receives its motion from the little pinion C, carried on the shaft which also supports D. D reverses the motion of E, which is driven from the drum F. E also drives G, which gives motion to the lower roller. The fly-wheel H is borne at the extreme end of the shaft which carries F and E. The fillets are rested on J while being *flatted*, and are, after flatting, placed in the trough K, from which they are taken to a rolling mill in the drag room, of precisely the same construction as that exhibited at [Fig. 7](#), to be passed twice through at equal pinches, with a view to render them still more accurate than they were when leaving the rolling room, as well as to reduce them to the exact thickness at which the trier has found they will produce the best work at the draw-bench. The rolling mill in the drag room was provided with steel rollers. Steel rollers are of somewhat recent invention, and seem to have received a high character from those whose opinion may be modified by further experiments more accurately made. My opinion, founded on experience, is to the effect that they are not worth their extra cost; but that their usefulness may be more fully developed when they shall have been fitted with the arrangements proposed for the gauging mills before explained.^[19] After these alterations have been made, it may fairly be questioned whether steel rollers will, under circumstances every way similar, produce better work than is produced by the ordinary chilled-iron rollers; in other words, I believe that steel is not a better substance for rollers than chilled cast-iron. Those who have to sell, and those who have to use, have, of course, different motives; he who has to use a machine should judge calmly, and not be led away because the invention is new. It is amongst these considerations that I am convinced that steel rollers do not save money; for if they wear longer without getting out of order, they also require a longer time to put them again in order. Rollers, made from whatsoever substance, cannot reasonably be expected to produce a fillet from every part of which blanks of equal weight can be struck, because it is not yet possible to produce a compound of equal hardness throughout; but if the construction of the mill be altered, steel rollers may approach nearer to

that perfection which is gained by the draw-bench, but they can never replace it. The fillets are taken from the mill to the draw-bench.

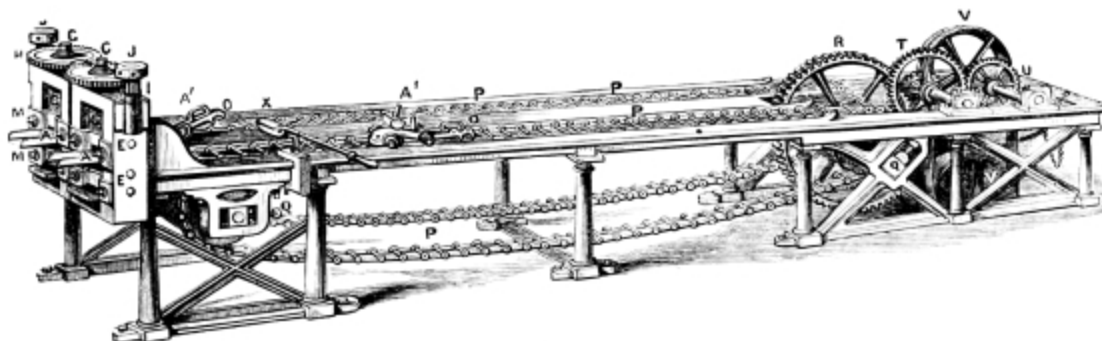


Fig. 11.—Draw-bench.

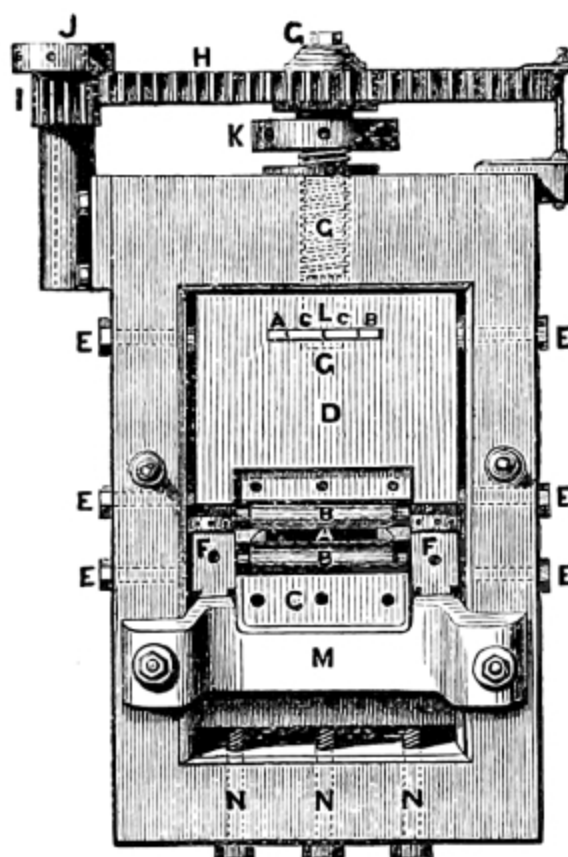


Fig. 12.—Head of Draw-bench.

[Fig. 11](#) represents the *draw-bench*, the name of which is retained, as being in fact its only appropriate one. The *flatted* end of each fillet is passed into the opening shown at A. The *dog* A' is then run up till its teeth seize the fillet. The lever is depressed until one of the hooks O catches a bar of the circulating chain P, which in its onward motion drags the dog, and causes it to bite the fillet and *draw* it through the opening at which it has been entered. P gets its motion from a notched cam, the axle of which is shown at Q. There are two distinct chains to each draw-bench, and there are two distinct draw-benches, so that one description does for both double ones. R is a cogged wheel, the shaft of which, Q, carries two notched drums, and each drum gives motion to a chain, so that both chains travel at the same pace. R is set in motion by the pinion S, on the shaft which is driven by the wheel T. T is driven by U, which is on the shaft driven by the [strap and drums V. Fig. 12](#) is a representation of the head of the draw-bench, and in studying this engraving it will be well to refer at the same time to [Fig. 11](#). The dog takes its name from its resemblance to the head of a bull-dog. It consists of a pair of levers, whose long arms extend beyond the axle-tree of the wheels nearest to O, and whose shortest arms are formed by the passing of the other axle-tree through the lever. The teeth are set at the front of the short arms. The axle-tree near O is fixed to the bars forming O, and runs loosely between the long arms of the lever, so that when O is pulled forcibly it causes the axle-tree to open the long end of the levers, and thereby to close the short end or teeth of the dog, the more rigidly in proportion to the pressure exerted at O. Directly the fillet has passed through the cylinders the dog springs slightly by the elasticity of the fillet, and thus releases itself from the chain; at the instant of release the weight over the foremost wheels falls, and by its fall lifts the hooks O so high as to admit of their escaping contact with the circulating chain P. The position of the teeth of the dog is [shown by A in Fig. 12](#). The flatted part of the fillet is just so thin as to admit of its passing easily between the cylinders B until seized by A, but the part which is not thinned comes against the cylinders B, and requires considerable force to drag it between them. The cylinders B do not rotate; in fact, they may be considered as forming part of a solid mass.^[20] The lower cylinder is laid on the bed C, and is clamped there by a cheek fastened on to C by three screws, the holes for which are shown on C; the upper cylinder is fixed to the mass D by a precisely similar arrangement. The beds C D are held perpendicularly by the points of the

screws E; and we may now view the cylinders as secured to, and forming part of, their beds. The distance between the cylinders is regulated by the capstans F, which separate the beds of the cylinders, and so separate the cylinders. The accuracy of this adjustment is all-important, because the distance between the cylinders determines the thickness of the fillet which passes between them. The bed D of the upper cylinder is required to be movable at pleasure; it is therefore provided at G with four wedges, two of which, *c c*, are cut so that if looked upon from the top a round hole shows itself, and through this hole the end of the screw, which at this point L is plain, and has a neck turned in it, passes, with its head beneath the wedges and against the lower G. So soon as this is effected the wedges A B are pressed into their places, and these holding *c c* together, cause them to secure G by its neck; if, therefore, G be now caused to rise, the block D must rise with it, but the head of this screw rests on the solid block D, while its neck is just so long as to admit of this without itself being pressed against the wedges *c c*. G is a very fine-cut screw which fits into a female screw cut in the frame of the head of the draw-bench; it is moved to any distance varying from the hundred-thousandth part of an inch (0·00001) and upwards by the wheel H, which receives a very minute motion from the pinion I by means of a lever fitting into the capstan head J. K was originally intended to be used to set or fix the screw G when it had been brought to its proper position, but it is not used; for, in fact, the cylinders wear away appreciably by the passage of the fillets, so that they constantly require to be brought nearer together to make up for this wear. With some species of gold the friction is so great that, although oil is used, the cylinders become so hot as to render the gold pasty; in such case a kind of welding takes place, which causes the tearing of the fillet. If this extreme point be not reached, as indeed it seldom is, the cylinders become of varying temperatures, and so great is the effect of this, that in order to compensate for it, the upper cylinder has to be continually raised or depressed. The beds which carry the cylinders become worn by the strain and fret, and require grinding out at intervals; therefore, to allow for the difference which this would make, screws N are provided, by which the cylinder in the lower bed can always be raised to its proper position. We cannot but admire the ingenious productions of inventive minds; and surely if ever there were a marvellous machine for assisting the coiner, it is this; indeed, it may be doubted whether a more admirable instrument for its purpose can be

contrived. Sir John Barton, who invented and directed the making of it, took into consideration every circumstance which could possibly arise, but he never saw practically the full advantage of his conception. Under my direction, this machine was so used that the average work produced was very favourably compared with the trials recorded by Sir John Barton, and details of which were given to me by the late Mr. W. H. Barton. There are some persons who smile at the draw-bench, but it is one of those inventions which will outlive its detractors, at least, so long as economy and perfection are points to be studied in coining. Foreign Mints are said to have found no advantage in the use of the draw-bench. It is to be regretted that they have not found a man with sufficient intelligence to use so accurate an instrument; it is surely not the fault of the sun if men are blind to its splendour. Mr. J. Martin, of the Paris Mint, has recently made some very accurate experiments with the draw-bench, and has produced results every way in accordance with those obtained by me in the Royal Mint, and is convinced that the draw-bench may be considered as the coiner's right hand.

Against all sound advice the Master had been induced to buy a pair of steel rollers, at a cost of £800 (I state the sum on the authority of Mr. W. H. Barton, who, being Comptroller at the time, probably knew the actual cost), and with these I made an elaborate series of experiments, extending over many weeks. The results were wholly adverse to the new rollers—others arrived at the same conclusion—and I reported to the Master, in writing, on the facts as I found them, showing that the average gave an advantage of 19·92 per cent. in favour of the draw-bench, and on this ground I maintained the superiority of that machine. He replied by a peremptory order, directing me “to prohibit now and for ever” the “use of the draw-bench, and to take measures for its removal from the Mint.” Thus absolved from responsibility, I—after making a written protest—obeyed that which I could no longer withstand. The result was that the rejected blanks, which had averaged 3·60 per cent., now advanced to 23·52 per cent. I was unable to avoid this great expense, and when I spoke to him about it he was not very amiably disposed; however, he finally wrote me the following note, on which I remarked to him “that steel rollers, whether driven by cogged wheels or by straps, would still be steel rollers.”

“7th June, 1861.

“DEAR SIR,

“In the cutting and adjusting room you may return to the use of the draw-bench till the gearing of the steel rollers is altered. I had overlooked the circumstance that in the United States Mint cogged wheels are not used for that purpose.

“Yours truly,

“THO. GRAHAM.

“G. F. Ansell, Esquire.”

It is satisfactory to me to find that Mr. Fremantle confirms the opinion at which I had arrived. He says (“European Mints,” [page 9](#)):—“The adjustment of the fillets after rolling is in some European Mints performed by the draw-bench, as in England. In others it is performed by carefully adjusted rolling mills only. The experience acquired both in this country and abroad tends to show that it is advisable to retain the draw-bench—at any rate, until some more accurate method is found of equalising the fillets and reducing them to their correct thickness.” If Mr. Fremantle will investigate for himself instead of being guided by the opinions of others, he will rigidly follow the correct conclusion at which he has arrived; if, however, he will look in the drag room in the Royal Mint, he will discover evidence of the former existence of a series of rollers which were replaced by the draw-bench. Yet one of his co-travellers, who is manifestly willing to make experiments, says of the draw-bench, “Still it has yet to be proved that it can compete in sustained accuracy with a well-constructed rolling mill.” Quite true; but I think the answer holds good, and any one who would replace the draw-bench by rolling mills—which have hitherto failed—should, as Sir John Barton did, prove his proposition otherwise than at the Government expense, and then on success obtain fair remuneration. Before leaving the study of this instrument it is right that I should mention that Mr. John Murray, of the Royal Mint, has added immeasurably to the value of it by inventing a most ingenious machine for grinding the cylinders ([see B, page 28](#)) with accuracy, so that those cylinders which have been used can with ease be, as it were, repaired. In justice to Mr. Murray I should add that I omitted all mention of his invention in any former issue, because some principal contractors were to my knowledge intending to make him an offer

of a sum of money for it, and under such circumstances I could not think it right to give a description of it. A like feeling induces me to postpone to another occasion details of this clever contrivance; its simplicity is such that a mere outline would put a maker of instruments in possession of means for its construction.

The fillets which have passed between the draw-bench cylinders are cut into lengths of about 18 inches by the shears shown at X, in [Fig. 11](#), and are then sent to the trier, who by means of a hand-press similar to, but more delicate than, that shown at [Fig. 7](#), punches out one or two blanks from each length of 18 inches, and weighs it, or them, against a standard weight, and in accordance with his judgment directs the fillets to be cut by the boys or men at the machines. The trier allows 0·20 grain on the pound of blanks for the loss which afterwards occurs by annealing.^[21] If, however, the bars, as previously suggested, were cast uniformly of the thickness of 0·50 inch, this allowance, which is equal to a value of £34 8s. 7½d. on each million, need not be made; but in fact, as will be seen, even this is not sufficient to cover that loss under present circumstances.

Not only has it become a habit to smile at the draw-bench, but every unfair attempt has been made to get rid of it. Officers were abused because they would not report against it, and the machine itself was submitted to usage never contemplated by the inventor. The then chief coiner, who had formerly seen soap used in calico works in Manchester, insisted on using soap-suds, whereby the beds for the cylinders became rusted, the cylinders destroyed, and the whole machine thrown out of gear, so that 30 per cent. of all the work was rejected: these perverse trials, extending over weeks, appeared to throw discredit on the draw-bench. Having been a calico-printer, that officer endeavoured to engraft on to the processes of coining some of the practices adopted in that business; but had he duly appreciated the principle on which the draw-bench acts, he could not have attempted the use of soap-suds, or, at least, on its complete failure would have assented to its disuse. The very first object of the draw-bench is to produce just so much friction as will cause the lateral displacement of the molecules of the metal then under its operation. Soap, when it adheres to the bullion, permits the fillet to pass without friction, and when it is scraped off by the cylinders the gold becomes so hot from friction that the hand cannot with safety touch it; hence there are alternating parts of the fillet very hot and absolutely cold,

and, as a consequence, no two blanks can be obtained of equal weight. Sir John Barton not only made the machine, but determined also that oil was the only fit lubricator for metal made to pass through the cylinders of the draw-bench. Forgetful of the immediate wear on the surface of the metal, others have attempted to replace the cylinders by wedges of steel, so made as that a constant stream of cold water should keep them cool when in operation. This also failed from the formation of rust, while the work produced never equalled that obtained from cylinders. The proper use of the draw-bench is to reduce fillets to an equable thickness in every part, and this can only be done when the trier, after studying the quality of his metal, has determined how much he can remove by once passing through the cylinders; then by keeping up a constant succession of fillets, properly smeared with oil, so that the equable friction always maintains them at the same temperature. Under such circumstances, and with a remedy of 0.24 grain, the rejected should not exceed 2 per cent., and this on the whole day's work should contain half its weight of light and the other half heavy blanks. This also was a curious instance of want of knowledge on the part of that controlling officer, who for years maintained that if the work were properly conducted *all the rejected* should be on one side of the remedy.^[22] The fillets, notwithstanding the draw-bench, cannot be brought to perfect accuracy, and to meet such variations as arise, a difference is made in two of the cutting-out punches by altering their diameters to such an extent that a blank cut by them from a standard fillet would vary in weight from a blank cut by a standard cutter from the same fillet. One cutting-out punch is so altered that a blank would be 0.125 grain, and the other that a blank would be 0.250 grain heavier. This admits of a fillet otherwise too thin being used; but if the fillet be found to err on the other side, it is passed once more either through the draw-bench, or through the mill at a spring-pinch. The trier, Mr. William Fenton, is a peculiarly steady man, possessing a calm judgment, with considerable energy, and upon him depends the accuracy of the whole process of coining; he has not unfrequently so managed his work that upon 5,000,000 of sovereigns coined, he has been within one sovereign of the calculated value.

The fillets, having been thrown by the trier into the receptacles which indicate the particular cutting-out punches to which they are to be taken, are fetched by a man, who wipes off the oil, and then carries them to the cutting-out room, where the fillets are cut into blanks and scissel. In this

place it may farther be noticed that soap-suds cannot be used in the place of oil, because the soap could not be removed without washing,—a practical impossibility,—and would therefore remain, and add indefinitely to the weight of the gold, and thus open the door for speculation. The cutting-out presses used in the Royal Mint are very cumbersome, and when in operation are terribly noisy: it is therefore hoped that at no distant period they may be replaced by some of a far more simple construction; but it is believed that under all circumstances it will be found wise to adhere to the plan of cutting out a single blank at each descent of the punch.^[23] For bronze it is well to obtain five or more blanks at each blow, but the limited variation of weight allowed by law on blanks of the precious metals would render this false economy. [Fig. 13](#) represents one of the twelve cutting-out presses, which are all driven by the wheel A, provided with a series of cams on its outer rim; one of these cams, B, is in the act of striking the friction roller C, which is attached to, and forms part of, the lever D. D is fixed to an upright shaft E, which at F is cut with a screw thread working into a female screw fitted into the main shoulder of the press at G. If now the cam B strikes C, and throws it outwards, it causes the shaft E to take a part of a revolution, and in so doing the screw F makes it rise and carry with it the block H, whose tendency to circular motion is prevented by a plug fitting into its groove, and fixed in the guide I. The lower end of H carries the screwed cap J, which supports the cutting-out punch, so that when E rises it carries the cutting-out punch through just the same distance that itself travels upwards. The cutting-out punch is now ready for action, and is released by the continued revolution of the wheel A, as will be seen by the figure; but as it could not fall with sufficient force of itself, assistance is rendered by the pressure of the atmosphere, as will be seen by the following arrangement. The lever D is provided, near its junction at E, with a loop of iron travelling on a screw, so that it may be moved farther from, or nearer to, the centre of action, and thus admit of the increase of power. This loop of iron is represented by K, and is continued by a rod of iron across the upper part of the room and through a hole in the wall to a system of levers L, from which a rod is suspended, the lower end being connected with a piston working in the chamber M. The chamber M is an hermetically-closed vessel secured to a stone firmly fixed in the floor. The piston works in this chamber, and is covered with about two inches of oil, which prevents the access of the atmosphere by leaks to any part beneath the piston. If the

piston, therefore, be raised from the base of the chamber, a vacuum is produced in that portion from which the piston is removed, and consequently the atmosphere presses on the surface of the oil, which in its turn presses on the piston, and carries it down (the reverse of the action explained by Mr. James Napier in “European Mints”); in its fall the piston pulls down the cutter, which has been raised; for the same blow which raises the cutter also raises this piston; therefore, when the cutter is raised the workman places the fillet N, from which blanks are to be cut, on the bolster, shown at O, and holds it firmly while the cutter descends and punches out a blank, which falls through the bolster into the drawer P. By the time that the down stroke of the cutting-out punch is complete, the wedge Q has entered a slit in the spring R, and strikes the spring, thus throwing the machine back, and preparing it to start when the cam B shall strike the friction roller C. The point at which Q may strike R is determined by a screw near the left-hand T. The upright shaft E, which is partly hollow, terminates at a flat cogged wheel, and the upper part of F is made to pass into E; while at S is an arrangement which serves to detach or connect these pieces, that the cutter may upon occasion be used by the hand by means of the lever T, as well as to admit of the necessary alterations, as the punches, by regrinding, become shorter. The cutting-out punch, when it rises, carries with it the fillet from which the blank has been punched, until the fillet comes against the guard W, which detaches it.

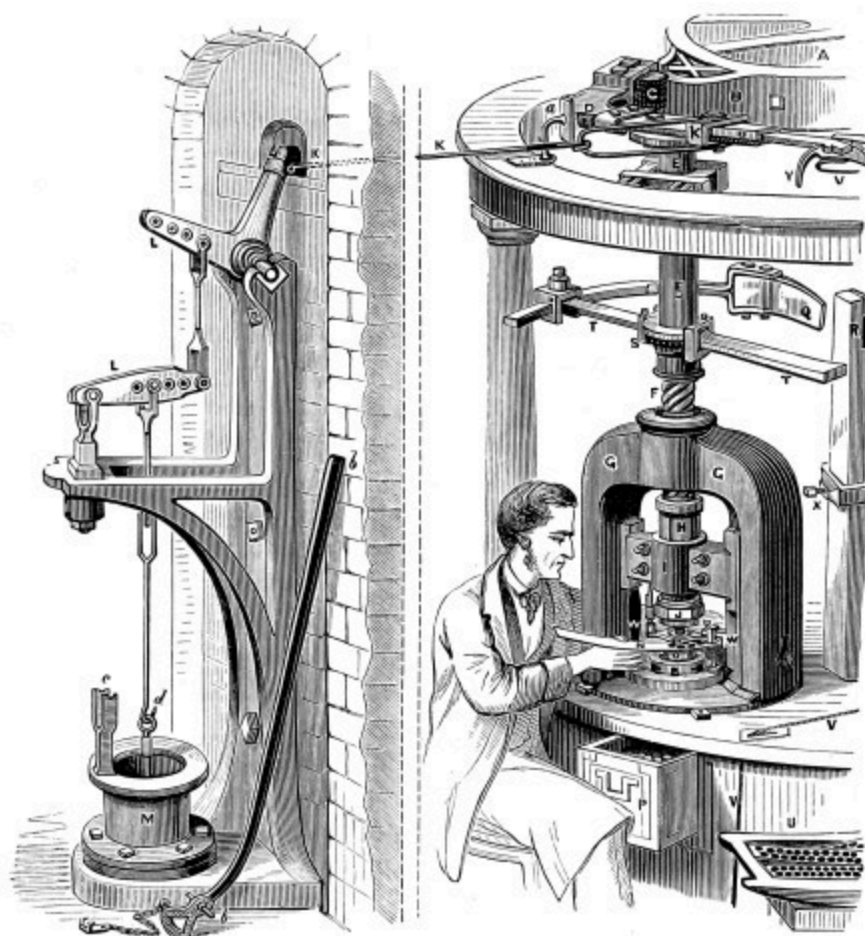


Fig. 13.—Cutting-out Press.

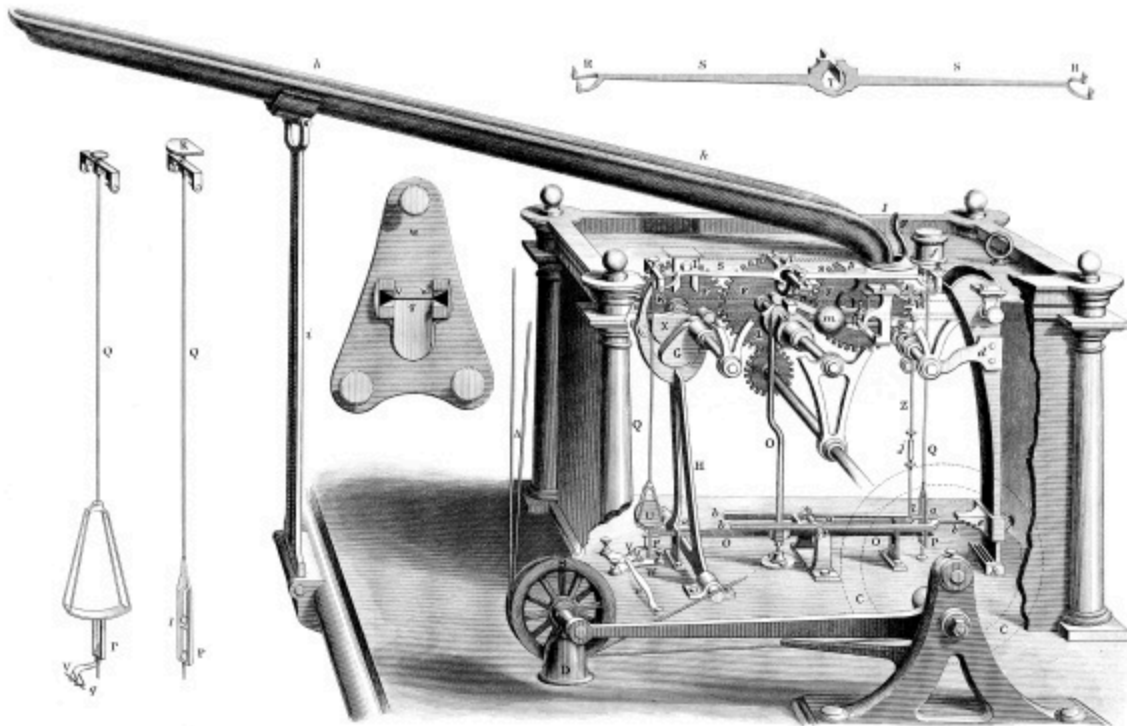
The fillets from which the blanks have been punched have the appearance of ribbons perforated with round holes, and are now called *scissel* (from the Latin *scindo*, to cut). These are thrown into a tray, U, from which they are taken at intervals, and bound up by strips of the same into bundles of 180 ounces—in the case of silver 360 ounces—ready for remelting. The cutting-out press is set at liberty to start by the workman pressing his foot on a lever in connection with the line and spring indicated by V; and so long as he keeps this lever down, the press is worked continuously, but when he releases it the spring catches the extreme end of D, and motion is arrested. The blanks which accumulate in the box P are collected at frequent intervals and examined, to see that their edges are smooth; if they be ragged, as may happen from the wear or fracture of the edge of either the cutter or bolster, a loss would be entailed in after

processes which would cause the coin to be outside the prescribed limit, and to pass at an illegal weight into circulation, as the rough edges would be removed after the weighing of the blanks had been effected. According to the quality of the work—the character of the gold—under operation, the trier tests more or less frequently the variations of weight in a given number of blanks. This process is called POUNDING, and is, next to the *trying*, the most important of his duties; if such an expression can be admitted, he has, by *trying*, fired his shot, and here determines if he has hit the bull's-eye, all depending on his own unaided judgment. The gauge, [Fig. 8](#), is found of great service in detecting irregularities as to diameter and thickness, which would not be, and are not, detected by weight, for the weight may remain equal, although both diameter and thickness may vary. All these points require considerable care on the part of those whose duty it is to attend to them; for the quality of a coinage is determined in this room: blanks which once leave it cannot be afterwards altered. The subsequent operations, being purely mechanical, would be quite as well performed by automaton machines.

The completing processes form undoubtedly the prettiest and most interesting part of the operation of coining; nevertheless the processes already described constitute its most essential features. The blanks are weighed from this room in drafts of about 720 ounces, and placed in bags; each bag, therefore, contains four *journeys* of about 180 ounces each. The term *journey* is said to be derived from an old French word, but circumstances render it probable that this derivation has been applied by others than the original inventor of the word, for it manifestly was not used by the coiner of it to imply a “day's work.” I cannot find an author who gives a time at which 720 ounces of silver became a journey, while 180 ounces of gold were also called by the same name. It must have taken far longer to coin 720 ounces of shillings or groats than to coin 180 ounces of sovereigns or half-sovereigns; hence the same word could never have been intended to mean “a day's work.” Be this as it may, it is time that such names gave place to more appropriate and expressive terms. It is to be hoped that 500 ounces will become the standard maximum weight for bags of either of the precious metals; such a system would be the means of reducing the chance of error, it would have an actual meaning, and would be a convenient weight for a man to lift to and from a scale-pan. When such

a measure shall be convenient to the officers of the Bank of England, that institution will subserve the interests of the Mint by adopting it.

In a paper recently published by the Commissioners on the International Coinage (at page 228) it is stated that “in the British Mint 15 pounds troy of standard gold are coined into 701 sovereigns nearly (15 lbs. = £700 17s. 6d.).” This statement does not exactly convey the truth, and is calculated to mislead, because such an operation is not, and never has been, conducted in the British Mint, where the invariable rule is to abide rigidly by the law, and to coin 20 pounds troy weight of standard gold into 934·50 sovereigns, as specified in the Mint Indenture, now replaced by the Act of Parliament, 33 Vict., cap. 10. Of these sovereigns so coined, 701 are placed in a bag, and called a *journey*; but by a Mint fiction the journey was considered by the old moneyers, and yet more ancient weigher and teller, 15 pounds troy exact, because the parts of an ounce were difficult of addition; and this was maintained until 1860, when a disputed weight with the Bank of England as to the value of the deliveries corrected that which I had previously pointed out as an error. It is now the custom to consider a journey as consisting of 180·03125 ounces, instead of, as it is, 180·03210 ounces, an error of ·00085, which has a money value of about 6s. 7½d. on each delivery to the Bank.



COTTON'S PATENT AUTOMATON BALANCE.
WITH PILCHER'S IMPROVEMENTS.

The bags of blanks should^[24] be carried forward to the room in which the weighing of the individual blanks is effected by Mr. Cotton's [automaton weighing machines](#). Mr. Cotton's weighing machines form perhaps the most elegant and clever invention of modern times. They effect the process of weighing far more accurately than man can hope to do, and with extraordinary accuracy determine the weight of about twenty-three blanks per minute. Peculiarly admirable as are these machines, their perfection has been greatly increased by improvements suggested by Mr. Richard Pilcher, who has immediate charge of those in the Royal Mint. Mr. Pilcher, whose inventive genius is only equalled by his desire to give to the public the benefit of his inventions, has rendered these automaton balances serviceable to the Mint; whereas, when they left the hand of Mr. William Cotton, they were of great service to the Bank of England alone, for there only two determinations, or in fact one determination is necessary. In that institution it is required to show that the coins issued by it are not below the legal

weight for circulation, whereas the Mint must guarantee that coins leave its works neither above nor below the limits fixed by law: hence the necessity for the incorporation of Mr. Pilcher's improvements with Mr. Cotton's beautiful invention.

After reading the letter of Mr. James M. Napier in the *Times* of September 2nd, 1869, I can but express my surprise that that gentleman has not seen fit to acknowledge Mr. Pilcher's modifications instead of claiming for himself all the credit, when a part only—and that undoubtedly an important one—is his due.

The steel engraving exhibits a view of [COTTON'S automaton machine](#) as used in the Royal Mint, where seventeen such machines are employed. For the purpose of illustration, the brass side of the machine has, in imagination, been torn away, as also has the top of the machine. To obtain the highly-finished drawing from which this engraving was made was a matter of no ordinary difficulty, but it was accomplished by the skill and assiduity of Mr. E. S. Gibson, to whom my thanks are due, as well as to Mr. J. W. Lowry, for the pains he has bestowed in executing the engraving.

The whole theory of this balance rests on the fact that the centre of gravity and the centre of action are in one line: either being disturbed, the balance is no longer equal. The machine gains its motion from a shaft fixed to the ceiling of the room. Steam contained in a boiler exists under an ever-varying pressure, arising from the amount of work which it may be necessary for the engine to perform, or from the irregular combustion of the fuel, as well as from other causes. Since it is of the utmost importance that the automaton balance should be made to maintain a uniform pace, Messrs. Napier and Sons found it necessary to drive the shaft which gives it motion by a small atmospheric engine, which is placed in the weighing room. It has been found that a chamber may be kept equably exhausted of air, if the atmosphere be admitted to it by a uniformly weighted valve. Such a chamber—to be described presently—is used in the Mint for other purposes, and Messrs. Napier conducted from this a pipe, by the agency of which the atmospheric engine is worked with a far more even and steady motion than could be obtained by steam; in fact, the exhausted chamber becomes a regulated spring, which softens down the variations in the motion of the steam-engine. The shaft supported by the ceiling conveys its motion to the weighing machine by a line A, which, passing over the

friction wheels B, circulates round the stepped wheel C, which runs loose on the shaft communicating with E. The line A is maintained with sufficient rigidity by a weight D, which is suspended at the end of the lever carrying the friction wheels B. The weight D is just sufficient to insure the continuous working of the machine, but is so light as to permit the line A to slip on C in the event of anything going wrong in the works of the balance. When the machine is to be set in motion, a kind of cheek is made—by screwing—to touch the face of the wheel C, and thus, by friction, C gives motion to the wheel E. This is an elegant mode of meeting a chance of accident, for in the event of the weight D proving to be too heavy, any extra force simply disconnects this cheek from the face of C, and so stops the machine. The machine having been set in motion, E, by communication with the wheels F, all of which are driven by it, causes the cam G to push forward the lever H, which, terminating at I, pushes forward the flattened continuation of I indicated by dotted lines, until it moves a blank placed in the collar J, at the bottom of the hopper *h*, on to the scale-pan K, which, for the sake of clearness, is isolated, and will be seen behind the machine and under the extreme end of the hopper. So soon as the blank coin has been placed on the scale-pan K, the cam L lowers a lever *n*, the office of which is to permit the opening of the forceps M, and thus to release the rod, Q, dependent from K upon the knife-edge R. The forceps are closed by the cam L, which raises *n*, and by it compresses an attached spring. The forceps are intended to hold *this* rod Q while the blank is placed on K, because the friction caused by the placing of the blank would have a tendency to push K from the knife-edge on which it is suspended, and thus blunt its delicate edge. While the forceps are opened the cam N, by its partial revolution, lifts the rod O, which is steadied in its motion, by a pin rising from it, and entering the inverted arch *o*; its lower extremity working into a socket on the table on which the whole frame of the machine stands.

Towards its lower extremity the rod O will be seen to branch out right and left, until each end passes through a kind of step in the rods Q, indicated by P. The office of this rod is to bring the beam, from which the rods Q are dependent, to a dead level, as well as to release both ends of the beam by one action. At the moment that the forceps M have released the right hand rod Q, the cam N, by O, releases both the rods Q, by rising from the steps P, thus permitting the beam to determine the weight of the blank placed on K. A close inspection of the steel engraving will show that the

rods Q are suspended from and rest upon the knife-edges R of the beam S, which has a centre knife-edge T, by which the whole mass is supported and poised. The knife-edges are made to find their own planes or resting-places upon curved or hollow pieces of steel, thus securing the smallest point of contact with a certainty of the smallest amount of friction. In ordinary balances the substance to be weighed is placed in a pan, which is on the same level as the pan which contains the counterpoise; but in Mr. Cotton's balance this condition is of no consequence, so that the counterpoise rests ultimately upon a point which is at the same distance from the centre of action as the point upon which the matter to be weighed rests. The counterpoise U is placed in a kind of cage, and any variation from this standard is at once indicated, even if it reach only to the thousandth part of a grain. By law, the weight of a coin may vary to a certain extent from a standard weight;^[25] the variation or latitude allowed is called *remedy*, from the Latin *ad remedium*; and in weighing, this remedy is taken advantage of by a contrivance much simplified by Mr. Pilcher. The weight of a sovereign is 123·274 grains, but it may by law be either 123·474 grains, or it may fall to 123·074^[26] grains. Mr. Pilcher therefore reduced the counterpoise to the minimum allowed, thus avoiding the placing of a remedy-wire for the light side; and he then made the remedy-wire *q* (shown in the first enlarged portion on the left-hand side of the steel engraving), which is placed on the stand W, upon a peculiarly-formed point indicated by V, so heavy that any blank which would not raise it and the counterpoise must be within the remedy on the *heavy* side. In accordance with this arrangement, the continuation of the rod Q is terminated by a cage at V; so that if a blank be so *light* as to be unable to raise the counterpoise U until the stirrup comes in contact with the remedy, it is too light to make a legal coin. This fact being determined, the motion of the machine causes the cam X to bring back the rod I (indicated by dotted lines), that it may be ready when required to push forward another blank, and the forceps M to grasp the rod Q, while the cam Y permits the falling of the rod Z, which is nearly counterpoised by the ball *m* (the precise length of the rod being regulated by the screw *j*), until its finger *a* rests upon the indicator *b*. The depth to which *a* shall fall is fixed by the step *l* (shown by Q in the second enlarged portion of the steel engraving). This is, of course, determined by the forceps, securing Q at the position indicated by the weight of the blank. The indicating finger *b* having come to rest, the continued motion of the machine causes the cam *c* to

permit the shoot *d* to fall until one of its steps *e* comes into contact with the indicating finger, when the lower part of the shoot must be exactly over, and form part of, one of the tubes *k*, which terminate in boxes labelled respectively “heavy,” “light,” “medium.” The shoot having taken its position, the continued motion of the machine causes the cam *G* to induce the placing of another blank on the scale-pan *K*, and this blank, by advancing, pushes off the one just weighed, which, falling into and through the shoot, passes to the compartment reserved for it. Suppose the newly-placed blank to be too heavy for forming a legal coin, the same operation goes on, but it now lifts not only the counterpoise, but also causes the stirrup at *V* to lift the remedy-weight *q*. This is a most delicate operation, for if a blow, however light, be given, it would cause unsteadiness in the beam. Mr. Pilcher therefore determined to make the carriage *W*, which supports the remedy-wire *q*, stand upon micrometer screws, by which the remedy-wire is made just to touch the stirrup, without pressure, while it also rests upon the most minute points, formed by cutting away every part of *V* which is not actually required—in fact, shelving it out. The scale-pan *K* is protected from draught by the lantern *f*, while the blanks are directed into the collar *J* by the guard *g*, as they slide down the hopper *h*, which at its centre is supported by an upright, *i*. The blocks *p* are the supports which hold all the machinery to the roof of the machine.

Some new machines have recently been supplied to the Mint by Mr. James Napier, but it is not certain that these are cheaper than the old ones. That they are lower in price may be admitted, and they are certainly more convenient in use, from the fact that the wheels shown at *F* are placed at the back. It is, however, to be regretted that Mr. Napier did not introduce into these new machines the recent inventions of Mr. William Bradshaw, which are supremely simple, and, now that they are effected, one cannot but wonder that so many minds having been engaged on these machines, improvements such as these have not earlier seen the light. To Mr. Bradshaw, particularly, great credit is due, for he was obliged to overcome, not only innate difficulties, but difficulties of position, and these latter were of no mean kind. It is to the credit of the Mint authorities that they finally adopted these improvements, and ordered the necessary alterations to be made in *all* the machines.

Mr. Bradshaw's improvements enable the automaton balances to weigh 30 per cent. more coins in the same time; yet they permit each coin to occupy a longer space of time in being weighed. This would appear to be a mechanical contradiction, but if reference be made to the plate it will be seen that the cam G pushes forward the lever H, which ultimately, by the slide, pushes the blanks on to the scale-pan K. By altering the shape of the cam G, Mr. Bradshaw causes it to do its work more rapidly, and thus leaves the piece on K longer than before; but that this time may not be lost, he alters also the shapes of the cam L and of the cam N, so that the forceps M are opened, and the beam is released more rapidly: thus the balance would be longer in action; but to utilise the time ready to be saved, he alters also the driving pulley E, causing the machine to make 30 per cent. more determinations per minute, and that this may be effected with greater certainty he reduced the depth of the step, shown at *e*, to just one-half, so that the beam has to travel only half the distance it formerly travelled to determine the position of the light, heavy, or medium piece. He next proceeded to alter the rod O, because he observed that it received a tilting motion arising from its being lifted by a shoulder, as shown in the plate. To overcome this defect he made the rod straight up to a certain point, where he divided it into a kind of loop which passed over the cam N, which, as it revolves, lifts this rod perpendicularly. He still found that the cam N had a tendency to push this rod against its bearings, especially at *o*, so he placed a spring of brass between the top of its loop and the cam N, and secured at one end to *p*, and by this contrivance made the cam to lift the brass, and that in its turn to raise the rod O. There was yet another difficulty, caused by an occasional stoppage of the action of the spring which softens the descent of the bearing of the rod O into its lower socket; this he removed by cutting a hole in the side of the socket, so that it can be seen at once if the spring be in action. I am not one who approves his suggestions as to the remedy-wires. These may be the means of saving time, but they are wrong in principle, and I do not describe them.

Great inconvenience arose from the collection of dust, spangles of bullion, and other foreign matter on parts of the balance. This Mr. Bradshaw overcame by two simple contrivances; first, he pierced the bottoms of the grooved trays, so that these substances might in a great measure fall through while the blanks were being arranged in rouleaux, thus separating the largest pieces, which, falling down the shoot, would stop its action; and,

secondly, he placed a glass shelf midway between the table and the beam, that is, beneath the hopper *h*, where it terminates at J, and above the beam S, where it is seen near the forceps M, thus protecting the beam and forceps. These may appear to be trifling alterations,—they are nevertheless to Cotton's balance what the compound metal balance-wheel is to a watch. Mr. Pilcher was granted the *magnificent* sum of £40 for his invention of the file (to be described), but it is hoped that Mr. Bradshaw will be rewarded with a far more open-handed generosity; for if, by judicious liberality, encouragements are offered to those in the Civil Service to give to the country the benefit of their inventions, we may still hope that even our Government institutions will bear comparison with ordinary commercial manufactories as regards the development of inventive genius. Nor is this a small matter, for all the working improvements in the Cotton's balance have been made by those whose duty it is to attend to their working; yet I am not aware that any acknowledgment, either direct or indirect, has been made to the officers concerned. It is said that their salaries cover all their time and energy. This may be true, but such a policy is not calculated to result in many great steps towards perfecting either machines or processes.

It is well to state that the beam in Mr. Cotton's balance is 8·90 inches in length, and that its weight is 288·41 troy grains.

Those blanks which are neither too light nor too heavy are called *medium*, and are weighed and put into bags for future issue as coins. The *light* are reserved for the melting-pot. The *heavy* blanks should be reduced by means of a peculiar kind of file to the weight of medium blanks, and thus saved from being melted, so avoiding further loss to the Mint. This file was invented by Mr. Pilcher, who, being the officer of this room, considers his duty to be neglected if there be any improvement capable of being made, but which is left undone; he never tires till the invention is complete and the machine made. Mr. Albert Barre, the distinguished engraver to the Paris Mint, declares this file to have surmounted all the difficulties he has met with in this part of the process. In his own words, "it leaves the face of the blank untouched, and free to develop the work of the engraver, which no other file does, or can effect, because any metal which is *ploughed* out from the face of a blank, leaves a hole which is not filled up in coining."

The opinions which I have here expressed have so far prevailed on two of the travellers to European Mints, as to induce them to glance, though

rather shyly, at the necessity for reducing heavy blanks; but Mr. Napier writes a passage which I quote, as it demonstrates that he, who is admittedly one of the most successfully ingenious men of the age, has thought over and studied the words I wrote at page 60 of the last edition—he took that book in his hand to the Mint on the day after its publication—with that attention to which he thought they were entitled. I beg the reader to ponder over Mr. Napier’s words, they are most important; but I hope those who will be called upon to vote the necessary money will also read my remarks on this extract from Mr. Napier’s report on European Mints. He says at page 44:—

“Treatment of Too-heavy Blanks.—Notwithstanding the employment of the best mechanical appliances for rolling the bars, adjusting the fillets to gauge, and cutting out the blanks, and the most intelligent and painstaking workmen to carry on the operations, still, owing to unavoidable imperfections in the means, the blanks, when cut from the fillets, will not be found equal in weight, nor, indeed, so nearly equal that they can be recognised as of equal value, and passed on to be manufactured into coins destined to be put into circulation, and to represent one value. The irregularities will necessarily be more or less, according to the care and ability of the workmen, and the condition of the machinery; but under the most favourable circumstances only a certain proportion of the blanks will be found to come within the limits which it has been considered desirable to set down to restrict the quantity of error.

“This being so, the following questions present themselves: Are the too-heavy and too-light blanks to be returned to the crucible and remelted, or to be passed on and detected only in the finished coin, and then sent to the crucible? Or are too-light blanks to be avoided altogether, and all the blanks to be made either of sufficient weight or too heavy? and in such case are the too-heavy blanks to be submitted to some process for reducing their weight to within the legal limit? Or is a more exact means to be preferred, by which they will be equalised to that perfection which cannot practically be questioned, and by which they will become, for all purposes, one and the same weight?

“The facts collected are, that at nearly all the European Mints the reduction of too-heavy blanks, both of silver and gold, is practised, to avoid the loss attendant upon their return to the crucible, and that the endeavour is to make all the blanks either standard weight or heavier, so as to admit of reduction, to err, indeed, on the heavy side, even to the extent of necessitating the reduction of all the blanks. So great is the importance generally attached to this department, that it is calculated that, in some instances, nearly as many hands are employed in the service of it, including the weighing, as in all the other working departments combined. In Berlin, where particular care is devoted to the weight of the coins, between 200 and 300 men are employed in times of pressure in the weighing and equalising of the blanks, which operations are in this instance performed by the workman with a hand shaving apparatus, fixed to the table at which he sits, and a pair of scales also placed near to him. The work done by one man is revised by a

second, and the file is used for giving the finishing touch, so that the blanks are brought to great perfection in weight. But although the extent of the means shows the great importance attached to the results in the establishments visited, and the large expenditure which, at least in the opinion of the directors of these Mints, may be permitted with profit in this department, the serious consideration of the employment of so many hands would militate against the carrying out of the system in the New Mint. It may be satisfactory, therefore, to know that, in case the desirability of an exact coinage should be entertained, means for assuring it are at hand without the employment of manual labour for the purpose. The subject has, indeed, for many years occupied attention, and, after much labour and expense, the difficulties have been overcome, and a machine has been constructed to the order of the Government of India which fulfils all the required conditions for equalising the coinage to that perfection that no appreciable difference need exist between the standard weight and the coin blank which has passed through the machine. This machine differs essentially from those employed at the Mints visited, and which, it has been explained, only take a fixed quantity off each blank presented to the cutting tool, without reference to the weight of the blank, the operation being repeated or not, according to the result; or, in other words, no matter what be the weight of the blank, whether nearly light enough, or much too heavy, the cutting tool of the machine treats all in the same manner.

“The equalising machine referred to, on the contrary, not only takes from the too-heavy blanks, each in succession, a quantity of their surplus weight, but it ascertains how much it is necessary to take off each blank to reduce it to the standard weight; and it deducts that amount only; so, although the blanks which are supplied to the hopper, or reservoir of the machine, enter the machine of unknown weight, they are equalised by it in once passing through, and are delivered into a receptacle for the finished work, all of one weight, the standard, or so near to it that good ordinary scales, such as are used for weighing blanks and coins, will not detect a difference.

“Means can therefore be provided for the purpose, should it be decided to reduce the blanks to equal weight, or only to within the pale of accuracy instituted by Parliament, the basis of which legislation must be taken to have been the then practicable approach to perfection; for it is to be presumed that, with a sufficient assurance that accuracy akin to perfection was attainable ALONG WITH ECONOMY, the wisdom of Parliament would have enacted accordingly.

“The first outlay for self-acting coin blank equalising machinery WOULD NECESSARILY BE LARGE, but it is believed at the same time that the saving would be considerable, especially upon gold.

“The question of the amount of economy would have to be more fully gone into, and the expense of the machinery accurately estimated. Extensive and continuous experiments with the machine have also STILL TO BE MADE TO ACQUIRE A KNOWLEDGE OF ITS USE, and to ascertain, after a *lengthened* probation, whether any modifications in the details are advisable: the limited experiments already made leave no doubt, however, of the successful result.

“The arguments in favour of an *exact coinage* chiefly rest upon an economical basis, and branch out beyond the walls of the establishment into questions of currency, wear, and recoinage, which it is not thought necessary to treat of here; at the same time it may be stated that it seems desirable that the coinage should be made as exact to weight as possible CONSISTENT WITH ECONOMY, and that an *exact coinage* would be found more economical as a circulating medium, would be more just, and a better starting-point from which to legislate for the withdrawal from circulation of worn coin than a coinage issued from the Mint with acknowledged errors, or differences in weight so substantial as to be represented in value, in the case of gold coins, by some of the lesser denominations of the coins of the realm.”

The reasons which Mr. Napier gives are worthy of grave consideration—they are mine wholly—except so far as regards those to which I have drawn attention by printing them in small capital letters, and which show that even with his vast experience he dare not answer for the success of his machine, while the suggestion of the necessity “to acquire a knowledge of its use” should make a desire to recur to a successful machine paramount. If, indeed, I rested on my own opinion, I should be content to see Mr. Napier’s machine tried, although I think its failure would be established; but it so happens that I can quote the opinion of the late Master of the Mint on the machine under consideration, expressed at a period when the efficiently economical conduct of the Royal Mint had received his utmost consideration. In his letter to the Treasury, dated 7th June, 1860, he says:—

“I may be allowed to call to your recollection that a Parliamentary grant of £1,100 was obtained by the Mint in 1856 for the purchase of two automaton filing and adjusting machines, which it was proposed to have constructed by Messrs. R. Napier and Sons, the eminent engineers. No part of this grant has been appropriated. The rapid amendment in the blanks which took place immediately afterwards led me to suspend the order for these machines, and ultimately to abandon the idea of any great expenditure for the object contemplated. In the meantime also, a machine of a much more simple character was contrived by Mr. R. Pilcher, of the weighing room, and was constructed in the Mint, with no assistance from without. Pilcher’s adjusting machine has proved sufficiently effective, costs nothing for labour, and has now been in constant operation for two years. The cost of making a pair of machines such as we now possess is estimated at £60, a sum which was saved to the public by the mode in which the work was executed. Trusting that the Lords Commissioners of her Majesty’s Treasury will be disposed to consider favourably the merit and practical value of such an invention, I venture to solicit their Lordship’s sanction to the application of £60, the sum just stated, to the benefit of the inventor, jointly with an ingenious mechanic on the establishment, who gave material assistance in the construction of the machine. I would propose the following awards, if agreeable to their Lordships:—To Mr. R. Pilcher, £40; to Meredith Jones, £20.”

It was upon advice tendered by myself that Mr. Graham thus excluded the machine now recommended by Mr. Napier. I maintain that a simple machine should be preferred before a complicated one, and the Mint authorities should give Mr. Pilcher's £60 machines—admittedly perfect in operation—a fair trial against Mr. Napier's £1,100 machines, and see if they prove *equally effective*. If, on the other hand, Mr. Napier's fails, then it is but fair to the public that that machine which is undoubtedly a success, and for which the inventor has been *publicly rewarded, although in a most inadequate manner*, should be used.

The machine thus spoken of was used with the utmost advantage until 1866, when its use was abolished by the advice of Mr. John Graham, who subsequently, in 1868, broke it to pieces that *his improvements* might be rendered permanent, thus causing an expense of 50 per cent. on the total cost of manufacture for *rejected coins*,^[27] because the use of the file having been abolished, the coined money alone was weighed, thus rendering it impossible to reduce the heavy pieces.

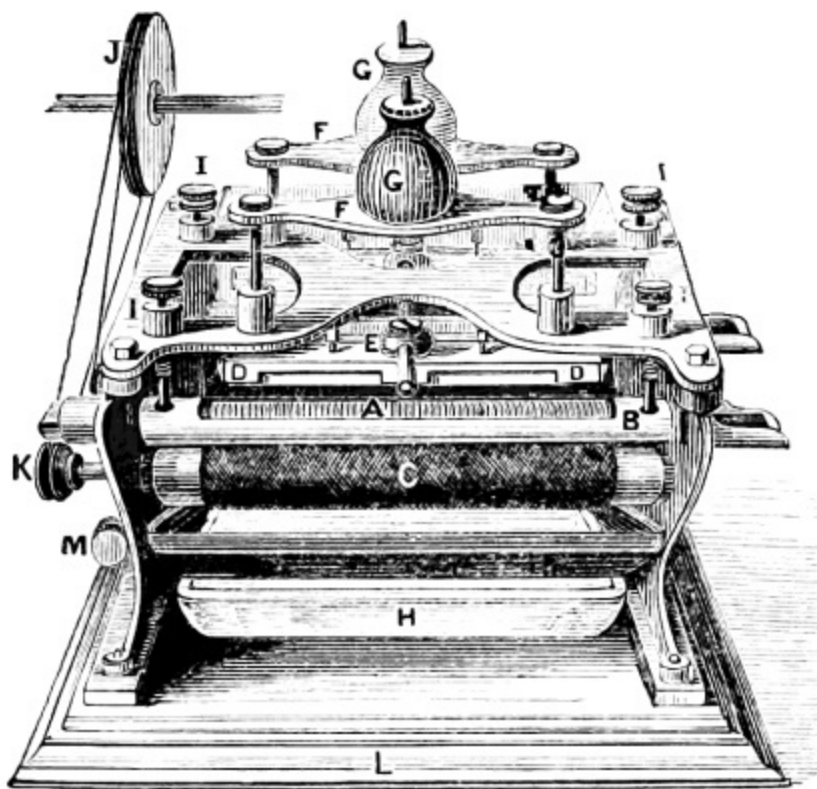


Fig. 14.—Pilcher's Filing Machine.

Although the use of this file has thus been stopped, it is greatly to be hoped that it will again be employed, and thus enable the Government to produce coined money at the cheapest possible^[28] rate, because it enables the Mint authorities to obtain the largest per-centage from the bars. Feeling that this much-desired re-introduction will take place, it is thought wise to give the following description of Pilcher's file, which, if not desired for the benefit of our own Mint, will be adopted by other coining countries:—[Fig. 14](#) is a representation of this compact machine. The blanks A are placed in rouleaux in a tube B, which is open at top and at bottom; through the opening at the bottom the blanks rest their EDGES on the file C, which, as it revolves about 1,000 times per minute, files off metal from the edge of the blank. Each machine has two tubes, and when both have been filled the rod D, which carries a triangular knife-edge, is released by the lever E, and the knife-edge resting upon the upper edge of the blanks A,—with the intention of offering resistance to their rotary motion,—enables the operator to remove MUCH OR LITTLE metal from their edges at pleasure by increasing the resistance which D offers by adding a weight G on to the gallery F. H is a glass dish into which the dust, as it is removed from the edges of the blanks, falls, thus insuring a perfect *separation of the dust* from the blanks. The blanks in B are kept in their position by small blocks of ebony, which are secured by the thumb-screws I. Motion is given by the wheel J, which communicates with the pulley K by a cord or catgut. The whole machine stands on a block of mahogany L, secured to a table of oak. M is a screw which is used to tighten the cords between J and K. Between C and H is a hopper of brass to catch flying particles of the precious metals. Each file reduces 250 sovereign blanks per minute.

The objections which were raised to the introduction and use of this machine were curious, and amongst others it was firmly maintained that it caused the coining press to make brockages, because the blanks which had been filed were smaller than those which had not been thus treated. The amusing part of this objection was that *all* the blanks were afterwards reduced to a uniform diameter by the edges being compressed *before* being coined.

The medium blanks—now unfortunately the coined moneys—are rung by boys to detect any which may be dumb or cracked, and which are rejected for melting. Dumb or cracked pieces arise when bubbles of air are

enclosed in the bars at the time of pouring the fluid metal into the moulds in the melting-house.

Besides this source of dumb work may be mentioned another of large occurrence in silver, and occasionally met with in gold, which results from an imperfect mixture of the alloy at the time of melting, and develops itself at the draw-bench, where whole fillets, of six feet long, may be seen to separate into two complete layers of metal, the inner surfaces being coated with a thin film of copper, frequently quite pure, but sometimes in the form of suboxide. It would thus appear that a globule of copper becomes enveloped in a volume of fluid gold or silver, and, in the act of pouring, this globule is drawn out into a kind of wire, perhaps extending some inches in length, enclosed in precious metal. When this is rolled it is all flattened together, but there is no adhesion between the surfaces, so that when the fillet passes through the draw-bench the lateral motion given to the atoms of the metal causes the slip which finally separates the two surfaces of metal, and the eye at once detects the existence of the fault. In the coinage of 1857, no less than 141·38 ounces of gold fillets thus imperfect were returned to the melter in one day, and on the whole of that coinage these fillets averaged nearly 0·50 per cent. ([see page 92](#)). In the case of silver this defect is of less consequence, but of greater occurrence. It is the duty of the melter to see that such bars are neither produced nor forwarded, and there can be but little doubt that in a properly conducted Mint the melter would be directed to re-melt such fillets without payment to the men.

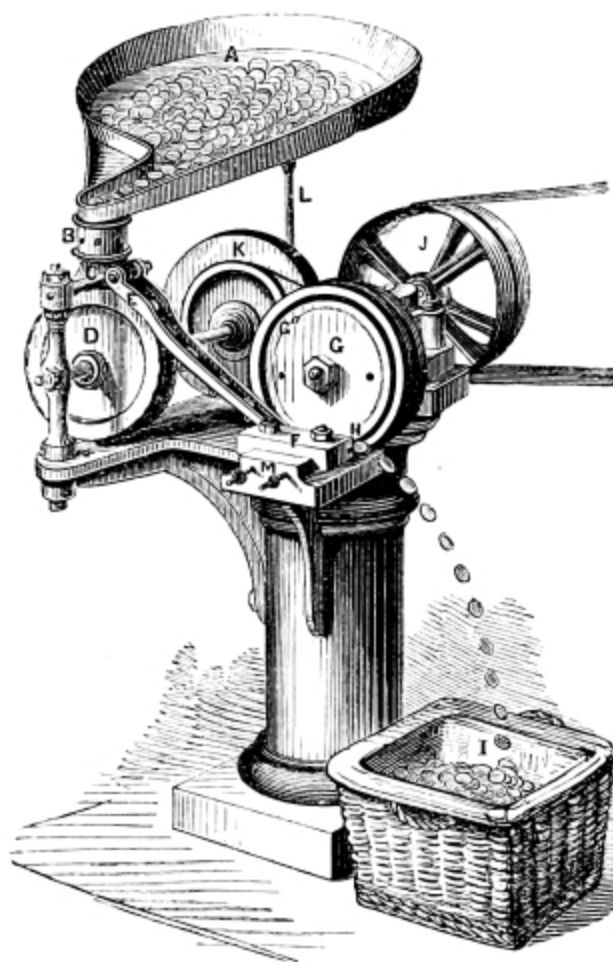


Fig. 15.—Jones's Edge-Compressor.

The medium blanks which would now be fit for coining should be weighed up in bags of about 500 ounces, instead of 180 ounces, as is the custom, and sent forward to the room in which is placed a machine which has to diminish the diameter of each blank by compressing its edges. This is called a *marking machine*, but such a name being obviously improper, it is preferred to call it the EDGE-COMPRESSOR. Up to 1861 the best machine for this purpose was that invented and patented by Messrs. R. Heaton and Sons, of the Mint, Birmingham; but in the early part of 1858 the subject of the varying diameter of blanks came under consideration as effecting the production of brockage. Mr. Pilcher was consulted by me, and proposed a machine, of which he gave the outline, which I submitted to the Master of the Mint on the 31st of March, 1858. This machine was to be fed from the

top, but was abandoned because it did not meet the desired minimum of 1,000 pieces per minute. Mr. Pilcher therefore proposed to erect on the same spindle five revolving discs, and to face these five discs with as many cheeks. At a given time it was determined to manufacture a marking machine, and one of five which Mr. Pilcher proposed was to be made. It was preferred to ask Mr. Meredith Jones to assist in its manufacture. Mr. Jones had also a design for a machine for this purpose, and it was arranged that as Mr. Jones made Mr. Pilcher's file he should make his own machine. Mr. Jones proposed, and Mr. Pilcher accepted, the following proposition. Mr. Pilcher was to retire from the machine, and in consideration of his so retiring, and allowing the machine to be called Jones's marking machine, Mr. Jones was to give Mr. Pilcher one-half of whatever sum might be paid by the Mint for the use of the machine. [Fig. 15](#) represents Mr. Jones's machine. The blanks are placed in the hopper A, and fall by an incline into a tube B until they rest at C, on a notched wheel D. As D revolves each of its notches carries away the bottom blank of the pile from the tube B, and leaves it to slide down the tube E till it reaches the block F. The block F is cut with a narrow groove, which exactly corresponds with the groove G^a on the plate or disc G. The disc G revolves, and as the blank slides down and comes with some little impetus against the groove in the block F, the groove G^a catches it, and causes it to take two revolutions between the disc G and the block F, finally permitting its escape at H, when it falls into I, being now reduced in diameter and thicker on the edge, but its centre remaining as it was before. The machine is driven by J, the shaft of which carries a reduced rigger for the driving of K. The hopper is supported by the rod L. The distance between F and G represents the diameter of the blank after its edge is compressed, this distance being determined by the screws seen at M. Blanks of all diameters may be compressed at this machine if the block F and the plate or disc G be removed and replaced by others, neither operation taking many minutes. The edges of the blanks are compressed at the rate of 700 per minute. A boy of fourteen could work this perfectly well, and with more convenience now that the hopper is replaced by such a one as is used to supply Mr. Cotton's weighing machine. The edge of the blank is compressed with a view to prepare it for the *crenating* to be given by the collar in the after process of conversion into a coin.

The blanks are conveyed from the edge-compressor to the annealing room, where they are placed in rouleaux in iron boxes, the tops of which are

luted on with clay, the boxes are then placed on iron carriages, and run into reverberatory furnaces, heated by Juckse's smoke-consuming apparatus, where they remain for an indefinite period. The furnaces are like that represented in the rolling room for annealing the fillets in copper tubes. To anneal the blanks, the temperature should be raised rapidly until the boxes attain a full red heat; and the time allowed for the operation should be from twenty to twenty-five minutes at the utmost. After the heating, the boxes are withdrawn from the furnace and placed on the floor of the room, till the blanks are assumed to have become of a low red or black heat, the point below that at which copper combines rapidly with oxygen; they are then opened. If the boxes be opened while the blanks are still of a full red heat, there is what is called too much loss by annealing, or, in other words, too much of the copper has become oxidised, and thus rendered soluble in the dilute sulphuric acid into which they are presently removed. The box having been opened, the blanks are turned out into a copper tray, and on this carried to a cistern of cold water, in which a colander stands; they are thrown through the water into this colander, and thus cooled rapidly. When cool, they are taken in the colander to a leaden cistern of nearly boiling dilute sulphuric acid, into which they are placed, and allowed to remain for about three or four minutes. By means of this bath of sulphuric acid a thin skin of oxide of copper is removed; the gold on the surface from which the copper has been washed out is of a sponge-like form, and gives a beautiful bloom to the new coin when it is struck. This process is called *blanching*, and is of great interest to the coiner, as it is here that his loss by coining takes place. The bars of gold sent to the coiner are found to be standard within certain limits,^[29] therefore the blanks produced from those bars must be of the same fineness; but by this process of blanching, copper is removed; hence the standard, or the amount of pure gold contained in the blanks, is proportionably raised, and this tells on the million to an amazing extent. When the annealing is conducted for too long a period, the copper becomes oxidised to too great an extent during the time of annealing, for there is no atmosphere more capable of oxidising than is that of one of these furnaces; it is for this reason that the heat should be gained rapidly, before that oxidising atmosphere shall have time to permeate the luting and attack the copper of the standard gold. At the trial of the pyx in 1861 the gold coined by Mr. Thomas Graham, the late Master, was found to be "4 grains on the pound too fine;" that is to say, containing in 5,760 grains 5,284 grains of

pure gold in the place of 5,280 grains, which it should have contained. This is well for the people, but it is bad for the Master of the Mint, because he has lost 4 grains of pure gold on each pound that he has coined, which amounts to no less than £757·65870 in value upon every million of sovereigns. This loss may well be avoided, either by adding extra alloy to allow for this removal of copper, or, as would be far wiser, by abolishing the process of annealing the blanks, as was done in 1859, when more than a quarter of a million of sovereigns were coined from unannealed blanks obtained from brittle gold which was much harder than any ordinary gold to be met with. It must be remembered that the saving of bullion would leave a large margin for the increased destruction of dies, and in addition the coin would wear longer in circulation.

By his elaborate experiments Mr. Charles Hatchett proved not only that the British standard gold was the best alloy for coins, but that this alloy should be so manipulated as to avoid either the extreme of ductility or of hardness, and when this medium was attained the coins produced wore longer and better than any others.

It will be of importance in this place to consider the question of annealing the blanks previously to coining them, and experience teaches that this process should be omitted. The gold alluded to as coined in 1859 from unannealed blanks was the same gold that was spoken of as being so toughened in the form of bars when the fluid metal had been poured through a stream of coal gas.^[30] The gold contained antimony, arsenic, and lead, so was predisposed to become brittle; but by pouring through the air it absorbed oxygen, and became certainly the most brittle gold that has passed through my hands. It was toughened by reducing the oxide of copper, and worked remarkably well, producing the most workable fillets, regular and uniform blanks, which yielded an average of 5·67 per cent. rejected. The blanks were tough, soft, and very malleable, capable of being cut with a chisel without fracture, and ringing musically on an iron block, thus satisfying all the requirements of good coining gold; but directly these blanks were annealed they showed 30 per cent. of pieces so brittle that they could be broken by a moderate pressure of the finger and thumb. After coining, the whole mass became so brittle that less than 2 per cent. remained sound after twice ringing on an iron block, as is the usual practice. These brittle coins, when again annealed, and suddenly cooled by plunging

into cold water, became tolerably tough, and withstood ringing to 8 times, when they again became brittle. It now became evident that none of this gold should be issued to the public,—indeed, I may state that this same gold had been rejected by all the coining countries as unfit for coin. It was therefore ordered to be melted and returned to the Bank as unfit for coin, its value being £167,539. After every possible obstacle had been placed in the way, I overcame the aversion to change, obtained permission to try some experiments on this gold, rewrought it, and coined it all without annealing. The coin thus produced was so tough that an ordinary man could not break a sovereign even by the aid of a pair of pliers.^[31] Yet when these toughened coins were annealed, they became so brittle that a child could break them readily. In the Royal Mint some of these coins in each state are preserved. Mr. Graham, then ever anxious to secure to the Mint any advantage, was pleased to address to the Treasury a letter on the subject of this gold, in which he said, “A correspondence between the Governor of the Bank and myself on the brittle property of the gold occasionally imported into the Mint to be coined was formerly brought under your notice in my letter of the 21st March, 1857. The evidence of the evil in question was not confined to the Royal Mint, but has, I believe, been felt at all other mints, and also by goldsmiths generally, since the recent gold discoveries. It was traced to the presence in the gold of a minute portion of antimony or arsenic (often not greater in quantity than one-tenth of a per cent.), which escapes the observation of the assayers. But no remedy in dealing with such gold then presented itself, except the expensive one of having the brittle gold refined.

“An unusually large proportion of the gold received for coining last year was of this defective character, but the whole of it was successfully coined notwithstanding, and no part returned to the Bank to be refined, as on former occasions.

“This improvement in the practice of the coining department is the result of a laborious investigation made in the coining department, of which I am happy to assign the chief merit to Mr. Ansell.

“It now appears that the antimony or arsenic acts injuriously when the gold blanks are allowed to cool gradually, but not when cooled suddenly, after annealing, the gold appearing to have time to crystallise and become granular under the influence of the antimony particles in the one case, but not in the other. The improvement may be justly represented as one of

considerable value. It saves entirely much extra labour hitherto applied to brittle gold without any beneficial result. It will also lead to the diminution of waste, of which brittle gold was always a fruitful source.”^[32]

I do not accept the conclusion arrived at by Mr. Graham, for my experiments showed clearly that this peculiar gold could not be heated even to a black heat after coining without becoming absolutely brittle under all circumstances. Notwithstanding the experience gained by the coining of this brittle gold, the annealing of coined money was effected by the present authorities at the Royal Mint in 1868, when a vast amount of brittle gold was coined, of which considerably more than half a million sterling in coined pieces were annealed and plunged into cold water, just as was done in 1859, and with precisely similar effects; that is, the production of brittle pieces, as soft as lead and perfectly rotten. These pieces were sent to the Bank of England, where they have given great dissatisfaction. They should have been melted, as were those of 1859. I, being an officer of the Mint at the time, offered to coin this gold as I had coined that in 1859; my offer was declined, and a vast mass of very brittle coin was, in addition to that of which I have spoken as having been annealed after coining, issued to the public, besides many thousands of ounces uncoined having been returned to the Bank of England by the Mint authorities, who found themselves unable to coin it; these facts being, as I think, little to the credit of the official ability.

The Right Honourable Lord Kinnaird, with a view to relieve the Mint authorities, suggested, on the 22nd of March last, from his place in the House of Lords, a proviso “that such gold”—specified in clause 8 of the Coinage Bill then under discussion—“should be free from lead, antimony, and other substances, so as to admit of its being coined without previous refining.” The Government promised to consider this suggestion, and, after consideration, rejected it, although the right thus proposed to be established has always been claimed by the Mint but disputed by the Bank of England. His lordship then drew the attention of the House of Lords to the issue of the brittle sovereigns, to which I have alluded, as having given so much dissatisfaction at the Bank, and on the 25th of March, the following letter appeared in the *Times*:—

“THE COINAGE BILL DEBATE.

“To the Editor of the Times.

“SIR,—In your full and very correct report of what I stated in the House last night on the Coinage Bill, there is an error which I think must have occurred in the printing—‘sweat’ should be ‘sweep.’ I think what I said in regard to the £640,000 sent by the Mint to the Bank would have been clearer if I had added—‘Its brittleness was increased by being annealed after coining.’

“Your obedient servant,
“KINNAIRD.

“50, Avenue Road, Regent’s Park,
“March 23rd.”

Attention being thus forcibly drawn to a great evil, the Marquess of Lansdowne, in reply to Lord Kinnaird’s remarks on the appointment of Mr. Roberts, stated that “Mr. Roberts was a very eminent chemist, who had proved his capabilities by devising a system for depriving gold of the obnoxious properties on which the noble lord recently commented.” Lord Kinnaird has already pointed out that this process, which the eminent Mr. Roberts has “devised,” was discovered by Dr. Percy, and that the process was published by that gentleman so long since as 1848. Dr. Percy conceived the elegant idea that if a stream of chlorine gas were passed over an ore containing gold, the chlorine would combine with the gold and dissolve it out from its matrix, as water would do sugar. This it does with complete success, forming a chloride of gold which is readily soluble in water, and from which it can be obtained with the utmost ease.

With regard to the invention of the process for which Lord Lansdowne gave Mr. Roberts credit, the facts of the case stand as follows. Dr. Percy discovered that by means of chlorine gas he could separate gold from its ores. He described his invention in the Transactions of the British Association in 1848, and in the Philosophical Transactions in 1850. Plattner carried it out practically about 1852 in Silesia, and I, in January, 1856, conducted a long series of experiments upon it for the Chancellorsville Gold Mining Company. The exact process now proposed or “devised” by Mr. Roberts was patented by Mr. Miller, the present assayer to the Sydney Mint, in 1867—No. 1767. When Lord Lansdowne finds himself thus deceived, he will probably in future think it necessary to examine statements suggested to him before making them from his place in the House of Lords.

This *new* process consists in forcing chlorine gas through melted gold, and it is assumed that the chlorine will carry off the silver, lead, antimony, tin, and arsenic, which render the gold brittle. On the face of the proposition, there are very grave reasons for believing that it will fail if tried. This also is the deliberate opinion of the most eminent metallurgical chemist of this or any other time; indeed, it has been tried and found a failure. It has failed of its object, and instead, attacks the gold, which, with the chlorine, will leave the melting-pot and pass into the chimney, creating a use for the proposed chimney chambers of Mr. Roberts.

Experience at the Mint is to the effect that the run of gold ingots produced by the melting of worn and light coins culled from those in circulation is found to be worse than standard by 1-500th, or two in each thousand parts. There can be no doubt but that this is mainly due to the fact of the loss of copper by the processes of annealing and blanching. In actual practice extending over many millions, it is found that 1,000,000 sovereign blanks lose 5,708 grains of copper: now, since this copper is removed from the surface, it leaves a sponge of pure gold coating that surface. Pure gold is very soft, and rapidly wears away, so that when the coins from which 5,708 grains of copper have been taken go into circulation, this film of gold, which amounts to £555·62 on each million pieces, is removed by friction. It will probably be argued that the copper being taken first and the gold afterwards, leaves the coin of the same standard as the original bars;^[33] if, however,

this proposition be examined, it will be seen that the bars are assayed and found to be below standard 0·55562 on the thousand coins; but the blanks obtained from the bars are, by annealing and blanching,^[34] brought to standard before they are coined, and those coins assayed WHILE NEW are found to be standard by the jury of goldsmiths at the trial of the pyx, because they retain this film of gold; let them, however, go into circulation and lose their film of gold, and they will be reduced to the original composition of the bar; that is, worse than standard 0·55562 in the 1,000. The trial of the pyx^[35] is supposed to be a great protection to the people against deterioration of the coinage. This opinion is not entertained by those who have witnessed the whole operation, for then it is manifest that the copper is sufficiently oxidised while melting to insure the alloy containing enough gold to enable the jury to find it “Fine, 4 grains on the pound,” or any other degree of fineness. There can be no doubt that the jury should be composed of men knowing something practically of assaying, and who would not hesitate to publish every detail of the process adopted, stating specifically the amount of loss incurred by melting the ingot produced from the selected coins. That the jury is composed of honourable men—goldsmiths—is beyond doubt; but that these goldsmiths should know how to assay is also an important matter, and should be a *sine quâ non*. It would be well if some independent member would move the House for a return of the assays by the Mint of ingots produced from worn and light coin received from the Bank of England.

Having thus specifically stated in my previous editions the facts as they exist of our coinage, I waited a refutation of my statement. I was informed that the Bank of England had made careful assays, and proved that the worn gold coin was not below the standard of fineness. I was also informed that bullion merchants in London had obtained newly-coined sovereigns, and by means of a scratch-brush had removed the film of pure gold of which I spoke, and upon a re-assay no difference had been detected. My reply was, and continues to be, the assays were either made by bunglers or were carelessly made, while with regard to the statement related *for* the Bank of England, I simply refused credence because in my own knowledge that Institution has been greatly puzzled for years as to the reason of the lower assay, and it was myself who pointed out the cause to Mr. Thomas Graham on the 4th January, 1860. That gentleman mentioned my suggestion to the then chief cashier, who preferred to think “it was due to the base coins which found their way into the Bank.” I said then as I think now—“I believe the officers of the Bank of England are far too sharp to take one base coin in five hundred.” This reply settled the discussion.

That the light and worn coin is habitually below the standard is conclusively proved by the fact that if any man should take, as the great Duke of Wellington did, a thousand sovereigns to a jeweller’s to be made into plate—and his Grace took coined money that he might be sure to have plate of the fineness of 22 carats, or standard gold—he would find, as the duke found to his cost, the plate “cut” at the Goldsmiths’ Hall as being below standard. That the case still exists is so well acknowledged by jewellers that they invariably add “fine gold” when they melt coin for plate which is to be stamped with the Hall mark.

“Monetarius,” who wrote to the *Times* from Malvern—where, singularly, Mr. Thomas Graham was then residing—and Sir John Herschel, quoted in their letters to the *Times* the report of a Belgian chemist to prove that our coinage is invariably standard. Such authorities seemed to quiet public opinion; but Lord Kinnaird moved on two occasions in the House of Lords for the “Returns of assays by the Mint of ingots produced from worn and light coin received at the Mint from the Bank of England.” The Government firmly refused these returns on the alleged ground that they were “too bulky;” Lord Kinnaird, however, explained the true cause of the refusal when he said that these returns would conclusively prove that the gold coinage, when worn, is habitually below

the standard, as is also shown by Mr. Ernest Seyd in his letter of the 20th, inserted in the *Times* of the 21st August, 1869:—

“... The British public are under the impression that English sovereigns stand highest as far as correctness in quality is concerned. In reality, the British gold coin falls much more below its standard fineness than French, American, and Russian coin.... But the best proof of the greater inferiority of the British gold coin is given by the action of the Bank of England. The Bank of England cuts light gold coin, and so renders it unfit for further circulation. The holder of such coin thus loses, in the first place, the value of the gold worn away by abrasion, and the cut sovereigns are handed back to him as being now only bullion (the loss so far amounts to from 1*d.* to 4*d.* per sovereign); the bullion now remaining is the metal of which the coin had been made, presumably at the rate of 77*s.* 10½*d.* per ounce, and worth that much, or, at the Bank rate for buying gold, at least 77*s.* 9*d.* per ounce. Yet the Bank of England will not give more than 77*s.* 6½*d.* for it. Worth at Mint price, 934½*d.*, the Bank pays but 930½*d.* per ounce—a deduction of 0·43 per cent.”

I need not express an opinion upon such a case so stated; but I can quote one of far higher value, for Mr. J. G. Hubbard, who is a profound authority, says, in his letter in the *Times* of September 4th, 1869:—

“This inaccuracy of our Mint assays was pointed out by Mr. Seyd in his intelligent and well-informed letter of the 20th ult., and I can corroborate his observations by my own experience of sovereigns sent to Russia, where the Mint, more exact than our own, ranked sovereigns at fully ⅓ carat grain worse than standard.”

Let us now see what steps are taken by the authorities, and then judge of their motive for introducing a reform so greatly needed, and which would never have been attacked had not I unceasingly urged it since 1860. In his Report on European Mints, Mr. C. W. Fremantle says:—

“I have therefore received with much satisfaction the expression of Mr. Roberts’s opinion that the process of blanching may, as far as gold coin is concerned, be discontinued without detriment to the appearance of the coin. *It is this process which leads to the deterioration of standard found to exist in gold which has been for some length of time in circulation, and in consequence of their Lordships’ decision that light gold coin shall FOR THE FUTURE BE RECEIVED BACK BY THE MINT FOR RECOINAGE, the question becomes at this moment one of PECULIAR IMPORTANCE.*”

I have emphasized some of the passages as showing that this decision is very like that of the unjust judge—not so much from conviction of right as from the necessity of the case. I do not charge the Mint with yielding to importunity; but being made to bear the loss, they are willing to amend that which they *now* openly admit to have been a wrong. Why is the opinion of Mr. Roberts quoted? why not some of the great officials at the Mint? Silence on this head is significant. Mr. Roberts will hardly desire to claim a discovery which he found in my last book, for he, too, bought one of the first copies which were sold. That I may not appear to be claiming that which is not my own, I quote the words from Mr. Roberts’s report:—

“It is evident that bars slightly below standard may be allowed to pass to the subsequent operations of coining, on the assumption, justified by calculation, that the blanching would be attended with a sufficient elevation of standard to bring the coins within the limits of fineness prescribed by law. *The film of soft pure metal, however, is removed by wear; and the coins, when remelted, will consequently be found to be below standard fineness.*

“This is a question of some importance NOW that the Mint has undertaken the operation of re-coining light pieces, and I would recommend, therefore, that the process of blanching gold coin should be altogether abolished.”

The italics are my own, and if the words so pointed out be read with reference to those I have persistently used, there can be but one conclusion, Mr. Roberts having had but a few months’ experience in the Mint, notwithstanding Mr. Fremantle’s reliance on his judgment.

As regards the trial of the pyx, useless and mischievously deceptive as it is shown to be, the Act of Parliament, 33 Victoria, cap. 10, makes it imperative to give this excuse for a dinner at the Goldsmiths’ Hall at least once a year. Had the Government assented to the proposition of Lord

Kinnaird, and nominated the Professor of Metallurgy for the time being in the Royal School of Mines, one of their own officers, and who at present is the most distinguished of metallurgists, foreman of the jury, the country could have relied upon the verdict now given honestly enough; but upon inadequate knowledge, indeed, if the reader will refer to [pages 10-12](#), it will become clear that so delicate a process as assaying should be watched by those who can appreciate its refinements, or its results can possess no judicial value.

I attended the trial of the pyx, which was held on the 17th July, 1860, with an especial view to, and an avowed intention of, publishing details of the operation. With a desire to further this object Mr. W. H. Barton, the then Deputy-Master, gave me the following particulars, and I, for myself, observed that the coin when melted was placed in an open crucible, without flux, and for part of the time without a cover: the ingot produced was of a GOOD BLACK colour from the oxidation of the copper. The objections to such a melting can hardly be exaggerated, and I am of opinion that however fairly selected—and they were honestly taken—1·310 ounces is not a sufficient bulk on which to form an opinion as to the purity of £24,654,849 of gold coin.

Coined Money.	Name of Master when Coin was Pyxed.	Value of the Coined Money Pyxed.	Weight of Pieces taken from the Coined Money for the Pyx.
Gold	SIR J. F. W. HERSCHEL, BART. THOMAS GRAHAM, ESQ.	£ s. d. 2,977,190 12 2	Ounces. 1052·200
		24,654,849 0 9¼	8300·087
Silver	SIR J. F. W. HERSCHEL, BART. THOMAS GRAHAM, ESQ.	100,450 7 0	121·899
		2,331,666 19 6	2811·543

Coined Money.	Name of Master when Coin was Pyxed.	Weight of Coins taken to melt into an Ingot for the Assay.	Loss by Melting to obtain the Ingot.	Weight of Bullion actually taken from the Ingot for the Assay.
Gold	SIR J. F. W. HERSCHEL, BART. THOMAS GRAHAM, ESQ.	Ounces. 9·116	Ounces. 0·020	Ounces. 0·766
		51·500	0·077	1·310
Silver	SIR J. F. W. HERSCHEL, BART. THOMAS GRAHAM, ESQ.	3·137	0·046	1·287
		67·881	0·652	1·781

The authorities of the Mint, having assented to part of my proposition, have yet stopped half way, and blindly considering that the blanching *alone* was the cause of loss to and fictitious value of the coinage, they will find that their loss is greater now than it was when they unwisely permitted blanching. I do not point out to them the reason why—that is now their business; but I confess that I am a little amazed at Mr. Roberts's recommendation—one that was given up by the moneyers before he was born, yet now gravely made by that gentleman in these words, "I should prefer that in order to prevent needless wear to the dies, the blanks should be annealed, after every trace of oil has been removed by washing with the aid of alkali, the utmost care being taken to avoid oxidation by exposure to air." One seems to see Mr. Roberts in the presence of men who would dwell on his instructions as would a soldier on those of his general, directing the filling of a

copper tube with gold blanks, previously divested of “oil by alkali,” so as to leave a film of lime soap neatly protecting their surface, and covering them with powdered charcoal. The tube so filled will be duly placed in the furnace, duly heated, and duly plunged into the cistern of cold water by the foreman, who, having taken an insurance ticket from the Accidental Death Company, feels quite safe, when, suddenly, a mass of copper and some pounds weight of gold blanks occupy the space from which it is hoped his head may have retired in time. Quite seriously, I have seen the cap of a tube blown suddenly off, and the blanks projected from the tube so plunged into water—from the same cause that bursts a kitchen boiler when cold water is admitted to one already red hot—with such violence as would certainly kill a man if he chanced to be in the line of fire. Do they not at the Mint understand the true cause of loss? An officer who tells his men to “go and do,” is not so likely to effect his object as if he were to show them “how to do” the work required of them; neither “bullying” nor making the men pay for losses will stop those losses of bullion. Knowledge alone is power. Knowledge alone can direct. Knowledge alone is required in the Royal Mint. There are plenty of men there who are most desirous to carry out any instructions; but, if the bugle give an uncertain sound, who can obey?

The effect of the annealing having been thus dwelt upon and explained, I will continue the description of the subsequent operations, stating what was formerly done and what had better be recurred to till experience is gained by those who are entrusted with the management of the Mint. After about three minutes’ boiling in dilute sulphuric acid, the blanks in their colander were washed under a stream of cold water, to remove the sulphate of copper and the sulphuric acid from their surfaces, otherwise these substances would deface the dies used for coining, as well as give the coin a dirty appearance. From the washing they were taken to a sieve of sawdust, [A, Fig. 16](#), into which they were thrown to be dried by friction with the hand. A revolving drum of brass lined with wood has lately been introduced for this purpose. The objections to this form of apparatus appear to be that the blanks get hardened by blows, through falling against each other, lose more in weight, and become indented on the surface, thus producing a less perfect coin. It had been wiser to have introduced a sieve with an eccentric motion, which, if filled with sawdust, would shake that dust between the blanks and dry each one perfectly, when, by the removal of a false bottom, the sawdust could be allowed to sift itself from the blanks by one or two turns of the sieve, and the latter would be as soft as they invariably were before the introduction of the brass drum. This would be a matter of importance if, as is to be hoped, the authorities are induced to abolish the process of annealing and blanching, for then it would still be necessary to remove the film of oil which is left on the blanks in the cutting-out process, and this could be done more quickly and effectively in the proposed sieve than by *any other process*. The sawdust absorbs the water from the surface of the blanks just as would a piece of sponge; but the surface is not all, for it is found that by the removal of the copper the gold on the face of the blank has been brought almost into the condition known as *frosted*; it is, in fact, to a small extent, honey-combed. The gold represents the comb, and the copper the honey, but as the copper is removed, water takes its place, and has to be dislodged from these interstices by heat. The blanks, having been partially dried in the sieve A, are put into the tray B, and from this they were shot into the colander oven C, which was closed, and then thrust into the heated chamber D, and twisted gently round and round at intervals during ten minutes. The blanks were then turned out into a sieve E, by which any particles of dust were separated; the blanks were finally, by the aid of the tray B, placed in a bag. The sawdust fell from the sieve A on to a plate of iron heated by the furnace E, by which it was dried, and became ready for another operation. The chamber D was heated by a small furnace beneath it.

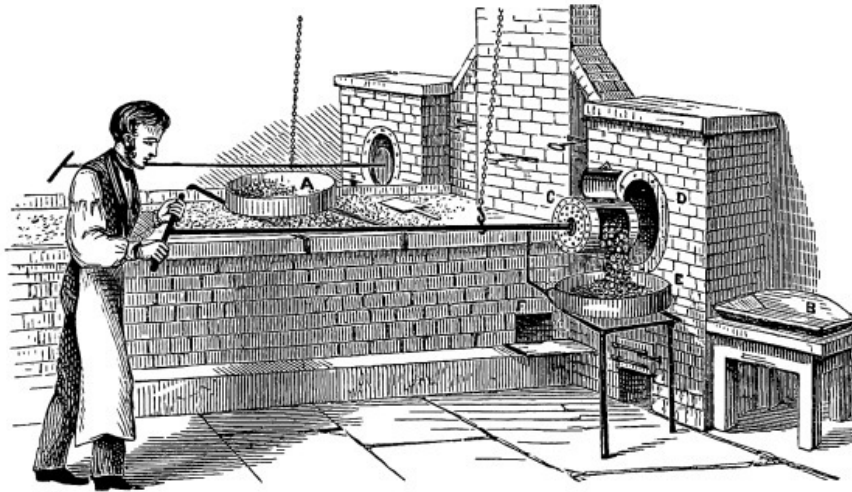
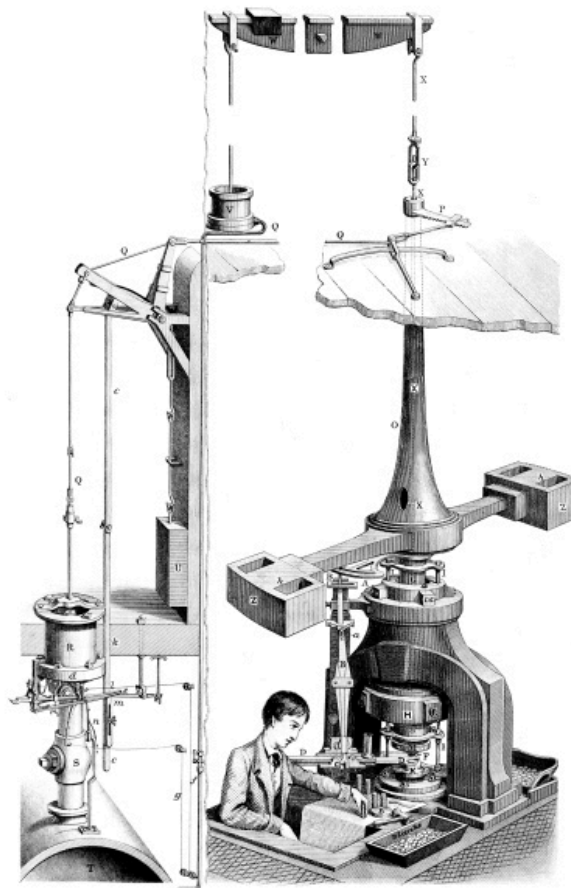


Fig. 16.—Drying Room.



BOULTON'S SCREW-COINING PRESS.

AS USED IN THE ROYAL MINT.

The blanks, having been thoroughly cleaned after annealing, are taken to the coining press, which at the present time is, as to efficiency, about what the old water-wheel is to the steam-engine; yet, as a relic of the past, it is a magnificent conception. It is perhaps the most interesting in its work of any of the coining machinery; its action is truly pleasing to watch, but one must feel that the totally deaf would enjoy the watching it at work more than do those who are endowed with the sense of hearing, because the noise of the presses would not disturb their thoughts. The noise is really painful. In these opinions Messrs. Fremantle and Napier express entire concurrence. The blanks taken from the colander oven are each by a single blow at this machine converted into coins possessing the obverse and reverse impressions, as well as the *crenated* edge, which is one of the means employed to protect the coin from the peculations of the clippers, those enemies to coin of all ages and all countries, but whose business has departed, not so much from the crenated edge as from the better balances placed in the hands of almost every man—certainly within reach of every man. The crenated edge is known to be no protection against the plan called “sweating,” and which is effected by shaking the new coins in bags, when perhaps an ounce of gold may be obtained from 1,000 new sovereigns. The sovereigns thus treated are passed, and the operator makes his profit, but the light gold is detected by the balance, not by the eye. This is not the place to discuss^[36] such a question, so that we pass on to a description of the machine which is used to give the image and superscription to coins which will be current in accordance with the law.

The steel engraving illustrates Mr. Boulton’s [screw-coining_press](#). The blank is laid by the automaton hand D on the lower die F; D retires, and the collar then rises and encloses the blank, while the upper die, fixed to the main screw of the press by the securing apparatus M, comes down with a blow estimated to be about forty tons, and, striking the blank, causes its particles to rearrange themselves, and to assume the form given by the engraving on the dies and the crenated collar which surrounds them;^[37] in other words, the plain blank becomes by one blow a coin in every way complete. The following description will convey to the reader an explanation of the processes as they arise. The press having been set at rest, with the fullest space between the dies, is called *up*—that is to say, the upper die has been raised from the lower die, and in this position the automaton hand D has conveyed a blank from the tube E to, and holds it over, the lower die F; upon the first motion of the press downwards, the eccentric wheel or cam A causes the lever B, which works on the pivot *a*, to withdraw D by the pin C. The lever B may be lengthened or shortened at pleasure by an arrangement against the lower B. The first motion towards the withdrawal of D causes its finger or hand to open, and to release the blank, which falls upon the face of the lower die. The opening of the finger, or hand, is effected by a pin which works in a slit in the movable finger of D, near to C. The continuation of the motion which draws back D, causes the rods I, which are carried by the main screw G, to release the collar K, by the levers J. The collar K, being thus relieved from downward pressure, rises by the elasticity of the springs L until it wholly encloses the blank which has been left on the lower die. The rods I pass through the shoulder of the frame of the press, and are destined for another service besides that just described; for the main screw G, which travels through a female screw fixed in the frame of the press (as may be seen by the dotted lines between G and H), thereby receives its power to rise or fall upon each part of a revolution, whereas the upper die must strike the blank a blow just such as would be struck by a hammer, and without a twisting motion; therefore that motion of G is lost just below H, where G fits into a cup suspended rigidly on, and is prevented from twisting by, the rods I, as shown at N. To the lower part of the cup N is fastened M, which carries the upper die. So that, just at the moment that the levers J have permitted the collar to enclose the blank, the upper die reaches the full force of its blow, and comes upon the blank. The force of the blow, by converting the blank into a coin, causes an instantaneous recoil of the screw G, which is assisted in its rise by

a kind of balance just equal to its weight. The nature of this arrangement will be better understood by reference to the engraving, where the funnel-shaped tube O is seen to be traversed by a rod X, represented by dotted lines, and which terminates in the head of the screw G near the lowest X; the opening in the funnel being for the convenience of fixing this rod to G. At Y is a swivel, where the twisting motion of X, given by G, is lost. X is connected at its upper end with a balanced beam W, the other end of which is in communication with the chamber V by a rod which carries a piston working in V. The chamber V is not wholly exhausted, but is in connection by a tube with the partially-exhausted chamber T, an arrangement which, while it becomes a counterpoise to the weight of the press, is used as a regulator of the blow to be given, so that it is quite possible to coin blanks of differing denominations by the agency of this chamber without altering the vacuum in the large vessel T.

The motive power of the press, that which brings down the die with force, is gained by the pressure of the atmosphere upon a piston in the cylinder R, and is conveyed to the press as follows:—The lever P is fixed firmly on to the trumpet-shaped tube O, and is connected by a series of rods, Q, with the piston in R, from which the tube S communicates with T. The large chamber T is kept in a constant state of partial exhaustion by the action of a thirty horse-power steam-engine, through a well-made iron tube of considerable length. The boy who works the press starts it by first pulling a cord *f*, terminating in the box in which he sits; the cord *f*, by its spring *n*, releases the valve *d* at *i*, and permits it to act when required; he next pulls the cord *g*, which by its loose lever opens the valve *j*, through which the atmosphere passes freely across the bottom of the piston in the cylinder R into the tube S, and so into the partially-exhausted chamber T; this would cause a rush of air through the valve *d*, but *d*, at the instant of its release by the line *f*, is closed by a spring. Atmospheric air expands instantly if the pressure be removed from it, so that immediately on the opening of the valve *j*, the air beneath the piston in R expands, and part of it, entering the chamber T, produces an unequal pressure upon the upper and lower sides of the piston, and the result is that the piston is forced down by the weight of an atmosphere it is unable to support. By the time that the piston has completed its stroke, the rod *c* has brought a button, which it carries on its farther side, and situated at *k*, down so low that it strikes the lever *l*, and closes the valve *j*, thus stopping the expansion of air below the piston, while the button *m* has at the same instant permitted the falling of the lever *e*, and the consequent opening of the valve *d*; when the recoil of the screw G takes place, the piston in its descent expels (by compression) the small amount of air remaining in the cylinder R through *d*, thus opening it so that air can rush in at this valve and permit the rising of the piston. The rising of the piston insures the removal of the button at *k*, and so soon as *k* is removed, *l* is forced up by a spring at its farther end, and opens the valve *j*, thus giving the means of consecutive action to the press. If it be desired to stop the press, the boy loosens the line *f*, and so permits the spring *n* to keep the valve *d* permanently open. The boy, before supplying blanks to the tube E, piles them on an iron block *b*, and flattens each blank in the pile by one or two blows of a hammer on the top of the pile.

In “European Mints” are some apt words, with which I conclude my remarks on this great invention of Mr. Boulton—great beyond compare at the period of its birth, great beyond admiration if its results be estimated faithfully and gratefully by a thoughtful people. We in our day are indebted to Mr. Boulton for his press; yet its age has passed, and as a park loses its timber, so must our Mint lose—sorrowfully to all who love the past—its glorious mementos of men who dwell in history to make present men mere manikins. Original minds are not now tolerated; subservience if you will; but invention—not if a man values or depends upon his office. Mr. Napier says truly, “The press just mentioned and its apparatus are complicated, scattered, and unhandy, reaching into three apartments, and require a foundation of great solidity, and a special

construction of building. The noise and vibration accompanying their performance are disturbing to a greater extent than in the case of the blank cutting screw press to which reference has been made.”

It has been conceived that coins are made of two discs of metal soldered together, and that the crenated edge is intended to hide the join. This notion has arisen from the occasional appearance of a cracked coin in circulation; the true explanation of the cracked coin being that at the time of pouring the fluid metal into the mould, an air bubble has been enclosed, and this air bubble has rent the fillet asunder at the time of its being rolled; but the separation does not exhibit itself until the final annealing, when it is too late for its discovery: recently, however, it has become a custom to ring the coined gold before it goes into circulation, and thus to detect and stop such defective pieces.



Fig. 17.—Petition Crown.



Fig. 18.—Inscription on edge of the Petition Crown.

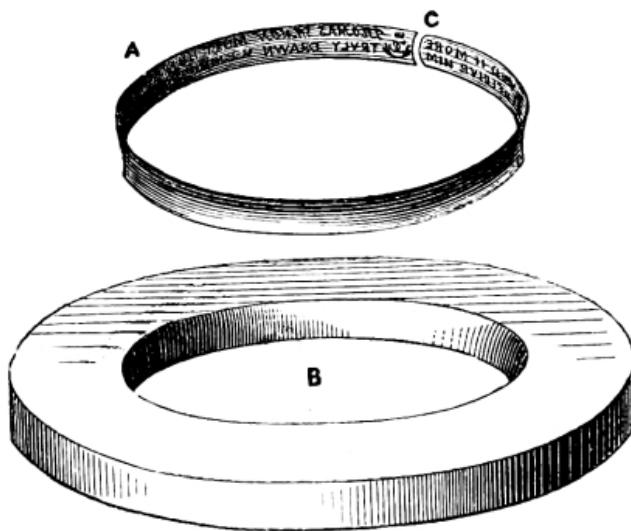


Fig. 19.—Fillet and Collar.

The origin of the crenated, commonly called the “milled,” edge (thus giving the name of the instrument to the substance operated upon), was the desire to make any clipping of the coin easily

noticeable. Many devices were used, but John Evelyn suggested the motto, *Decus et tutamen*,^[38] and this was retained for some coins until 1854. The crenated edge is now produced, without exception, and it is really the best, as admitting of easier detection if counterfeited. The distinctive marking of the edges of coins is of very early date, and was performed by various species of hand labour, each in its turn easy of falsification. At length this was effected by machinery, until finally the coin was formed by striking the blank while enclosed in a collar. The use of the collar, and its first invention, are somewhat doubtful; Mr. Hawkins, however, appears to have established the fact of its having been used at so early a date as William the Conqueror (*vide* “Ruding,” vol. i. p. 158); but the earliest coin which gives a clear notion of the use of a collar is the celebrated Petition Crown of Simon, which he coined for and presented to Charles II. [Fig. 17](#) represents this very highly-prized coin, one of which was sold a few years since for £275. This wood engraving is by Messrs. Cheshire and Dickenson, from a fac-simile drawing taken by Mr. E. S. Gibson; it is an exact copy of the coin in the Mint collection. It is surrounded on its edge by a petition in raised letters, of which [Fig. 18](#) is a fac-simile. The late Mr. W. H. Barton thought that this coin must have been struck while surrounded by a collar, or it could not have been produced at all. This opinion induced a further search for the collar which was used, and which is believed to be in existence, and to have been seen by men now living,—it is probably amongst the collection of curiosities of coining left by Mr. W. H. Barton. Mr. Barton suggested that the petition must have been engraved on a fillet of steel, which was then coiled up in a collar, as indicated by [Fig. 19](#), and that the blank to be coined was then placed in the coiled fillet of steel. The coin having been struck, the fillet with the coin was knocked out from the collar, just as medals are now knocked out; and, once free, the fillet sprang off from the coin. This conjecture was so strongly confirmed by minute examination, that it may be interesting to represent the mode of operation by means of [Fig. 19](#), where A is the engraved fillet coiled up and just ready to drop into the collar B; the opening C which is left is copied from the petition on the coin, which shows the metal to have been forced through this part of the fillet, where its ends, which were cut to fit each other, did not quite meet. The head of this protruding part has been filed off, the file marks still remaining visible on this part of the edge of the coin. This fact renders Mr. Barton’s conjecture almost a certainty. The coin weighs 517·10 grains; its average diameter is on the obverse 1·5784 inches, and on the reverse 1·5837 inches, both measures having been taken on the extreme edge. The petition is somewhat sunken and rounding inwards; if therefore it be measured from its lowest part, the centre of the edge of the coin, the diameter of the coin is 1·5741 inches, giving 0·0043 inch which must be worn away before the top of the letters of the petition can be touched in ordinary wear; thus the coin would circulate for centuries, and its petition still remain untouched. The coin being 0·0053 inch greater in diameter on its reverse than on its obverse side, admitted of its being more readily punched out from its collar, which was doubtless cut to admit of this design being carried out. The engraving on this coin is the cause of its great value, Simon having so perfectly carried out the idea that coins should represent sculpture, that this specimen stands wholly unrivalled as a coin. It has led to the impression that so good artists are not to be found at this day as flourished then. The fact is that artists seek such employment as remunerates them for their study; and until their conditions of employment are satisfied we must abstain from the desire to see coins such as were produced in ages past, and which gave an honourable name to the country producing them. It may be hoped that this means of preserving the history of the age in which so much has been done may be continued. It was probably from this collar of Simon’s that Sir Isaac Newton gained his idea for that which he is said to have invented. The collar of the present day leaves nothing to desire, for it gives so distinctive a mark to the edge of the coin that any mutilation must be detected; while forgers are unable to produce coins to imitate those struck in it, unless they use the same means, which would incur too great an expense. The coinages of Victor Emmanuel illustrate beautifully

how easily coins may be struck, so that, after years of wear, they still represent the original more than creditably. To take an instance, a bronze coin, 5 centesimi, has been purposely subjected, for nine years, to more than ordinarily rough usage. This coin is little worn; the whole of the features are still perfect, except that the hair is worn off above the ears, and part of the moustache is removed. The secret of success in this coin appears to be that the ear and adjacent parts are slightly sunken, while the whole of the work and inscription are kept well below the protecting edges, and there is not an indistinct figure or letter on the coin.

In the British Mint false notions lead its authorities to believe that apparent saving is real economy; so far from such being the case, if our coins carried a better design they would not only look better, but wear longer, and still maintain their weight. While it is my intention to allude in another place to the large loss occasioned by our silver coinage, I submit to the reader the facts arrived at by Mr. William Miller, the late chief cashier in the Bank of England, who devoted great attention to the rate of wear of coins. He found that coins which had lost their protecting edges wear far more rapidly than those with a protecting edge, and showed me tabular statements demonstrating that fact. It was the intention of Mr. Miller to publish these tabulated results. However, his painfully early removal has, I fear, prevented that benefit being given to the public. His brother has no knowledge of their having been published; but Mr. George Forbes, the chief cashier in the Bank of England, has courteously given me a table closely allied to those of which I speak, and another to which I shall refer hereafter. I now, with permission, print this table, showing the rates at which coins of the different reigns have worn, and which rates all tend to prove clearly Mr. Miller's opinion to have been correctly formed. The fact of the increasing rate of wear would appear to be explained by the larger surface exposed to friction by the removal of the protecting edge. When a little later I come to the discussion, on [page 155](#), of the second table, I shall have occasion to say a few more words on this, which seems to find a fitting place on the preceding page, but contains matter which will cause too great a digression if dwelt upon now.

A STATEMENT SHOWING THE RESULTS OF SOME EXPERIMENTS MADE UPON A LARGE NUMBER OF SILVER COINS, OF WHICH 169,000 WERE EXAMINED SINGLY, WITH THE VIEW OF ASCERTAINING THE PRESENT CONDITION OF THE SILVER CIRCULATION, AND THE LOSS BY WEAR.

It should be observed that the average of loss by wear in a century, given in the Table, is founded upon the wear which the several coinages have undergone up to the present time; but as the rate at which a coin wears is an increasing rate, it is obvious that the actual loss in a century would be much greater than that here given:—

	Amount Coined since 1816.		Withdrawn from Circulation.	Remaining in Circulation.
		£	£	£
CROWNS	Victoria	117,414		117,414
	William IV			
	George IV	140,726		140,726
	George III	321,750		321,750
HALF-CROWNS	Victoria	1,043,251		1,043,251
	William IV	380,556		380,556
	George IV	1,113,848		1,113,848
	George III	2,387,088	252,870	2,134,218
FLORINS	Victoria	1,541,161		1,541,161
SHILLINGS	Victoria	2,609,504	490	2,609,014

	Amount Coined since 1816.		Withdrawn from Circulation.	Remaining in Circulation.
	William IV	412,038	49,944	362,094
	George IV	878,922	350,622	528,300
	George III	3,304,224	867,674	2,436,550
SIXPENCES	Victoria	1,182,339	298	1,182,041
	William IV	281,994	34,614	247,380
	George IV	81,378	47,738	33,640
	George III	919,314	312,250	607,064
£ 16,715,507			1,916,500	14,799,007

LEGEND:

(A) = Average of *Loss by wear* in 100 *Years*, at the rate at which the Coinages of the several reigns have worn, up to the present time.

(B) = Loss by wear on Amount withdrawn.

(C) = Loss by wear on Amount in Circulation.

(D) = Number of Coins examined singly.

		(A)	(B)	(C)	(D)
		Per Cent.	£	£	
CROWNS	Victoria	5·50			
	William IV				
	George IV	7·04		15,776	4,000
	George III	7·46			
HALF-CROWNS	Victoria	14·17			
	William IV	15·30			
	George IV	15·80	13,546	250,270	20,000
	George III	15·92			
FLORINS	Victoria	15·00		10,147	5,000
SHILLINGS	Victoria	26·55			
	William IV	26·82			
	George IV	28·74	167,684	413,044	80,000
	George III	29·96			
SIXPENCES	Victoria	37·00			
	William IV	39·00			
	George IV	44·80	72,420	221,380	60,000
	George III	45·00			
			253,650	910,617	169,000
				253,650	
			£ 1,164,267 total loss by wear.		

BANK OF ENGLAND, *December, 1860.*

W. MILLER.

As further showing the rate of wear, I append a note which I have copied from the eighth volume of the *Journal of the Society of Arts*, written by Mr. John Miller, to whose courtesy I have

often been indebted, and who shows that the general rate of wear is rather higher than would appear from the above facts, for his figures prove a loss by wear to the extent of £4 11s. 8d. on a hundred sovereigns in a hundred years, as will be seen in his letter which follows:—

“BANK OF ENGLAND, GOLD WEIGHING ROOM,
“29th August, 1859.

“SIR,—According to your wish I collected one hundred sovereigns of the date one thousand eight hundred and twenty, which, on weighing, I find have lost in value one pound six shillings and sevenpence.

“As these sovereigns were taken from parcels sent in from different places, and at different times, during a long period, I think the experiment is as likely to give a correct estimate of the loss which the coin undergoes by wear as any that could be tried.

“I have the honour to be, Sir,
“Your very obedient servant,
(Signed) “J. MILLER.

“A. Latham, Esq., Deputy Governor.”

That I may convey as accurate information as possible I also give the following tables from the same volume of the Society of Arts. These tables seem so clear as to render detailed explanation unnecessary. It would appear that coins wear pretty much in proportion to the kind of usage they receive; for Professor Jevons, in his pamphlet “On the Condition of the Metallic Currency of the United Kingdom,” shows indisputably that at the West End of London they suffer less than at the East End, where they are thrown with force on rough and often sandy benches while at the West they are treated with more consideration.

RESULTS OF SOME CAREFUL EXAMINATIONS MADE IN 1858 OF LIGHT GOLD AND SILVER COIN OF THE EARLY YEARS OF THE PRESENT SYSTEM OF COINAGE, 1817 TO 1825, TAKING 1822 AS THE MEAN OF THOSE YEARS.

LEGEND:

- (A) = Mint Weight of each Coin.
- (B) = Mint Weight of £100 worth.
- (C) = Weight of £100 worth of old Coin.
- (D) = Loss in 36 Years, 1822 as the Mean.

	(A)	(B)	(C)	(D)
	Ounces.	Ounces.	Ounces.	Ounces.
Sovereign	·25682	25·682	25·360	·322
Half-Sovereign	·12841	25·682	25·070	·612
Half-Crown	·4545	363·636	345·850	17·786
Shilling	·1818	363·636	315·575	48·061
Sixpence	·0909	363·636	296·950	66·686

The above data applied to 100 years give the following results:—

LEGEND:

- (A) = Mint Weight of £100 worth.
- (B) = Loss of Weight of £100 worth in a Century.
- (C) = Loss of Value in a Century.
- (D) = Loss of Weight of each piece in a Century.
- (E) = Loss of Volume from 1,000 parts in a Century.

	(A)	(B)	(C)	(D)	(E)
	Ounces.	Ounces.	£ m.	Ounces.	
Sovereigns	25·682	·895	3·485	·00895	35·
Half-Sovereigns	25·682	1·700	6·620	·00850	66·
Half-Crowns	363·636	49·406	13·587	·06176	136·
Shillings	363·636	133·503	36·713	·06675	367·
Sixpences	363·636	185·239	50·941	·04631	509·

(Signed) W. MILLER.

BANK OF ENGLAND, *8th October, 1859.*

In this interesting table Mr. Miller curiously shows that the loss of value and of volume bear equal relations to a coin, from which I infer that he went so far as to measure determinately the amount of water displaced by such coins as are referred to in these very valuable researches; that he possessed a rigidly honest mind I know from occurrences in my own dealings with him, and therefore I am sure his figures are the results of actual experiment, and so, entirely reliable.

These facts show that before it is determined to withdraw our coins when they have reached the age of eighteen years, the Mint authorities should produce coins with a more perfect protecting edge, as was done at my suggestion for shillings and sixpences, and then by abolishing the annealing of blanks give to our coins a medium hardness so that they will be able to bear comparison with the Italian as regards wear and tear, and retain a juvenile appearance after the exercise, instead of their present smooth worn surface.

The method of manufacturing the collar with the interior crenated circle, which is employed to give the protecting edge to the finished coin, is very simple:—A piece of flat, soft steel is bored with a hole of the required diameter, and is fixed in a lathe so that it may be made to revolve rapidly at pleasure. While the partly-formed collar is revolving in the lathe, the slide-rest is made to place a kind of cogged wheel, carried on a rod of steel, into the hole which has been bored; by a screw in the slide-rest, the cogged wheel is brought down until it touches the inside of the partly-formed collar; at the moment of contact, the cogged wheel is turned round by the friction of the collar against it, and its continued pressure upon the inside of the collar causes the latter to receive an imprint from the cogged wheel. When the imprinting is complete, the collar is hardened, and it is then fit for use at the coining press. The system of manufacture is supposed to be the most perfect; but it is manifest that no two collars are produced of exactly the same internal diameter; hence the coin issuing from each collar of the same denomination must, as it does, vary in diameter; but this is of small consequence, the difference being so minute that it is not detected unless by very accurate measurement; indeed, coins coming from the same collar must vary in size, since the collar wears away by use to a certain extent. The diameter and consequent thickness of a coin are not determined by law, but by its suitableness for the public to whom it is to be issued; consequently the usual mode of arriving at the diameter of a coin is to make it of such a size that it will emit the most musical sound it is capable of. This rule was not sustained in the case of the small florin—better known as the “godless florin;” hence, coupled with the omission of the *Dei Gratia*^[39] in its superscription, the coin fell into disfavour.

The following measurements, taken from a set of proof coins of the present reign, will convey an idea of the probable size of any coin of the realm; but from the reason before stated, a man need not be disappointed should he find the diameters differ from any he may examine by an accurate gauge. While stating the diameters (which never vary beyond a few thousandths of an inch), it is

thought proper to give the legal weight and legal tender^[40] of each denomination of coin current in Great Britain:—

Denomination of Coin		Diameter.	Weight in Troy.		Legal Tender.	
		Inches.	Grains.	Ounces.	To the value of—	
Gold.	Sovereign	0·8680	123·2744	0·2568	The highest sum known.	
	Half-Sovereign	0·7622	61·6372	0·1284		
Silver.	Crown	1·5048	436·3636	0·9090	Forty shillings sterling.	
	Half-Crown	1·2714	218·1818	0·4545		
	Florin	1·1826	174·5454	0·3636		
	Shilling	0·9296	87·2727	0·1818		
	Sixpence	0·7648	43·6363	0·0909		
	Fourpence	0·6456	29·0909	0·0606		
	Threepence	0·6383	21·8181	0·4545		
	<i>Maundy.</i>					
	Fourpence	0·6957	29·0909	0·0605		
	Threepence	0·6383	21·8181	0·4545		
	Twopence	0·5294	14·5454	0·0303		
	Penny	0·4388	7·2727	0·0151		
Copper.	Penny	1·3502	291·6666	0·6076	One shilling sterlg.	Proclaimed illegal since 31st December, 1869.
	Halfpenny	1·1155	145·8333	0·3038	Sixpence "	
	Farthing	0·8575	72·9166	0·1519	Sixpence "	
	Half-Farthing	0·6953	36·4583	0·0759	Sixpence "	
Bronze	Penny	1·2000	145·8333	0·3038	One shilling "	
	Halfpenny	1·0000	87·5000	0·1822	One shilling "	
	Farthing	0·8000	43·7500	0·0911	One shilling "	

The press for coining is still defective: one of its defects is the production of imperfect coins, of which [Fig. 20](#) is a common type. It is produced by the failure of the automaton hand to place the blank exactly on the lower die, so that the collar, when it rises, fails to enclose it; hence, when the upper die descends to strike the blank, only that part of it which is over the die is impressed, the collar being forced down by the pressure of the other part of the blank upon its upper surface. On other occasions the automaton hand drops its blank too early, or fails to bring it forward at all: on such occasions it sometimes receives a severe pinch, and retires with an impression produced on its end by the dies; and on either of these occurrences, the dies may come forcibly together, and disfigure each other, so that the next coin which is produced carries with it on each side a faint impression of both dies as well as its proper impression. This could never happen if the levers which carry the weights Z on the screw of the press performed actually the whole of the functions for which they are designed. A contrivance has been adapted to the coining presses to stop their action should the automaton hand fail to bring forward a blank for coining. This, after months spent in attempting its use, is set aside because it increases the evil it was intended to remedy. [Fig. 21](#) is a representation of an imperfect half-crown which had been submitted twice to the process of coining, and by oversight passed into circulation in 1818. It had been current, and was taken in ordinary change in July, 1861; it is much worn, but on each side may be clearly traced the impressions of the obverse and of the reverse, singular effects being produced where the crown of the reverse fits so neatly on to the back of the head of his Majesty; and where, on the obverse, the shield displaces his Majesty's face. This could hardly have been an accidental occurrence; it would seem that the workman of the press must have struck the coin properly, and then taken some pains

to fit it into a larger collar and give it a second blow. If the collars were all of precisely the same size such an act would be impossible; but there are sufficient variations in the size of the collars to admit of this explanation. Some accidental productions have been called by ignorant writers *incused coins*, and conjectural histories of them have been published; when, in fact, their history is simply this: it so chanced that the coin last struck remained upon the upper die, and gave its lower-side impression to the upper side of the next blank which was coined. This is no uncommon occurrence, and the effect of it would be that such a coin would have the same device on each side; but of course one would present the design in relief and the other in intaglio.



Fig. 20.—Brockage.



Fig. 21.—Brockage.

Such being some of the imperfections of Mr. Boulton's [coining press](#), it may perhaps be assumed that they can be readily overcome; but I am not aware of the existence of any press which is free from them. The French press has, however, many advantages, and the Mint authorities have recently had a new one made, from which extraordinary results were anticipated, but it now appears that this is simply another improvement of the usual type, and that it has cost about £1,500 and remains on the premises of the manufacturers.

As the coined money issues from the press it is collected in trays and examined; all imperfect coins, curiously termed “brockages,” are picked out; and the good coins weighed into drafts of 701 sovereigns, equal in weight to about 180 ounces: these are at present sent to the weighing room, where they are examined for imperfect coins by passing over a kind of blanket, so arranged on a series of rollers that each coin lying on its surface can be seen as the blanket revolves. The blanket covered with coins is carried round a set of rollers, and thus produces them on the upper side of a lower blanket, when the reverse side can be seen. The effect of this machine is to worry the eye, if constant watching be enforced, for no one can watch a string of coins in continuous motion *and see each one*. A very minute alteration of this machine would cause it to stop for a few seconds at

intervals, and during that stoppage the accustomed eye could readily detect and select the imperfect coin.

After this overlooking, the coins are rung as blanks used to be, and then weighed separately, *all* the rejected going to the melting-pot, by which a waste^[41] of 50 per cent. is incurred at the extreme end of an elaborate process; but this unwise course, it is to be hoped, will soon be abandoned when its expensive and perfectly useless extravagance is considered. After the various operations of the weighing room the coin is collected and weighed into separate bags, each containing 701 sovereigns; the exact weight of the contents of each bag is noted; and, the bags having been placed in a truck, are taken to the Mint Office, where they undergo what is called *pyxing*, which is simply the selecting from each and every bag a pound weight, from which two coins are taken; each coin is weighed and its weight recorded. Of these coins one is placed in the hands of the assayer to determine its value as to per-centage of gold, and the other is sealed in a packet, which is placed in a *pyx* for the trial of *THE PYX* at Westminster—an ancient process now useless, because any skilled man can detect by assay a deterioration of the coin. These particulars having been taken, the coin is in due course delivered to the officers of the Bank of England, who conduct it in amounts of about £140,000 to the Bank in a waggon.

The law enacts that 20 lbs. weight troy of standard or crown gold shall be made into 934·50 sovereigns, and this proportion gives the means of determining the theoretical weight of one sovereign; for if the 20 lbs. troy weight produce 934·50^[42] coins, it is only necessary to divide by that number the number of grains in 20 lbs. troy, and the quotient will represent the weight of a single sovereign, viz., 123·2744783306581059 troy grains; therefore the journey of 701 sovereigns should weigh 180·032102728731942215 troy ounces, and a million 256821·829855377 troy ounces, equal to 7·8618927506797 tons avoirdupois; hence the War Indemnity of France will weigh 1572·37855 tons.

Since the coins of a people must represent food in proportion as they are of specific weight, it is to be regretted that the law does not fix the weight (with a remedy) of the individual sovereign, instead of fixing the number of coins in 20 lbs. weight, and giving a remedy^[43] of 12 grains on the pound troy. Irrespective of law, the practice of the Royal Mint is to apportion the estimated amount of remedy (or latitude for error) to each coin; but in America the pound weight alone is studied; hence the individual coins vary so considerably in weight that it pays as a commercial speculation to select the heavy coins, and to sell them as bullion; thus leaving the light coins in circulation, to the dishonour of the nation, because if a man take 1,000 American gold coins to the Bank of England they will not be received at their nominal value, but as bullion; hence the loss becomes personal, and so the coins of America cannot, and do not, stand on a par with those of England.

The paragraph above has stood unaltered since the first edition, and I reproduce it because its history is perhaps curious. Mr. Graham, soon after I entered the Mint, commenced urging reasons why the system of weighing coins individually, which he considered expensive, should be abandoned; as well as I was able I met his proposition by firm but adverse arguments, until at last he “ordered me to omit the weighing of individual pieces, and to *pound the work* as is directed by law.” I therefore asked him, as the matter was a very grave one, to *write the order*. This he refused to do, and thought it hard of me to require a written order, and so the matter dropped; when, however, Mr. John Graham was appointed, Mr. Graham told me that he could now carry out his wishes without a “written order.” In the meantime he had asked me “to write a treatise on coining for Tomlinson’s Cyclopædia—in fact, I have already referred Mr. Tomlinson to you, and he will call; your difficulty will be to compress your information into a sufficiently small space.” After so long an attempt to preserve the accuracy of our coinage, I felt unwilling to allow such an

opportunity to pass, and therefore placed my opinions in print, and thereby gave Mr. Graham so deep an offence that he refused to place my book in the Mint library, although I gave him a copy for that purpose; it was still absent from those shelves when I left the Mint. In August, 1864, Mr. John Graham, acting under the "Master's orders," directed "that the contents of only one bag in three should be weighed," and this was carried out till pieces which bore the device of florins found their way to the Bank of England, and amongst them were found some of the intrinsic value of 1s. 6d., while others reached so high as 2s. 9d. Mr. Graham was greatly distressed about this occurrence, and, notwithstanding his brother's position, directed me to stop this new system at once. He then attributed the innovation to myself, and blamed me for all that had happened. I therefore told him that he, and he alone, was to blame, for I had persistently refused to do it unless fortified with a written order. This caused, as I felt at the time, our final separation; but had I been willing to accept blame which belonged, as I told him, to himself first, and in a lesser degree to his brother, he would have made it a fair ground for my removal from the Mint—a result he had earnestly endeavoured to effect. I was so far free from blame that I was actually absent from the Mint part of the time, and during the remainder refused to enter the room where the blanks were cut, such action on my part resulting from a desire to protest firmly, by deed as I had done by word, against what I knew to be a fatal step. The wisdom of my proposition to apportion the remedy to the individual piece was so apparent that the Legislature incorporated it in the Act 33 Victoria, cap. 10, and as detailed information may be of special value to many, I quote the first schedule to that Act on the opposite page.

That others concur in my thus claiming the origination of the apportionment of the remedy to the individual piece, is, I think, clearly demonstrated by the subjoined letter, which appeared in the Money Market and City Intelligence of the *Times*, on Thursday, March 10, 1870.

"The following relates to the coinage question:—

"March 8th.

"SIR,—In your article of this day you draw attention to Mr. Ansell's new work, 'The Royal Mint,' which I have studied carefully. I shall feel obliged if you will permit me, through you, to direct the attention of members of Parliament to the suggestion thrown out by Mr. Ansell in his 'Treatise on Coining' in 1862, and repeated in the book above alluded to.

"In the New Coinage Bill, which is to be discussed on the 10th inst., the Chancellor of the Exchequer would appear to have adopted in part the proposal of Mr. Ansell, to make the remedy on the individual coin, instead of on the pound weight of coins, as is now the law. Mr. Ansell gives very strong reasons, besides the expense, why 0·2568 grain should be enforced as the legal remedy on the individual coin, and surely, while the legislature is invited by Mr. Lowe to adopt 0·20 grain, it would be wise that those reasons should be considered, for the Act, once obtained, will be difficult to repeal, and the Mint seems capable of improvement.

"I am, Sir, your obedient servant,
"NUMMORUM FAMULUS."

I may add that Mr. Fremantle has purchased a copy of my last impression for the Mint Library.

The coinage of silver is precisely the same in every detail as the coinage of gold; but there are one or two points that require to be dwelt upon. Weight for weight, silver is more bulky than gold, and is coined in larger quantities; hence it has become important to expedite its coining by every means. Thus, in the process of annealing, it is simply placed unprotected, except by a lid of iron, upon an iron truck, and run into the annealing furnaces, from which, after twenty minutes, it is withdrawn and plunged at once into cold water. In the annealing of the silver blanks it is important to have the copper of the alloyed metal removed from their surface; this is effected by annealing the blanks for from seven to ten minutes in open pans, when the copper becomes oxidised, and is removed by the process of blanching, described under the coining of gold. ^[44]

FIRST SCHEDULE.

Denomination of Coin.	Standard Weight.		Least Current Weight.	
	Imperial Weight. Grains.	Metric Weight. Grams.	Imperial Weight. Grains.	Metric Weight. Grams.
Gold:				
Five Pound	616·37239	39·94028	612·50000	39·68935
Two Pound	246·54895	15·97611	245·00000	5·87574
Sovereign	123·27447	7·98805	122·50000	7·93787
Half-Sovereign	61·63723	3·99402	61·12500	3·96083
Silver:				
Crown	436·36363	28·27590	—	—
Half-Crown	218·18181	14·13795	—	—
Florin	174·54545	11·31036	—	—
Shilling	87·27272	5·65518	—	—
Sixpence	43·63636	2·82759	—	—
Groat or Fourpence	29·09090	1·88506	—	—
Threepence	21·81818	1·41379	—	—
Twopence	14·54545	0·94253	—	—
Penny	7·27272	0·47126	—	—
Bronze:				
Penny	145·83333	9·44984	—	—
Halfpenny	87·50000	5·66990	—	—
Farthing	43·75000	2·83495	—	—

Denomination of Coin.	Standard Fineness.	Remedy Allowance.		
		Weight per piece.		Millesimal Fineness.
		Imperial Grains.	Metric Grams.	
Gold:				
Five Pound	Eleven-twelfths fine gold, one-twelfth alloy; or millesimal fineness 916·66.	1·00000	0·06479	0·002
Two Pound		0·40000	0·02592	
Sovereign		0·20000	0·01296	
Half-Sovereign		0·10000	0·00648	
Silver:				
Crown	Thirty-seven fortieths fine silver, three-fortieths alloy; or millesimal fineness 925·	1·81818	0·11781	0·004
Half-Crown		0·90909	0·05890	
Florin		0·72727	0·04712	
Shilling		0·36363	0·02356	
Sixpence		0·18181	0·01178	
Groat or Fourpence		0·12121	0·00785	
Threepence		0·09090	0·00589	
Twopence		0·06060	0·00392	
Penny		0·03030	0·00196	
Bronze:				

Denomination of Coin.	Standard Fineness.	Remedy Allowance.		
		Weight per piece.		Millesimal Fineness.
		Imperial Grains.	Metric Grams.	
Penny	Mixed metal, copper, tin, and zinc.	2·91666	0·18899	None.
Halfpenny		1·75000	0·11339	
Farthing		0·87500	0·05669	

The weight and fineness of the coins specified in this Schedule are according to what is provided by the Act fifty-six George III., chapter sixty-eight, that the gold coin of the United Kingdom of Great Britain and Ireland should hold such weight and fineness as were prescribed in the then existing Mint indenture (that is to say), that there should be nine hundred and thirty-four sovereigns and one ten-shilling piece contained in twenty pounds weight troy of standard gold, of the fineness at the trial of the same of twenty-two carats fine gold and two carats of alloy in the pound weight troy; and further, as regards silver coin, that there should be sixty-six shillings in every pound troy of standard silver of the fineness of eleven ounces two pennyweights of fine silver and eighteen pennyweights of alloy in every pound weight troy.

In annealing the precious metals, it is intended to use large retorts of Stourbridge clay instead of the naked flame, as is the present practice. The advantage of using retorts will be the protection they will afford to the substances to be annealed, and the uniform temperature at which they may be kept for any required time. So great are the obstacles to improvement in the British Mint, that, although this suggestion was made by me, and plans were prepared, in June, 1861, the necessary alterations have not yet been attempted; indeed, it has been reserved to Mr. P. F. Comber, the intelligent chief coiner of the New Mint in Melbourne, to adopt this suggestion. The furnace for his use has been made by Messrs. Deane, of Arthur Street East, London Bridge, who, from experience, know this kind of furnace to be successful; indeed, it has been long used to gain higher temperatures than are required for metals in the annealing of glass, pottery, &c. I have recently been permitted to make a long series of experiments at the works of the Patent Fuel Company, on their wharf at Deptford, and these have convinced me that their mode of converting hydrocarbons into vapour, and burning that vapour with a proper supply of air under pressure, could be applied in the Mint with great advantage to the heating of the annealing furnaces, especially if the authorities of that department should adopt the retorts proposed.

In my previous editions I have suggested the use of these retorts, and I am glad to see that the Royal Mint is likely to adopt them, for Mr. Napier recommends them in his Report, and Mr. Roberts not only coincides in that recommendation, but specifically says, “It is a question for consideration whether muffles of clay instead of iron, and gas as fuel, might not be used with advantage.”

The coinage of bronze is somewhat new; that is to say, it has, after centuries, been re-introduced; and as little is known respecting it, it will be well to consider one or two circumstances connected therewith. The dimensions of the bars are previously given,^[45] those measurements were determined by me after much troublesome experimenting, and the bars are found to be capable of producing the best blanks, as regards equality of weight one for another, with the least labour. In the rolling of bronze there are some singular facts to be noted; for instance, the finding the same metal at one time ductile, and at another absolutely brittle; yet if the bronze has been properly melted, with due precaution to avoid the access of atmospheric oxygen, it is uniformly malleable and ductile, and may be rolled from such bars as described without once annealing. It is sometimes preferred not to use the knowledge gained, and then complications commence. The following mode of operating will meet all cases:—The bars may be rolled down to half their thickness, and then will anneal perfectly well in an open furnace on an iron truck. The

heating should be conducted rapidly, and when the fillets get to a full red heat they should be withdrawn, thrown out singly on the floor, and allowed to remain till perfectly cold. Should any bars be annealed in the rough state, they must be kept away from water. A little water thrown upon the end of a bar when red hot causes it to become rotten throughout, and on submission to any pressure it will fall to pieces as would gingerbread; yet if the bars be partly rolled to a certain proportion of their thickness, they may be plunged at once into cold water without damaging them for work. After annealing, the fillets should be blanched in diluted sulphuric acid, containing one per cent. of the ordinary commercial acid. It is well to save time by blanching two or three tons at a time; but a few minutes effect all that is required—that is, the separation of the film of oxide from the surface. The film is then removed by a mop made of rag or cotton waste, and with little labour a few boys may clean many tons in a day. Fillets which have been blanched before being rolled produce clean and bright blanks. The blanks require somewhat different treatment. It is better to anneal them in copper tubes. The bottom of the tube should be covered to about the depth of an inch with charcoal dust, then the tube filled with blanks, except allowing for another layer of coarse charcoal dust, and the top put on to the tube. The annealing should not occupy more than thirty-six minutes; the highest temperature should be that at which the tube looks a full reddish white, and this should be gained as rapidly as possible. The tube, after removal from the furnace, should be allowed to remain at rest till perfectly cold. The charcoal is intended to combine with the oxygen, which would otherwise combine with the metals of the bronze during the heating and the cooling of the blanks in the copper tubes. Having regard to the production of perfect coins, the blanks should be cooled in an atmosphere of ordinary coal gas, by which every particle of oxygen is prevented from access, and a great part of the oxides already existing in the alloy reduced by the combination of their oxygen with the hydrogen of the coal gas, thus leaving the blanks somewhat porous, and comparatively soft, so that, when coined, the metal wholly fills the work on the dies, and the coin is produced with a good protecting edge. Bronze as used at the Royal Mint consisting of 95 of copper, 4 tin, and 1 zinc in 100 parts—zinc should be omitted because it causes useless labour—may be coined to great perfection if the blanks be cooled in coal gas after annealing: whereas bronze of a far softer nature cannot be made to fill the work of the dies satisfactorily by any other method yet known, unless the engraving on the dies be very shallow. The bronze coins, which have replaced the old copper money, have what is considered to be an innovation in the inscription by the repetition of the T in the abbreviation of Britanniarum, which is simply the classical mode of expressing in an abbreviated form the number of the possessions which together form Great Britain; it is equivalent to MS., which is the abbreviation for manuscript, while MSS. represents the word manuscripts; so BRITT. represents the cluster of islands or possessions called Great Britain. The objection to BRITT., on the ground of its being an innovation, is singular, as this word occurs on the shillings of George III., coined in 1816. It is also curious that the original dies for the bronze coinage were made to carry BRIT. only, until a coin found its way into the hands of the Right Hon. W. E. Gladstone, who at once pointed out the error, and thus caused the re-introduction of the classical BRITT. Mr. Gladstone also suggested the issue of octagonal bronze coins, which undoubtedly would have been a mistake had it been adopted. Perhaps it may not be uninteresting to record that the pattern penny in bronze, which was submitted to and approved by her Majesty, was lost in its passage through the post, for the postman opened the letter and destroyed both it and the penny in a closet in the Royal Mint. Copper and bronze money are merely tokens; it is therefore well to reduce such tokens to as low a weight as is consistent with the rigidity of the coin.

By Royal Proclamation, dated at Windsor, 13th of May, 1869, the old copper moneys are declared illegal; or, in the words of the proclamation, “No copper moneys whatsoever (other than and except such bronze moneys as are now current by virtue of our proclamation bearing date the

17th day of December, 1860, or any proclamation dated subsequently to the said 17th day of December, 1860) shall be allowed to pass or be current in any payment whatsoever within the kingdom of Great Britain and Ireland after the 31st of December, 1869.” So that since that date no copper moneys have been legally paid away. This, however, will not affect those who wish to preserve copper coins of this or former reigns as specimens.

As dies are the means by which metal becomes COIN, it will be material to add the processes of their manufacture. Experience alone can indicate the proper steel to be used, so that it is useless to attempt a description. But the form of the steel is a matter upon which die-makers differ. Some use square bars of that metal, and cut off pieces, which are forged into the proper form for dies; however, the more experienced die-makers use round bars of steel, which are cut by a tool somewhat of the shape of the bar, so that when the tool is struck every part of its edge begins to cut. The tool is so shaped as to leave the end of the bar from which the piece has been cut quite square, so that it may form the bottom for the next piece cut off. The piece of steel thus cut off is taken to a lathe, where its upper end is turned somewhat into the form of the top of a sugar-loaf; the depth of this form being determined by the style of work to be impressed upon the steel when it is to be formed into a die. For the original matrix the piece of steel is turned to a flat end, and is usually strengthened by driving a carefully-welded ring of hot iron on to it: although some prefer thus to simply shrink on a ring of iron, the more experienced choose welding, because that which is shrunk on is never so solid, and is apt to enclose air bubbles, which prevent the mass becoming one, thus permitting an undue yield in a faulty spot when pressure is applied. In the making of a matrix, the steel is softened to different extents by different engravers, but all soften more or less. The steel having been softened, the engraver cuts with a kind of hand chisel, by a motion of the wrist, such design as may be required, and after many months of work produces a finished matrix, from which he requires to produce many copies, because this one matrix would itself soon wear out, and by frequent use would be more exposed to risk. It is, therefore, hardened to admit of its being submitted to such pressure from another piece of steel as to impart its design to that steel without itself being distorted by the pressure. For this purpose the matrix is fitted with a ring of iron round its upper part, which, when fitted on, gives the engraved part the appearance of being the bottom of a cup; this cup is filled with a paste made of beer grounds and fine charcoal dust, or, better still, with that small charcoal which is found behind the flue of the forge—a kind of soot produced by the spent sparks which fly upwards, and which is so much in request by the file-makers. Thus protected, the matrix should be placed in the naked fire wholly enclosed with fuel, and heated as rapidly as possible, keeping it constantly turned round and round, so as to insure each particle of its mass being equally heated; a quarter of an hour should be long enough to produce the necessary temperature. It must be borne in mind that the great principle is to avoid the decarbonisation of the steel, therefore the less time that is occupied the better. The temperature of the matrix should be that rich and peculiar full red which for a few seconds is visible before white heat supervenes, for a white heat once gained causes the steel to scale off, and renders the matrix partially worthless. This peculiar temperature once gained, the matrix must be permitted to remain for three minutes in the fire without any blast. If it should get too hot it must be cooled by adding exhausted fuel from the hearth. At the end of three minutes’ quiescent heating it may be assumed that the centre of the matrix has become of the same temperature as its outer particles; it is then taken quickly from the furnace and placed in a kind of colander, so that water may wholly envelope it, and yet flow freely past it while its under surface is wholly exposed. The matrix C is placed in the [colander D, Fig. 22](#), bottom upwards, and is then brought immediately under the pipe dependent from a kind of funnel A, which should hold at least a hundred gallons of cold water; the bottom of the pipe must be secured by a plug of wood B, capable of being knocked out instantaneously. This is the most important process in the hardening of the matrix, and requires

considerable steadiness with great rapidity of motion. The matrix being held in the colander about half an inch above a large surface of water E, and immediately under the centre of the plugged pipe from the funnel, the plug B is knocked out from the pipe, and the water falls in a continuous stream wholly unbroken until striking the very centre of the base of the matrix, thus securing the commencement of the process of hardening FROM THE CENTRE. So soon as this first shock of the flowing water has been effected, the matrix must be sunk into the vessel of water below, until it is uniformly covered to the depth of half an inch, and must be held there until the whole of the water from the funnel has run out; then the matrix must be sunk to the bottom of the vessel of water, and allowed to remain till absolutely cold. If it be removed before it be cooled throughout, there is great danger of a crack on its surface being produced by the expansion from within. The plug is preferred to a tap in the pipe from which the water is to flow, because the turning of a tap gives a motion to the water, which causes it to fall in a hollow stream, and thus obviates the very effect which it is desired to produce. This process has been followed for more than thirty years, and during that time the manufacturer who uses it—one of the most eminent engravers and die-makers in Europe—has never met with any accident, nor has he found any failure. I have recently visited a manufactory in South Yorkshire in which many thousands of dies are made annually on this system, without an instance of failure. Die-makers, after the hardening, proceed as follows:—Remove the paste of charcoal from the face of the matrix, and then clean the face of the die with hydrochloric acid. After this it is tempered—that is to say, it is raised to a temperature which somewhat relaxes the rigidity of its particles; this is commonly judged of by the COLOUR which the face of the matrix assumes; but since no two men are likely to distinguish the same tint, nor can any one actually judge of colour unerringly, it is wise to adopt another standard, the one most preferred being to temper the matrix until the engraver can make a good graver *bite* while the die is hot. The tempered die may be cooled by plunging it either into oil or into water as soon as the tempering has reached the desired point. If at this instant the tempering cease, when the die is dead cold the graver will no longer touch the steel, which is now just so hard as to resist a blow without fracture. It is next polished, and is then fit for use; that is to say, is fit for the multiplication of dies from itself.

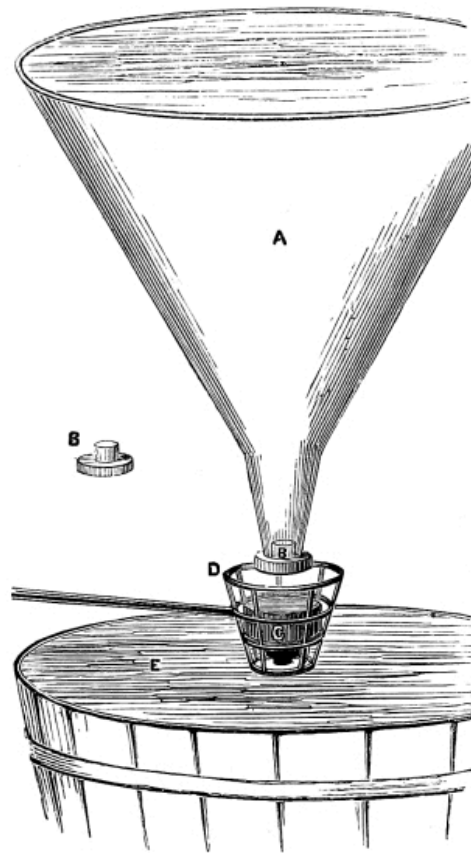


Fig. 22.—Die Hardening.

The multiplication consists in the taking a set of punches, each being an exact copy of the matrix. The steel, having been formed into masses with sugar-loaf tops in accordance with the depth of the engraving, is ready to receive the impression from the matrix, just as hot sealing-wax is ready to receive the impression from a signet. The matrix is placed beneath a press, and on its face is placed the sugar-loaf end of the annealed mass of steel; a very light blow is given by the press, so as to obtain simply the sinking of the first portion of the cone, for if a violent blow be given at this stage it is manifest that a large mass of steel is forced into a smaller space; in other words, the particles of steel which formed the cone are pressed into a dense mass on the surface of the intended punch, and prepare a crack, which by-and-by developes itself. After this light blow, the partly-formed punch is placed in a covered pot filled with charcoal, and heated in a furnace just as was described for the hardening of the matrix, bearing in mind that it is the cooling, not the heating, which softens the steel. So soon as the heating has been raised to the point before indicated, the pot containing the punch is removed, and bedded in hot ashes, and there left to cool as slowly as possible; it should be forty hours before it becomes cold. When cold it is cleaned, receives a second very light blow under the press, examined, afterwards receiving a third very light blow, and is then a second time annealed, &c. The number of blows required to bring out to perfection on the punch the work on the matrix, depends entirely upon the size of the matrix and the depth of the work engraved upon it. For these reasons, no accurate instructions can be given; but for all cases light blows should be made, because all metals mould more readily under gentle treatment.

From the punch is obtained, by processes which should be in every way similar, a series of dies for the use of the coining press. Each die from the punch is examined, and has the figures for the date impressed upon it before it is hardened. The dies are of two kinds: one, C, the obverse, with a long neck, upon which the collar B fits freely, and a second, the reverse, with a short neck, so that the face of the die just enters the collar sufficiently to meet and compress the blank. [Fig. 23](#) demonstrates these two dies, and exhibits the collar, showing at B the milled interior circle, which gives the crenated edge to the finished coin at its formation. For the past few years the reverse die has been made to carry, in addition to its recognised device, a small number, with a view to determine at which coining press, and on what particular day, that die was used, that bad work might be traced to an individual. It need hardly be said that the accidents encountered in daily work overcome the object thus sought, for a die may last either minutes or days in wear owing to the irregularities of the steel from which it is made, and besides, the boys who work the presses are of necessity changed at irregular intervals.

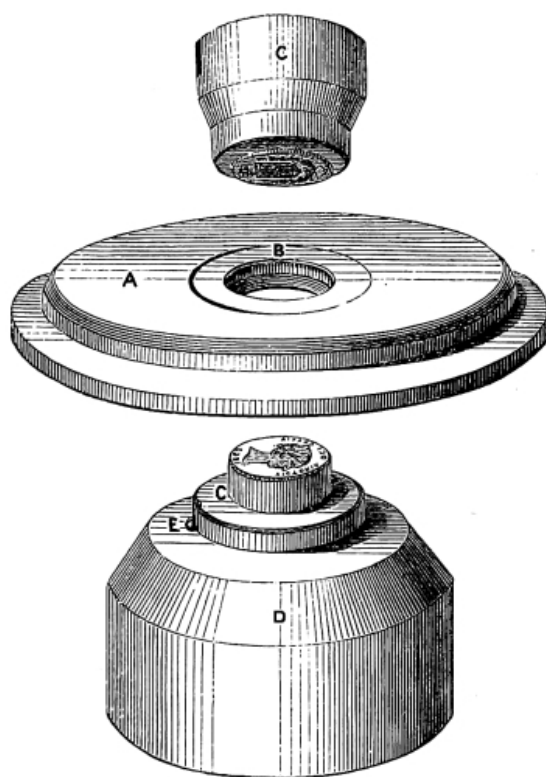


Fig. 23.—Coining Dies and Collar.

The long-necked die is fitted into a bolster D, as shown in [Fig. 23](#), and is prevented from shifting its position in that bolster by a plug of iron indicated at E. The bolster, with its die, is secured in a kind of dish beneath the screw of the press by a series of screws, which admit of adjusting the bolster so that the die it carries shall be placed immediately beneath the upper die, which is carried on the main screw G of the press, as may be seen at [page 60](#).

There is little doubt but that certain stages of die manufacture would be rendered more effective if more reasonable treatment were employed. It has yet to be demonstrated that carbon steel is the best metal that can be employed, for it seems more than probable that some alloys

would render a metal which would possess within itself more of the properties which are necessary for the perfecting a die, and for its duration when in use. For the elimination of such an alloy we must await the time when the non-resident assayers are replaced by resident chemists who, during the leisure times in the Mint, will be willing and glad to experiment on some of the compounds of iron with the rarer metals, for in that field will surely be found a combination which will be definite, and therefore homogeneous and unvarying. There can be no doubt that the cause of the uncertainty which prevails as to the endurance of steel for dies arises from its unequal composition.

Gold possesses as many colours, and exists in as many conditions, as any substance ordinarily met with. Those various changes are due to the existence of impurities in the metal, which are neglected in assaying, because the only business of the assayer is to determine that the sovereign actually contains the lawful weight of pure gold, and also so much alloy as is specified: what this alloy shall be is, in the abstract, a small matter so that it is not silver, because if it be silver it is worth the expense to separate it, and thereby destroy the coinage. Yet, although the existence of such metals as lead, bismuth, tin, antimony, and arsenic, as well as palladium and osmium-iridium is not illegal, it becomes necessary to separate them, because these metals affect the coin in various ways. Palladium and platinum alloy with the gold only after considerable time, therefore these metals may generally be seen in the finished coin, frequently as a slight bar across its surface, sometimes as spots on its crenated edge. Osmium-iridium causes trouble, inasmuch as it is so infusible, that it simply floats about in the melted gold, and several crystals agglomerate; where this takes place, the difficulty of manufacture is so great, that it is impossible to produce the desired amount of coin within the limits of remedy as to weight. The simplest means for the removal of these substances is to allow the fused metal to remain for a considerable time, when they settle to the bottom, and the clear metal can be poured off; the dregs, if so they may be called, are then sent to the refiner, but the process of separation is too long, and hardly adapted for a description in this place.

As a practical matter, the other, or volatile metals, and lead, may be, and generally are, removed by means of corrosive sublimate. This is an expensive measure, and does not answer its intention, for some of the mercury combines with the gold and renders it brittle. In practice, the best process is that recommended by Mr. Warrington, who proposed to add to gold found to contain tin, antimony, or arsenic, 10 per cent. of oxide of copper. The gold is fused in a Payen crucible, and the oxide of copper is then added and stirred into the gold by a rod made of the same substance as the crucible. This operation is easy of performance, because the infusible oxide of copper stirs up with the gold readily, and each particle of antimony, arsenic, or tin, as it comes into contact with the oxide of copper, reduces it to the metallic state, the volatile metals combining with the oxygen, while an equivalent of copper alloys itself with the gold. The impurities, as they oxidise, mix or combine with the oxide of copper. One or two experiments tell how much copper is thus added to the gold; consequently, how much fine gold is to be added to keep the alloy standard—and subsequent treatment by the oxide of copper simply removes the volatile metals. Gold treated by this process is perfectly workable on a large scale, and there is no trouble. A condition of success is, that no reducing agent, such as carbon, be present, and therefore plumbago pots cannot be used. In practice, half an hour is the best time to keep the metal in contact with the oxide of copper. The objection is frequently expressed, that to refine gold is not the business of the coiners. This opinion, however, will bear modification, for the expense of the oxide of copper is trifling as compared with the cost of labour in treating brittle gold.

I now advert to a most important matter in relation to coining, viz., that of the loss sustained, its cause, and prevention.

It appears that the Royal Mint should sustain less loss than any other mint by the coining of gold and silver. If allowance be made for the sale of the sweep or dust^[46] which results from a coinage, the total loss, inclusive of every operation in coining, should be so small that it might be passed without notice; in fact, there ought to be a minute increase of weight from traces of oil which are left on the fillets to enable them to pass through the cylinders of the draw-bench,^[47] by melting^[48] there seems to be too great a loss of metal: this should reach about £100 per million coined, and such loss would be wholly explained by the refining, which takes place through the removal of copper by oxidation; although this is minute, still it is enough to explain the loss which I indicate. If the assays be closely watched there can be no loss, for the trial of the pyx invariably shows the gold coin to err on the side of purity, so that if the Master of the Mint should determine to issue gold of exact standard, and refuse to avail himself of the latitude allowed, he may fairly cover every source of loss, and coin money WITHOUT WASTE OF METAL. Each grain that is found in excess of the standard upon the pound weight of gold causes a loss of about £180 upon each million coined. The moneyers asserted that they sustained a loss of £700 for each million coined, such loss being exclusive of melting. This has never been believed to have been a truthful statement, nor has the loss by coining alone often reached so high an amount, although so large a sum as £373 per million would seem to have been determined, by some careful experimenters, as the necessary loss. Where the gold went to was not stated; but supposing such a loss to accrue, it is manifest that either the gold volatilised or the floors must be paved to many inches in thickness with gold, but it is an established fact that matter once existent cannot be annihilated; yet it has required many arguments to convince those most interested in coining that the mere cutting up of bullion into small pieces does not alter its actual weight: the present Mint Authorities entertain the fallacious notion that large loss of gold is, for metallurgical reasons, NECESSARY. Notwithstanding Mr. Graham's opinion that a loss of £206 on a million *was too high*,^[49] they in their "Reports on European Mints" have arrived at the conclusion "that the 'waste' shown to have existed of late years in the English Mint *has not been excessive*."

To leave, then, the realms of speculation, and for a short time to examine figures, a matter which is little understood will be reduced to one of easy comprehension. Mr. W. T. Brande, in connection with others, carried on a series of experiments between 1851 and 1856, the results of which he gave me in writing, and which induced him to arrive at the conclusion that by coining alone there was a loss varying from 1·20 grains on the pound, to 3·10 grains on the pound troy, or a mean of 2·15 grains on each pound troy coined. This loss was thought to be a necessary consequence of coining, and by a simple rule-of-three it tells as follows:—

A loss of	1·20	grains	on the lb.	=	£208·33	on each	£1,000,000.
"	3·10	"	"	=	£538·18	"	£1,000,000.
Mean loss	2·15	"	"	=	£373·26	"	£1,000,000.

Mr. Brande and his colleagues, by his own admission, did not know the value of the sweep recovered, therefore this loss was believed to be absolute. We will next examine the accounts and see the fruit of that belief, and then determine the effect of a contrary faith. And in the following statements, given in a tabular form, I would observe that the facts recorded were compiled by the Mint officials for the Master at my suggestion, and that copies were given to me by the late Master, who indeed invariably, when such statements were prepared, handed them to me to copy and examine for my own information, and to see that they were accurate.

It is a fact that the Mint books do not record the loss upon the gold coined in November and December, 1851; or, if they do, these accounts have been only quite recently balanced. In the

following statement I omit the coin of those two quarters, because the late Deputy Master assured me that the accounts were not made up. This, I am aware, lays me open to a correction; I therefore state that the coined money of these two months is not included in my calculations.

**STATEMENT SHOWING THE RATE OF COIN OBTAINED AND THE LOSS OF METAL INCURRED
IN EACH QUARTER FROM MARCH, 1852, TO DECEMBER, 1857.**

(A) = Proportional Loss of Metal on each Million Ounces Coined.

Quarter ending:—	Total Amount of Gold Coined per Quarter.	Rate per Centum of Coin obtained from Rough Bars.	(A)
	Ounces.	Ounces.	Ounces.
March 31st, 1852	300260·201		446·94
June 30th, "	501032·558		399·17
Sept. 30th, "	615753·337		500·50
Dec. 31st, "	828159·837	50·51	639·37
March 31st, 1853	1113414·454	47·28	691·60
June 30th, "	1223454·550	49·40	472·34
Sept. 30th, "	248434·554	39·34	1130·64
Dec. 31st, "	484331·447	42·81	784·42
March 31st, 1854	769791·029	54·21	730·61
June 30th, "	101611·604	49·65	787·21
Dec. 31st, "	194968·684	47·76	583·07
March 31st, 1855	515639·229	47·42	447·05
June 30th, "	501708·378	47·01	456·20
Sept. 30th, "	283944·316	43·53	420·18
Dec. 31st, "	1012329·518	44·57	509·08
March 31st, 1856	360409·244	38·39	489·05
June 30th, "	818465·116	40·36	509·57
Dec. 31st, "	362599·792	41·74	406·28
March 31st, 1857	73638·878	39·64	233·84
June 30th, "	19872·998	40·52	57·86
Dec. 31st, "	1154606·290	54·69	18·45

There will be occasion to refer to the next table for further information; but as the facts in this table are not strictly comparable with any others that will be given, it is better to explain this as it stands. Of the rate per centum of coin obtained from the bars there will be occasion to speak at a later period. These figures are placed here with a view to save repetition, which would be necessary when that subject is discussed.^[50] Although the figures in the first column represent ounces, each item can at pleasure be converted into pounds sterling by multiplying it by £3·89375, which is the decimal expression for £3 17s. 10½*d.*, the price per ounce of standard gold. In such case it will only be necessary to multiply the weight of the coinage by that value, and then to consider the loss of metal as pounds sterling—thus, 300,260·201 ozs. + £3·89375 = £1,171,318 5s. 3¼*d.*, which sum lost weight by coining at the rate of £446 18s. 9½*d.* per million coined. If calculation be made by addition and division of the figures in the third column, it will be observed that the average loss per million between March, 1852, and December, 1856, is £577·96,^[51] and this was believed to be absolutely wasted in coining; but this amount may be fairly reduced by the

deduction of £72 10s., the average value of sweep recovered, when it will remain £505·85. It will also be seen that the rate per million missing—for that is the fact—varied a good deal according to circumstances.

It has been asserted that these losses were mere matters of account, and resulted from the removal of “fat or oil” in the various operations. These explanations shall be admitted to their fullest extent, but that they do not contain the cause for all the losses is to be gathered from the fact that gold has been returned to the Mint by the brickmaker, who found it in the ashes which he had bought. That there were cases of actual theft I am myself aware; for I, by the Master’s sanction, dismissed two men who were detected. Eleven others shared the same fate, but no prosecution was attempted or permitted, although strongly urged. As bearing on this point the following, in relation to “peculation,” is told and believed in the Mint:—In 1856, when the men had formed a strong opinion as to the honesty of one particular person, they took their own means to watch him, and to see how and when he abstracted the metal. A clear-headed man made it his business to do this duty. The suspected man was assisting the officer at the scales in the drag room, when the watcher saw him, after the bullion was weighed, take a piece stealthily and secrete it in the palm of his hand. The watcher, thinking to secure detection, requested the officer at the scales to re-weigh that draft; but this gentleman refused to comply. The watcher then told the officer what he had seen, and that if he would open the man’s hand the piece of gold was still in it, when the officer, addressing the suspected person, said, “Is it all right, John?” John replied “Yes,” and no more passed. On his road home “John” stumbled, and a piece of uncoined gold fell on the pavement.

That metal was actually removed from the premises is beyond doubt, for a late officer picked up a piece of gold in the courtyard wrapped in brown paper, and the police brought to the Master a “flat end”^[52] of gold, which had been offered for sale. The police inspected all the men in the Mint employ; but the person who abstracted the gold did not personally offer it for sale, and thereby escaped detection.

On the 2nd of December, 1853, the cutting room book records a loss of 87·26 ounces of gold, for which no satisfactory explanation is given. This book also contains some significant remarks by the officer who had charge of that department.

The Mint books adopt the word “Wasteage” as explanatory of these losses, which were believed to be a necessary result of coining; so generally, indeed, was this firmly-implanted belief entertained, that at the bottom of each day’s account the word loss was printed, so that the officer had only to write the amount which was missing.

If indeed proof that unnecessary wasteage^[53] took place be required, it will be but necessary to quote from the letter of the Master of the Mint to the Treasury, dated 7th June, 1860. This letter was placed in my hands on the 1st June, 1860, by the Master, who desired me to copy such parts as related to loss and were of service to me, and that I might confirm the accuracy of his figures. He said, “It appears by the table that the average loss on the gold coinage of the first six years is £784 0s. 0¾d., and on the gold of the last three years £172 8s. 11½d. per million coined, showing an improvement of £611 11s. 1¼d. per million coined. On the twelve millions of gold coined during the last three years the saving exceeds twelve thousand pounds.” The three years here referred to were contained in the period during which Mr. Graham entrusted me with the management of the coining department.

That the actual amount of loss by coining may be got at, it will be necessary to view the matter in a different form, and in that way to show the amount of money recovered by the sale of sweep;

that is, by the sale of the ignited rubbish accumulated during a coinage, and in the statement given on [page 90](#) the money value of each item is specifically represented. It need not be insisted that the utmost care should be exercised in preserving the sweep, when it is stated that this yields, at the end of a coinage, from 15 to 20 ounces of material bullion called *scrap or supply*, when picked over before sale, for only such as will pass through a fine sieve is sold; yet this sweep used to stand about in open, unlocked boxes from month's end to month's end.

It would be wrong to state that there is no *apparent* loss by coining gold. If, however, the whole circumstances be examined, it will be found that, after deducting every legitimate waste, an appreciable gain should be exhibited. In annealing sovereign blanks the loss is 54·12 ounces on each million. This then would appear to be an absolute waste; but starting on the undisputed fact that "*matter cannot be lost*," let us see what becomes of these 54·12 ounces apparently lost. 54·12 ounces contain 25,977·60 grains troy. Now, since careful analysis extending over more than thirty millions of gold annealed has determined that each million lost 5,708 grains of copper, and this copper is washed away as sulphate of copper, it may be admitted that so much is lost, thus leaving 20,169·60 grains unaccounted for. These 20,169·60 grains are just 42·02 ounces, and if proper care were bestowed I could show how every particle should be recovered at the end of the coinage.

That there should be a gain is demonstrable from the fact that the gold as received into the coining department is quite clean; when wrought it becomes coated with oil, and this being returned on the scissel^[54] to the melting-house is really charged as gold. In 1858 Mr. W. H. Barton, experimenting independently, arrived at the conclusion that there were 200 ounces of oil on 765,370 ounces of gold scissel, while on the same scissel I, by careful experiments, determined that there were 118·39 ounces of oil, the remainder being dust and other foreign matter. Further experiments, conducted by Mr. Barton for his own satisfaction, brought him to adopt my figures as accurate. This proportion would give 154·55 ounces to each million ounces of scissel; but since good work returns only 35·80 per cent. of scissel on the rough bars, it becomes manifest that this same proportion is but 101·22 ounces in each million coined.

Now the MONEYERS^[55] used to make an allowance of 347 ounces to each million ounces of scissel, and with such sharp business-like people it is barely possible that they allowed too much. We may therefore assume that the scissel of the present day is much less covered with oil than it used to be; and this is indeed the truth, for whereas the trucks were wont to be actually spotted with oil which dropped from gold scissel, they are now unstained; and it may be mentioned that, to test the minute amount on the scissel, the Master has upon occasions wiped it with a white cambric handkerchief, by way of settling a disputed point, without perceptibly soiling the handkerchief. Admitting then a loss of 42·02 by annealing and blanching, we get, by deducting this from the gain by oil, an absolute gain of 59·20 ounces on each million ounces coined. In a little while, when examining the table, we shall see how this theory worked out in practice; for, to be perfect, it should be capable of demonstration. The question will naturally arise, If the oil be now so small in proportion, why should there not be an increased loss by melting? And here indeed arises a very nice point, but one which is proved to demonstration—that whereas large amounts of oil *cause* loss, small quantities of oil PREVENT loss by melting, and for the following reason. Standard gold consists of copper 2 parts, gold 22 parts. So soon as this gold approaches red heat it changes to a purple colour on its surface, owing to the oxidation of the copper. If, however, a trace of oil be present, the heat, while it warms the gold, volatilises the oil, and thus places the gold in an atmosphere free from oxygen, so that the standard gold remains metallic until it liquefies; hence the saving of loss, for any oxide of copper would be lost in the pot. This matter

of the oil has been dwelt upon because it is a vexed question, and when the melter, by want of care, makes a large loss, he invariably, by a human weakness, attributes the loss to the oil on the scissel. It was on such an occasion that the assistant-melter produced the following amusing parody on the three witches of Shakespeare. The illustrative picture is clever, but unfitted for these pages.

First Witch. Round about the journeys go,
In the dirt and rubbish throw;
Extraneous matters, small and great,
Everything to make the weight.

Second Witch. Stone swept up and pounded small,
Pieces of the stuccoed wall,
A bit of saffron-coloured brick,
Odds and ends of broken stick;
Here's the sweeping of the floor,
Weighing full an ounce or more.

All. Hubble bubble, toil and bubble,
Give the melter every trouble.

Third Witch. Copper by annealing got,
Send largely to the melting-pot;
A pennyweight of leather strap,
A piece of old brown paper cap,
Some cotton fluff, ten grains of coal,
Then OIL to saturate the whole.

All. Hubble bubble, toil and bubble,
Give the melter every trouble.

MACBETH (*as Justice*). How now, ye secret, black,
and cunning rollers!
What is't ye do?

All. A deed without a name.

24th November, 1862.

J. G.

Upon consulting the table—[see p. 90](#)—it will be found that, by deducting the value of the sweep recovered, from the value of the metal lost, between the periods November, 1851, and March, 1857, the actual loss was £19,930 18s. 0d., which sum, divided by the amount of gold coined in that period—viz., £40345185·450, gives £494 as a *bona fide* loss on each million pounds coined, when the belief was to the effect that there must be a loss. The great fact, *that matter once existent cannot be lost*, was never applied to the operations of minting until, in 1856, I demonstrated to the Master that the floors not being paved to many inches thick with gold, the bullion which was stated to be “wasted” in coining must have been *absolutely volatilised*, the proposition exposing me to more than ridicule at the time, and to positive hatred and

misrepresentation thereafter; but when once the determination was formed to arrest what I saw clearly was unnecessary loss, it was carried out firmly, and in November, 1856, I commenced my determinations of the *necessary* amount of loss in each operation, as well as a series of experiments to elucidate other points requiring reform and explanation.

In furtherance of this firm resolve I assembled the men together, and informed them that I had now taken charge of the departments over which I had been placed, and that I should not permit any man to leave the Mint until the account was satisfied, and I had received from them the bullion to its uttermost part which I proposed to entrust to them. That I might effect this determination I weighed out to them, in their presence, a limited weight of gold, and then directed their foreman, on their behalf, to satisfy himself that the weight I charged was fair and honest; and then, with a firmness which surprised them, I followed this gold step by step, never for an instant permitting it to leave my sight, and at *each step in its passage from bullion to coined money* I carefully weighed it, and made the foreman again check me, all the men being witnesses. At the end of the operations a **FIXED GAIN** exhibited itself. I was called upon for an explanation, which I gave, and each man was thus convinced that it was no use to attempt to cheat me, and was compelled to acknowledge that unless by abstraction there could be no loss. Many arguments followed for months, but **LOSS** became a matter of history so long as I was present,—not so in my absence, even at that time, as the books will show.

Great firmness was required, but I persevered, and received not only thanks, but very valuable assistance from many of the men. With a view to keep the Master supplied with information as I gained it, I adopted a plan of reporting to him daily in writing the various changes which took place in the bullion under my care as regards differences of weighing, loss or gain, the amount per cent. of coin obtained, and other such details, and at the end of each coinage I summarised the whole of such facts into a tabular form, and thus rendered to him a complete history of the operations conducted. This proved of immense service, for upon the occurrence of an unusual event, attention was at once drawn to it, and means adopted to discover the cause, while at the same time a constant study of these reports could not fail to induce for myself a more minute inquiry into the various branches of my subject. I feel it cannot be unfitting to give, before proceeding further, one of these statements or reports relating to gold, and when I come to the discussion of silver I will, in its proper position, place such another table, because these seem to convey much practical information which will be useful to those engaged in coining.

The report placed on [page 92](#) is a copy of one I made to the Master, so I give it *in extenso*, and it will be found that by it bullion may be traced from its first admission to the coining department to its final issue as coined money, while each item of its history finds a place in the immediate department in which the bullion may be under operation.

STATEMENT OF LOSSES INCURRED AND OF SWEEP RECOVERED.
THIS REFERS WHOLLY TO GOLD IN THE COINING DEPARTMENT.

Date—Financial Year.	Weight of Coinage.	Value of Coinage.			Weight of Lost Metal.
	Ounces.	£	s.	d.	Ounces.
Nov. 1851 to March 1852	351837·830	1,369,968	11	0	134·206
April 1852 to March 1853	3058360·186	11,908,489	19	5½	1851·888
" 1853 " 1854	2726011·580	10,614,408	11	9¾	1804·559
" 1854 " 1855	812219·517	3,162,579	14	10½	424·092
" 1855 " 1856	2158391·456	8,404,236	14	7½	1039·798

Date—Financial Year.				Weight of Coinage.	Value of Coinage.	Weight of Lost Metal.
”	1856	”	1857	1254703·786	4,885,502 17 2	581·615
”	1857	”	1858	1256586·572	4,892,833 19 4	17·784
”	1858	”	1859	736150·582	2,866,386 6 6¾	16·603
”	1859	”	1860	980070·679	3,816,150 4 2	74·857
”	1860	”	1861	861205·000	3,353,316 19 5	263·568
”	1861	”	1862	1872214·638	7,289,935 14 11	
”	1862	”	1863	2351444·336	9,155,936 7 8	27·480
”	1863	”	1864	1807332·130	7,037,299 9 8	47·054
”	1864	”	1865	1949438·200	7,590,624 19 10	43·491
”	1865	”	1866	1256347·104	4,891,901 10 9	57·612
”	1866	”	1867	175838·208	684,670 0 5	5·037
”	1867	”	1868	109580·129	426,677 12 1¾	3·357
June	1868	to Sept.	1868	280993·740	1,094,119 7 6	*19·87

Date—Financial Year.				Value of Lost Metal.			Value of Sweep.		
				£	s.	d.	£	s.	d.
Nov.	1851	to March	1852	522	11	3	89	1	3
April	1852	to March	1853	7,210	15	9¼	716	11	11½
”	1853	”	1854	7,026	10	0¼	707	7	3½
”	1854	”	1855	1,651	6	1¾	222	14	3
”	1855	”	1856	4,148	14	3	635	4	1½
”	1856	”	1857	2,264	13	3	422	13	10
”	1857	”	1858	69	4	11	192	14	10
”	1858	”	1859	64	12	11½	209	16	7
”	1859	”	1860	291	9	6	254	4	9
”	1860	”	1861	1,026	5	4	545	10	5½
”	1861	”	1862						
”	1862	”	1863	+ 107	0	1	360	1	11
”	1863	”	1864	+ 183	4	4	459	0	1
”	1864	”	1865	169	6	10	295	9	3½
”	1865	”	1866	224	6	6	350	3	9
”	1866	”	1867	19	12	3	Nil.		
”	1867	”	1868	13	1	5	Nil.		
June	1868	to Sept.	1868	77	7	7	Nil.		

Date—Financial Year.				Value of Lost Metal. per Million Coined.			Value of Sweep. per Million Coined.		
				£	s.	d.	£	s.	d.
Nov.	1851	to March	1852	381	8	10¼	65	0	2½
April	1852	to March	1853	605	10	3¾	60	3	5¾
”	1853	”	1854	661	19	6½	66	12	10
”	1854	”	1855	522	2	1¾	70	8	5
”	1855	”	1856	481	14	10¾	75	11	7½
”	1856	”	1857	463	10	11¼	86	10	4¼
”	1857	”	1858	*14	3	0	39	7	10½

Date—Financial Year.				Value of Lost Metal. per Million Coined.			Value of Sweep. per Million Coined.		
”	1858	”	1859	✱ 22	11	0½	73	4	0½
”	1859	”	1860	76	7	7	66	12	5
”	1860	”	1861	96	8	6	51	5	1
”	1861	”	1862						
”	1862	”	1863	+ 11	3	8	39	6	7
”	1863	”	1864	+ 26	0	8	65	4	5
”	1864	”	1865	✱ 22	6	2	38	18	6
”	1865	”	1866	✱ 45	17	1	71	11	8
”	1866	”	1867	28	12	10		Nil.	
”	1867	”	1868	30	12	8¼		Nil.	
June	1868 to Sept.		1868	70	14	5¾		Nil.	

NOTES.—The sign + indicates that on those special coinages there was an actual gain by weight to the extent specified, which will be explained on [page 98](#).

The * is intended to draw the attention of the reader to these figures when a little later they are explained at pages [89](#), [93](#), [94](#), [95](#), [98](#).

To obtain the true amount of loss, deduct the value of the sweep from the value of the gold lost.

**RETURN TO AN ORDER OF THE HOUSE OF LORDS
DATED 17TH FEBRUARY, 1870.
THE LORD ROSSIE. No. 1.**

**FROM THE COINING DEPARTMENT
OF THE ROYAL MINT.**

A STATEMENT of the Weight and Value of the Gold Moneys coined in each Financial Year from 1851 to 1869 inclusive, exhibiting the Weight and Value of the Loss or Waste sustained in each Year, as well as the Value of Sweep recovered, and the average Proportion of such Loss or Waste and Sweep recovered to each Million Pounds Sterling coined.

Date.				Weight of Coinage.	Value of Coinage.		
				Ounces.	£	s.	d.
Nov.	1851	to March	1852	351837·830	1,369,968	11	0
April	1852	to March	1853	3058360·186	11,908,489	19	5
"	1853	"	1854	2726011·580	10,614,407	11	10
"	1854	"	1855	812219·517	3,162,579	14	10
"	1855	"	1856	2158391·456	8,404,236	14	7
"	1856	"	1857	1254703·786	4,885,502	17	4
"	1857	"	1858	1256586·572	4,892,833	19	4
"	1858	"	1859	736150·582	2,866,386	6	7
"	1859	"	1860	980070·679	3,816,150	4	2
"	1860	"	1861	861205·000	3,353,316	19	5
"	1861	"	1862	1872214·638	7,289,935	14	11
"	1862	"	1863	2351444·336	9,155,936	7	8
"	1863	"	1864	1807332·130	7,037,299	9	8
"	1864	"	1865	1949438·200	7,590,624	19	10
"	1865	"	1866	1256347·104	4,891,901	10	9
"	1866	"	1867	175838·208	684,670	0	5
"	1867	"	1868	127485·816	496,397	17	11
"	1868	"	1869	1270778·550	4,948,093	19	7

Date.				Amount of Waste.	Value of Waste.			Value of Sweep recovered.		
				Ounces.	£	s.	d.	£	s.	d.
Nov.	1851	to March	1852.	134·206	522	11	3	89	1	3
April	1852	to March	1853	1851·888	7,210	15	9	716	11	11½
"	1853	"	1854	1804·559	7,026	10	0	707	7	3½
"	1854	"	1855	424·092	1,651	6	2	222	14	3
"	1855	"	1856	1039·798	4,148	14	3	635	4	1½
"	1856	"	1857	581·615	2,264	13	8	422	13	10
"	1857	"	1858	17·784	69	4	11	192	14	11
"	1858	"	1859	16·603	64	13	0	209	16	7
"	1859	"	1860	74·857	291	9	6	254	4	9
"	1860	"	1861	263·568	1,026	5	4	545	10	5½
"	1861	"	1862							
"	1862	"	1863	!+ 27·480	107	0	1	360	1	11
"	1863	"	1864	+ 47·054	183	4	4	459	0	1
"	1864	"	1865	43·491	169	6	10	295	9	3½

Date.				Amount of Waste.	Value of Waste.			Value of Sweep recovered.		
"	1865	"	1866	57·612	224	6	6	350	3	9
"	1866	"	1867	5·037	19	12	3	Nil.		
"	1867	"	1868	3·357	13	1	5	Nil.		
"	1868	"	1869	45·585	177	9	11	538	3	1

Date.				Value of Waste per £1,000,000			Value of Sweep per £1,000,000		
				£	s.	d.	£	s.	d.
Nov.	1851	to	March 1852.	381	8	10	65	0	3
April	1852	to	March 1853	605	10	4	60	3	6
"	1853	"	1854	661	19	7	66	12	10
"	1854	"	1855	522	2	2	70	8	5
"	1855	"	1856	493	12	11	75	11	8
"	1856	"	1857	463	10	11	86	10	5
"	1857	"	1858	14	3	0	39	7	10
"	1858	"	1859	22	11	1	73	4	0
"	1859	"	1860	76	7	7	66	12	5
"	1860	"	1861	96	8	6	51	5	1
"	1861	"	1862						
"	1862	"	1863	11	13	8	39	6	7
"	1863	"	1864	26	0	8	65	4	5
"	1864	"	1865	22	6	2	38	18	6
"	1865	"	1866	45	17	1	71	11	8
"	1866	"	1867	28	12	10	Nil.		
"	1867	"	1868	26	6	7	Nil.		
"	1868	"	1869	35	17	5	108	15	2

! Note.—The financial years 1862/1863 and 1863/1864 exhibit a *gain* in the Coining Department.

C. W. FREMANTLE, DEPUTY-MASTER OF THE MINT.

ROYAL MINT, 2nd March, 1870.

**STATEMENT OF WORK PERFORMED IN THE COINING DEPARTMENT FROM
OCTOBER 2ND, 1857, TO DECEMBER 17TH, 1857, INCLUSIVE.**

The Coinage was completed in 53 Days.
The Value of the coined Gold was £4,495,748 4s. 10d.

IN ROLLING ROOM.		
RECEIVED.		
Weight of Gold Bars for Sovereigns by Mint Office Beam		2110962·00 ozs.
Weight of the same Bars by Rolling Room Beam		2110939·71 "
Showing a minus difference upon the Mint Office weight of		22·29 "
RETURNED.		
	As by Rolling Room Beam.	As by Mint Office Beam.
	Ounces.	Ounces.

IN ROLLING ROOM.		
RECEIVED.		
Fillets	2016337·80	
Stopped Pots	1256·15	1256·15
Brittle Bars	12142·36	12142·36
Ends	81207·18	81208·24
	2110943·49	94606·75
Deduct “received”	2110939·71	
Increase during work.	3·78	
The Ends were 4·49 per cent. on Bars.		

IN CUTTING ROOM.		
RECEIVED.		
Gauged Fillets for Sovereigns by Rolling Room Beam	2016337·80 ozs.	
Gauged Fillets for Sovereigns by Cutting Room Beam	2016331·10 ”	
Showing a minus difference upon the Rolling Room weight of	6·70 ”	
RETURNED.		
	As by Cutting Room Beam.	As by Mint Office Beam.
	Ounces.	Ounces.
Cut Blanks	1250996·04	
Scissel	755829·53	755830·74
Dumb Fillets	9540·00	9540·00
	2016365·57	765370·74
Deduct “received”	2016331·10	
Increase during work.	34·47	
The Dumb Fillets were 0·45 p. c. on Bars.		
The Scissel was 35·80 ” ”		

IN WEIGHING ROOM.		
RECEIVED.		
Cut Blanks for Sovereigns by Cutting Room Beam	1250996·04 ozs.	
Cut Blanks for Sovereigns by Weighing Room Beam	1250989·11 ”	
Showing a minus difference upon the Cutting Room weight of	6·93 ”	
RETURNED.		
	As by Weighing Room Beam..	As by Mint Office Beam.
	Ounces.	Ounces.
Good Blanks	1163502·34	
Rejected Blanks	87488·31	87488·57
	1250990·65	87488·57
Deduct “received”	1250989·11	
Increase during work.	1·54	
The Rejected was 4·14 per cent. on Bars.		
” ” 6·99 ” Blanks.		

IN ANNEALING AND PRESS ROOMS.		
RECEIVED.		
Good Blanks for Sovereigns by Weighing Room Beam		1163502·34 ozs.
Good Blanks for Sovereigns by Annealing Room Beam		1163487·44 "
Showing a minus difference upon the Weighing Room weight of		14·90 "
RETURNED.		
	As by Press Room Beam..	As by Mint Office Beam.
	Ounces.	Ounces.
Coined Sovereigns	1154590·87	1154606·29
Pyx Pieces	1903·92	1909·21
Brockages	6926·03	6925·90
One Sovereign missing.	·25	
Loss of Weight by Annealing and Blanching	57·05	
	1163483·12	1163441·40
Deduct "returned"	1163487·44	
Loss during work	4·32	
The Brockages were 0·33 on Bars. The Coined Money was 54·79 on Rough Bars. " " 57·26 on Clean Bars.		

Paid for Wages of Men and Boys.

Rolling Fillets,	£252·402225
Cutting Blanks,	£260·624165
Annealing Do.	£145·437875
Coining Do.	£120·269875 = £778 14s. 8d. for £4,495,748 4s. 10d. Coined Money.

**Memorandum of
Coined Gold Moneys
delivered into Mint Office.**

	Ounces.
October	252030·274
November	594084·366
December	308491·650
	<u>1154606·290</u>

$$4495748·2 : 778·7 :: 1000 : 0·173208 = 3s. 5\frac{1}{2}d.$$

$$4495748 : 36·64 :: 1000 : 1·955d.$$

The Wages paid for Melting 2,110,962 were £36·64
= 1·955d. on each 1,000 Sovereigns coined.
= 0·001955 on each Sovereign coined.

1,000 Sovereigns cost 3s. 5½d. to coin
= 0·04156992 of a Penny for each Coin.

To the MASTER OF THE MINT.

GEORGE F. ANSELL, *January 20th, 1858.*

Reference to the table at [page 84](#) will demonstrate the fact that, by March, 1857, the average loss had fallen to £233·84, while in June of the same year it had been reduced to £57·86, finally reaching so low as £18·45 in

December, 1857, the sums being in every way strictly comparable with all those of that table, because the value of the sweep is not taken into account in any case. As is usual in all discoveries, the merit was soon found to be worth claiming, but that the credit was due to me the Master himself admitted; and it is a fact that I was the only new officer, while all those who had been previously employed were still retained; and I feel that the Master confirmed his opinion when, in consequence of the reforms I had effected, he obliged all officers to follow such directions as I gave. In all cases where time would permit, I made it a rule to obtain the Master's sanction before giving orders; but in cases of necessity I was empowered to act independently, and did so act, he confirming such orders as I gave. It should also be mentioned that so soon as I entered the Mint, Mr. Graham gave me positive instructions not to allow either of the then chief officers to interfere with the coining department. Nor would he permit a gold coinage to be conducted in my continued absence. If, however, the business of the Mint called me away, he, to induce the belief that I was in town and on the spot, would pay my expenses to and from London from his own pocket, and this he did on several occasions, because he found that "in my absence unnecessary loss re-appeared." The opposition offered to the coining of bronze in the Royal Mint was so great, that the Master wished me to go to the mint of Messrs. Heaton and Sons at Birmingham, and there make such experiments as were necessary to confirm the opinions I had formed on that subject. This firm, with a generosity ever their own, permitted me to make, during a considerable period, and free of charge, such experiments for the Mint as I deemed necessary, rendering every possible assistance; but my investigations and agreeable visit were brought suddenly to a close by the following note from the Master:—

"ROYAL MINT, 9th January, 1861.

"MY DEAR SIR.

"Owing to renewed pressure for gold coin, we find it necessary to begin rolling on Friday next. I am sorry to shorten your holiday, but your presence is very desirable as soon as gold enters the rolling room, and will relieve me of much anxiety.

"The pyx pence received from Mr. Gausby to-day appear to be of fair quality. We should like to have a ton or two.

"Very truly yours,

"THO. GRAHAM.

"G. F. ANSELL, ESQUIRE,

"At Messrs. Heaton & Sons, Birmingham."

To demonstrate, then, the correctness of the theorem I had proposed, it will be necessary to refer once more to the table on [page 90](#), where it will be seen that these results exhibit themselves in the financial year April to March, 1857-58, when the loss in each million coined was £14 3s., and the sweep recovered in that year was sold at the rate of £39 7s. 10½d. for each million coined. If, therefore, the proportionate loss be deducted from the value of sweep sold, we have for the first^[56] time in the experience of coiners a GAIN BY COINING to the extent of £25 4s. 10½d. on each million coined, thus approaching to the calculated gain of £59 4s., while in the next financial year it will be observed that there was a loss of £22 11s. 0½d., with a value of sweep recovered £73 4s. 0½d., giving an absolute gain of £50 13s., practically the result calculated upon. To comprehend, then, the reason, why in the following years a loss is again permitted, it will be necessary to enter into matters of the inner life of the Mint. By the year 1859 the hatred excited against me in consequence of these exposures had culminated to the extreme. The Master, then desirous of finding a permanent position for his brother,^[57] listened to and encouraged any stories against myself, notwithstanding the facts above related, and instead of consulting me and sending necessary orders through me, as usual, sent written orders openly by messengers, whose instructions were to read them to each officer. These styled by himself "circulating orders," were intended to disgust all the officers who had assisted in the reforms, and it was apparently determined by these and other like means to bring back the management of the Mint into the former chaos of irregularities, that he might make his brother's appointment appear necessary.

The disrespect thus exhibited to myself, and to the officers generally, had to a certain extent its designed effect, for in 1859-60 the loss rose to £76 7s. 7d., the sweep selling for £66 12s. 5d., showing an absolute loss of £9 15s. 2d. It will now be necessary to refer to a statement of losses and gains upon each *quarter*, because the accounts were not officially balanced, the next statement being for the years April, 1860, to March, 1862,^[58] but the quarterly account below^[59] shows that the loss rose to 65·444 ounces, progressing rapidly in June, 1861, to 105·823 ounces, while in March, 1862, it had reached the ugly figure of 157·745 ounces. Before, however, this vast loss had been reached, the officers consulted together with a view to devise a means of arresting the

downward course. About this time a piece of gold was discovered secreted in the drag room tunnel. I felt that this was a fact on which I could legitimately appeal to the Master against his new system, and this I did on the 31st December, 1861, when I represented to him that the reappearance of the losses was wholly due to his withdrawal of support from his officers, and that if he would not take the steps which were necessary, he alone must bear the blame, for I had now represented the truth to him. He asked my advice; so I told him boldly that he must come down officially to the works, as of old, and show himself a supporter of honest men. After much persuasion he assented to this advice, and came down. From that day the losses ceased, yet the effects, so far as the accounts are concerned, are exhibited till the quarter ending March, 1862; but the following quarter, ending June, 1862, exhibits a natural gain of 17·942 ounces. This table is of value, as showing the alternations of the apparent gain or loss; for in some quarters the loss appears to be greater than in others, arising from the fact that the bullion picked from the sweep is in some cases included; but this is always included in the return of the financial year.

**STATEMENT SHOWING THE LOSSES AND GAINS
UPON EACH QUARTER,
FROM JUNE, 1856, TO SEPTEMBER, 1868.**

Quarter ending:—	GOLD.		SILVER.	
	Loss.	Gain.	Loss.	Gain.
	Ounces.	Ounces.	Ounces.	Ounces.
June 30th, 1856	417·070			
Dec. 31st, "	147·325	—	268·460	
	—	—	9·550	
March 31st, 1857	17·220	—	690·314	
June 30th, "	—	—	94·026	
Sept. 30th, "	—	—	315·903	
Dec. 31st, "	21·316			
March 30th, 1858	—	4·673	500·335	
Sept. 30th, "	—	—	315·114	
Dec. 31st, "	—	—	211·250	
March 31st, 1859	16·603	—	259·082	
June 30th, "	9·417	—	185·358	
Sept. 30th, "	—	—	240·890	
Dec. 31st, "	—	—	403·593	
March 31st, 1860	65·444 ^[60]	—	69·096	
Sept. 30th, "	—	—	427·238	
June 30th, 1861	105·823 ^[61]			
Dec. 31st, "	—	—	606·765	
March 31st, 1862	157·745 ^[62]	—	—	2·501
June 30th, "	—	^[63] 17·942	170·965	
August 31st, "	5·671			
Dec. 31st, "	5·584	—	132·984	
March 31st, 1863	—	20·793		
June 30th, "	—	—	458·672	
Dec. 31st, "	13·767			
March 31st, 1864	—	60·821	48·845	
June 30th, "	—	13·843	9·042	
Sept. 30th, "	—	—	783·347	
Dec. 31st, "	39·679			
March 31st, 1865	17·660	—	147·780	
June 30th, "	—	—	480·346	
August 31st, "	—	1·020	195·546	
March 31st, 1866	58·632	—	230·515	
May 30th, "	5·037	—	136·014	

Quarter ending:—	GOLD.		SILVER.	
	Loss.	Gain.	Loss.	Gain.
Sept. 30th, "	—	—	631·037	
March 31st, 1867	—	—	260·352	
June 30th, "	3·562	—	158·951	
August 31st, "	—	0·205	43·033	
March 31st, 1868	—	—	271·540	
June 30th, "	—	—	222·545	
Sept. 30th, "	19·873	—	19·897	

These are grave charges, but an inquiry could not fail to establish their truthfulness. To return, then, to the table at [page 90](#), it will be observed that the loss of 1859-60 increased in the years 1860-62 to £45 3s. 5d.; but the stimulus applied by the renewed support of the Master reproduced its legitimate results, and in 1862-63, instead of a loss of bullion, there was a gain of £11 3s. 8d., which, with the sweep, made £50 10s. 7d., and this increased in 1863-64 to £91 5s. 1d.

STATEMENT OF LOSSES INCURRED AND OF SWEEP RECOVERED.

THIS REFERS WHOLLY TO GOLD IN THE MELTING DEPARTMENT.

Date—Financial Year.				Weight of Coinage.	Value of Coinage.		
				Ounces.	£	s.	d.
Nov.	1851	to	March 1852	351837·830	1,369,968	11	0
April	1852	to	March 1853	3058360·186	11,908,489	19	5½
"	1853	"	1854	2726011·580	10,614,407	11	9¾
"	1854	"	1855	812219·517	3,162,579	14	10½
"	1855	"	1856	2158391·456	8,404,236	14	7½
"	1856	"	1857	1254703·786	4,885,502	17	2
"	1857	"	1858	1256586·572	4,892,833	19	4
"	1858	"	1859	736150·582	2,866,386	6	6¾
"	1859	"	1860	980070·679	3,816,150	4	2
"	1860	"	1861	861205·000	3,353,316	19	5
"	1861	"	1862	1872214·638	7,289,935	14	11
"	1862	"	1863	2351444·336	9,155,936	7	8
"	1863	"	1864	1807332·130	7,037,299	9	8
"	1864	"	1865	1949438·200	7,590,624	19	10
"	1865	"	1866	1256347·104	4,891,901	10	9
"	1866	"	1867	175838·208	684,670	0	5
"	1867	"	1868	109580·129	426,677	12	1¾

Date—Financial Year.				Weight of Lost Metal.	Value of Lost Metal.			Value of Sweep.		
				Ounces.	£	s.	d.	£	s.	d.
Nov.	1851	to	March 1852	91·456	356	2	1½	133	0	2
April	1852	to	March 1853	1332·474	5,188	6	4¾	708	10	0
"	1853	"	1854	728·088	2,834	19	10½	709	9	8¼
"	1854	"	1855	223·811	871	9	3¼	209	2	4¾
"	1855	"	1856	867·277	3,376	19	2¼	805	5	4¾
"	1856	"	1857	515·298	2,006	8	9¾	469	1	5¾
"	1857	"	1858	275·452	1,072	10	10	262	18	5¾
"	1858	"	1859	248·804	969	15	7¼	231	2	1½
"	1859	"	1860	259·069	1,008	15	0	240	16	3
"	1860	"	1861	17·668	3,183	15	11	729	19	10½

Date—Financial Year.				Weight of Lost Metal.	Value of Lost Metal.			Value of Sweep.		
”	1861	”	1862							
”	1862	”	1863	636·395	2,477	19	3	590	2	7
”	1863	”	1864	629·815	2,452	6	10	507	11	0
”	1864	”	1865	464·080	1,807	0	8	509	10	0
”	1865	”	1866	585·622	2,280	5	4	535	5	0
”	1866	”	1867	37·440	145	15	0	Nil.		
”	1867	”	1868	—	—			71	10	6

Date—Financial Year.				Value of Lost Metal per Million.			Value of Sweep per Million.		
				£	s.	d.	£	s.	d.
Nov.	1851	to March	1852	259	18	9½	97	1	9¼
April	1852	to March	1853	435	13	7¾	59	9	10¾
”	1853	”	1854	267	1	9¼	66	16	10
”	1854	”	1855	275	11	1	66	2	5½
”	1855	”	1856	401	16	3¾	95	16	4
”	1856	”	1857	410	13	10¼	96	0	3¼
”	1857	”	1858	‡ 219	4	1¼	‡ 53	14	8½
”	1858	”	1859	‡ 337	19	7	‡ 80	12	6¼
”	1859	”	1860	264	16	8	63	2	0
”	1860	”	1861	297 10 1			68 11 9		
”	1861	”	1862						
”	1862	”	1863	270	12	9	64	9	0
”	1863	”	1864	348	9	6	72	2	5
”	1864	”	1865	238	1	2	67	2	5
”	1865	”	1866	466	2	6	109	8	2
”	1866	”	1867	212	18	6	Nil.		
”	1867	”	1868	—			167	12	6¾

This mark (‡) is intended to attract the attention when at pages [98](#), [103](#), these figures are discussed. To obtain the true amount of loss, deduct the value of the sweep from the value of the gold lost.

**RETURN TO AN ORDER OF THE HOUSE OF LORDS,
DATED 17TH FEBRUARY, 1870.
THE LORD ROSSIE. No. 2.**

FROM THE MELTING DEPARTMENT OF THE ROYAL MINT.

A STATEMENT of the Weight and Value of the Gold Moneys coined in each Financial Year from 1851 to 1869 inclusive, exhibiting the Weight and Value of the Loss or Waste sustained in each year, as well as the Value of Sweep recovered, and the average Proportion of such Loss or Waste and Sweep recovered to each Million Pounds Sterling coined.

Date.				Weight of Coinage.	Value of Coinage.		
				Ounces.	£	s.	d.
Nov.	1851	to March	1852	351837·830	1,369,968	11	0
April	1852	to March	1853	3058360·186	11,908,489	19	5
”	1853	”	1854	2726011·580	10,614,407	11	10
”	1854	”	1855	812219·517	3,162,579	14	10
”	1855	”	1856	2158391·456	8,404,236	14	7
”	1856	”	1857	1254703·786	4,885,502	17	4
”	1857	”	1858	1256586·572	4,892,833	19	4

Date.				Weight of Coinage.	Value of Coinage.
"	1858	"	1859	736150·582	2,866,386 6 7
"	1859	"	1860	980070·679	3,816,150 4 2
"	1860	"	1861	861205·000	3,353,316 19 5
"	1861	"	1862	1872214·638	7,289,935 14 11
"	1862	"	1863	2351444·336	9,155,936 7 8
"	1863	"	1864	1807332·130	7,037,299 9 8
"	1864	"	1865	1949438·200	7,590,624 19 10
"	1865	"	1866	1256347·104	4,891,901 10 9
"	1866	"	1867	175838·208	684,670 0 5
"	1867	"	1868	127485·816	496,397 17 11
"	1868	"	1869	1270778·550	4,948,093 19 7

Date.				Amount of Waste.	Value of Waste.			Value of Sweep recovered.		
				Ounces.	£	s.	d.	£	s.	d.
Nov.	1851	to March	1852	91·456	356	2	2	133	0	2
April	1852	to March	1853	1332·474	5,188	6	5	708	10	0
"	1853	"	1854	728·088	2,834	19	11	709	9	8¼
"	1854	"	1855	223·811	871	9	3	209	2	4¾
"	1855	"	1856	867·277	3,376	19	2	805	5	4¾
"	1856	"	1857	515·298	2,006	8	10	469	1	1¾
"	1857	"	1858	275·452	1,072	10	10	262	18	5¾
"	1858	"	1859	248·804	968	15	7	281	2	1½
"	1859	"	1860	259·069	1,008	15	0	240	16	3
"	1860	"	1861	817·668	3,183	15	11	729	19	10½
"	1861	"	1862							
"	1862	"	1863	636·395	2,477	19	3	590	2	7
"	1863	"	1864	629·815	2,452	6	10	507	11	0
"	1864	"	1865	464·080	1,807	0	3	509	10	0
"	1865	"	1866	585·622	2,280	5	4	535	5	0
"	1866	"	1867	37·440	145	15	8	Nil.		
"	1867	"	1868	54·460	212	1	0	71	10	6
"	1868	"	1869	701·084	2,729	16	11	788	7	4

Date.				Value of Waste. per £1,000,000.			Value of Sweep per £1,000,000.		
				£	s.	d.	£	s.	d.
Nov.	1851	to March	1852	259	18	10	97	1	9
April	1852	to March	1853	435	13	8	59	9	11
"	1853	"	1854	267	1	9	66	16	10
"	1854	"	1855	275	11	1	66	2	6
"	1855	"	1856	401	16	4	95	16	4
"	1856	"	1857	410	13	10	96	0	3
"	1857	"	1858	219	4	1	53	14	8
"	1858	"	1859	337	19	7	80	12	6
"	1859	"	1860	264	6	8	63	2	0
"	1860	"	1861	297	10	1	68	11	9
"	1861	"	1862						
"	1862	"	1863	270	12	9	64	9	0
"	1863	"	1864	348	9	6	72	2	5
"	1864	"	1865	238	1	2	67	2	5
"	1865	"	1866	466	2	6	109	8	2

Date.				Value of Waste. per £1,000,000.	Value of Sweep per £1,000,000.
"	1866	"	1867	212 18 6	Nil.
"	1867	"	1868	427 2 6	144 1 9
"	1868	"	1869	551 13 4	159 6 6

C. W. FREMANTLE, DEPUTY-MASTER OF THE MINT.
ROYAL MINT, _2nd March_, 1870.

In November, 1863, Mr. John Graham was appointed; but he did not interfere in the coining department until about March, 1864. The disastrous effects of his mismanagement now became apparent, and the losses have again made, it is to be feared, an habitual appearance. In the first year of his efforts, the loss per million^[64] became £22 6s. 2d., the sweep selling for £38 18s. 6d., giving a gain of £16 12s. 4d. only; in the next year, a gain of £25 14s. 7d.; and it was now determined, notwithstanding a written protest, that the sweep and *scrap* were not worth collecting, so they were lost. The next two years exhibit absolute losses of £28 12s. 10d. and £30 12s. 8½d. per million respectively, while in September, 1868, a loss of 19·873 ounces is recorded as a balance; yet it is a fact that no gold was ever produced to satisfy this balance, so that in my opinion it will be correct to state that the loss is at the rate of £70 14s. 5¾d. on each million coined. Unless upon the existence of such facts, it is difficult to understand why the Mint authorities should not publish annually an official statement of the loss or gain made by coining gold and silver, as well as the amount coined; but, that the whole details may be complete, it is now necessary to enter upon the discussion of the melting-house accounts. The table on p. 96 exhibits the operations of that department upon the same gold as that just discussed. If, as in the case of the table relating to the coining department, the total loss between November, 1851, and March, 1857, be reduced by the value of the sweep sold, it will appear to be, on gold, £11,599·825, and this divided by £40,345,185·450, the total sum coined, will give an average loss of £280 7s. 8¾d. per million coined, when, as was admittedly the case, the gold scissel was saturated to dropping with oil. Let us now see if this loss be increased by the reduction of the amount of oil which was formerly sent to the melter on the scissel.

When in 1857-58 the loss by coining was put a stop to, and a gain was produced, it was roundly asserted that the melter's account would show that the *loss*, so far from being stopped, had only been transferred from one department to another; but by examining the table now under discussion, it will be seen that when, in 1857-58,^[65] the *gain* of £25 4s. 10½d. occurred in the coining department, so far from there being a CORRESPONDING LOSS in the melting-house, that which had averaged £280 7s. 8¾d., fell to £165 9s. 4¾d.,—as will be seen by deducting the sweep sold from the loss recorded in the year 1857-58. When, in the next year, the gain by coining rose to £50 13s., the melter's loss did not reach its former average amount, for in that year, when he had as large an amount of brittle gold to contend with as in 1858, his loss was £257 7s. 0¾d.,^[66] or still below the old average; whereas, if from this period the loss be estimated, it will be found that its average is £212 6s. 1d. per million coined, as against £280 7s. 8¾d., or a clear saving of £68 1s. 7d. effected by lessening the amount of oil on the scissel ([explained at page 87](#)); and this amount may fairly be added to the total sum saved per million coined, which will be for coining £494, for melting £68 1s. 7d., making together £562 1s. 7d. on each million coined. That the loss by melting should reach the extravagant sum of £212 per million coined—Mr. Mathison, the contractor under the Company of Moneyers, made an average loss of £112 only—can hardly be maintained even in the Royal Mint; that it does occur is indisputable. The cause for it is not far to seek; yet it is hardly within my scope in this place to point out the means by which it should be reduced to within the limit of £100, which is demonstrably the maximum limit it should be permitted to reach. This, then, is the state of affairs as regards loss by the coining of gold.

Thus far I have considered the rate of loss at a period when such loss was deemed to be *necessary*; its reduction when it was demonstrated to be *unnecessary*, as resulting from preventible causes, and, in continuation, the means whereby *loss should be prevented to the establishment of the absolute profit*, if a correct system be established throughout the Mint. I now proceed to show the re-establishment of the old system of UNNECESSARY LOSS—loss which should not and would not be permitted in any well-regulated institution; but when men are permitted to travel at the expense of the country to European Mints, with the simple object of finding reasons why loss should arise, and authorities for its existence, we must hush our surprise, and be thankful if that loss do not exceed the amount heretofore reached. To me it appeared singular that the Mint Officials should be present in the House of Lords and hear Lord Lansdowne state from his place that “upon

exact inquiry he had been assured that a loss of one shilling on £100 was not an unreasonable sum,” without subsequent protest against such a statement. It is more singular if one should compare that estimated loss with what those Officials state in their “Reports on European Mints;” but it is most singular, that by studying the returns made by those Officials to the House of Lords, it becomes apparent that throughout the whole affair there is exhibited a great disregard of the exact truth. To make clear my view of the case, I will adduce the three statements. Lord Lansdowne’s fixed sum of a shilling on £100, is equivalent to £500 on a million; while Mr. Fremantle, at [page 7](#), “European Mints,” states that careful enquiry has shown, that as regards melting “in all Mints, considerable allowance is made for such loss, and that the ‘waste,’ shown to have existed of late years in the English Mint has not been excessive.” The return shows that between 1866-69 it averaged £296 per million.^[67] Mr. Roberts says: “In the English Mint the amount of loss on gold by melting is considered to be 0·173 per mille, or a grain on the troy pound;” that is to say, £173 on a million. Why cannot these Authorities agree? Lord Lansdowne, of course, speaks on instructions received, probably from Mr. Fremantle, who thus exhibits his inability to agree with himself, and agreement with his fellow-traveller seems hopeless; for whereas Mr. Fremantle reduces Lord Lansdowne’s loss from £500 to £296, Mr. Roberts reduces the latter sum nearly 50 per cent., and states it at £173.

I will now refer to the returns made to the order of the House of Lords on the motion of Lord Kinnaird, and which, for this purpose, I have printed on pages [91](#), [97](#), opposite the tables obtained from the same source whence these returns were made. By the return ([see page 97](#)) the Mint Authorities clearly admit that whereas the loss by melting was in 1866-67 only £212 18s. 6d., it became, in 1867-68, £283 1s. 9d., and increased to £392 6s. 10d. in the subsequent year, as will be seen if the value of the sweep be deducted from that of the waste. It appears to me that such startling facts require attentive enquiry, especially if it be borne in mind that the loss by melting as shown in the return was so little as £165 9s. 5d. in 1857-58. Since, however, it is now the custom in the Mint to pass the bars for coining, directly from the melting-house to the coining department—in accordance with the recommendation made by me in my Report to the Master of the Mint dated 29th of January, 1859—so as to bring the whole operation of coining under one responsible officer, as I therein urged, it is but right I should give the Mint the benefit of any advantage that can be obtained by a retrospective treatment of accounts, and therefore I will refer to the table on [page 101](#) which shows the final result of operations conducted in the coining and melting departments. In cases where gain was made by coining I will deduct such gain from the loss by melting, and where loss by coining took place I will add it to that by melting. Thus treated then, the loss by melting and coining for 1866-67 becomes £241 11s. 4d., for 1867-68 £309 8s. 4d., and for 1868-69 £319 9s. 1d., or a steadily increasing amount, as will be seen subsequently on [page 104](#).

As regards loss on gold, enquiry would prove that a remarkable oblivion to all carefulness or the principles of economy exists in the minds of Mint Officials; for this noble metal, being one of the most dense as well as one of the most valuable, can be packed away into a small space; hence the utmost care, guided by knowledge of the subject, is necessary if waste is to be prevented. To start with that indisputable axiom, “matter cannot be lost,” is but to have the truth placed in unmistakable language; yet by the return^[68] made to Lord Kinnaird’s motion a most startling contradiction to this truism is brought to light. The Mint Authorities, from the time of the government occupation of the Royal Mint, have manifestly *omitted* to think that “matter cannot be lost,” and they have assumed that the mere fact of subdivision must entail loss; but this fallacy is at once dissipated if Lord Kinnaird’s idea be adopted. He, in his speech in the House of Lords, said: “If any of your lordships will place a thousand sovereigns in a box, and cause the box to be shaken until the sovereigns have reduced themselves to impalpable powder, and will then weigh the dust—gold-dust—produced, it will be found that the dust now weighs the exact weight of the sovereigns before shaking.” So apt an illustration would be spoiled by comment, yet we find that the return ([see page 102](#)) exhibits the fact that the Mint has paid to the Bank of England sums varying from £1,132 to as low as £161 for each million of gold coined into sovereigns. Now the processes necessary to the conversion of gold into sovereigns is merely one of subdivision—of less degree than absolute powdering—therefore it is quite clear there can be no loss of gold. If there be any missing, it is because it is misplaced. Misplaced gold is practically lost, because Mint experiences appear to be adverse to the finding of the place of secretion. By this return it appears that the Mint has actually paid so large a total sum as £59,084 1s. 4d., in seventeen years to the Bank of England for gold which should have been returned to that institution. The Government of Mr. Gladstone, professedly desiring to economise governmental expenditure, permits this!!!

While it is the opinion of the Government that LOSS IS REASONABLE, it is hopeless to expect a stoppage of that which should not be permitted; yet it is right to state that these losses are not only demonstrably unnecessary,

but that they have been proved by practice to be capable of great reduction, as shown by the following table, compiled from the returns before quoted on pages [91](#), [97](#).

STATEMENT showing the ultimate Loss by Coining Gold, by Melting Gold, and the Gain by Coining Gold, as also the absolute Loss resulting from both operations, from 1851 to 1869. Compiled from the Returns for which *Lord Kinnaird* moved.

AVERAGE RATE FOR EACH MILLION COINED.

Date. Financial Year.	Ultimate Loss by Coining £1,000,000 Gold.			Ultimate Gain by Coining £1,000,000 Gold.		
	£	s.	d.	£	s.	d.
1851-1852	315	8	7			
1852-1853	545	6	10			
1853-1854	595	6	9			
1854-1855	451	13	9			
1855-1856	418	1	3			
1856-1857	377	0	6			
1857-1858				25	4	10
1858-1859				50	12	11
1859-1860	9	15	2			
1860-1862	45	3	5			
1862-1863				51	0	3
1863-1864				91	5	1
1864-1865				16	12	4
1865-1866				25	14	7
1866-1867	28	12	10			
1867-1868	26	6	7			
1868-1869				72	17	9

Date. Financial Year.	Ultimate Loss by Melting Ingots, &c. for £1,000,000 Gold.			Ultimate Loss by both Operations.		
	£	s.	d.	£	s.	d.
1851-1852	162	17	1	478	5	8
1852-1853	376	3	9	921	10	7
1853-1854	200	4	11	795	11	8
1854-1855	209	8	7	661	2	4
1855-1856	306	0	0	724	1	3
1856-1857	314	13	7	691	14	1
1857-1858	165	9	5	140	4	7
1858-1859	257	7	1	206	14	2
1859-1860	201	4	8	210	19	10
1860-1862	228	18	4	274	1	9
1862-1863	206	3	9	155	3	6
1863-1864	276	7	1	185	2	0
1864-1865	170	18	9	164	6	5
1865-1866	356	14	4	330	19	9
1866-1867	212	18	6	241	11	4
1867-1868	283	1	9	309	8	4
1868-1869	392	6	10	319	9	1

**RETURN FROM THE ROYAL MINT TO AN ORDER
OF THE HOUSE OF LORDS,
DATED 17TH FEBRUARY, 1870.
THE LORD ROSSIE.**

A Statement showing the individual sums paid by cheque by the Master of the Mint to the Bank of England in consideration of any loss or waste by coining gold, by melting gold, and by assaying gold on each occasion of a balance of accounts by those institutions between 1851 and 1869 inclusive, specifying the proportional payments in relation to each million coined during the several periods.

Date of Payment.	Period during which Waste occurred.	Amount of Cheque.			Rate of Waste per Million.		
		£	s.	d.	£	s.	d.
1852, March 31st	From 11th July 1851 to 31st March 1852	890	1	5	649	0	0
" August 26th	From 1st April to 30th June 1852	1,926	10	5	951	0	0
" December 6th	From 1st July to 30th September 1852	2,051	16	9	882	0	0
1853, June 17th	From 1st October to 31st December 1852	3,209	8	2	995	0	0
" July 8th	From 1st January to 31st March 1853	5,270	1	3	1,132	0	0
" September 30th	From 1st April to 30th June 1853	4,564	0	0	1,026	0	0
" December 30th	From 1st July to 30th September 1853	1,147	5	2	1,112	0	0
1854, August 30th	From 1st October 1853 to 30th June 1854	4,583	0	0	878	0	0
1856, May 2nd	From 1st July 1854 to 31st December 1855	8,685	14	3	895	0	0
1857, June 12th	From 1st January 1856 to 30th May 1857	5,265	0	0	803	0	0
1858, May 5th	From 1st June 1857 to 31st March 1858	1,110	14	7	230	0	0
1860, March 12th	From 1st April 1858 to 30th June 1859	1,282	16	7	360	0	0
" June 27th	From 1st July 1859 to 31st March 1860	1,084	6	3	347	0	0
1862, August 29th	From 1st April 1860 to 31st August 1862	4,272	11	7	389	0	0
1863, March 30th	From 1st September 1862 to 31st December 1862	1,766	14	8	350	0	0
" June 3rd	From 1st January to 31st March 1863	606	19	4	161	0	0
1864, September 6th	From 1st April 1863 to 31st July 1864	2,584	11	11	281	0	0
1865, March 29th	From 1st August 1864 to 31st December 1864	1,320	16	9	419	0	0
" October 19th	From 1st January to 30th September 1865	438	6	3	169	0	0
1866, July 4th	From 1st October 1865 to 30th June 1866	2,633	13	2	485	0	0
1867, December 12th	From 1st July 1866 to 30th November 1867	227	5	20	457	0	0
1869, November 30th ^[69]	From 1st December 1867 to 30th November 1869	4,162	1	0	461	0	0
		£ 59,084	1	4			

C. W. FREMANTLE, DEPUTY-MASTER AND COMPTROLLER.

ROYAL MINT, 2nd March, 1870.

In this table ([page 101](#)) care has been taken to remove every item which can mislead, and therefore the ultimate or absolute loss to the Mint is alone put forward; and while the subsequent table ([page 102](#)) shows the money and its rate per million actually paid to the Bank for loss; this shows the internal accounts of the operative departments in the Mint only, so that every item which can reduce the loss has been admitted, and it is therefore necessary to point out that each sum exhibited in the last column as the ultimate loss by both operations, is the rate for each million coined, all being under precisely similar circumstances in every respect. I have excluded loss by assay.

If this table ([page 101](#)) be examined, it will be seen that between 1851 and 1857 there was an invariable statement of loss by coining; then between 1857 and 1859 a period of gain; to be followed in the next two periods, 1859 to 1862, by loss, which was again overcome; and for the next four years a varying gain, alternated by two years of loss, and followed finally by another statement of gain.

Enquiry may fairly be made why, if up to 1857 loss had been permitted, gain could subsequently be established. The reply is, that the management of the coining department was placed under myself by Mr.

Graham, and I acted firmly on my belief that “matter cannot be lost;” hence, if the accounts were short, I made the men hunt the floors—not the chimneys—carefully until the gold was found.

The Master of the Mint then discouraged my efforts, and the result was rapidly increasing loss, until I placed on the Master^[70] the responsibility of such loss, when there returned four years of gain, thus once more proving that under proper management loss would not occur; but from this period I was removed from responsible management, and the result was a return to the habitual loss. This would appear to have been overcome in the year 1868-1869, for in that year a gain to the extent of £72 17s. 9d. (see Return, [page 91](#)) is exhibited. It will be necessary to examine this gain to see how far it is real, and the proof is at hand that it is obtained under false pretences.

For special reasons which are not exhibited,^[71] the figures relating to 1851-1852 on the table at [page 96](#) are fictitious; the fact being that losses which should have been stated are excluded. If, however, the table from 1852 to 1869 be examined it will be observed that the ultimate loss by melting varies considerably, but that in the year 1857-1858 it fell to £165 9s. 5d., while in that year a gain was made by coining to the extent of £25 4s. 10d.^[72] This gain being deducted from the loss by melting, leaves a final loss of £140 4s. 7d., as will be seen in the last column at [page 101](#). To test then the gain shown in 1868-69 by the same rule, it will be observed that while the coining department (see [page 101](#)) made a profit of £72 17s. 9d.,^[73] the loss by melting was £392 6s. 10d., or higher in amount than it has been in *any previous year*, so that the gain claimed for coining is *false*—it has not arisen—it is a STATEMENT OF ACCOUNTS, obtained by sending inaccurate weights of bullion to the melter, and thus shifting to that department the loss which fairly belonged to the coiner. The same remark applies to the year 1865-1866, when it will be seen that the system of loss fairly reset in.

BONA FIDE GAIN BY COINING GOLD IN ITS FINAL OUT-TURN is only to be obtained by the possession of a firm belief in the impossibility of legitimate loss, and by entrusting the operations of coining to such men as appreciate the fact that it is atoms which build up the universe. I proved, and maintained my assertion for years, that there should be no loss, but on the other hand gain, by coining gold, and this I did with a reduction in the loss by melting to a lower point than has ever before or since been known, and with a total loss from both causes of only £140 4s. 7d. (see [pages 101-102](#)); and, had occasion served, it was always intended that I should direct the melting of gold as I had done that of silver,^[74] for as I had clearly shown that coining could be fairly conducted without loss, the Master was to have given me an opportunity to show the same facts in melting.

Of this loss of £140 4s. 7d. in the final out-turn of gold, it should be explained that at that date the system of weighing to and from the bank was such that there was on that coinage a difference of weighing *against the Mint* of £75 on each million, so that had just weighments been made, my loss from all causes being reduced by that amount would have been £65 4s. 7d. on each million coined.

With such facts established and stated in the return to the House of Lords, it is at least singular that Lord Lansdowne should support his assertion that 1s. on £100 is a moderate and fair loss, by stating that this is also the opinion of eminent refiners in London. In my experience refiners do not know what their losses are, for their customers bear them. By special agreement with Sir A. Rothschild, the Mint pays 4d. per ounce on the total sent for refining, and receives back all the gold and silver; but the general public, by previous agreement, submit to a deduction of weight, and thus pay the expenses of refining. Mr. Graham, in his paper published after his brother's failure to conduct the gold coinage as I had done, fixed the loss at £300 on a million. Lord Lansdowne now fixes it at 1s. on £100; that is, at the rate of £500 on a million; and this sum is being rapidly approached, for Lord Kinnaid's return shows that for the past three years it has averaged £464.^[75] And since such a loss is to be borne and pleasantly spoken of by Mr. Lowe, the Master of the Mint, as “going up the chimney,” it is well to see if that can take place.

The volatilization of gold and silver requires an extremely high temperature. This temperature is never reached in melting the precious metals for coinage; therefore vapour of gold or of silver cannot by any possibility be conveyed into the chimney; consequently, cannot be found there. That this statement is true, has been demonstrated satisfactorily in Brussels, where the chimneys were discreetly searched to find the silver which the contractors were informed would be found: the chief contractor assured me that they failed to find the most minute trace of either silver or gold. Yet Mr. Roberts, at [page 23](#), “European Mints,” says, “The precious metals actually volatilized may be arrested by condensation. This point has received much attention in the Roman Mint, where the flues of the gold melting furnaces have been placed in communication with condensing

chambers of simple construction, the adoption of which has been attended with very satisfactory results.” On [page 17](#) he has already said “I strongly recommend that a condensing chamber be attached to the flues of the gold melting furnaces. Such chambers have not hitherto been used in the Mint in this country, but their advantage has been proved in the Roman Mint, where they have been the means of effecting a considerable saving.”

It is with regret that one sees men entrusted with the control of so great a work as our coinage, making propositions on the authority of the Roman Mint, whose gold coinage is perhaps the smallest in Europe. If, however, Mr. Roberts should conduct on a large scale the plan *he has devised* for removing the obnoxious properties from brittle gold, these chambers may become useful. In the meantime the present Mint chimneys have the accumulations of more than sixty years, and it would be quite worth while to pull them down and obtain the £59,000^[76] which the Mint admits to have lost ([see page 101](#)). Are the chimneys lined with finely-divided gold, or does that metal accumulate in ingots? if the latter be the case, their draught must be slightly impeded, for fifty-nine thousand pounds worth of gold would more than fill an imperial bushel; in the popular view a “thousand sovereigns fill a quart measure.” If Mr. Roberts should fail to find any trace of gold, he can reconsider his proposition; should he, however, find the sum which has been supposed to be lost, that treasure will pay the cost of new chimneys.

If, instead of Lord Lansdowne appealing to Lord Kinnaird’s generosity to leave unsaid things which should be said, his Lordship would refute the facts and figures which Lord Kinnaird places before their Lordships’ House, it seems probable that the question would be settled; but while Lord Lansdowne simply pleads that “the person who supplies information to Lord Kinnaird is only a dismissed clerk,” it becomes manifest that the facts placed before Lord Kinnaird are awkward to contradict, and still more awkward to explain, except by the painful delusion Lord Lansdowne adopts—that “his lordship confuses the accounts.” It may fairly be admitted that the accounts furnished by the Mint Authorities to the House of Commons are stated in such a manner that bankers and merchants find it impossible to comprehend them, and to arrive at their meaning, more than one banker in London *has applied to me* to unravel those statements which have been characterised as “such that if our younger clerk were to present them he would be dismissed our service.” Still, that Lord Kinnaird has not fallen into this error has, it is hoped, been now made fully manifest. Lord Lansdowne, in continuation, stated that “the author of this book appears to have left unnoticed the losing operations in the Mint, and Lord Kinnaird seems to have fallen into the trap.” Had Lord Lansdowne looked into the book he thus criticized, he would have found that each process of gain and loss is treated at considerable length, but none with more detail than the processes of the melting both for gold and silver, as he will learn if he will refer to the book, now out of print, but a copy of which is in the Mint Library—for gold melting, at pages [69](#), [74](#), [81](#), [82](#), [83](#); and for silver melting, at pages [101](#), [104](#), [109](#), [111](#), where it will be clearly seen that these losses are dwelt upon and shown to be inexcusable. The returns upon which I have thus commented are also of importance as introducing a new system in accounts—that of altering such accounts without reason assigned or explanation given. If the return on [page 91](#) be examined for the year April 1867 to March 1868, it will be seen that a variation exists between it and my table on [page 90](#); for whereas I state the value of the coinage for that year at £426,677 12s. 1¾d. and the loss at 3·357 ounces, or at the rate of £30 12s. 8¾d. on the million coined, the return states the money coined at £496,397 17s. 11d.; and leaving unaltered the *amount of loss*, shows it to be at the rate of £26 6s. 7d. per million coined, a similar alteration has been made in the account of the melting department. Seeing that my figures were those supplied by the then Deputy Master to the late Master of the Mint, I cannot comprehend why those accounts should be issued by Mr. Fremantle, who has entered the Mint since that gentleman’s death, with a variation in the figures, but I shall presently show that such a proceeding is not confined to this one transaction.

While the subject of accounts is under notice, I think it right to observe upon a passage in Mr. Fremantle’s “Report on European Mints,” at [page 7](#), in which that gentleman says—

“I propose, also, that the ‘sweep,’ under which term are comprised dust collected from the floors, carbonaceous deposits from flues, ground-up crucibles, &c., should be treated in the Mint, instead of being sold, as at present, to the highest bidder. It will be important to determine what metallurgical process should be adopted with a view of carrying this intention into effect; but the step cannot fail to be advantageous to the Mint, not only as tending to economy, but also more especially as enabling the *department to substitute a real for a fictitious statement of ‘waste,’ the regulations of the public service NOT ALLOWING THE PROCEEDS OF THE ‘SWEEP,’ when sold to the public, to be brought to account AS A SET-OFF AGAINST THE GROSS AMOUNT OF LOSS.*”

that while I quite appreciate his thus carrying out the suggestion I never ceased to urge, I do object to his stating that which is hardly true as indicated by the passages I have marked. Such a statement must have been a

slip of the pen, as it is but necessary to refer to the returns made to the House of Lords^[77] to exhibit the fact that these accounts are now kept and actually published to Parliament, and in my own knowledge, those returns are accurate—not false—for I saw much of the money paid for the sweep, which I delivered to the purchasers on the authority of Mr. Barton. There is no sense in which these returns can be said to be false unless in the one year I have pointed out. If they be false, is it not a serious matter for any one to sign his name and submit them as accurate to Parliament?

Let us, then, turn to the accounts, and see how far the other expenses are unnecessarily inflated—and why? In such a consideration it will be necessary to revert, as in the case of losses, to the earlier periods of the Government occupation of the Mint; and, without going too far into details, it may be stated that the average amount of *rejected work* (see [page 44](#)) reached, in some instances, 70 per cent.; but if the averages for sovereigns be taken as extending over long periods, a good general knowledge may be gained, thus:—

In 1855, between June 6th and August 13th, the highest was 40·51 per cent., the lowest 13·13 per cent., the mean of 41 days giving 21·49 per cent.; while, at that period, the coined money was 43·53 per cent. on the clean bars.

In 1856, from January 2nd to January 22nd inclusive (18 working days), the highest was 21·48 per cent., the lowest 12·72 per cent., the mean of 18 days giving 18·51 per cent.; while, at that period, the coined money was 38·39 per cent. on the clean bars.

In 1856, during a short coinage I reduced its average amount to 13·78 per cent., the coined money increasing to 41·74 per cent. on the clean bars; but, as shown at [page 92](#), this improved in 1857 to 6·99 per cent., while the coined money rose to 57·26 per cent. on the clean bars.

In 1860, from January 4th to February 18th inclusive (41 days), the highest was 7·71 per cent., the lowest 2·33 per cent.; the mean of 41 days was 4·11 per cent., which by the file was reduced to 2·07 per cent.,^[78] while, at that period, the coined money was 57·43^[79] per cent. on the clean bars.

Before arriving at definite conclusions as regards the amount of rejected work, it will be necessary to understand that up to the 2nd of August, 1861, the remedy^[80] allowed in practice was 0·30 grain, and that in proportion as the remedy is reduced, the amount of rejected is increased; it is also the case that when work is good all the blanks or coins approach the standard weight, but when it is bad they as invariably go to the side or margin of the remedy. The following table exhibits the amount per cent. of rejected work^[81] which took place, with the various remedies; and these facts once determined, are to be remembered, because the rejected will in all cases bear these proportions to each other. In this table sovereigns are alluded to.

REMEDY IN PARTS OF A GRAIN.

	0·30 Grain.	0·25 Grain.	0·24 Grain.	0·23 Grain.	0·22 Grain.	0·21 Grain.	0·20 Grain.
Average in 1860	4·00	8·00	...	...	...	...	12·00
" 1861	...	10·00	13·00	15·00	15·00	18·00	19·00
" "	5·10	10·50	...	...	...	...	19·66

These considerations had no weight when the Coinage Bill was being hurriedly passed—at late hours, so as to avoid discussion on its clauses—by the House of Commons. It in due course reached the House of Lords, where Lord Kinnaird moved a clause—which would have been greatly to the benefit of the Mint had it been adopted—to the effect that the remedy on the individual piece should be 0·2568 grain; that is—the remedy from the old scale under the Mint Indenture proportioned to one coin. This Coinage Bill introduced a new principle, and it would have been wise had it given the Mint every chance of success. Instead, however, of proceeding cautiously, his lordship's motion was rejected and the remedy declared fixed at 0·2000 grain, thus necessitating the use of a working remedy of 0·1700 grain, or perhaps 0·1500 grain, and the consequent rejection for remelting of “only 15½ per cent. of the whole of the money coined.” My experience tells me, more probably of about 35 per cent.; but upon this head I will allow Mr. Napier to speak. He says, “Indeed, roundly taken, the returns to the crucible may be put at nearly 50 per cent.; so that to produce a given weight of finished coin, something like double the weight of metal must be melted. So long as this state of things lasts, the process up to

the formation of the blank must be regarded as imperfect.” It is not the *process that fails*, it is the want of power *to conduct that process* which causes so candidly admitted a continuation of the fatal system re-introduced by Mr. John Graham.

The proportioned legal remedy was 0·2568 grain. Now we can only wish to be within the law. If, therefore, 0·25 grain had been adopted as a *working remedy*, the automaton balances are so accurate that no piece could ever be found beyond the remedy, on either side, and an immense expense would be saved in labour alone, for the rejected goes to the melting-pot; yet, owing to a mistake on the part of the Bank of England, 0·20 grain was used in 1861, but under my urgent representation this was relaxed because of the extravagant expense, and for some years 0·24 grain was used, in accordance with the following order from the Master:—“Mr. Ansell is requested to increase the remedy on sovereigns from 0·20 grain to 0·24 grain, and the half-sovereigns in a similar proportion.—14th of August, 1861. (Signed) THOMAS GRAHAM.” Under other management, since the early part of 1868, the working remedy has been reduced to 0·15 grain. Such useless waste, it is to be hoped, has only to be pointed out that this clause of the New Coinage Act may be repealed.

Following these considerations, the next table will show the rate per cent. of rejected sovereigns and half-sovereigns, as well as the remedies practically employed in regular work. They are, in all cases, the average results of whole coinages of each specific year mentioned.

Date.	Sovereign Blanks.	Half-Sovereign Blanks.	By Filing Metal from the Edges of the Heavy Blanks they were reduced to the Standard Weight when the original rate per cent. was reduced to, on		Sovereigns coined (that is, weighed after the Coin was struck, so the heavy were melted).
			Sovereigns.	Half-Sovereigns.	
	Per Cent.	Per Cent.	Per Cent.	Per Cent.	Per Cent.
1855	21·49	32·29	...	...	...
1856	14·50	20·04	...	...	...
1857	7·44	16·13	...	...	...
1858	4·50	11·16	...	...	...
1859	5·67	8·86	4·14	5·52	...
1860	3·62	10·08	1·69	4·65	...
1861	5·75	...	3·36	...	...
”	21·43	...	12·22	...	...
”	16·07	...	6·68	...	...
1862	9·48	17·06	3·37	11·53	...
1863	9·68	...	4·25	...	...
”	18·76	18·08	...	11·69	...
”	18·76	...	6·76	...	...
”	...	28·79	...	12·86	...
1864	15·15	28·27	5·07	12·79	...
”	...	...	...	...	5·74
1865	...	...	...	...	6·61
1866	...	...	...	...	9·87
1867	...	...	...	...	...
1868	...	...	...	...	17·82

Date.	Half-Sovereigns coined (that is, weighed after the Coin was struck, so the heavy were melted).	Remedies employed on Sovereigns on each piece.				Remedies employed on Half-Sovereigns on each piece.	
	Per Cent.	Parts of a Grain.				Parts of a Grain.	
1855	...	...	...	...	0·30	...	0·15
1856	...	...	...	...	0·30	...	0·15
1857	...	...	...	...	0·30	...	0·15
1858	...	...	...	...	0·30	...	0·15
1859	...	...	...	...	0·30	...	0·15
1860	...	...	...	...	0·30	...	0·15
1861	...	...	...	...	0·30	...	...
"	...	...	0·20	...	...	...	...
"	...	...	...	0·24	...	...	...
1862	...	...	...	0·24	...	0·12	...
1863	...	...	...	0·24	...	...	...
"	...	...	...	...	...	0·12	...
"	...	...	...	0·24	...	...	...
"	...	...	...	0·24	...	0·12	...
1864	...	...	...	0·24	...	0·12	...
"	14·76	...	...	0·24	...	0·12	...
1865	12·60	...	...	0·24	...	0·12	...
1866	15·64	...	...	0·24	...	0·12	...
1867	20·28	...	...	...	...	0·12	...
1868	...	0·15	...	...	...	...	...

It will thus be seen that of the whole mass of gold coined in 1868, 17·82 per cent. of sovereigns and 20·28 per cent. of half-sovereigns went respectively to the melting-pot—a state of things which no contractor would tolerate for a week; yet this waste has continued throughout 1870. Now, there can be no reason why the bullion should not again be weighed in the form of blanks, and the heavy blanks reduced by the file, as they were in former days; for it must be remembered that that file^[82] has met with approval and adoption by the highest authorities on coining in their respective mints, and there is manifest folly and waste in sending fully 50 per cent. more rejected work to the melter than is absolutely necessary. These remarks will be more impressive if the table of rejected work be examined for 1861, where it will be seen that, with a remedy of 0·30 grain, the rejected was 5·75, reduced to 3·36 per cent., whereas it immediately went up to 21·43, reduced to 12·22 per cent., when, on the 2nd of August, the remedy was reduced to 0·20 grain; and that on the 14th of August, when the remedy was increased to 0·24 grain, the rejected became 16·07, and was reduced to 6·68 per cent., these being the figures obtained by the total amounts passed through all the automaton machines. We will next, then, see the effect of this extravagance in manufacture on the total out-turn of coin, because this is the final test of the cost of manufacture, it being readily understood that if 17 per cent. of the whole coinage be remelted, whereas 2·07 per cent. used to suffice, there must of necessity be a clear waste of labour to the extent of 15 per cent. Therefore, not only is unnecessary labour exacted, but also a large additional loss by melting is—under the present Authorities—incurred, as well as the loss by coining (see pages [101](#), [103](#)). Having considered this, it will be fitting that the subject of wages paid to workmen shall receive attention, as a means to arrive at the cost of coining a sovereign, and which, it may be here stated, has been by others estimated at far too high a sum.

In discussing, then, the amount of coined money obtained from gold bars, it will be necessary to enter into details of figures, and to show not only the rate per centum of coin obtained from bars as forwarded by the melter, but also the rate obtained from the clean bars—that is, from bars whose ends have been sheared off, and from the total weight of which the brittle bars and stopped pots have been deducted, thus leaving none but solid workable bars called, in practice, “clean bars,” as against the rough bars, which, indeed are rough enough to deserve that title.

**STATEMENT SHOWING THE RATE PER CENTUM OF MIXED COIN
FROM ROUGH BARS OF GOLD, AND THE RATE PER CENTUM OF
SOVEREIGNS AND HALF SOVEREIGNS FROM THOSE RESPECTIVE
BARS. THE BRITTLE BARS AND STOPPED POTS ARE DEDUCTED.**

Coinage conducted between the Periods:—	Mixed Bars, “Rough.”	Mixed Coin.	Rate per cent. on Mixed Bars.
	Ounces.	Ounces.	
November 1856 to April 1857	1,103,403	456,575	41·37
October 1857 to March 1858	2,280,033	1,238,171	54·24
November 1858 to June 1859	1,955,961	915,530	46·80
January 1860 to March 1860	1,460,209	802,847	54·28
January 1861 to April 1861	1,707,098	862,329	50·51
June 1861 to March 1862	4,448,353	1,874,885	42·14
May 1862 to March 1863	4,534,377	2,354,805	51·93
October 1863 to June 1864	4,946,697	2,323,238	46·96
October 1864 to December 1864	1,981,150	958,354	48·87
January 1865 to August 1865	1,129,772	518,673	45·91
November 1865 to May 1866	3,580,017	1,396,567	39·29
April 1867 to June 1867	305,418	127,576	41·77
July 1868 to August 1868	678,976	280,993	41·38

Coinage conducted between the Periods:—	Sovereign Bars, “Rough.”	Sovereigns Coined.	Rate per cent. on Sovereign Bars.
	Ounces.	Ounces.	
November 1856 to April 1857	480,704	288,592	60·03
October 1857 to March 1858	2,133,366	1,168,227	54·29
November 1858 to June 1859	1,341,862	592,325	44·14
January 1860 to March 1860	1,185,227	657,362	55·46
January 1861 to April 1861	1,402,538	717,010	51·12
June 1861 to March 1862	4,448,353	1,874,885	42·14
May 1862 to March 1863	4,237,267	2,220,713	52·40
October 1863 to June 1864	4,557,216	2,166,112	47·53
October 1864 to December 1864	1,713,659	847,357	49·44
January 1865 to August 1865	779,345	373,340	47·90
November 1865 to May 1866	2,676,226	1,041,926	38·11
April 1867 to June 1867	...	...	...
July 1868 to August 1868	678,976	280,993	41·38

Coinage conducted between the Periods:—	Half Sovereign Bars, “Rough.”	Half Sovereigns Coined.	Rate per cent. on Half Sovereign Bars.
	Ounces.	Ounces.	
November 1856 to April 1857	622,699	167,983	26·97
October 1857 to March 1858	146,667	69,944	47·68
November 1858 to June 1859	614,099	323,203	52·63
January 1860 to March 1860	274,982	145,400	52·87
January 1861 to April 1861	304,560	145,319	47·71
June 1861 to March 1862			
May 1862 to March 1863	297,110	134,092	45·13
October 1863 to June 1864	389,481	157,125	40·34
October 1864 to December 1864	267,491	110,986	41·48

Coinage conducted between the Periods:—	Half Sovereign Bars, “Rough.”	Half Sovereigns Coined.	Rate per cent. on Half Sovereign Bars.
January 1865 to August 1865	350,427	145,332	41·47
November 1865 to May 1866	903,791	354,640	39·23
April 1867 to June 1867	305,418	127,576	41·77
July 1868 to August 1868			

**STATEMENT SHOWING THE GOLD BARS USED IN EACH COINAGE
FROM NOVEMBER, 1856, TO AUGUST, 1868, AND THE RATE
PER CENTUM OF ENDS, OF MIXED GOLD COIN, AND OF
SOVEREIGNS AND HALF SOVEREIGNS PRODUCED FROM
THOSE BARS.**

Coinage conducted between the Periods:—	Mixed Bars, “Clean.”	Ends cut from Bars.	Sovereign Bars. “Clean.”
	Ounces.	Per Cent.	Ounces.
November 1856 to April 1857	1,057,776	4·13	460,851
October 1857 to March 1858	2,192,798	3·82	2,051,872
November 1858 to June 1859	1,798,717	4·41	1,210,447
January 1860 to March 1860	1,414,516	3·13	1,148,130
January 1861 to April 1861	1,619,251	5·14	1,330,448
June 1861 to March 1862	4,215,722	5·22	4,215,722
May 1862 to March 1863	4,351,531	4·03	4,066,506
October 1863 to June 1864	4,566,032	[83] 7·60	4,261,432
October 1864 to December 1864	1,851,724	6·48	1,602,533
January 1865 to August 1865	1,047,901	7·24	722,921
November 1865 to May 1866	3,336,536	6·80	2,494,243
April 1867 to June 1867	291,696	4·50	..
July 1868 to August 1868	612,857	9·57	612,857

Coinage conducted between the Periods:—	Half Sovereign Bars, “Clean.”	Mixed Coin, including Pyx Pieces.	Sovereigns from “Clean” Bars.	Half Sovereigns from “Clean” Bars.
	Ounces.	Per Cent.	Per Cent.	Per Cent.
November 1856 to April 1857	596,925	43·16	62·62	28·12
October 1857 to March 1858	140,926	56·39	56·93	49·63
November 1858 to June 1859	588,269	50·89	48·93	54·94
January 1860 to March 1860	266,386	56·75	56·43	54·54
January 1861 to April 1861	288,803	53·18	53·88	50·03
June 1861 to March 1862	...	...	44·47	
May 1862 to March 1863	285,025	54·03	54·60	47·39
October 1863 to June 1864	304,600	50·88	50·82	51·25
October 1864 to December 1864	249,191	51·75	52·87	44·53
January 1865 to August 1865	324,980	49·49	51·64	44·72
November 1865 to May 1866	842,293	41·80	41·77	42·10
April 1867 to June 1867	291,696	...	...	43·70
July 1868 to August 1868	...	...	45·84	

The half-sovereigns shown in this table ([page 111](#)) at 26·97 per cent. were coined in November, under the old system, but they are placed here that these returns may be accurate. The effect of the reduction of the

remedy from 0·30 to 0·20 grain (which has been explained at [page 107](#)) is evidenced in the amount of coin obtained in the period—June, 1861, to March, 1862; yet it is fair to admit that some of this gold was exceptionally bad, for the Bank of England, finding that the bad gold of 1859 had been coined, paid the Mint the compliment to send at this period some of a singularly rotten character, with an appearance almost woolly; and if reference be made to the table of rejected at [page 109](#), this gold will be seen to have produced, with a remedy of 0·24 grain, 16·07, reduced to 6·68 per cent., over the remainder of the coinage, thus making the disastrous effect of the reduced remedy more apparent. As has been before stated, the gold of 1859 contained, besides a vast amount of brittle gold, 0·45^[84] per cent. of its whole weight of dumb fillets. Then, coming to the period, October, 1863, to June, 1864, we find a continued state of bad work, influenced by the remedy of 0·24 grain to a certain extent but to a much more marked degree by the new system which was now introduced; and by referring to the next table, it will be seen that the ends at one jump went up from 4·03 to 7·60 per cent., because a mistaken opinion led to the shearing off the ends from the bars in the rolling room. The order for the re-introduction of this abandoned custom was in the following terms, so could not be set aside by those who saw its *unwisdom*:—"In the practice of the rolling room, Mr. John Graham is requested to cause not less than 3 inches from the hollow end of each gold bar to be cut off before beginning the rolling, with a view of keeping back the doubtful portion of the bar.—(Signed) THOMAS GRAHAM. 7th December, 1863." Grave as was this error, it was surpassed by a real blunder, for at this period it was determined to stop all fillets that appeared to exhibit any signs of brittleness, thus hopelessly rejecting an immense bulk of really good work lest it should contain any bad; whereas, had the proper course been adopted, the good from each fillet should have been selected, as was the *invariable* custom under the *MONEYERS* and the small per-centage of really bad fillets rejected. By this system the men were thoroughly disheartened. To such a length was this mismanagement at last carried, that the officers and men allowed things to take their course, feeling sure that time would demonstrate the folly of the present system. However, the extravagance has continued down to this day. Yet another source of waste of labour was left; and even this was utilised. In the working of the draw-bench, as well as the mill, there are at starting many fillets so varying in thickness that it is not fair to make the trier send them to the *cutters* as regular work. It was the custom at that period to detain these fillets to the end of the day, when they were especially treated, cut at a special cutter, and the blanks sent to be weighed—the good were reserved, but the bad were returned to the trier, yet not charged to him as bad work, thus enabling him to save at least 70 per cent. of, perhaps, 5,000 ounces per diem. Under the new *régime* this was not permitted, from sheer want of knowledge, although its stoppage was condemned alike by officers and men. Causes such as these reduced the average amount of coined sovereigns, from rough bars, from 51 to 44 per cent., as will be seen by examining the table of coined money obtained from *rough bars*.^[85] By now referring to the accompanying statement, the effect of this mismanagement will be seen, for instead of an average of 52·40, only 48·48 per cent. of coin was obtained from bars^[86] that had been stripped of every fault that could interfere with their producing power. Instead of yielding, as these should have done, a larger proportion of coin, they really gave less than the coin previously obtained from rough bars, as will be manifest if this statement be consulted. But, that these facts may be more clear, this abstract is made from the two previous tables, showing the averages of coin obtained and the waste of labour by manufacture under the present system, so clearly to exhibit the amount of loss in money value by labour alone.

From	On Rough Bars.			On Clean Bars.			
	Mixed Coin.	Sovereigns.	Half Sovereigns.	Mixed Coin.	Sovereigns.	Half Sovereigns.	Ends.
November 1856 to March 1863	48·75	51·34	45·49	52·40	53·98	47·44	4·26
October 1863 to August 1868	44·03	44·87	40·85	48·48	48·58	45·26	7·03
Waste per Centum	4·72	6·57	4·64	3·92	5·40	2·18	2·77

STATEMENT SHOWING THE AMOUNT OF MONEY COINED AND OF PYX PIECES BETWEEN NOVEMBER, 1856, AND AUGUST, 1868.

LEGEND:

- (A) = Money Coined:—Sovereigns and Half Sovereigns.
- (B) = Coined Money:—Sovereigns
- (C) = Coined Money:—Pyx Pieces.
- (D) = Coined Money:—Half Sovereigns.
- (E) = Coined Money:—Mixed Pyx Pieces.

Coinage conducted between the Periods:—	(A)	(B)	(C)
November 1856 to April 1857	Ounces. 456111·568	Ounces. 288314·570	Ounces. 277·857
October 1857 to March 1858	1236713·674	1166819·294	1407·777
December 1858 to January 1859 ^[87]	450090·524	410119·648	584·990
March 1859 to June 1859 ^[88]	464407·847	181414·181	207·494
January 1860 to March 1860	801722·890	656425·814	936·549
January 1861 to April 1861	861205·000	715989·212	1021·293
June 1861 to March 1862	1872214·638	1872214·638	2670·720
May 1862 to March 1863	2351444·336	2217447·814	3265·725
October 1863 to June 1864	2320041·256	2163027·402	3085·452
October 1864 to December 1864	957057·966	846150·074	1207·216
January 1865 to August 1865	518037·315	372808·744	532·079
November 1865 to May 1866	1394809·101	1040421·949	1504·848
April 1867 to June 1867	127485·816		
July 1868 to August 1868	280608·540	280608·540	385·200
Coinage conducted between the Periods:—	(D)	(C)	(E)
November 1856 to April 1857	Ounces. 167796·998	Ounces. 186·436	Ounces. 464·293
October 1857 to March 1858	69894·380	49·947	1457·724
December 1858 to January 1859 ^[89]	39970·876	28·404	613·394
March 1859 to June 1859 ^[90]	282993·693	210·844	418·338
January 1860 to March 1860	145297·076	103·747	1040·296
January 1861 to April 1861	145215·788	103·618	1124·911
June 1861 to March 1862			
October 1863 to June 1864	157013·854	111·964	3197·416
October 1864 to December 1864	110907·892	79·094	1286·310
January 1865 to August 1865	145228·571	103·618	635·697
November 1865 to May 1866	354387·061	253·461	1758·309
April 1867 to June 1867	127485·816	90·265	
July 1868 to August 1868			

To show the extra cost of coining, these figures have but to be multiplied into the sums coined. Before, however, entering upon this field, it will be well to explain that the rates per cent. of coined money obtained, as exhibited in the last two tables, are inclusive of the pyx pieces, so as to show, as far as possible, the best results. The pieces taken for the Mint trial are excluded, because these do not ultimately find their way into circulation; but that the vast money loss may be seen, I have added, in a tabular form, a statement^[91] which will show the weight of coined money produced in each coinage from 1856 to 1868. It may, at first sight, appear that if the work be so badly conducted as to yield an average of 3·92 per cent. less coined money from a hundred ounces of bars, the expense incurred is increased to an equivalent amount; but such is not the case, for all depends on the point at which the extravagant workmanship may have taken place. If, for instance, the bad work be produced in the rolling room, it simply induces the expenditure of so much wages in the melting, because the men are paid for the bars produced; but if the rejection shall have taken place after the money is coined, it becomes a more serious matter, because, in such a case, the loss and expense attendant on each operation subsequent to the melting have been incurred. It is, therefore, extremely difficult to give a general rule, but an instance will suffice. The coinage for 1857,^[92] inclusive of every *manufacturing* expense, cost 6s. 5d. per thousand pieces; while that of 1866 cost 13s. 2d. per thousand pieces. That the exact amount of waste may be estimated for these two years, then, it is only necessary to multiply the weight given previously ([refer to page 114](#)) for the coined money and pyx pieces, which, in the period from November, 1856, to April, 1857, would be 456,111·568, added to 464·293, and the sum of these, multiplied by £3·89375,—the decimal expression for £3

17s. 10½d., when the total value of that coinage will be obtained, and found to be £1,777,796·15. Since each thousand coins cost 6s. 5d., the total sum was coined at an expense of £570 7s. 6d.; but, had the rate of expense been the same as for the coinage of 1866, the total cost would have been £1,170 7s. 7½d., or exactly £600 0s. 1½d. more than it should have cost for actual working expenses. So by the rule of proportion the actual expense for the coinage of 1866 may be obtained, and this table is valuable as showing the weight of each coinage, but will naturally be more serviceable to those who seek practical information than to general readers, and for the benefit of such it is added.

On examining this table ([see page 114](#)), it will be observed that the coinages of December, 1858, to June, 1859, and against which is placed, are bracketed, with a view to call attention to the fact that the accounts of those two coinages became inexplicably mixed. Since the present and the late Prime Minister have stated from their places in Parliament that the Royal Mint is found wanting in administrative ability, it may not be out of place to suggest that there is abundant room for reform in the Office division of that Department, for it is seldom that the accounts can be obtained for many months after the work is finished, and this circumstance, as is apparent, is a serious inconvenience to the operative department.

Following these considerations on the extravagance of manufacture, and reflecting on what the processes should be, it appears that this is a fitting point at which to study the cost of producing gold coins, and, consequently, the rates of wages paid to the men for labour. Therefore, it may be well to state that there were two systems of payments to workpeople: one by which they were paid for *time*, that is, by the hour, for cleaning the working departments, machinery, &c., and for the more important coining of bronze; while for the coining of the precious metals they were paid by a scale of *piece-work*. These systems produced nothing but dissatisfaction amongst the workpeople, for at times they were receiving very good wages, and at others none at all; and to meet such contingencies the men and boys who were on the establishment received a kind of retaining fee, which was called subsistence, and in bitter irony no more fit name can be given to the miserable pittance which, under this form, was paid to them. It commenced at 6s. a week, and, after twenty years, rose to 10s. a week; and if, as often happened, the Mint were idle for months together, this was all the poor fellows had to subsist upon; and, to reduce the value of this pittance to its smallest proportion, they were not entitled to it unless they should each day present themselves, and, by a fiction, ask for leave of absence for that day, the longest period for which leave can be granted. After many attempts to compel the men to comply with these terms, they were wisely permitted to go home, and “wait further orders;” so that, by a kind of tacit resistance, they were able to overcome a regulation which would make it impossible to obtain employment elsewhere, because most other work commences before eight, at which hour they were required to present themselves. If, instead of this unsatisfactory state, the men were paid fixed wages, and, in addition, so much for each 100 lbs. of coined money produced, they would be satisfied, while the coined money would be produced at a cheaper rate, for each man would be concerned to see that every exertion was made for the final event; whereas, under the then arrangement, with a specific amount of gold to be coined, it is manifest that, with management such as is now under discussion, the sum for wages might be doubled, because the men were paid at the following

Rates for Piece-work.

LEGEND:

- (A) = Rolling Fillets.
- (B) = Adjusting and Cutting.
- (C) = Marking and Annealing.
- (D) = Coining in Press Room.
- (E) = Melting. (Ends are not deducted.)

For each 100 lbs. Troy of Good Work produced.		(A)	(B)	(C)	(D)	(E)
		s. d.	s. d.	s. d.	s. d.	d.
Gold	Sovereigns	3 0	5 0	3 0	2 6	5
	Half-Sovereigns	5 0	10 0	6 0	5 0	5
Silver	Florins and upwards	1 6	2 6	2 0	1 8	5

For each 100 lbs. Troy of Good Work produced.	(A)	(B)	(C)	(D)	(E)
Shillings	2 0	3 6	3 6	3 0	5
Sixpences	4 0	7 0	7 0	6 0	5
Fourpences	5 0	8 0	14 0	12 0	0
Threepences	6 0	10 0	14 0	12 0	5

And, to view this matter in its practical light, we can demonstrate its unadvisedness. Thus, in the coinage of 1857,^[93] there were 2,110,962 ounces of gold bars wrought, which produced 2,016,337·80 ounces of fillets (equal to 95·51 per cent.). These fillets produced 1,163,502·34 ounces of good blanks, from which 1,154,590·87 ounces of coined sovereigns were obtained. Now, if these various weights be reduced to wages—the men were paid for the produce of each operation—by the table above given, they will yield as follows:—

	£	s.	d.
For melting	36	12	9½
” rolling	252	8	0½
” cutting and adjusting	260	12	5¾
” annealing and blanching	145	8	9
” coining in press room	120	5	4¾
	£ 815	7	5½

So that for producing 1,154,590·87 ounces, or £4,495,748, of coined money, the men were paid £815 7s. 5½d., which sum was equally divided amongst the whole body, except that paid to the melters, for they are still paid, by a curious fiction, as packers and tellers. This, then, is the sum paid for wages when the coin produced averaged 54·79 per cent. on the bars; but let us see its amount compared with another statement put forward by the late Master of the Mint when the sovereigns produced averaged—the coinages concerned terminated in December, 1864,^[94] and May, 1866^[95]—49·44 and 38·11^[96] per cent. on the bars respectively—in other words, let us compare this rate of pay with that given in the Report of the Commissioners on the International Coinage, from which book the figures for 1864 in the following Table are taken.

**STATEMENT SHOWING THE COST OF PRODUCING
1,000 COINED SOVEREIGNS AND HALF SOVEREIGNS.**

Date of Coining	Denomination of Coin.	Total Number of Coins in the Coinage.	Average Number of Pieces per Week.	Total Sum paid to Workmen as Wages.
1857	Sovereigns	4,495,748	497,625	£ s. d. 778 14 8
1858	”	47,549		8 4 2½
1858	Half Sovereigns	544,312	272,156	91 10 3
1864	Sovereigns	5,663,656	514,878	2,011 11 7½

Date of Coining	Cost of 1,000 Pieces in Wages.	Cost of 1000 pieces in Salaries and contingent expenses.	Cost in Loss of Metal by Coining.	Total Cost of producing 1000 Gold Coins.
1857	£ s. d. 0 3 5½	£ s. d. 0 8 1¼	£ s. d. 0 2 9½	£ s. d. 0 14 4¼
1858	0 3 5¼	0 17 5½	0 4 1¼	1 5 2
1858	0 3 4½	0 18 3¾	0 4 1¼	1 5 11¼
1864	0 7 2	0 12 9¼	0 6 0	1 5 11¼

Date of Coining	Cost of 1,000 Pieces for melting.	Total Cost of producing one Gold Coin.
	£ s. d.	£ s. d.
1857	0 0 1·955	0 0 0·174
1858		0 0 0·302
1858		0 0 0·309
1864		0 0 0·311

By which it is demonstrably shown that whereas at that period 1,000 coins cost for wages 7s. 2d., at the periods above given, and shown as 1857-8, the cost was 3s. 5½d. for the same number of coins, thus showing a clear saving for wages, by the proper system of management, of 3s. 8½d. on each 1,000 sovereigns coined, or on that total quantity no less than £1,050 2s. 8½d. While such extravagance is not only tolerated but approved, miserable savings are effected at the cost of the helpless. But to treat of the wrongs to which men in this Department are submitted would take volumes which none would read.

Since, however, the appearance of the last edition of this book, the Mint Authorities have adopted in part the plan of wages payment I therein suggested, and which I had submitted to the late Master of the Mint in my Report dated 29th January, 1859. Indeed, I believe the system now in force was recommended to the Treasury, in the first instance, by Mr. Thomas Graham. I am sorry they did not give full effect to my propositions, still they have improved the positions of the workmen by the alterations made, but for the reasons I shall state I cannot concur in what they have done. I leave my original proposition to speak for itself, as I reproduce it a few pages hence, and here content myself with exhibiting the system of payment to workmen now in force in the Royal Mint. *Vide* “Mint Reports,” No. 7, 1870.

“We propose that for the future all piece-work in gold and silver coining shall be paid for at the reduced rate of 1s. 9d. per 1,000 good pieces, and all bronze coining at the rate of £2 10s. per ton for pence, £3 10s. per ton for halfpence, and £7 per ton for farthings.

“In addition to their wages for piece-work, we propose that the men should be entitled to a uniform payment of £1, and the boys to a payment of 10s., and after three years’ service, to 15s. per week. The only exception to these arrangements would be the payments made to certain overmen, which would be in one case £1 10s., and in five other cases £1 5s. per week. These payments would continue to be made when the Mint is at work as well as when it is unemployed, and to the boys as well as to the men. All the men and boys would thus receive sufficient weekly wages to maintain them, both during a cessation of work and while the Mint is in full operation.

“By this arrangement an average saving of £100 a year only will be effected.”

To the principle involved in this system I take exception; for the manufacturers, who gave the idea for it, never coined the precious metals, so that the whole operation of the system could not have been present to their minds when they recommended it. By this plan the men are paid for simple numbers, and not as they should be, in proportion to the labour and anxiety incurred.

That this fault may stand out clearly, I submit in the following table a fair week’s work on each coin, and we will assume that the one kind of coin follows the other, as indeed would really be the case in actual operation, only at greater intervals.

GOLD	
Sovereigns.	Half Sovereigns
600 journeys per week. = 420,600 pieces at 1/9 per 1,000.	300 journeys per week. = 420,600 pieces at 1/9 per 1,000.
Wages £36·80	£36·80

SILVER.			
Florins.	Shillings.	Sixpences.	Threepences.
240 journeys per week. = 475,200 pieces at 1/9 per 1,000.	180 journeys per week. = 712,800 pieces at 1/9 per 1,000.	108 journeys per week. = 855,360 pieces at 1/9 per 1,000.	60 journeys per week. = 950,400 pieces at 1/9 per 1,000.
£41·55	£62·37	£74·84	£33·16

BRONZE.		
Pennies.	Half Pennies.	Farthings.
30 cwt. per diem, or say 9 tons a week at £2 10s. per ton.	25 cwt. per diem, or say 7·50 tons a week at £3 10s. per ton.	12 cwt. per diem, or say 3·60 tons a week at £7 per ton.
£22·50	£26·25	£25·20

It is reasonable to suppose that men are more exposed to temptation when gold is within their reach (and that this is Mr. Fremantle's view may be gathered from one of his reasons for urging a removal of the Mint, viz., to avoid "opportunities for speculation") than when they are operating upon silver or bronze; yet we find by the above demonstration that sovereigns and half-sovereigns yield almost *the worst wages* to the men, for the sums specified are divided amongst the whole body of them, who probably number now, as when I left the Mint, 27 men and 11 boys. The contrast is stronger if the payment for *florins* be compared with that for *THREEPENCES*, which, with a tithe of the labour, yield just *double* the remuneration. The Reporters give a reason why they altered the system, which after I have pointed out the above facts, will seem to be curious. They state that the *inequalities of wages* induce "the improvident to contract liabilities, and, on the other hand, the more skilful workmen are induced to accept employment elsewhere, and the Mint loses their services." In my own experience, the men prefer regular wages of a smaller amount to an irregular income of greater value, because they can then regulate their household expenses and save money by fixed weekly sums, but that this reason never entered the heads of the Reporters is manifest from their own admission. They urge "*this arrangement because an average saving of £100 a year will be effected,*" which saving is to come out of the pockets of men already underpaid. I would say to the Reporters, "Muzzle not the ox that treadeth out the corn."

The total cost, then, of producing a sovereign in 1857 under proper management was 0·174*d.*, while under Mr. Graham's system it would appear to have been 0·311*d.* in 1864. If, however, the facts be examined, and stripped of the sensational effect of the cost of one coin, it will be found that the difference is far from infinitesimal, for whereas the total absolute cost per 1,000, inclusive of everything, was 14*s.* 4¼*d.* in 1857, it had reached £1 5*s.* 11¼*d.* in the years 1864-66 ([see page 117](#)). The increased expense thus incurred amounts to 11*s.* 7*d.* on each 1,000 coins, or, on the whole amount of that coinage, to £3,280 4*s.* This, then, is the state of cost at a selected period, when a large amount was coined weekly. But what would have been the cost had it been taken on the total coinage? For then we should have found an increase in the amount per 1,000 pieces for salaries and contingent expenses, whereas the figures for 1857 include the whole coinage—beginning, middle, and ending; so that the average produce of coins per week, although appearing to be smaller than that of 1864, is really far greater, for whereas the largest number reached in any one week in 1864 was 585,899^[97] pieces, in 1857 the largest number in any one week was 915,506^[98] pieces, or larger by 329,607 pieces, the average per week being 17,253 pieces less, because the coinages of 1864-1866 referred to, do not contain the beginning and "tailing-off" of the coinage; in simple fact, they are figures written *for effect, not for information*. That the complete coinage costs more for salaries and contingent expenses is self-evident, because those expenses remain the same if no coins be struck, but that this is a fact is demonstrated by the little coinages of 1858 in the same table, ^[99] where it will be seen that, although the payments for wages nearly approach the cost of 1857, the cost for salaries and contingent expenses is greater than it was in 1864. Another point is here also proved to

demonstration—that whereas the loss by coining is clearly proved to be unnecessary, the late Master of the Mint states it at 6s. per 1,000, or £300 per million coined. That this should have been so stated is of ill omen,^[100] for when the belief of Mr. Brande and his colleagues led them to the conclusion that loss was necessary at a mean rate of £373 per million, they exceeded £500; and, by rule of proportion, we may expect the Mint to make, now that the belief of the late Master has found expression, a loss of £452.^[101] Be this as it may, the difference between 6s. and 2s. 9½d. is sufficient to make men think, for even this trifle of 3s. 2½d. on 1,000 coins amounts to £908 2s. 6½d. on that single coinage. That the loss on the coinages of 1858 should be so great is explained by the fact that in a small coinage the first loss by gilding the machinery is as great as in a large one; but in the case of a great coinage this is so distributed over the mass as not to appear. Had it been determined, however, to exhibit the actual facts by the system of selection followed by others, not only would this loss of metal have disappeared, but a gain would have been shown, for on those particular coins there was an actual gain of 1·39 ounces, which is at the rate of £9 18s. 10½2d. per million coined, or 2·386d. per 1,000 pieces coined, so reducing the cost of these actual coins to £1 0s. 10½d. and £1 1s. 6d. per 1,000 respectively; but as these matters are dwelt upon for information alone, it is preferred to place the fair proportion of the loss on this coinage on this portion of it. It should be observed that the contingent expenses for 1857 do not include £1,100 voted for the new files,^[102] because that sum was never appropriated to that purpose; but that they do include the subsistence paid to the men, and, indeed, every possible sum other than weekly wages paid to the men by the piece-work scale above quoted. I have not questioned, nor have I investigated, the figures used by the late Master of the Mint, for as he wrote those papers, and invited the Signatories to give their names after the copies were fairly made, it is but just to suppose that that gentleman satisfied himself of their accuracy.

The Chancellor of the Exchequer has proposed to reduce the value of the gold coinage by removing one grain of gold from the sovereign, a reduction of 0·81119 per cent, of its value. This matter has been so fully discussed, and its error so completely demonstrated, that little need be here said; but there is one consideration which should not be omitted. The new standard of value, for such it will be, will be worth only its value in gold immediately after it has quitted our shores. To travellers, therefore, it is a tax of about one per cent., for each of Mr. Lowe's pounds will fetch but 19s. 10d., and, whether the coins be exchanged here or abroad, the reduction must be borne; whereas, if it were determined to charge the *importer* for the cost of coining, and for that alone, no tax could fall on individuals—a system which must be unjust, and the coin would, by this burden, be restricted in its power of sale as bullion, so small a sum as one-eighth of one per cent. having frequently determined the sending of coin rather than of bullion, because, when the exchanges are nearly equal, merchants thus save the cost of assaying, which, inclusive of loss of interest, &c., amounts to 2s. 6d. per £100. To fix, then, the sum proper to charge for the coining of gold is practically a simple matter, if it be desired, as should be the case, to protect our coinage from conversion into bullion, to the profit of special merchants, but to the loss of the nation. In the following proposition the exact cost can be determined, and that should be the maximum limit of a Mint charge or *mintage*, for if that charge be so enlarged as to insure a profit, the fears of Mr. J. G. Hubbard as to the illicit coiners relieving the Mint may be expected to be realised.

Between April, 1855, and December, 1865, £59,581,957 were coined, or an average of £5,958,195 in each year. If, however, six millions be accepted as the average yearly coinage of gold, it will give data for the following calculations.

That the cost of coining may be always the same, let there be thirty-six efficient workmen and twelve boys, and let these be paid for wages in the following manner:—To each man £1 a week, whether at work or not, and to boys a sum beginning at 14s. per week, to be increased 1s. per week for each additional year of service, until the age of twenty be reached, when they should receive the same sum of £1 as is paid to the men, but the “rating as men” should be stopped till a vacancy occurs, that the number of men, inclusive of melters, shall not exceed thirty-six, as then no injustice will be done the men, and the boys are sufficiently provided for to enable them to wait. When, however, work is in progress I would pay for piece-work as follows; for—

	£	
Sovereigns	0·806	per 100 lbs. troy of coined money.
Half Sovereigns	1·571	”
Florins	0·555	”
Shillings	0·800	”
Sixpences	1·348	”

Threepences	2·330	”	”
Bronze pence	7·500	per ton of	coined money.
” halfpence	10·000	”	”
” farthings	15·000	”	”

This sum should be divided by a simple system equally amongst the workpeople, so that men should take one whole share, boys who have served more than three years and a half two-thirds of a share, and boys of less service than three years and a half one-third of a share. When the fair share of piece-work exceeds the permanent amount of fixed wages, that sum should be deducted from the total sum allotted to each, so that the permanent wages would become a fixed charge secured on the piece-work to be performed. This being performed intelligently, each workman would participate in the benefit, therefore each would do his best; whereas no increase can take place in the cost of production, for any carelessness in the various operations simply causing so much waste labour, would bring no remuneration, and the wages would thus bear proportion to the anxiety incurred and be equalized.

To pursue, then, this proposition further, it will be convenient to assume that 420,600 sovereigns are coined per week, this being a fair average amount to take if the complete coinage be conceived to be 6,000,000 of finished coin. In such a case fourteen weeks would be required to effect the operation, and its total cost will stand thus:—

	£
For assays by out-door assayers	998·4
” wages to workmen	1,035·0
” salaries and contingent expenses	2,843·0
” loss of metal by coining	1,200·0
	<u>£6,076·4</u>

In estimating salaries and contingent expenses the estimates of 1857 have been taken, because the pressure for bronze has passed away, and the establishment has, it is supposed, gone back to its then dimensions. At that period it was usual to estimate the cost of coining gold, inclusive of assaying, at the rate of one-sixth per cent. In the above calculations the cost of assaying is included, as well as the salaries and contingent expenses, from which, however, has been deducted the fixed wages paid to the workpeople for fourteen weeks, because they are supposed to have been earned as piece-work.

Allowance at the extravagant rate of £200 per million coined is also made for loss of metal by coining, so that under this proposition the total cost of producing six millions of coined gold becomes £6,076·40. This must be the absolute sum, for the cost of dies, police, &c., is included in the amount for salaries and contingent expenses; it therefore follows that each £100 would cost for manufacture £0.101273, or about 2s. 0¼d. Here, then, are exhibited the grounds on which the calculations are based, and unless there are errors on the face of the figures it is demonstrated that the total cost of coining, even with an extravagant estimate, may be reduced to about one-tenth of one per cent. Why, then, should a profit of nine-tenths of one per cent. be desired or granted? If an ultimate decision should be formed to charge £1 for each thousand sovereigns coined, a stimulus will be given to the Mint authorities to investigate the cause of their losses, the means whereby these may be stopped, and to how great an extent useless officers may be parted with, thus to make a minute but legitimate saving out of the allowance, for there is no substantial reason why we should coin free of charge; nor, on the other hand, is it right that the cost of that operation should fall on the tax-payer, while a just rate for mintage would, unless under very exceptional circumstances, protect the coinage from illegitimate conversion into bullion, for it must be remembered that there is already in reality a tax of 1½d. per ounce for the conversion of bullion into coin, paid by the *importer* to the Bank of England, and the charge for mintage will be in addition to that tax.

A very strong reason against an exorbitant charge may be found in the fact that the French Mint is content to fix its rate at about 6 francs 70 centimes for each kilogramme of their standard gold, which is coined into pieces of the value of 3,100 francs, or about 0·216129 per cent., say 4s. 3¾d. for each £100 coined, and this under a system of contract which gives a profit to the contractors, so that it becomes clear, coining is practically done AT A FIXED RATE IN FRANCE, that rate being above the actual cost of our coinage in 1857, but below the charge proposed by the Chancellor of the Exchequer by six-tenths of one per cent., irrespective of the deduction of 1½d. per ounce made at the Bank when gold is bought.^[103] This being the case, it is reasonable to suppose that

all the gold for coining will go to France; whereas, if it be desired to convert the coin of France into *our proposed debased coinage*, each £1,000 so converted will produce but £990·275, because the $1\frac{1}{2}d.$ per ounce is equal to 0·16051 per cent., and to this tax is added 0·81119 per cent. by the deduction of the one grain proposed by the Chancellor of the Exchequer, for it must be understood that the coined money of France will still be treated as bullion. Perhaps a still plainer view of the case may be obtained, if it be assumed that a man has a million pounds' worth of gold, which, for cheapness' sake, he gets coined in France, where he will pay £2,161·29 for coining it. Circumstances, however, make it desirable that this million in French gold coins should be converted into coin of the British standard as coined at the Royal Mint, where the owner will find that, for the conversion of his bullion, he has to submit to a deduction equal in value to £9,725, which, added to the sum charged in France for the first coining, makes a total of £11,886·29. How many persons will be willing to sacrifice such a sum for the convenience of others? The result must be that, unless under the most exceptional circumstances, no gold except for currency can be coined in England, it will inevitably be sent to France, because coined gold will there always be 0·81119 per cent. more valuable than in England.

In discussions recently earned on in the newspapers, the liability of gold coin to be sifted of its pieces which are heavier than the theoretical standard has been a good deal dwelt upon; but there is one view which, perhaps, has not occurred to those who are unacquainted with the manufacturing details. It is true, then, that the standard weight of a sovereign deduced from its proportional weight to 20 lbs. troy is 123·274478 grains, and that this piece may vary so far in weight as to be either too light or too heavy by no less than 0·2000 grain, and yet be a perfectly legal tender. With careful manufacture the coins issued should be so apportioned that there shall be half the number on the light and the other half on the heavy side of the standard weight.^[104] By referring to the papers furnished for the guidance of the jury at the trial of the pyx it will be found that the gold coins reported on at the last two trials were on the light side of the theoretical weight, for those submitted to trial in 1861 were by number 24,655,335 pieces, and weighed only an equivalent to 24,654,849 sovereigns, consequently the Bank of England received exactly £486 more in coined moneys than their bullion was worth; therefore this was a profit to that Institution. And on the occasion of the trial which took place in 1866 there were by number 34,927,188 pieces, which weighed only an equivalent to 34,927,008 sovereigns, so that the Bank gained exactly £180. These facts demonstrate the fairness with which the Mint coins and issues its coined moneys, not in accordance with the letter, but with the spirit of the law which governs its actions; yet the evidence thus given is to the effect that the bulk of the coins are on the light side. The current weight of the sovereign, as authorised by Act of Parliament 33 Vict., cap. 10, and published in the *London Gazette*, 12th August, 1870, is 122·50 grains, and below this weight the Bank of England will not receive it in payment of twenty shillings; but the half-sovereign, being a coin of convenience, is allowed to circulate till its weight, has fallen to 61·125 grains, below which it is not received at the Bank in liquidation of a debt of ten shillings. Thus, then, while a sovereign may be legally coined so that its weight may be either 123·474478 grains or 123·074478 grains, it is still permitted to circulate, and is a legal tender at the current weight of 122·50 grains, or a difference of 0·574478 grain below the minimum weight permitted by the New Coinage Act for its issue from the Mint. If, as will be found to be the case on an average of years, the rejected coins by weight amount to 16 per cent. when the remedy is 0·20 grain, and to 9 per cent. when the remedy is 0·25 grain, it is evident that the chance of obtaining profit by picking coins is not greater than 3·5 per cent., because the difference between a fifth and a quarter of a grain makes an increase of 7 per cent., and of these rejected pieces rather less than half are on the heavy side, because, as has been shown, the whole deliveries to the Bank of England are on an average light pieces; therefore it will be perfectly safe to consider that each hundred sovereigns contain at the extreme three coins which may be two-tenths of a grain heavier than the standard weight of 123·274478 grains; but it is by no means certain that any hundred pieces would contain one single coin heavy by this amount even if the remedy were made to be 0·25 grain, because the automaton balances of Mr. Cotton are so accurate as never to permit the issue of pieces beyond the limits assigned, and any piece which at its passage through the machine equalled the maximum weight would be, and is, inevitably reduced—infinitesimally, if you will—in weight, by abrasion against other pieces in the act of falling, as well as when put into its bag previously to going to the Bank of England. When the remedy was 0·30 grain the case was different, but it may now be assumed that it is impossible for any one to make profit by selecting heavy pieces,—this irrespective of the fact that the bankers find it worth their while to select heavy pieces for transmission to the Bank of England, because they then obtain the coins which are of current weight, yet not intrinsically worth twenty shillings;—whereas the recent public discussion makes it appear that it is worth the risk to certain unknown individuals in Brussels to buy our newly-coined sovereigns, pick from them such as exceed 123·274 grains, and melt them into ingots for sale as bullion, and return to us the

light pieces. The only modes by which it pays such persons to buy new sovereigns for the purpose of such gain is to shake them together in canvas bags, or to submit them to the electrotype process—as is often done by chemical students—and by these means obtain from each a specific amount of gold, sending the sweated coins into circulation, having reduced each of them to the lowest current weight by a very short rough usage. It is different when coins, instead of being individually weighed, are simply *pounded*, that is, weighed *en masse*, just to determine that a certain number are in a given weight: in such circumstances many coins most unequable may be issued, as was the case when silver coins were thus treated in the Royal Mint in August, 1864, against all sound principles. At that period the then Master, under false notions of economy,^[105] determined not to weigh individual pieces of silver money, because silver coins were mere tokens, but when coins bearing the device of FLORINS, and ranging in intrinsic value from 1s. 6d. to 2s. 9d. irrespective of extrinsic worth, were returned to the Mint, this economy was soon abandoned.

I have dwelt at considerable length upon the losses incurred by the coining of gold; such losses I now find are defended by high authorities, but I hope some effort may be produced by my words, yet not mine only, for I will quote those used by Mr. Graham in his letter to the Treasury^[106].—

“This gives in a million coined a net loss of £257 7s. 0¾d.; ... but after making allowance for the circumstances just stated, the loss of gold in melting remains higher, in my opinion, than it ought to be; and it will, I trust, be found to admit of some further reduction in future years.

“A correct estimate of the whole waste in coining is obtained by combining the returns of the two departments:—

	£	s.	d.	
Loss of melting department	257	7	0¾	per million.
Surplus of coining department	50	13	0	”
Difference	206	14	0¾	”

It thus appears that in 1858-59 the net loss in minting £1,000,000 of gold was £206 14s. 0¾d.’

He then proceeds to show that “the loss on the gold coinage for the (*then*) last three years was £172 8s. 11½d.,” yet Mr. Fremantle’s return to the House of Lords No. 30, 1870, shows a loss of £464 as the average in the last three years, and he in “European Mints,” uses these significant words: “The waste shown to have existed of late years in the English Mint has not been excessive.”

I reproduce these words because the Chancellor of the Exchequer has accepted contracts for coinages for foreign countries, and I propose to study the matter in relation to his probable profit.

With such losses accruing as are here exhibited, it appears to be a step worthy only the present Chancellor of the Exchequer to advertise for coinage contracts for foreign countries. At what price will they be granted to him? It is perfectly clear that, to make a profit, he must charge more than it will cost, so it will be well to examine his chance of success. We may assume that it will cost him 2s. 6d. to coin £100 in sovereigns as the expenses of salaries, wages, and loss; in addition to these he will find it necessary to pack his coined money for exportation, whither he must pay carriage, insurance, &c. Should he feel that he need not charge for wear and tear of machinery, he must still remember that it will require renewal at intervals, and every expense must be included in a total sum of 4s. 3¾d., or the foreign contractors (whom he was sanguine enough to suppose would give him information) will be sure to beat him in the tender for the contract. As to silver, it is already certain that his loss by coining would alone make any profit hopeless even to a Chancellor of the Exchequer who knows how to get five quarters’ taxes in one year. Lord Kinnauld has shown that the Mint loses by manufacturing silver coin, by sheer loss of metal, at the rate of 3s. 9¾d. for each £100 coined ([see page 146](#)), and in addition to this tax he has to pay 1s. 9d. for each 1,000 coins manufactured for labour, besides the contingent expenses, including packing, insurance, &c. All expenses must come within 15s. for each £100, because this is the sum at which the foreign contractors gladly accept coinages. While saying thus much, it has been necessary to reserve facts which would be indispensable to a contractor, because

such information should be rendered to a contractor only on terms such as another would be willing to offer for it, and to publish it here would be hardly fair to those whose business it is to live by coining.

The Royal Mint, suffering thus from faults which its officials profess themselves unable to remedy, it seems an eccentric proceeding that while the Government Institution bears good-naturedly all the loss, the Bank of England should obtain all the profit, “as a set-off against the stock of bullion it is compelled to keep.” It is to be remarked that the Chancellor of the Exchequer stated that it was “necessary to expedite the coining of gold, because it bears interest while under coinage,” thus exhibiting an absence of knowledge, for the Mint pays no interest. It is no concern of the Bank whether its bullion be reserved as ingots or coin, because it is compelled by Act of Parliament to keep a specified amount to meet the excess issue of its notes, and this circumstance alone enables the Bank of England to coin gold without loss of interest—gold at the Mint is still considered as reserved by the Bank. So far, indeed, from the Bank losing money by importing gold for coining, it makes a clear profit, as is shown by the following figures—this even under its new system of assay—

BANK OF ENGLAND PROFIT.

By the charge of 1½ <i>d.</i> on each ounce of gold	£1605·100
By the system of absorbing assay fractions	166·666
By the turn of the scale on purchase	71·339
On each million coined	<u>£1843·105</u>

Nor is this all; for the Bank exacts from the Mint the last ounce or fraction of an ounce of all it sends, and besides receives gold invariably coined on the light side of the standard, to the average extent of £11·50, and makes an additional profit by the turn of the scale, which gives them from the Mint £75 on each million, so that their total gain comes to be £1,929 12*s.* on each million sent to the Mint for coining, that Department good-naturedly finding alloy into the bargain. That I do not exceed the facts of the case would seem to be conclusively proved by Messrs C. W. Fremantle and C. Rivers Wilson, for those gentlemen in their Reports state the amount at £3,458, and affirm (see Reports on the Mint, page 4) that on £5,000,000 the Mint would make a profit of £8,000. How they reconcile their figures I do not know, for by their own showing it should be £17,290. They had better accept my figures, which would produce a profit of £9,648 on the same amount.

I feel that I have produced sufficient evidence to oblige me to withdraw the opinion I have hitherto held, and which I thus expressed—

“If, as I can clearly show, the Mint can still be advantageously retained under Government control, it would appear to be a pity to place the coinage in the hands of contractors, for it is evident that they must be paid such a price as will yield them a profit, and this sum had better be saved to the nation.”

On the contrary, I now think it is the duty of the Government TO AT ONCE COIN BY CONTRACT, as the cost of such process will be definite, whereas it is now illimitable. Instead of the tax-payers paying for New Mint buildings, *which are not required*, contractors would gladly buy the present Mint just as it stands, and in three years amass colossal fortunes. If Mr. Lowe were to hand over the coinage of gold, silver, and bronze—by which the Government now loses so large a sum—for one year, with proper security, to an unofficial person, who should pay all the costs and reap all the profit, I am sure the annual vote for the expenses of the Mint and coinage may be remitted and the work will be better performed. At a period when the Government has seen fit to send Commissioners to European Mints to learn the mystery of coining, it seems a perversion of reason to adopt the opinions of those learners, and upon their recommendations build new Mint premises. I am aware that it is thought by the sanguine that the old Mint site will sell for a vast sum, which will pay for the new site; but is it in the experience of the public that such transfers are made without loss? Indeed the probability of vast expense is so great that it becomes a settled question. The Commissioners—the chief of whom says,

“... While I must not be understood to be in a position to offer positive recommendations on many points, and more especially as regards machinery, while the question as to the future site of the Mint is still pending, yet I trust that I have shown the importance of the information which we have acquired, and the possibility of applying it with advantage to many Mint arrangements in this country”—

recommend machinery and experiments with which two of their number are admittedly unacquainted, and their Reports show throughout that they are really writing upon subjects to which their minds have not yet become accustomed, and which consequently they cannot fully appreciate. Instead, therefore, of the Legislature building new premises on the advice of such officers, it will surely be wise to give the contract system a fair trial, say for a year, in accordance with the suggestion of Lord Kinnaid. It is not the premises, it is not the machinery, it is not the workpeople, but the utter want of experience in the chief officers that gives rise to the state of muddle and dissatisfaction amongst the officials, and causes loss of bullion, which does not go into the chimney, and other expenses which will not bear the light of day.

Mr. Fremantle recommends new premises, on the ground that such should be built expressly for Mint purposes. The present Mint was so built, and is now

capable, with very limited expenses, of adaptation. It would be cheaper to burn it down and rebuild the operative departments, than to remove it to another site, where the same want of experience will surely give rise to precisely similar evidences of mismanagement as now exist. It is because the Government prefers to put the wrong man in the wrong place—square men in round holes—that it is thought necessary to transfer the Mint, really that those gentlemen may be nearer their own comfortable homes and friends.

The East End of London is recognised as a disagreeable part to dwell in; but the Mint is essentially a manufactory, and, as such, is not fitted for the westerly parts of the metropolis, and those who object to the neighbourhood should relinquish their offices. Would it be tolerated, if proposed, to remove the Enfield factory to the Thames Embankment, so that it “should be more immediately associated with the Treasury offices?” In its present site the Mint has supplied the coinage, and could issue multiples of its past productions; economy being the order of the day, why incur unnecessary expense?

In the meantime, the old rule holds good—a bad workman complains of his tools; and those who demonstrate their inability to conduct the Mint in such manner as to derive the revenue which they admit should accrue and which they profess themselves unable to secure, should gracefully assent to place the matter in the hands of contractors till the Government shall see fit to put the department in the charge of some person who will practically develop the truth, instead of hunting in the chambers of the chimneys for that which they must know full well has left the Mint by another means.

The Commissioners seem to have gone out of their way to show their want of knowledge on vital points in the manufacture of money. Thus Mr. Fremantle recommends the substitution of graphite for iron pots in silver melting, on the authority of Mr. Roberts, whose experience is expressed in these words—“Graphite crucibles permit of the total contents being poured into moulds, and this enables the accounts to be adjusted daily.” Truly the accounts may be *adjusted* daily; but most commercial people object to *adjustment*, as that process commonly leads to a minus account, and frequently an employment of the detective police. To balance an account is a very different proceeding; but is it true that graphite crucibles expedite the process? No, it is not true, as witnessed by the [table on page 102](#), where it will be seen that the amount of gold which is *not* “*poured into the moulds*” varies from £1,132 down to £161, and has, at the last return, risen steadily to £461, although the gold referred to was MELTED IN GRAPHITE CRUCIBLES. At a later period I will show these facts in greater detail. ^[107]

So much interest does not attach to the coining of silver as to the coining of gold; yet, as the amount of profit to be made by this coinage depends on the length

of time which silver coins can be kept in circulation, it is necessary to consider whether the Royal Mint has at any time made, or does now make, so much profit as it should, and whether the coins when made are so manufactured as to be fit in the highest degree to bear the wear and tear to which it is intended that they shall be submitted. These questions, although they may appear to be of little interest, are really important, because the Mint, if properly managed, should, by its silver coinage, pay all its expenses, and by its bronze coinage, if the latter be discreetly conducted, render an actual profit to the Government. I propose presently^[108] to touch upon the bronze coinage, when inquiry will demonstrate that this is conducted in such a manner as that a large per-centage of the fair profit is sacrificed annually.

In those countries where silver is legally the standard of value the coins contain the amount of silver which is equivalent to a fixed sum of gold, and in all such cases, if silver should happen to rise in price, bullion merchants buy large quantities of new silver *coins* and export them, because these remain at a fixed price, although all other silver has risen in value. In England this contingency is avoided by giving to coined silver an *extrinsic value*; that is to say, by raising silver whose natural market price varies^[109] between 5s. per ounce and 5s. 2d. per ounce to an artificial value of 5s. 6d. per ounce when it is coined into money. It is the practice of the Mint to buy silver only when it is at its lowest market price, and then in parcels of the value of £50,000. Upon an average of years it is found that the profit thus accruing^[110] should amount to about 8 per cent. Even with perfect management the whole of this 8 per cent. would not be secured, because all worn and light silver culled from circulation is returned to the Mint by the Bank of England for recoinage into new money, piece for piece, and many of the worn^[111] pieces represent only half their original extrinsic value. What we call a shilling is intrinsically worth, when coined, about 10¾d., but when it is culled from circulation it is often worth only 6d., therefore a loss of 4¾d. is incurred by re-coining it, quite independently of the labour and loss of metal. It should, however, be stated that the Mint makes a small profit by receiving the unreported fractions as well as the excess of weight^[112] invariably given when ingots are purchased. In France the five-franc piece is coined so that it contains its intrinsic value, the importer paying 75 centimes for each 100 francs in value. Since this sum yields a profit to the contractor, we may be certain that it covers every possible expense of production. If this cost be proportioned to our money, it will appear to be at the rate of 15s. for each £100. Let us see, then, if the silver coinage of Great Britain is produced at so cheap a rate as is that of France, and this shall be determined by discussing the unnecessary losses which are permitted, in a similar manner to, but less elaborately than, that adopted for the gold coinage. The tables which will be exhibited are those given me by the late Master of the Mint, under precisely the

same circumstances as described when speaking of the tables relating to gold. I will then briefly treat of these losses as they were, as they are, and as they should be. Those who are habituated to the working of gold feel, when that work ceases, a kind of relief, which is perhaps natural if the relative value be considered. The Mint, then, suddenly changes from this metal to one of just about a fourteenth of its value, for to the Mint people gold at £3 17s. 10½*d.* is as 14·16 is to 1 of silver, which they view as of a value of 5s. 6*d.* per ounce; again, it is more bulky and unmanageable. These causes are to be admitted to their full share of importance, but should not induce us to accept them in explanation of facts which are not otherwise reasonable, and such seems to be the case with silver in the Royal Mint. Mr. Seyd, in his admirable work on Bullion and Exchanges, at page 560 says, "We should like to see a return moved for in Parliament of the wastage of gold at the British Mint from 1850 to the present year; it might show whether there has not been at certain periods of the management of the operative department much less loss than at others, and whether there has not been occasionally a small gain even, as there certainly ought to be with proper management." Mr. Seyd has shown himself a master of the subject he has undertaken, and perhaps the facts stated in the earlier pages of this book may reply to some of his questions. He then proceeds to say, "A return of the wastage of silver might also prove interesting, as well as an account of the imperfect work returned to the melting-pot during each year." When an author who has so successfully studied a subject invites such information, one can but feel that it is wise to give some details. I propose, however, to give only such facts as will bring us up to the year 1861, because the following years vary but little, and if I were to give another eight years it would require more space than it is deemed wise to occupy.

In the following tabular statements will be found every particular relating to silver arranged in a precisely similar manner to the facts regarding gold, and it is hoped that they are given with sufficient clearness to render unnecessary a lengthy explanation. I have given other details at [pages 136-142](#).

In explaining the [table on page 132](#) it should be noted that the £1,315 9s. 4*d.*, given as the value of the loss per million in 1852-53, is not the true value, for, as in the case of gold, the account of the loss was not kept. The loss of those years is, therefore, reduced by sharing it with the coined money of the financial years, June, 1851, to March, 1853; if this sum be, as it should be, excluded, the loss will be £1,695 17s. 6½*d.* The same allowance must be made for the melting-house account in the table on [page 134](#), when that sum will become £1,375 12s. 4¾*d.* instead of £1,067 0s. 11¼*d.* It is deemed fair to reduce this loss as far as possible; therefore the whole sum coined between June, 1851, and March, 1853, has been made to bear the one loss of these 1845·449 ounces. The same test being applied to these tables as was applied to that relating to gold, ^[113] it will be found that the average

loss by coining silver between June, 1851, and March, 1857, was £1,013 11s. 2¼d., whereas in the next period, between April, 1857, and March, 1860, it fell to £365 15s. 10½d. This reduction then shows that the former loss, taken at its lowest, gives evidence of *unnecessary* wastage to the extent of £647 15s. 4d. on each million coined; but this sum is not that shown by the Mint books, for, as above stated, it is below the truth. Great, then, as were the amounts missed, we shall find them far greater if the state of the melting-house be considered, and to do this is necessary, because it was asserted that the loss was arrested in the coining department by sending more oil and dirt to the melting-house. Let us then see how this stands in regard to the truth, for while in the period 1851 to 1857 the oil literally *dropped from the scissel and drained through the trucks on to the floor*, in the latter period the trucks were not soiled, and the use of oil in the rolling room was rigidly prohibited, while in the drag room only so much as was required to lubricate the fillets at the drag bench was permitted to be used, and this was as far as possible removed by careful wiping with cotton waste. It is also a fact that whereas there had been an HABITUAL LOSS in the rolling and drag rooms when oil was used very largely, there was now a GAIN IN BOTH ROOMS; in the former to the extent of one in ten thousand ounces wrought, and in the latter to the extent of two in ten thousand. These facts must be of importance while considering the following statement ([see page 134](#)).

**STATEMENT SHOWING THE AMOUNT OF SILVER COINED AND
OF THE LOSS ATTENDING IT IN THE COINING DEPARTMENT.**

Date—Financial Year.				Weight of Coinage.	Value of Coinage.		
				Ounces.	£	s.	d.
June	1851	to	March 1852	314689·050	86,539	9	9¼
April	1852	to	March 1853	1088196·550	299,254	1	0¼
"	1853	"	1854	2135071·750	587,144	14	7½
"	1854	"	1855	501459·800	137,901	8	10¾
"	1855	"	1856	1138355·000	313,047	12	6
"	1856	"	1857	1792800·000	493,020	0	0
"	1857	"	1858	1420560·000	390,654	0	0
"	1858	"	1859	1449360·000	398,574	0	0
"	1859	"	1860	2052720·000	564,498	0	0
March	1861	to	June 1861	650160·000	178,794	0	0

Date—Financial Year.	Amount of Loss by Coining.	Value of the Silver Lost.		
	Ounces.	£	s.	d.
June 1851 to March 1852				
April 1852 to March 1853	1845·449	507	9	11½
” 1853 ” 1854	2943·025	809	6	7½
” 1854 ” 1855	657·362	180	15	5¾
” 1855 ” 1856	1632·957	449	1	3¼
” 1856 ” 1857	2191·367	602	12	6¼
” 1857 ” 1858	910·264	250	6	5½
” 1858 ” 1859	834·679	232	0	2¾
” 1859 ” 1860	898·837	247	3	7¼
March 1861 to June 1861	1357·375	373	5	5½

Date—Financial Year.	Value of the Sweep Sold.	Lost Silver per Million Coined.	Sweep per Million Coined.		
	£ s. d.	£ s. d.	£	s.	d.
June 1851 to March 1852	31 12 8				
April 1852 to March 1853	90 4 10½	1,315 9 4	315	18	3
” 1853 ” 1854	160 12 4	1,378 8 4¾	273	11	1¼
” 1854 ” 1855	69 19 2¼	1,310 17 11	507	6	3¼
” 1855 ” 1856	104 5 7½	1,434 9 9¼	333	3	3¾
” 1856 ” 1857	149 10 9¼	1,223 6 3¾	303	6	2¾
” 1857 ” 1858	98 1 10	640 15 6¾	251	1	11
” 1858 ” 1859	125 8 4	582 2 1	314	13	3¼
” 1859 ” 1860	10 18 3	437 17 6¼	19	6	7¼
March 1861 to June 1861	24 14 4½	2,087 14 6¼	138	5	0½

To obtain the true amount of loss or waste, deduct the value of the sweep from the value of the silver lost.

**RETURN FROM THE ROYAL MINT TO
AN ORDER OF THE HOUSE OF LORDS,
DATED 17TH FEBRUARY, 1870.
(THE LORD ROSSIE.)
COINING DEPARTMENT.**

A STATEMENT of the Weight and Value of the Silver Moneys coined in each Financial Year from 1851 to 1869 inclusive, exhibiting the Weight and Value of the Loss or Waste sustained in each Year, as well as the Value of Sweep recovered, and

the average Proportion of such Loss or Waste and Sweep recovered to each Million Pounds Sterling coined.

Date.				Weight of Coinage.	Value of Coinage.				Amount of Waste.
				Ounces.	£	s.	d.	Ounces.	
August 1851	to	March 1852		314689·050	86,539	9	9	1845·450	
April 1852	to	March 1853		1088196·550	299,254	1	0		
”	1853	”	1854	2134891·750	587,095	4	8	2943·025	
”	1854	”	1855	501459·800	137,901	8	11	657·362	
”	1855	”	1856	1138355·000	313,047	12	6	1632·957	
”	1856	”	1857	1792800·000	493,020	0	0	2191·367	
”	1857	”	1858	1420560·000	390,654	0	0	910·264	
”	1858	”	1859	1449360·000	398,574	0	0	843·679	
”	1859	”	1860	2052720·000	564,498	0	0	898·837	
”	1860	”	1861	625714·000	172,071	7	0	1032·502	
”	1861	”	1862	769680·000	211,662	0	0		
”	1862	”	1863	532147·610	146,340	11	10	811·466	
”	1863	”	1864	835920·000	229,878	0	0		
”	1864	”	1865	1936800·000	532,620	0	0	1100·169	
”	1865	”	1866	1880640·000	517,176	0	0	906·407	
”	1866	”	1867	1739520·000	478,368	0	0	891·389	
”	1867	”	1868	961200·000	264,330	0	0	473·724	
”	1868	”	1869	597600·000	164,340	0	0	280·764	

Date.				Value of Waste.			Value of Sweep recovered.		
				£	s.	d.	£	s.	d.
August 1851	to March 1852	April 1852	to March 1853	507	10	0	121	17	6½
”	1853	”	1854	809	6	8	160	12	4
”	1854	”	1855	180	15	6	69	19	2¼
”	1855	”	1856	449	1	3	104	5	7½
”	1856	”	1857	602	12	6	149	10	9¼
”	1857	”	1858	250	6	5	98	1	10
”	1858	”	1859	232	0	3	125	8	4
”	1859	”	1860	247	3	7	10	18	3
”	1860	”	1861	283	18	9	46	7	1½
”	1861	”	1862						
”	1862	”	1863	223	3	1	Nil.		
”	1863	”	1864						

Date.				Value of Waste.	Value of Sweep recovered.
”	1864	”	1865	302 10 11	15 6 7
”	1865	”	1866	249 5 3	16 9 4
”	1866	”	1867	245 2 8	Nil.
”	1867	”	1868	130 5 6	121 3 3½
”	1868	”	1869	77 4 2	Nil.

Date.				Value of Waste. per £1,000,000.	Value of Sweep per £1,000,000.
				£ s. d.	£ s. d.
August 1851 to March 1852				1,315 9 5	315 18 1
April 1852 to March 1853					
”	1853	”	1854	1,378 10 9	273 11 7
”	1854	”	1855	1,310 18 1	507 6 3
”	1855	”	1856	1,434 9 8	333 2 3
”	1856	”	1857	1,222 6 4	303 6 2
”	1857	”	1858	640 15 4	251 1 11
”	1858	”	1859	582 2 1	314 13 3
”	1859	”	1860	437 17 5	19 6 7
”	1860	”	1861	739 18 8	120 16 0
”	1861	”	1862		
”	1862	”	1863	593 3 0	Nil.
”	1863	”	1864		
”	1864	”	1865	568 0 9	28 15 7
”	1865	”	1866	481 19 4	31 16 9
”	1866	”	1867	512 8 8	Nil.
”	1867	”	1868	492 16 11	458 7 8
”	1868	”	1869	469 16 2	Nil.

C. W. FREMANTLE, DEPUTY MASTER AND COMPTROLLER.
ROYAL MINT, 2nd March, 1870.

**STATEMENT SHOWING THE AMOUNT OF SILVER COINED
AND OF THE LOSS ATTENDING ITS MANIPULATION
IN THE MELTING DEPARTMENT.**

Date—Financial Year.	Weight of Coinage.	Value of Coinage
	Ounces.	£ s. d.

Date—Financial Year.				Weight of Coinage.	Value of Coinage		
June	1851	to March	1852	314689·050	86,539	9	9¼
April	1852	to March	1853	1088196·550	299,254	1	0¼
"	1853	"	1854	2135071·750	587,144	14	7½
"	1854	"	1855	501459·800	137,901	8	10¾
"	1855	"	1856	1138355·000	313,047	12	6
"	1856	"	1857	1792800·000	493,020	0	0
"	1857	"	1858	1420560·000	390,654	0	0
"	1858	"	1859	1449360·000	398,574	0	0
"	1859	"	1860	2052720·000	564,499	0	0
March	1861	to June	1861	650160·000	78,794	0	0

Date—Financial Year.				Amount of Loss by Melting	Value of the Silver Lost.			Value of Sweep Sold.		
				Ounces.	£	s.	d.	£	s.	d.
June	1851	to March	1852					36	0	1
April	1852	to March	1853	1496·945	411	13	2¼	111	19	4½
"	1853	"	1854	1814·808	499	1	5¼	143	2	4
"	1854	"	1855	882·870	242	15	9½	187	19	7
"	1855	"	1856	519·069	142	14	10½	144	0	2¼
"	1856	"	1857	2912·149	800	16	9¾	151	12	4¼
"	1857	"	1858	1001·676	275	9	2½	100	13	6¼
"	1858	"	1859	+733·101	+201	12	0¾	115	4	11
"	1859	"	1860	199·288	54	16	1	101	8	8
March	1861	to June	1861	238·495	65	9	10¾	49	4	4½

Date—Financial Year.				Value of Lost Silver per Million Coined.			Value of Sweep per Million Coined.		
				£	s.	d.	£	s.	d.
June	1851	to March	1852						
April	1852	to March	1853	1,067	0	11¼	383	11	1
”	1853	”	1854	849	19	11¾	243	15	0
”	1854	”	1855	1,760	11	11¾	1,363	2	10
”	1855	”	1856	456	19	7½	460	0	5¾
”	1856	”	1857	1,624	7	1¾	307	10	6¾
”	1857	”	1858	705	2	6½	257	14	2¾

Date—Financial Year.				Value of Lost Silver per Million Coined.	Value of Sweep per Million Coined.
”	1858	”	1859	+505 16 2½	289 2 10¾
”	1859	”	1860	97 1 8¼	179 13 9
March	1861	to June	1861	366 6 3¼	275 5 7¼

To obtain the true amount of loss, deduct the value of the sweep from the value of the silver lost.

**RETURN FROM THE ROYAL MINT TO AN
ORDER OF THE HOUSE OF LORDS,
DATED 17TH FEBRUARY, 1870.
(THE LORD ROSSIE.)**

MELTING DEPARTMENT.

A STATEMENT of the Weight and Value of the Silver Moneys coined in each Financial Year from 1851 to 1869 inclusive, exhibiting the Weight and Value of the Loss or Waste sustained in each Year, as well as the Value of Sweep recovered, and the average Proportion of such Loss or Waste and Sweep recovered to each Million Pounds Sterling coined.

Date.				Weight of Coinage.	Value of Coinage.			Amount of Waste.
				Ounces.	£	s.	d.	Ounces.
August	1851	to March	1852	314689·050	86,539	9	9	1496·945
April	1852	to March	1853	1088196·550	299,254	1	0	
”	1853	”	1854	2134891·750	587,095	4	8	1814·808
”	1854	”	1855	501459·800	137,901	8	11	882·870
”	1855	”	1856	1138355·000	313,047	12	6	519·069
”	1856	”	1857	1792800·000	493,020	0	0	2912·149
”	1857	”	1858	1420560·000	390,654	0	0	1001·676
”	1858	”	1859	1449360·000	398,574	0	0	484·490
”	1859	”	1860	2052720·000	564,498	0	0	199·288
”	1860	”	1861	625714·000	172,071	7	0	*+20·516
”	1861	”	1862	769680·000	211,662	0	0	
”	1862	”	1863	532147·610	146,340	11	10	+48·918

Date.				Weight of Coinage.	Value of Coinage.			Amount of Waste.
”	1863	”	1864	835920·000	229,878	0	0	
”	1864	”	1865	1936800·000	532,620	0	0	1516·266
”	1865	”	1866	1880640·000	517,176	0	0	3296·589
”	1866	”	1867	1739520·000	478,368	0	0	2923·255
”	1867	”	1868	961200·000	264,330	0	0	1526·136
”	1868	”	1869	597600·000	164,340	0	0	599·329

Date.				Value of Waste.			Value of Sweep recovered.		
				£	s.	d.	£	s.	d.
August	1851	to March	1852	411	13	2	155	19	5½
April	1852	to March	1853						
”	1853	”	1854	499	1	5	143	2	4
”	1854	”	1855	242	15	9	187	19	7
”	1855	”	1856	142	14	11	144	0	2¼
”	1856	”	1857	800	16	10	151	12	4¼
”	1857	”	1858	275	9	3	100	13	6¼
”	1858	”	1859	133	4	8	115	4	11
”	1859	”	1860	54	16	1	101	8	8
”	1860	”	1861						
”	1861	”	1862	5	12	10	203	8	10½
”	1862	”	1863						
”	1863	”	1864	13	9	1	120	6	2
”	1864	”	1865	416	19	6	278	1	1½
”	1865	”	1866	906	11	3	104	5	8
”	1866	”	1867	803	17	11	Nil.		
”	1867	”	1868	419	13	9	347	4	3½
”	1868	”	1869	164	16	3	31	0	11

Date.				Value of Waste per £1,000,000.			Value of Sweep per £1,000,000.		
				£	s.	d.	£	s.	d.
August	1851	to March	1852	1,067	0	10	404	5	9
April	1852	to March	1853						
”	1853	”	1854	850	1	4	243	15	4
”	1854	”	1855	1,760	11	8	1,363	2	10
”	1855	”	1856	455	19	8	460	0	3
”	1856	”	1857	1,624	7	2	307	10	6

Date.				Value of Waste per £1,000,000.			Value of Sweep per £1,000,000.		
”	1857	”	1858	705	2	7	257	14	2
”	1858	”	1859	334	5	6	289	2	10
”	1859	”	1860	97	1	8	179	13	9
”	1860	”	1861	14	14	0	530	3	3
”	1861	”	1862						
”	1862	”	1863	35	15	2	319	15	7
”	1863	”	1864						
”	1864	”	1865	782	17	4	522	1	0
”	1865	”	1866	1,752	18	2	201	12	9
”	1866	”	1867	1,680	9	11	Nil.		
”	1867	”	1868	1,587	14	9	1,313	11	3
”	1868	”	1869	1,002	17	6	188	18	2

Note.—The Financial Years 1860/1861 1861/1862 1862/1863 1863/1864 exhibit a *gain* in the Melting Department.

C. W. FREMANTLE,
DEPUTY MASTER AND COMPTROLLER ROYAL MINT,

2nd March, 1870.

By similar treatment to that applied to the [table on page 132](#) this will yield an average loss between June, 1851, and March, 1857, of £698 13s. 3d. per million coined; whereas in the period from April, 1857, to March, 1860, there is a gain by melting of £87 13s. 3d. per million pounds sterling coined. To look a little more closely into these figures, it will be observed that in the year 1858-59 there was a positive gain of no less than £505 16s. 2½d. on each million coined; this is not to be explained by any known fact, but it seems probable that in that year a circumstance occurred that had happened many times previously, and from the prevalence of the same folly. At the Mint Office it is the custom to weigh the silver from the coining department to the melting department in drafts of 1,440 ounces, or for occasions of 720 ounces. Now it not unfrequently happens that the weigher leaves the counting of the number of drafts to his scalesman, and so an error creeps in. Some days afterwards the coining department finds itself short, say 720 ounces, and by searching its books finds it has been credited with too little on a specific day; after investigation the error is admitted, and the old receipt altered. Now such an error happened on the 3rd of August, 1858, when 1,440 ounces, or two journeys, were delivered, but not credited; however, on the 16th August this was discovered, and the receipt altered at the Mint Office. It is probable that the melter never made a corresponding alteration in his books, and if that be the case, the gain of 1858-59

will disappear, to be replaced by a loss of £487 0s. 8¼*d.*, which would appear to be more natural. Should these lines meet the eye of the officer concerned, he could at once see if this be the explanation of so large a gain, and which has caused him much anxiety. That a gain can be made by melting is not a fact, although, if the statement be examined for the year 1855-56, it will be seen that a gain of £3 0s. 10¼*d.* appears to have been made; and once more, in 1859-60 a plus of £82 12s. 0¾*d.* per million seems to have been realised. These facts, taken as they stand, would mislead, and for reasons which shall now be stated. Sufficient is, however, here shown to prove that the stoppage of the use of oil effected a reduction in the habitual loss of the melting-house, while this fact renders it impossible to show that the stoppage of loss in the coining department was due to the more free use of oil and dirt ([see page 131](#)).

In July and August, 1858, the melter being absent on leave, I was directed by written order of the Master to take his duties. Silver being under operation, the following quantities were melted, and with the results recorded:—

Given to Melt.

	Ounces.
Ingots and alloy	434,393·548
Scissel	457,032·810
Clean ends	1,440·000
Extra alloy	499·539
Plus on weight charged by Mint Office	51·680
	893,417·573

Produced by Melting.

	Ounces.
Bars	889,974·690
Ends	2,154·750
Grains	462·050
Assays	114·230
Loss by melting	711·853
	893,417·573

To analyse this loss—for by weight it was absolute—will give much information, and the details of proportionate loss were as follows:—

	On each Million Ounces Melted.
On ingots of silver (produced from lead by Pattinson's process) and alloy for the manufacture of medals	947·91
On ingots, 5-franc pieces, and alloy	263·85

On ingots, 5-franc pieces, scissel, and alloy	418·40
Worn coin (culled from circulation) and scissel	710·06
Scissel melted by itself	1,036·45
Mean Loss	675·34

Thus, then, it appears that upon this weight of silver and alloy there was, when melted, a loss of 711·853 ounces, or a mean loss of 675·34 ounces in standard silver; yet, on two undoubted occasions, the table above given^[114] exhibits a gain upon the work of two whole years; why, therefore, should the loss just stated have been permitted to take place, when every atom of metal was weighed in the presence of witnesses before being placed in the pots, and was not lost sight of by the officers concerned until it was returned to the scales to be again weighed? The reply is simple: the gain in 1855-56 and 1859-60 was not made, and its appearance is simply a matter of account, for the melter was not accustomed to, and probably to this day does not, weigh silver into his melting-pots—he is, or was, satisfied with the Mint Office weights, which are so far from accurate that no ironmonger would accept them.

On this melting alone, including 29 days' receipts from the Mint Office, there were differences to the extent of 82·11 ounces plus their weight, and 30·43 ounces minus their weight, so that if the one be deducted it still leaves an absolute plus on the weight charged by the Mint Office of 51·68 ounces. Here, then, is a means of gain;^[115] but if the increase of weight derived from this source be taken on the ingots and five-franc pieces bought, it is at the rate of 203·60 in a million ounces purchased. Besides this, the melter is not charged with the extra alloy he puts into his silver, and this on an ordinary average amounts to 1,126 ounces on each million ounces wrought; so that this work, which, in fact, did yield a loss of 711·853 ounces, or at the rate of £796 on the million pounds sterling, would, by the old system of computation, have lost £201. Since, then, it is thus demonstrable that a difference of £591 arises when it is known that the whole of the day's work is weighed into as well as from the melting-pot, it is but reasonable to believe that a greater difference may have arisen when it was known by the Mint Office weigher that his weights were not checked, and, therefore, in weighing ingots to the melter, he probably just satisfied himself that they were as heavy as they were charged by the bullion dealer, and allowed the plus weight which is invariably given by that merchant to go to the melting-house not charged into the account. It is a fact that if the weigher finds ingots sent by the bullion dealer lighter than the weight charged, he sends them back, this practice not being rigidly dependent on the actual weight of the ingots as a whole; for unless each one is distinctly heavier than the weight set against it, it is returned, or the dealer must send silver to make up the required weight. And this case has more than once arisen; so that in practice, if an ingot be

charged as 1,000 ounces, it must weigh indefinitely beyond that weight up to 0·50 ounce, and is then received as 1,000 ounces; but if it weigh in addition 0·60, then it is received as 1,000·50 ounces. These “drafts,” as they are called, amounted on 402,573 ounces of ingots to 82·11 ounces, or at the rate of 203 per million ounces bought. When calculating the proportion of metal for melting, it is the custom to add 1 part of extra alloy to the thousand ounces of silver, and thus to allow for the loss which takes place by annealing and blanching. The extra alloy is not charged to the melter, as before said, but is weighed to him, and thus goes really to reduce his loss; in practice its proportionate use is increased, for after several re-meltings of scissel ([see page 35](#)) the bars become richer in silver than is required by the law, and then again extra alloy is added. In the case of the silver melted in 1858 it must be stated, that if from the loss exhibited the value of the sweep be deducted, that loss will still be too high, because the silver had not yet been recovered from the melting-pots; but with this allowance made, if we divide the total value of the sweep from 1851 to 1857 by the total sum coined, we shall arrive at a fair average price of that sweep, which may be taken at £404 2s., ^[116] thus leaving a loss of £392 on each million pounds sterling coined. But if, by the old style of calculation, the melter’s loss be estimated on the weight charged to the melter by the Mint office, it is £201, and this sum deducted from the average value of the sweep will give a gain of £203, where it is perfectly certain, from the extreme care which was taken, that the real loss was £392, which, minus £54 4s. 7d., subsequently explained, becomes an absolute loss, by melting weighed quantities, of £337 15s. 5d. per million pounds sterling melted. Allusion has been made to the silver left in the iron melting-pots: in the case which has been so much dwelt upon cast-iron pots from Glasgow were used, these absorbing an exceptionally large amount of silver. As, however, the object is to arrive at a fair average, it will be wiser to take into consideration the ordinary wrought-iron pot, which, although it costs more at first, is cheaper in the long-run, as it will stand from fifty to sixty meltings, whereas the bulk of the cast-iron pots used in 1858 leaked in the first melting, and no one of them stood a whole day’s work.

In treating of the melting of silver, it has been explained that charcoal is placed in the pot, to prevent the oxidation of the alloy. This, to a measured extent, it effects, but in the process it becomes burned, and its ash forms a slag with the oxide of iron of the pot, and this slag, always floating on the top of the fused silver, gradually covers the inner surface of the pot as the silver is poured out, and in so covering the pot secretes globules of silver in the “pin-holes” of the iron. At the end of the useful existence of the pot this silver is, or rather should be, obtained. The practice is to “sweat the pot,” and to fuse in it some common table salt—chloride of sodium. It would seem that the used-up wrought-iron pots yield by this process, on an average, 12 ounces of silver. But all the silver is not obtained at this

operation, for the pots stand out of doors, in a yard to which many persons have free access, for months before they are treated professionally, and the result of the preliminary sweat which they undergo is not a source of profit to the Mint.

In 1858 two wrought-iron pots were locked up when their active period had passed, and these were subsequently treated with great consideration, with a view to ascertain their capacity for retaining silver. They yielded under my hands:—

	Ounces.
By simple sweating	20·00
By detaching the slag and oxide from the pot and fusing it with salt	24·70
By separating the remaining silver from the pin-holes in the pot by means of diluted acid	4·60
	49·30

In the whole period of that melting I used nine wrought-iron pots, of which three were entirely used up, and six cast-iron pots, all of which leaked—some two on one day: it would therefore seem fair to assume that of these fifteen pots eight were worn out. We may then consider that had all the eight pots been carefully treated, and each made to yield its fair proportion of silver, we should have recovered silver at the rate above shown, or 24·65 ounces from each pot, and this weight, multiplied by eight, as proposed, will give 197·20 ounces, which, at 5s. 6d. an ounce, would be worth £54 4s. 7d., and by this amount may that loss of £392 be reduced, when the actual loss will be £337 15s. 5d., as determined by most carefully-conducted operations on weighed quantities. This loss, however, is not final, although absolute, because the sweep was sold for £289 2s. 10¾d.;^[117] and, deducting this, the final loss is £48 12s. 6¼d. on each million pounds sterling melted. There can be no doubt but that this is at least one-third in excess of the loss which should take place; yet, between 1851-57, it was, after all deductions for sweep, recoveries from worn-out pots, &c., had been made, no less than £698 13s. 3d. This excess might be accounted for either as wastage in the melter's department, or as legitimate loss by melting caused by the excess of oil sent on the scissel; but since the coiners permitted unnecessary loss to accrue, it will be right to charge this sum to their account, because, if they received metal free from oil, and returned it to the melter sustaining too much oil, it is evident that they received from their men bullion and oil in place of bullion. I therefore add £650 0s. 8¾d., the difference between £48 12s. 6¼d. and £698 13s. 3d., to the wastage previously explained,^[118] when that sum will be raised to £1,297 16s. 0¾d. on each million of silver coined.

In all former calculations the value has been determined on the million coined; but now that we approach the necessary considerations as to the loss by melting which should absolutely take place, it becomes necessary to state that, as a rule, 100 parts of silver bars should produce 57 parts of coined money, and in this proportion must the value of the sweep be reduced, that we may obtain its value proportioned to the silver lost by weight in melting. The sweep results, as in the case of gold, from the grinding of the burnt refuse of the melting-house, which consists of the sweepings from the floor, partly-used charcoal, and dirt—"matter in the wrong place," as Lord Palmerston so aptly designated it. At the end of the coinage this is sold, and its value is applied to the reduction of the apparent loss by melting. In the case now under discussion, we have found that there was an absolute loss by weight of 711·853 ounces on 893,417·573 ounces of bullion given to be melted. By proportion, this weight is equal to 796·764 ounces on each million. The sweep was sold for £289 2s. 10¾d. for each million coined; therefore, to obtain the specific amount that should be deducted by this source from the loss, we must assume that the coined money was 57 per cent. upon the bullion melted, when £164 16s. 3½d. becomes the value of the sweep belonging to each million melted. Since this sum is calculated by value, we will convert it into ounces, which is a simple process; for, as the other sums bear proportion to pounds sterling, we have but to consider it as ounces, when the sum we require will be 164·814. To this sum is to be added the amount of silver due from the melting-pots, as before explained, and which is 220·724 ounces, because it has been assumed that we should obtain 24·65 from each on the average number of pots used in melting, and this, calculated into the million, is as above. The total of these two sums, 385·538, deducted from 796·764, leaves a final loss of 411·126 ounces by weight in melting a million ounces of standard silver.

By experiments made with the utmost care, I arrived at the conclusion that, in melting silver bullion for a complete coinage, the loss by melting should not exceed £100 on each million sterling melted, and in this conclusion I am confirmed in a remarkable manner by Colonel Smith,^[119] who, with vast opportunities for careful every-day work and by absolute weighing, found a loss of 200 in each million for dirt and oxygen alone, and this he is willing to consider as 250 on each million of scissel; but he found his loss by melting ingots 30 on the million, and assuming that the total of a coinage would be half ingots and half scissel, this would give an average of 140 when he was "convinced that he was robbed," and that this was so he proved by compelling the naked workmen, by means of physic, to disgorge the silver they had swallowed. He, however, was probably still robbed, for black men rub pieces of bullion or coins together and secrete the dust obtained by that means in their hair, and thus abstract metal from the works of the Brazilian Mining Association, and probably from the mints in India. To such an extent is

robbery by this means carried on that the Association above referred to causes the head of each person to be washed before he leaves work, and thus recovers a large amount of gold dust.

If then, as has been shown, there is a final loss of £411 2s. 6d. on each million of silver melted, we must now determine that this sum amounts to £721 5s. 5½d. on each million pounds sterling *coined*; and since by coining a loss is sustained equal to £365 15s. 10½d., the total final loss is no less than £1,132 7s. 11½d. per million pounds sterling of silver coined. Such being the facts ascertained by the scales, we find that the loss by ACCOUNT is returned at £365 15s. 10½d. for the coining department, while, at the same period, the melting-house account shows a gain of £87 13s. 3d., which, deducted from the loss of the coining department, gives a final loss equal to £278 2s. 7½d., as against, in the former period, a loss by coining of £1,013 11s. 2½d., and by melting of £698 13s. 3d., or a total final sum of £1,712 4s. 5½d. The difference, then, between these two final sums, or £1,434 1s. 10d., shows the actual loss that the Mint authorities submitted to without explanation on each million of silver coined between June, 1851, and March, 1857. This sum is exactly 8s. 8d. for each £100 coined, while the French contract to coin each £100 for 15s., including loss, wages, salaries, &c. Although, for the sake of accurate demonstration, I have accepted the facts as stated, it is but fair that I should compare the results actually obtained with those stated by the usual methods, and then the contrast will become more apparent; for whereas the proportional loss by melting is shown to have been on weighed quantities £721 5s. 5½d., it is a fact that by the usual methods of account, that of neglecting to charge to the melter the extra weight added to the ingots by the system of purchase, and omitting the weight of the extra alloy, this loss becomes reduced to insignificant proportions, for it was exactly £180 0s. 1d., and as the sweep sold for £164 16s. 0½d. per million melted, the final loss would be but £15 3s. 11½d. Yet, for the sake of truth, I consider it better to charge to each department the metal by weight, and that weight should include every atom, because then, an exceptional loss having taken place, its significance is at once apparent; whereas, when unknown weights are placed in the melting-pot, the losses may vary to a considerable extent, and yet not be detected. By this plan fictitious gain becomes impossible—apparent loss inevitable; but *that loss is so determinate* that it can be calculated to a nicety, and any figure obtained beyond that *calculated* exhibits one of two facts,—that the melting or coining, or both, have by too great a degree refined the silver under operation, and this is demonstrated by the final assays, OR UNNECESSARY WASTE IS AGAIN PERMITTED, but both result from carelessness; therefore the Mint Master's duty is perfectly clear. While, then, absolute gain is impossible, and loss inevitable, it is, perhaps, well to exhibit the results of Colonel Smith's experience in the Indian Mints, because that gentleman states that there can be no loss, and that there must be a gain. Thus, then,

he and I are antagonistic on the plainest possible point; yet an examination of his statement will show that his gain, although absolute, is not legitimate, but results from a near approach to obtaining bullion without paying for it.

It must, then, be explained that in England, and in the Indian Mints, it is not the custom to assay bullion with a view to the determination of its actual value, but only to the nearest part of a pennyweight. Thus, standard silver is composed of 222 parts of silver and 18 of copper in 240. Assuming that it is the custom to determine the amount of silver to within half a pennyweight; and that, the assay showed 23 pennyweights and 11 grains, the silver ingot would be reported as containing 23 pennyweights, and the owner would lose the 11 grains, whereas it should be the custom to determine absolutely the real value of the ingot. Now, in India, Colonel Smith found these “unreported fractions,” when alloyed with copper, amounted to 1,230 parts on each million purchased, so that if he paid for a million ounces, he received 1,001,230 ounces. Here, then, was part of his profit. In addition to this gain he made another by the customary turn of the scale,^[120] but he has not told us its amount; yet his final result, from the receipt of the bullion to its return to the merchants as coined money, is at the rate of £1,165 gain on each million pounds sterling coined. Now it is clear that if so large a sum as £1,230 per million is obtained without purchase from the merchant, either that merchant should himself saturate those unreported fractions with alloy, or make Colonel Smith’s mints pay for them. It is asserted that this practice “exists in the Royal Mint, but not to so large an extent;” yet Colonel Smith states it to be at the rate of £932 per million coined. If this be the case, the Mint books should show for its official accounts a vast gain, instead of, as is the case represented by its purely manufacturing accounts, a loss. This may possibly be the case, but, so far as my experience goes, it is against the practice; for the bullion received is invariably greater in weight than that returned in coined money.

Colonel Smith has determined with great precision that from dirt alone there is a loss by melting of 1 part in each thousand melted. Before he paid for bullion he invariably insisted on its being pre-melted, because the Mint thus saved the loss occasioned by adhering dirt and extraneous matter; this was wise if the merchants would submit to it, but it is doubtful if English bullion dealers would first melt dollars and five-franc pieces before selling them to the Royal Mint.

With bullion thus manipulated I have shown that the Royal Mint also made a gain, but such gain is not real (see pages [136](#), [137](#), [138](#)), for it is evident that the silver having been bought with the intention of being converted into standard silver, each grain should be alloyed, and the whole amount of fine silver being thus alloyed should produce its calculated weight of standard silver; therefore in each department every atom should be weighed, and the total weight charged to any

particular operation returned or accounted for, and under these circumstances no gain can result.

Colonel Smith, however, has determined with great care the following facts from every-day work, and these are so valuable that it is deemed wise to quote them. He found, then, that he made an actual gain by coining from the above circumstances of 1,230 ounces on each million ounces of rupees produced as coin. In the blanching of the blanks that were used for the making of that coined money the loss by blanching was at the rate of 966 ounces per million blanching, and from the acid used in blanching he obtained 726 ounces of metallic copper, so that of the 966 ounces which disappeared only 240 ounces were a real loss, which resulted from oxygen and dirt. Here, then, are reliable figures; and, as he sold his sweep for a sum equal to 140 ounces of silver for each million ounces coined, it is clear that his real loss was just 826 on each million ounces coined. It must be seen that although the copper was recovered, it should in these calculations be neglected, because in the Royal Mint it is not in practice deemed wise (nor, indeed, is it the every-day custom in India) to separate this copper; therefore we will consider the whole matter lost to the Mint. Since, then, he has demonstrated that he received in his clean ingots more silver by 1,230 ounces than he paid for, it is only necessary to add this sum to the weight he charges to himself, which will then be 1,001,230 ounces; in manipulating this total weight he shows that he lost 554 ounces by melting and 826 ounces by blanching, or a total loss of £1,380 per million pounds sterling. This, compared with the practice in the Royal Mint, shows, as Colonel Smith says, that the Mints of India do not obtain their legitimate amount of profit; and he, in his own words, attributes the loss to what he considers its only fit cause—"peculation." For even in the Royal Mint—ill-managed institution that it is—the average annual loss, taken by the system of weighing, is only £721 5s. 5½d. Still, as showing the fluctuations, it is curious to observe that in the last period given, viz., from March to June, 1861, the recorded loss by melting is £91 0s. 8d., while that by coining is £1,949 9s. 6d., ^[121] or a final loss of no less than £2,040 10s. 2d.

Having given the returns relating to gold ordered by the House of Lords, I also give those exhibiting similar details for silver, and place that relating to coining on [page 133](#), while that demonstrating the facts as regards melting will be found on [page 135](#).

Mr. Fremantle assures us in his Report that "the 'waste' shown to have existed of late years has not been excessive," while Mr. Roberts tells us, "The apparent loss on silver melting is two grains on the troy pound, or 0·346 per mille, much of the metal being recoverable from sweep." 0·346 per mille is 346 per million, so we will take these two statements into consideration as regards the returns now to be noticed.

I propose to submit the following abstract from the return on [page 135](#), and thus bring into view *facts* recorded by the Mint Officials, that they may be contrasted with *opinions* expressed by those gentlemen, and for this purpose I deduct the value of the sweep from that of the waste, and thus arrive at the amount of the latter, and as “of late years” may fairly be taken to express the five last past, I will operate on those periods only.

Value of Waste per £1,000,000.		Value of Sweep per £1,000,000.		Final Loss by Melting Silver.
£ s. d.		£ s. d.		£ s. d.
782 17 4		522 1 10		260 15 6
1,752 18 2	Deduct	211 12 9	The	1,541 5 5
1,680 9 11	Value	Nil.	Result	1,680 9 11
1,587 14 9	of	1,313 11 3	is a	274 3 6
1,002 17 6	Sweep.	188 18 2		813 19 4
Total Loss divided by 5 years				4,570 13 8
Gives the Average Loss by Melting Silver				916 2 8¾

In other words, the loss which actually accrues is practically three times as great as Mr. Roberts finds necessary, and yet Mr. Fremantle approvingly says *it is not excessive*. I should advise Mr. Roberts to return to the use of wrought-iron pots and pay proper attention to the melting, FOR I HAVE PROVED that £100 on a million sterling is as much waste as should LEGITIMATELY arise, and Colonel Smith confirms my determinations, as will be seen at [pages 140-141](#). If these officers will pay sufficient attention to their business, they need not fear the variations of assay pointed out by LEVOL. These arise, it is true, but to an imperceptible extent only when melting is properly conducted, as I proved to the entire satisfaction of Mr. Graham in August, 1858, and who, in consequence, relinquished his intention of adopting the then proposed mode of taking silver for assay from the fluid metal by ladles. It is not by hunting hopeless hobbies that losses are to be stopped, nor will the silver be found in the chimneys—if so, some of it should be in those of the present Mint. Will Mr. Fremantle or Mr. Roberts state what has become of £916 worth of silver from each million coined in the past five years? The actual value lost is £4,580 14s. 6d., sufficient to make four immense ingots, which would surely stop the draught of any chimney in the Mint.

As this return has yielded matter for comment, I will refer briefly to that which exhibits the additional loss by coining, and shown on [page 133](#), the following abstract for the past five years will suffice.

Value of Waste per £1,000,000.			Deduct Value of Sweep.	Value of Sweep per £1,000,000.			The Result is a	Final Loss by Coining Silver.		
£	s.	d.		£	s.	d.		£	s.	d.
568	0	9		28	15	7		539	5	2
481	19	4		31	16	9		450	2	7
512	8	8		Nil.				512	8	8
492	16	11		458	7	8		34	9	3
469	16	2		Nil.				469	16	2
Total Loss divided by 5 years							2,006	1	10	
Gives the Average Loss by Coining Silver							401	4	4¾	

Whereas then the loss from 1851 to 1857 averaged, as I have shown on [page 131](#), £1,013 11s. 2¼d., and fell under my management in the period between 1857 and 1860 to £365 15s. 10½d., it has now again risen to £401 4s. 4d.—a steady increase at the rate of 10 per cent., which seems unreasonable.

A rigid Parliamentary inquiry into the circumstances of the Mint is as great a necessity at this time as it was shown to be in 1856; and such an inquiry might demonstrate not only a want of ability, but also whether all the officers are satisfied; whether there be not great heart-burnings, tending to a not very complacent style of performing their duty.

I quote the following from Hansard's Debates, as having occurred in the House of Lords, to show that the Government steadily resists inquiry into the affairs of the Mint; yet I hope the time will arrive when Parliament will assert its right and induce Ministers to yield an inquiry, however much "Mr. Lowe may deprecate it." On the 22nd March, 1870,

"LORD KINNAIRD said: I stated the other night that great robberies have notoriously taken place, and I maintain that these peculations and robberies still continue. If I had an opportunity before a Committee, I could prove this, but the returns which I have obtained will show the correctness of what I am stating. I was not surprised the other evening at a particular return being refused, for the officials of the Mint naturally dislike further exposure; but the returns before us clearly show that under the term 'waste,' which should rather be 'stolen,' large sums are lost in coinage. There ought to be no loss in coinage if the Mint was properly conducted. There might, indeed, be a small loss of £100 on the million in the melting department; but in the coining there should be a gain of at least £59, making the net loss £41. Instead of this, however, there was a loss, as those returns show, of £460 on every million coined. What becomes of that sum? It must go into some one's pocket, and in point of fact it is abstracted. Look, too, at the large sums which are paid to the Bank of England for loss every time there is a coinage; at one time there was a loss of £1,132 on every million for loss of gold abstracted and not returned in coinage. From 1851 to 1857 there was a loss; in the next two years there was a gain; the next two years there was a loss and the following year there was a gain, and after that losses. Now why should not the gain continue? Why was there a gain? Because the men were properly looked after, and could not abstract the precious metal.... I

hope your Lordships will agree to refer this bill to a Select Committee, or will hereafter agree to an independent inquiry into the Mint, conducted by other gentlemen than the Chancellor of the Exchequer's Private Secretary and the Deputy Master.... I move that the Bill be referred to a Select Committee."

I again quote Hansard's Debates, 24th March, 1870:—

"LORD KINNAIRD said: The noble Marquess, (the Marquess of Lansdowne) stated on Tuesday night that there was a considerable profit on silver, which no doubt ought to be the case; but the returns obtained by an Honourable Member in the House of Commons showed that, though in some years there was a gain, in nine years the total loss on the coinage of silver was £5,373. Bad workmanship, he might remark, made the coins much less durable, and therefore caused expense by rendering earlier repair of the coinage necessary. He hoped that, later in the session, the Government would concede an inquiry into the management of the Mint, when he would undertake to prove his allegations of mismanagement and speculation, and indeed the returns proved this....

"The MARQUESS OF LANSDOWNE thought that the noble Lord confounded two things. There was a loss incurred by the public in keeping the silver coinage up to its proper standard; but in its manufacture into coin, there was a considerable gain, amounting to something like £20,000."

Again, on the 18th July, I also quote from Hansard's Debates:—

The Management of the Mint.—Motion for a select Committee.

"LORD KINNAIRD: I rise to move for a Select Committee to inquire into the past and present management of the Mint, and although I very much regret that it is the intention of the Government not to assent to my Motion, I feel bound, nevertheless, to call your Lordships' attention to the mismanagement of the Department to which my Motion refers. It will be in your Lordships' recollection that when the Coinage Bill was under consideration I moved for certain returns connected with the gold coinage, my object being to show that considerable loss occurred in the process of coining. Some of those returns were granted, but others were refused on the plea that they were too voluminous to lay on your Lordships' table. But I have made inquiries, and find that refusal was made because the officials of the Mint did not wish for any searching inquiry, and because the returns would have exposed certain malpractices connected with the coinage. I also moved, when the Bill was in Committee, for certain returns connected with the coinage of silver and the transactions between the Bank of England and the Mint. I was informed on that occasion that no record was kept of those transactions; I saw at the time that was a subterfuge, and that the officials knew these returns would disclose certain malpractices at the Mint which they would prefer were not made public, for not only is a record of these transactions kept, but the returns could be granted with ease. I have found, however, that a return has been presented to the House of Commons, which gives me the principal part of the information I desired to obtain by the returns for which I moved. That return, I have heard said, is not strictly correct; but it is a return presented to Parliament, and, as far as the figures to which I wish to refer are concerned, I have reason to believe they must be correct. The noble Marquess (the Marquess of Lansdowne) who answered me on that occasion, admitted that as regards the gold there was great loss in coinage. He said that although the Government was not supposed to make money by the gold coinage, they made a considerable sum by the silver; and at another time (on the 24th March) he said that although there was a loss resulting from keeping the silver up to the

proper standard, through wear and tear, yet on the manufacture there was a profit of something like £20,000 a year. Now, I am not going to hold the noble Marquess to any statement he made on that occasion, because not being himself connected with the Mint he must have been dependent on the officials; but I can assure him that statement is entirely erroneous. It is quite true that a profit was made in the Mint in the year 1859 of £27,032 on the coinage of silver, and this arose from the large quantity coined and the very small purchase of worn coin. £647,064 worth of silver was coined in that year as against a yearly average of £341,018, and the purchase of worn coin amounted to only £8,096 as against the average of £14,633. But although this profit of £27,032 was made in this year, there was a loss of £5,373 during the next nine years. Now, my Lords, as regards the coinage of silver, there should be an uniform profit. The price of silver is 5s. 1d. or 5s. 1½d. per ounce, so that there should be a profit of 7½ per cent. upon the silver coin. In 1868 the sum purchased was £312,252, which should have yielded a profit of £23,368, but the return showed a loss of £10,896; so that, adding the loss on to what the profit should have been, we find that the Mint was £34,265 on the wrong side that year in its silver coinage; but inasmuch as the loss on the purchase of worn silver was £18,058 in that year, the net loss to the Mint through mismanagement was £16,207. Now, what became of that sum of money? How has it disappeared? The noble Lord admits gold was abstracted during the process of coining, and I say that the loss now arises from speculation. The Chancellor of the Exchequer himself contradicts the noble Lord, and admits that there is a loss on silver, for in the estimates I find he asked the House of Commons to vote £1,000 to make good the loss on silver, in addition to £15,000 for the loss occasioned by repairing that coinage for the coming year, which has to be paid out of the pockets of the people instead of the Exchequer being replenished by the legitimate profit of £20,000. But the most extraordinary loss is in the coinage of copper. One ton of copper costs about £100, and when converted into coin it is worth £358, giving an actual gross profit of £258; yet I find the Chancellor of the Exchequer asked for £1,000 last year to make good the loss on the copper coinage. I have been rather amused to find the Chancellor of the Exchequer speaking slightly of the loss on the gold coinage, saying in all probability it went up the chimney. I think it is more likely to have gone up "the spout." Now, I find another item in the estimates of £1,800 for the expenses of an inquiry into foreign Mints. This represents the expenses of three gentlemen who are going to inquire into the method of coining adopted on the Continent; but, on inquiry, I find that nearly all the machinery in use in foreign Mints went from this country, so that we could ascertain all that we want to know about the practice in foreign Mints, by making inquiries of certain engineering firms in this country. I may be told that the foreign Mints may have improved on the machinery supplied to them, but even this may be ascertained without going to this enormous expense. Moreover, most of the coining on the Continent is done by contract, and it is not likely that contractors will give a Government Commissioner, who is competing with them by advertisement, much information as to the way in which they carry on their work; so that altogether this £1,800 will be completely thrown away. And who are the gentlemen commissioned to make this inquiry? The Deputy Master is one of them—a most excellent, financial man; but, unfortunately, he knows nothing of the practical part of the business, and will gain very little by his trip. Then there is Mr. Roberts, lately appointed chemist to the Mint; ... and the third gentleman is Mr. Napier, one qualified in every respect for the duty. In 1858 a sum of £1,100 was set apart for the purchase of a filing machine in accordance with an estimate sent in by Mr. Napier for the purpose; but, before the order was executed, an officer in the Mint made a machine answering every purpose, which cost only £60, and nobody knows what became of this £1,100. I believe this Commission is going to Spain, but Mr. Napier manufactured the machinery in use there, so he is going to Spain to see how his own machines work, and he is to receive £2,000 for doing it. Perhaps, however, as the war has broken out

the trip will be postponed. Now, my Lords, what I say I am prepared to substantiate before a Committee, and I challenge contradiction. That an inquiry is needed even at this late period of the session is patent, for the House of Commons has only recently voted no less than £18,000 to make up the losses in the Mint. Mr. Fremantle and Mr. Rivers Wilson have made a Report on the subject of some of these complaints, and have suggested improvements; but these improvements would have no effect upon the working of the establishment, and would not touch the main defects in the system. One of the alterations made in consequence of that Report was a revision of the scale of wages. Formerly the workmen had a retaining fee varying from 6s. to 10s. per week when unemployed, and a certain wage beside when at work. But the pay was very small, and it was thought this might lead to peculation. The scale has accordingly been revised by the payment of £1 per week as retaining fee, but the revision is calculated to give the men a little less on the whole, a result to be expected when the arrangement was made by so skilful a financier as the Chancellor of the Exchequer, and the country saves £100 a year out of the workmen's wages. But the most extraordinary part of the revision is to come; and this affords another instance with the sugar duties, hawkers' licenses, and the gun tax, of the readiness with which the Chancellor of the Exchequer jumps to conclusions, without being at the trouble to inform himself upon the subject in hand. The workmen are by this new scale to receive less when coining gold than when coining silver, although the temptation to peculation is greater, so that at the time when their honesty should be secured by good payment, they are to be treated with a niggardly hand, and who can be surprised if they help themselves? The noble Marquess says there has been only one case of dismissal for peculation; but, if he inquires further, he will find there have been twelve. No prosecution has followed these dismissals, nor even an inquiry. Prosecutions would not be palatable to the Mint authorities, because they would expose the system. Now, I can assure your Lordships, that these statements I have made are true in every respect, and I challenge contradiction; and I trust the Government will think the facts really demand a searching inquiry. More than a year since, I brought the subject before the Chancellor of the Exchequer, but it was pooh-poohed, and I was told that a private inquiry would be made: but no inquiry has been made, and the system remains unamended. I therefore move for a Select Committee, to inquire into the past and present management of the Mint.

The MARQUESS OF LANSDOWNE: I regret that it is my duty to decline to accede to the motion of the noble Lord, but I feel sure your Lordships will admit, when I have made a brief statement of the facts of the case, that the Government is warranted in refusing the Committee. I must admit that two cases of peculation have occurred at the Mint; but the persons concerned were dismissed, and there has been no reason to suspect a recurrence of the practice; I am sure therefore, the noble Lord is under a misapprehension^[122] when he says there have been twelve cases of dismissal for peculation during the last few years. As regards the charge of wasteful management, I must repeat the statement I made on a previous occasion. The noble Lord, I fear, confuses two things—namely, the loss occurring from the necessity of keeping the silver coinage up to the proper standard, and the gain which accrues from the coining; and, on the whole, there is a distinct gain, although I may, on a previous occasion, have overstated the amount. There is, notwithstanding the information of which the noble Lord appears to be possessed, a gain of £5,000 a year on the silver coinage. The amount of waste on the gold coinage has, within the last few years, been extremely moderate: it is at present, 1s. per £100; and an eminent firm of refiners in the city has given an opinion that this is very satisfactory. So much for the past. With regard to the present, I must say that as the department has recently been reorganised, it would be hardly fair, before it had got into full working order, to institute an inquiry. The staff has been reduced, thus effecting a saving on this year's estimates of £5,000; the system of the payment of wages has been altered, resulting

in a small saving and in a marked effect on the staff; and the melting and coining departments have been united, avoiding that apparent confusion between the accounts of the two departments, which has, in some respects, misled the noble Lord. Lastly, I come to the Commission of Inquiry, which has been travelling on the Continent. Now, if the machines in use on the Continent have been mostly made at Birmingham, much may be learnt, as to their application and the character of the metals used, by the proposed inquiry. As regards the members of that Commission, Mr. Roberts is a very eminent chemist, and has proved his capabilities by devising a system for depriving gold of the obnoxious properties on which the noble Lord recently commented. The noble Lord did not fail to admit that Mr. Napier was eminently qualified for a post on the Commission; and, as regards the Deputy Master, he has exhibited great abilities in discharging the duties of his office, and I do not think the noble Lord has said a single word to show he has forfeited our good opinion.

Since I am desirous of drawing attention to the affairs of a Public Department, and to that only, I have omitted some passages in the foregoing debates because they did not affect the subject under discussion; still, I quote the remainder of the debate in so far as it was personal to myself in the Appendix at [pages 197-198](#). It will have been seen that in these debates in the House of Lords, Lord Lansdowne, on the part of the Government, made statements to which Lord Kinnauld, in his honoured letter to me, alludes, I therefore proceed to offer a few remarks.

That Lord Kinnauld does not confound *loss on the coinage* with loss occasioned by the repair or “keeping of the coinage up to standard,” is demonstrated by his lordship’s speeches, and by the return No. 157, 1869, presented to the House of Commons by Mr. Ayrton, which shows that, in 1859, £647,064 of *coined money* was obtained by *coining* £620,031 of silver purchased for coining, giving £27,033 as a clear profit for that especial year.

If, however, the circumstances be examined, it will be found that in 1859 only £8,096 was lost by the purchase of worn silver coin, as against the yearly average ([shown on page 5](#) in the return, but not inserted by me) of £14,633. Yet even this exceptional gain is not creditable, for, at the rate of 7½ per cent., it should have been £46,502 instead of £27,033.

The remaining nine years on the return—No. 157, 1869—show an average loss by the silver coinage as exhibited in the next table compiled from it. The Mint returns as presented to Parliament require explanation. I have therefore added two columns to demonstrate the profit and loss.

Date.	Total Value of Silver Coined.	Real Cost or Value of Metal.	Profit Realised.	Loss Sustained.
	£	£	£	£
1860	218,403	222,981		4,577
1861	209,484	215,029		5,545

Date.	Total Value of Silver Coined.	Real Cost or Value of Metal.	Profit Realised.	Loss Sustained.
1862	148,518	159,948		11,429
1863	161,172	171,855		10,688
1864	535,194	521,003	14,190	
1865	501,732	483,861	17,870	
1866	493,416	486,113	7,302	
1867	193,842	195,445		1,603
1868	301,356	312,252		10,896
	2,763,117	2,768,487	39,362	44,733
Deducting the Profit, £39,362, the Final Loss becomes				£5,371

Such final loss is indefensible. Yet Lord Lansdowne is led to believe that the seigniorage on silver yields a profit of £5,000 a year (modified from his first statement of £20,000), whereas this return, inclusive of the £27,033 before explained, exhibits a profit of £21,659 in ten years, or of an average of £2,165 per annum. This statement will, however, deceive those who rely on it, for two reasons. The year 1859, as already pointed out, is exceptional; and another return, No. 67, 1870, made to the House of Commons ([see page 150](#)) shows that, inclusive of this anomalous gain, the silver coinage has been a source of loss to the country to the exact average amount of 17s. 6d. on each £100 purchased for coinage. In printing this return I have made it up to 1868, omitting the year 1869, which alone shows a loss of £120,000, and must therefore be an incomplete return. It is to be regretted that the Government refused on two occasions a return of the account of losses on silver manufacture, because such a return would have included loss by assay as well as loss by coining and melting. The refusals were undoubtedly based on fear of exposure, because the return can be given readily—that they can be given, is proved by the fact that I have read the accounts in the bill books at the Mint, and have copies of such entries. Lord Lansdowne was instructed to state that no such accounts are kept. The subterfuge of the Mint Authorities consists in the fact that Lord Kinnauld has not asked for the return in the precise words in which the account is kept; it cannot, however, be again refused if moved for as “transferred from the account of the Mint expenses, for the satisfaction of the Audit Office to account for the waste or loss by coining silver.” If the Mint Authorities will place the bill book of the Royal Mint before me for a few hours, I will compile the table. Such a return would exhibit the fact that in the five years last past, the loss by misplacement and mismanagement exceeds £1,867 3s. 10¾d. per million coined; it would show that the average amount of loss reaches 3s. 9¾d. for each £100 coined, such sum being in fact about one-third of the total sum charged by the contractors who conduct the coinage in Paris for France. The contractor for the Paris Mint, in his charge, includes loss of silver by coining, melting, and assay, wear and tear of

machinery, wages to workmen for labour, salaries and contingent expenses; and while bearing these charges, he makes a handsome

**AN ACCOUNT OF THE SUMS ADVANCED IN EACH YEAR
FROM THE CONSOLIDATED FUND FOR THE PURCHASE OF
BULLION FOR COINAGE; AND OF THE SUMS PAID IN
EACH YEAR TO THE ACCOUNT OF HER MAJESTY'S
EXCHEQUER AT THE BANK OF ENGLAND, IN REPAYMENT
THEREOF.—(PURSUANT TO ACT 7 WILL. 4, C. 9.)**

Date.	Amounts Advanced from the Consolidated Fund.			Amounts Paid into the Exchequer, in Repayment of Advances.			Remarks.
	£	s.	d.	£	s.	d.	
1837	22,000	0	0				As given in detail in previous Returns.
1838	196,029	19	4	168,029	19	4	
1839	400,000	0	0	350,000	0	0	
1840	170,000	0	0	175,000	0	0	
1841	100,000	0	0	135,000	0	0	
1842	180,000	0	0	162,120	4	0	
1843	350,000	0	0	265,000	0	0	
1844	550,000	0	0	501,389	17	0	
1845	850,000	0	0	678,772	2	9	
1846	300,000	0	0	429,937	1	11	
1847	100,000	0	0	217,033	10	2	
1848				18,534	3	1	
1849	150,000	0	0	120,000	0	0	
1850	150,000	0	0	120,000	0	0	
1851	100,000	0	0	187,213	1	1	
1852	391,970	0	8	191,970	0	0	
1853	508,029	19	4	550,000	0	0	
1854	100,000	0	0	200,000	0	0	
1855	250,000	0	0	300,000	0	0	
1856	450,000	0	0	450,000	0	0	
1857	420,000	0	0	350,000	0	0	
1858	400,000	0	0	400,000	0	0	
1859	550,000	0	0	650,000	0	0	
1860	270,000	0	0	150,000	0	0	
1861	400,000	0	0	550,000	0	0	

Date.	Amounts Advanced from the Consolidated Fund.	Amounts Paid into the Exchequer, in Repayment of Advances.	Remarks.
1862	400,000 0 0	350,000 0 0	
1863	250,000 0 0	250,000 0 0	
1864	600,000 0 0	0 0	
1865	520,000 0 0	460,000 0 0	
1866	410,000 0 0	510,000 0 0	
1867	300,000 0 0	250,000 0 0	
1868	220,000 0 0	200,000 0 0	
32 Years	10,058,029 19 4	9,970,000 0 0 0 0	<i>Loss,</i> £88,029 19s. 4d. 17/6 <i>on the £100</i>

C. W. FREMANTLE, DEPUTY MASTER OF THE MINT.

ROYAL MINT, 18th February, 1870.

profit, his total sum for all being 15s. for each £100 coined. Is it probable that the contractor would tolerate a charge of 3s. 9¾d. for loss alone out of a total of 15s. Since the Government refuses figures, imagination alone can tell what is the loss in the Royal Mint; yet in the face of such facts Mr. Lowe accepts contracts as against the French contractor to whom he sends Commissioners—one of whom experienced “great difficulty in obtaining accurate information with regard to the amount of loss that accrues” on gold and silver—that he may instruct them in the cost of manufacture. Does a banker make known his secrets? He may permit you to stand at his counter and see cheques paid, but “thus far and no farther” is the guiding principle. Mr. Lowe’s contract notions have been already noticed.

To return, then, to facts admitted by the Government, and presented to the House of Lords in obedience to Lord Kinnaid’s motion; the return numbered 30, 1870, clearly demonstrates that there is, under the management of the present officers, and has been for the last five years ([see pages 144-145](#)), an average loss by coining of £401, and by melting, of £916, or a total of £1,317 on each million of silver coined. By way of contrast, it should be noticed that this return exhibits the fact that, when the Mint was under my management—between April 1857 and March 1860—the total average loss, under circumstances every way similar, was but £395 4s., inclusive of “melting and the losing operations,” for each million of silver coined. Well may the Government refuse a return which, at the smallest possible computation, would EXHIBIT AN UNNECESSARY WASTE TO THE EXTENT OF £922 ON EACH MILLION OF SILVER COINED, OR EXACTLY £3,144 ON THE SILVER CONCERNED IN THE RETURN No. 157, 1869, quoted at [page 149](#).

The unnecessary waste here displayed is not the only reason why silver yields no final profit or seigniorage—and the foregoing statements clearly demonstrate that Lord Kinnauld has not misunderstood the accounts. The great reason of the loss occasioned by the silver coinage, both by manufacture and by seigniorage, is want of knowledge on the part of the Mint officials, for it can require but little argument to show that that which costs 61*d.* and sells for 66*d.*, must, if properly manufactured, yield an absolute profit. So, to demonstrate more clearly the exact case, we will take the real cost of the silver for 1868 and see what profit this should have afforded, and then Lord Lansdowne will perhaps explain why the loss arises, for this return ([see page 154](#)) is perfectly free from all manufacturing details, and does not include wages, officers' salaries, or contingent expenses; it simply states the bald fact that so much silver was bought and converted into coined money, and by deducting one from the other there is demonstrated an actual loss of £10,896 12*s.* Now the amount bought was in real intrinsic value £312,252 12*s.*, and this produced in coined silver money £301,356. It has been shown in the return now under comment that silver fetched at that period (1859-1868) 61 $\frac{3}{4}$ *d.* per ounce, which, expressed in decimals, will be 61·375*d.*, and silver, by the Act of Parliament, is so coined as that each ounce is converted into money of a nominal value of 66*d.*, such extrinsic value being given to silver because it is a token currency, and as such is a legal tender to the limited extent of £2. The difference, then, between 66*d.* and 61·375*d.* is 4·625; if, therefore, 61·375 gives a profit of 4·625, it follows that 100 should give 7·535; but for the sake of simplicity we will consider this profit as at the rate of 7 $\frac{1}{2}$ per cent.; then £312,252 12*s.* at 7 $\frac{1}{2}$ per cent., should yield £23,418 18*s.* 10 $\frac{3}{4}$ *d.*, yet not only was the profit lost, but, in addition, there is a recorded—admitted—loss of £10,896 12*s.*, which, added to the loss of profit, brings that total to £34,315 10*s.* 10 $\frac{3}{4}$ *d.*; but that Lord Lansdowne may not be unfairly pressed, Lord Kinnauld expressed willingness to deduct from this sum the amount of loss entailed that year by recoinage (see * below) and which was £18,058 9*s.* 6*d.*, leaving thereby an irretrievable loss of exactly £16,257 1*s.* 4 $\frac{3}{4}$ *d.*, on that year alone.

**NO. 262.—RETURN TO AN ORDER OF
THE HONOURABLE THE HOUSE OF COMMONS,
DATED 30TH MAY, 1870.**

(Mr. AYRTON, [page 5.](#))

Worn Silver Coin Purchased for Re-coinage.

Date	Weight.	Nominal Value.	Mint Value at 5/6 per Ounce.	Loss by Re-Coinage.
	Ounces.	£ <i>s.</i> <i>d.</i>	£ <i>s.</i> <i>d.</i>	£ <i>s.</i> <i>d.</i>

Date	Weight.	Nominal Value.	Mint Value at 5/6 per Ounce.	Loss by Re-Coinage.
1860	357902·538	113,000 0 0	98,423 3 9	14,576 16 3
1861	308873·077	97,600 0 0	84,940 1 10	12,659 18 2
1862	425591·682	135,000 0 0	117,037 14 3	17,962 5 9
1863	324674·838	102,800 0 0	89,285 11 5	13,514 8 7
1864	388388·152	123,500 0 0	106,806 14 10	16,693 5 2
1865	292496·983	93,000 0 0	80,436 13 4	12,563 6 8
1866	361280·187	115,000 0 0	99,352 1 0	15,647 19 0
1867	376142·927	120,000 0 0	103,439 6 0	16,560 14 0
1868	388878·297	125,000 0 0	106,941 10 6	18,058 9 6
1869	325976·928	105,000 0 0	89,643 13 0	15,356 7 0
	3550205·609	1,129,900 0 0	976,306 9 11	153,593 10 1

The Yearly Average Price paid per Ounce for Silver Bullion purchased in the Market for Coinage in the Years 1860 to 1869, both inclusive, was as follows:—

1860	1861	1862	1863	1864	1865	1866	1867	1868	1869
d. 61¾	d. 61½	d. 61½	d. 61½	d. 61¼	d. 61	d. 61⅞	d. 60¾	d. 60⅞	d. 60⅞ ₁₆

The amount of Seigniorage paid into the Exchequer in the years 1860 to 1869, has been,—

1860	1861	1862	1863	1864
£ s. d. 27,134 1 8	£ s. d. 4,961 7 5	£ s. d. 11,227 6 2	£ s. d. 211 13 4	£ s. d. 6,160 6 8

1865	1866	1867	1868	1869	Total 1860-69
£ s. d. 57,645 17 9	£ s. d. 28,630 0 5	£ s. d. 4,331 10 7	£ s. d. 18,622 7 4	£ s. d. 12,345 4 10	£ s. d. 171,269 16 2

C. W. FREMANTLE, DEPUTY MASTER OF THE MINT.

ROYAL MINT, 31st. May, 1870.

Would Lord Lansdowne like to live on a *profit* of such a description? Should his lordship still maintain his statement that there is a gain from seigniorage by coining silver, perhaps he will be so good as to explain why the Chancellor of the Exchequer asked last year for £1,000 to “defray the anticipated loss,” as well as £15,000 to pay the expenses of the recoinage, or “keeping the silver up to the

required standard.” It has been stated that this return is false—it is surely a grave matter to make a false return to Parliament, in itself a crime which should be inquired into and punished.

The loss thus exhibited is entailed solely because the Master of the Mint habitually wishing to keep things quiet, refuses all Parliamentary enquiry, and contents himself with so-called retrenchment because he deducts £100 a year from the Mint workmen.

A continuation of the return to which Lord Kinnauld drew attention has been presented, No. 262, 1869, signed by Mr. Fremantle. Singular to relate, this is not merely *a continuation*, for it gives information in regard to seigniorage, and Mr. Fremantle has amended the account made by Mr. Graham; thus a return presented to Parliament has been altered by Mr. Fremantle without comment, and now exhibits results directly opposed to those shown for the same years by Mr. Graham. The words at the head of the column are, for some reason which does not appear, also altered, but the “real cost” of metal must mean the same as the “total amount paid” for it. I have, therefore, compiled from each a statement of the profit and loss; but I reproduce the figures presented in the one case by Mr. Thomas Graham, and in the other by Mr. C. W. Fremantle, BOTH RELATING TO THE SAME ACCOUNT, and both supposed to be accurate ([see page 154](#)).

It is necessary that I should state that Mr. Graham had been dead more than a year when his accounts were altered, in the manner now to be explained. If *his* returns were correct, Mr. Fremantle’s must be *incorrect*, or, “adjusted,” as Mr. Roberts would style it. If reference be made to the table on [page 154](#), it will be seen that by the return No. 157, Mr. Graham showed a loss to the extent of £5,373 4s. 6d. on nine years’ (1860-1868) silver coinage. Now, Lord Kinnauld explained that return fully to the House, and, to prove his accuracy, produced the return No. 30, which will be found on [page 150](#). Considerable astonishment was caused by such facts, but no reply was given. However, it now appears by the return No. 262, which I print under No. 157, and on the same page, that Mr. Fremantle has found the means to convert Mr. Graham’s *loss* into an absolute GAIN of so large an amount as £1,011,668 2s. 0d., and *that without any word of explanation*. The returns thus placed together are so contradictory, that I confess I utterly failed to unravel them, and therefore submitted them to more than one banker, several merchants, and, finally, to a professional accountant, not one of whom could suggest a key to the extraordinary process which has been adopted.

No. 157.—RETURN TO AN ORDER OF THE HONOURABLE THE HOUSE OF COMMONS, DATED 21ST APRIL, 1869, FOR ACCOUNT “OF ALL GOLD, SILVER, AND COPPER MONEYS OF THE REALM COINED AT THE MINT, FOR EACH YEAR FROM THE 1ST JANUARY, 1859, TO THE 31ST OF DECEMBER, 1868, &c.,” IN CONTINUATION OF No. 340, 1867-68.

(MR. AYRTON, [page 3](#))

	Total Value of Silver Coined.			Real Cost or Value of Metal.			Profit on Coinage.			Loss on Coinage.		
Date	£	s.	d.	£	s.	d.	£	s.	d.	£	s.	d.
1860	218,403	7	0	222,981	1	6				4,577	14	6
1861	209,484	0	0	215,029	0	10				5,545	0	10
1862	148,518	11	10	159,948	2	6				11,429	10	8
1863	161,172	0	0	171,855	15	0				10,683	15	0
1864	535,194	0	0	521,003	5	0	14,190	15	0			
1865	501,732	0	0	483,861	13	6	17,870	6	6			
1866	493,416	0	0	486,113	15	2	7,302	4	10			
1867	193,842	0	0	195,445	17	10				1,603	17	10
1868	301,356	0	0	312,252	12	0				10,896	12	0
	2,763,117	18	10	2,768,491	3	4	39,363	6	4	44,736	4	6
Deduct Profit, when there remains Absolute Loss										5,373	4	6

THO. GRAHAM, MASTER OF THE MINT.

ROYAL MINT, 19th April, 1869.

No. 262.—RETURN TO AN ORDER, &c., DATED 30TH MAY, 1870, &c., IN CONTINUATION OF PAPER No. 157, OF SESSION 1868-9.

(MR. STANSFELD, [page 3](#).)

	Total Value of Silver Coined.			Total Amount Paid for Silver Bullion			Profit on Coinage.			Loss on Coinage.		
Date	£	s.	d.	£	s.	d.	£	s.	d.	£	s.	d.
1860	218,403	7	0	141,112	7	11	77,290	19	1			
1861	209,484	0	0	100,327	11	0	109,156	9	0			
1862	148,518	11	10	51,165	19	11	97,352	11	11			
1863	161,172	0	0	53,134	12	7	108,037	7	5			
1864	535,194	0	0	445,191	11	9	90,002	8	3			
1865	501,732	0	0	416,524	6	9	85,207	13	3			
1866	493,416	0	0	282,247	12	6	211,168	7	6			
1867	193,842	0	0	181,000	17	11	12,841	2	1			
1868	301,356	0	0	80,744	16	6	220,611	3	6			
1869	76,428	0	0	135,083	0	11				58,655	0	11
	2,839,545	18	10	1,886,532	17	9	1,011,668	2	0 (11)	58,655	0	11

C. W. FREMANTLE, DEPUTY MASTER OF THE MINT.

ROYAL MINT, 31st May, 1870.

The return contains another enigma. For, whereas Mr. Fremantle, on page 3 of the return in which he has amended Mr. Graham's figures, exhibits profit to the extent of £1,011,668 2s. 0d. in place of Mr. Graham's loss of £5,373 4s. 6d.—but there being a loss of £58,655 0s. 11d. on his own year's coinage, the substituted profit(££) must be reduced by that amount, when it will be £953,013 1s. 1d. Again, page 5, he gives a tabular statement ([see page 152](#)), by which he shows that £171,269 16s. 2d. went to the Exchequer as seigniorage, while the loss by recoinage amounted to £153,593 10s. 1d., leaving a total profit of £17,676 6s. 1d. What has become of the difference between this profit and that in the amended return?—a small matter perhaps, but which yet amounts to £917,659 8s. 11d. in ten years. If it were made it can be shown to have been legitimately disposed of, or is it simply an “official statement,” such as those which public companies have of late so frequently exhibited. If, however, that sum has been realised, why does Mr. Fremantle show separately the amount of seigniorage? There lies below a grave matter for inquiry why a Public Department should place before Parliament an altered account without one word of explanation. If Mr. Graham has put forward a false return, one greatly against himself, it is right that the matter should be investigated. If, on the other hand, Mr. Fremantle has placed figures on paper which have no representatives in coin, it is indeed a grave matter. In either case it goes to prove the necessity for a searching inquiry into the affairs of the Royal Mint.

Lord Kinnaird does not stand alone in his view of the loss by the silver coinage, for Mr. William Miller, late Chief Cashier in the Bank of England, had, years since, arrived at the same conclusion, and produced at that time evidence to prove his statement. By the courtesy of Mr. George Forbes, at present Chief Cashier in the Bank of England, I am in possession of Mr. Miller's statement ([see p. 156](#)): its perusal induces me to wish that Government officials would put forward statements as lucid. Mr. William Miller, in this table, adopts a process which I am assured is entirely legitimate, and by it educes the fact that “*the Government owes the silver coinage a million and a half, and not as an account.*” It is perfectly true that the profit, if made, is not supposed to be placed at interest; but it is also the fact that the profit *has never been realised*, hence the loss of both capital and interest. His calculations are those of an accountant, and by them he argues that if the £6,493,000 of silver coinage had produced its legitimate gain, there would have accrued by it, and the interest thereon, a sum of **£3,880,000** clear profit: he then proceeds to show how that profit has been disposed of. Being himself aware of the amount of loss by worn coins culled from circulation, he states it at £255,000; and estimating the loss on that which remains current, to be eliminated in its turn, at £910,000, shows a loss by wear and tear of £1,165,000. The details for the complete study of this table will be found in [page 65](#), where every information is given in precise figures, while in this he adopts whole numbers, and proceeds to reason out the entire matter. That he may not press the Government, he assumes an extreme rate for the expense of coinage—varying from $2\frac{1}{4}$ to $3\frac{1}{2}$ per cent., while the real cost is about 0·75 per cent.—and adds to that amount the interest thereon, then deducting the total of these sums from the £3,880,000 which should have arisen, he shows a balance £1,411,000 which the Government owes to the silver coinage. While Mr. Miller's table on [page 65](#) shows an average loss by wear on the silver coinage current since 1816, at the rate of 24·31 per cent. in a hundred years, and on that of George III.—*the most worn*—of 98·34 per cent., the Mint—receiving its culled coin from the Bank of England—states its loss at the rate of 135·93 per cent. in the same period ([see return, p. 152](#)).

THE GOVERNMENT IN ACCOUNT WITH THE SILVER COINAGE.

COINED IN THE REIGN OF GEORGE III.:—

Crowns	£322,000
Half-Crowns	2,387,000
Shillings	3,304,000
Sixpences	919,000
	<u>£6,932,000</u>

Supposing the cost of the Silver to have been 5s. 1d. an ounce in the new money, there would remain 5d. per ounce as profit to the Government. That 5d. would be about 7·57 per Cent. on the amount coined.

7·57 per Cent. on £6,932,000	=	£525,000
Compound Interest thereon at		
3½ per Cent. for 43 Years, the		
true Average Number of Years		
as nearly as can be made out	<u>1,779,000</u>	
		2,304,000

COINED IN THE REIGN OF GEORGE IV.:—

Crowns	141,000	
Half-Crowns	1,114,000	
Shillings	879,000	
Sixpences	81,000	
7·57 per Cent. on	<u>£2,215,000</u>	= £168,000
Compound Interest at 3½ for 35 Years	<u>392,000</u>	
		560,000

COINED IN THE REIGN OF WILLIAM IV.:—

Half-Crowns	381,000	
Shillings	412,000	
Sixpences	282,000	
7·57 per Cent. on	<u>£1,075,000</u>	= £81,000
Compound Interest at 3½ for 15 Years	<u>111,000</u>	
		192,000

COINED IN THE REIGN OF VICTORIA:—

Crowns	117,000	
Half-Crowns	1,043,000	
Florins	1,541,000	
Shillings	2,610,000	
Sixpences	1,182,000	
7·57 per Cent. on	<u>£6,493,000</u>	= £492,000
Compound Interest at 3½ for 15 Years	<u>332,000</u>	
		824,000
		<u>£3,880,000</u>

LOSS BY WEAR:—

On amount remaining in Circulation	£910,000
Ditto withdrawn from Circulation	<u>255,000</u>

		£1,165,000
EXPENSES OF COINAGE:—		
Crowns and Half-Crowns at 2¼ per Cent.	124,000	
Shillings at 2·85 per Cent.	<u>205,000</u>	329,000
Compound Interest thereon, 30 Years at 3½ per Cent		<u>594,000</u>
		923,000
Sixpences at ¾ per Cent.	80,000	
Compound Interest thereon, 26 Years at ¾ per Cent.	<u>116,000</u>	
		196,000
Florins at 2½ per Cent	39,000	
Compound Interest thereon, 5 Years at ¾ per Cent.	<u>5,000</u>	
		44,000
7·57]per Cent. on the amount of Silver recoined		
at the Government's expense (£1,316,500)	100,000	
Compound Interest, at 3½ per Cent. for Average of 10 Years	<u>41,000</u>	
		141,000
	Balance	<u>1,411,000</u>
		<u>£3,880,000</u>

I communicate this Statement as my grounds for the Estimate that the Government owes the Silver Coinage a million and a half, and not as an Account.

December, 1860.

W. MILLER.

It has been proposed to reap a harvest of profit from the silver coinage, just as was done with the bronze coinage, but with this difference—that whereas the old *copper* pennies, when alloyed with tin and zinc, made TWO BRONZE pennies, thus making the profit by a diminution of weight, the present suggestion is to reduce the amount of silver in the silver coinage, and so to get profit by converting the old coinage into a new one of lower standard. So long as silver is only a token it can make no difference in matter of fact whether that token be intrinsically worth sixpence, eightpence, or tenpence, so that it represents and is legally payable for the twentieth part of a pound up to a fixed amount; but in another light the lowering of the standard has a very material interest to the people, and in this way. If those coinages which are made of different alloys from our own, that is, in different proportions of silver and copper, be examined, it will be found that none of them *wear* so well as our own. Elaborate experiments, carefully made, determined that silver, when alloyed with copper in the proportion of 222 of the former to 18 of the latter, formed the most fitting alloy. Tested by practice in the pocket, those experiments are demonstrated to have been justly performed, and that their results were accurate. It is worthy of remark, that in the reign of Edward III. these proportions formed the *Old Standard*, showing that long anterior to that period it had been used, although no record is preserved. It is also certain that these same proportions were used in many subsequent reigns, and we can but suppose that our ancestors had their wise reasons for arriving at such a standard. Indeed, Mr. Hatchett, in his elaborate experiments on gold and silver coins, arrived at the conclusion that metal of medium ductility was best fitted for the manufacture of coined money. He recommended the preserving of the present standard for gold on that ground, and fails to make any remarks as to silver; hence it may be fairly assumed that he considered it the best for its purpose. It is, however, certain that pure silver is too ductile, while silver alloyed with 20 per cent. is really as ductile as pure silver, and in our present standard we appear to have the medium impliedly recommended by Mr. Hatchett. The old

proportion was again adopted for our standard of silver coin when, by the Act Geo. III. cap. 68, sec. 4, it was enacted that 66 shillings shall be made from the pound troy of silver of the standard of 11 oz. 2 dwts. of fine silver and 18 dwts. of alloy. The alteration proposed has varied from a mixture of 10 per cent. to one of 20 per cent. of alloy. Now, one of these is a minute alteration, but in such matters it is surprising to the uninitiated to find how small a variation of proportion effects a great change in character; take, for instance, the difference in qualities of steel, where the proportion best fitted for each purpose must be used, or waste is incurred. In cases where the metal is required for cutlery a different steel is selected from that which is used for the manufacture of dies, and yet the difference consists in a minute proportion of carbon, to be detected only by analysis. So with the alloy of silver; and, if a profit be made by reducing the proportion of silver, a greater loss will take place by wear and tear, and the people pay for this, because the profit on a silver coinage depends on the length of the period during which the coins wear. Should a change be effected, it is to be hoped that it will not be based on the series of experiments lately conducted in the Royal Mint, and which were simply determinations of how much grinding each alloy could bear under different circumstances, there being no doubt that the proper mode is to let some thousands of coins be carried by careful observers in their pockets in the ordinary mode of carrying money, and at intervals to weigh those coins and see their loss, as well as to examine them for their deterioration in an artistic point. For it should not be forgotten that the act of coining is an important act of Sovereignty, and should be properly performed, that the Monarch may not appear to be held in light esteem. The ordinary wear encountered in the use of coins by shopkeepers could also be tested without risk or trouble. We as a people need not debase our coinage simply for the sake of a first gain, for, if that be the object, the Government might as well contract for advertisements, and thus make profit; but it may fairly be doubted whether our shillings would be improved in appearance if they were substituted for paving-stones,—the late favourite medium for advertising. Yet, if gain alone is to be the consideration, these points should be determined. It appears to be the object of a coinage to make a medal which shall best encounter wear and tear without loss or deterioration. This, we know, is best gained by employing artists of the highest talent, and coiners who know how to give effect to the artist's intention. It is almost indisputably the fact that this has never been so well performed as in the shillings of William Wyon, designed after the bust by Chantry, and which coins, to this day, far surpass those of subsequent coinages in every way. The remarks of Mr. Pepys seem to be quite fitting if applied to our present coinage, and demonstrate how truthfully it has been said that history reproduces itself; for on Feb. 20, 1660, he says:—"... where we met with Mr. Slingsby, who showed me the stamps of the King's new coyne; which is strange to see, how good they are in the stamp, and bad in the money for lack of skill to make them. But he says Blondeau will shortly come over, and then we shall have it better, and the best in the world." The protecting edge should be raised and thick, and the work of the die well sunk, as then by friction less surface is exposed, and the coin would not only last longer in circulation, but would during its whole existence carry the image and superscription. While this improvement should be effected, the note of Mr. Pepys, on the 23rd November, 1663, should not be forgotten, for he says, with great point:—"... with Alderman Backewell talking of the new money, which he says will never be counterfeited, he believes; but it is so deadly inconvenient for telling, it is so thick, and the edges are made to turn up." Mr. Backewell had become accustomed to the *hammered* money, and felt the thickness of the *milled*; but we have found the advantage of the milled money, and do not like it when, after a short existence, it becomes as thin as the hammered money was. The protecting edge not only protects the design on the coin, but it also makes the coin more difficult of bending, and bent coins are troublesome.

In former reigns unwise counsels have prevailed, and the coinages have been tampered with; but the invariable event has been disaster, as has been clearly shown by Mr. Lowndes, who says:—“Although the former debasements of coins by public authority, especially those in the reigns of King Henry the Eighth and King Edward the Sixth, might be projected for the profit of the Crown, and the projectors might measure that profit by the excessive quantities of alloy that were mixed with the silver or gold, and although this was enterprised by a Prince who could stretch his prerogative very far upon his people, and was done in times when this nation had very little commerce, inland or foreign, to be injured or prejudiced thereby, yet experience presently showed that the projectors were mistaken, and that it was absolutely necessary to have the base moneys reformed; the doing whereof was begun by King Edward the Sixth himself, carried on by King Philip and Queen Mary, and happily finished (though not without great charge, vexation, and trouble, the only offspring of such designs) by Queen Elizabeth, who (as is noted above) in the third year of her reign, when money was not plentiful, erected a distinct Mint in the Tower to convert the base (not counterfeit money) into sterling;” and continuing, he says:—

“Thirdly. Because making of base money will disgrace this Government in future generations, the critics in every age being apt to estimate the goodness or badness of ancient Governments by their coin, as hath been done, especially in the case of the Romans, and a temptation of this kind ought not to be left for future ages, to the prejudice of the honour of the present King.”

Surely, with such experience before him, the Chancellor of the Exchequer cannot wish to persevere with his proposal—which can hardly be called original—to debase our gold coinage, when such a measure has been so deliberately condemned by experience in former periods.

**STATEMENT OF WORK PERFORMED IN THE
COINING DEPARTMENT
BETWEEN OCTOBER 1ST, 1857, AND
MARCH 31ST, 1858, INCLUSIVE.**

The Coinage was completed in 63 Days.
The Value of the coined Silver was £234177·96875.

IN ROLLING ROOM.			
RECEIVED.		RETURNED.	
		As by Rolling Room Beam.	As by Mint Office Beam.
Bars for Florins.			
	Ounces.	Ounces.	Ounces.
As by Mint Office Beam.	761202·00	Florin Fillets	728342·56
As by Rolling Room Beam.	761212·67	” Ends	32956·95
Showing a plus difference.	10·67	Deduct “received.”	761299·51
		Increase during work.	86·84
Bars for Shillings			
	Ounces.		
As by Mint Office Beam.	528613·00	Shilling Fillets	504599·40
As by Rolling Room Beam.	528617·25	” Ends	24081·80
Showing a plus difference.	4·25	Deduct “received.”	528681·20
		Increase during work.	63·95

IN ROLLING ROOM.			
RECEIVED.	RETURNED.		
		As by Rolling Room Beam.	As by Mint Office Beam.
Bars for Sixpences. Ounces. As by Mint Office Beam. 175644·00 As by Rolling Room Beam. 175644·54 Showing a plus difference. 0·54	Sixpenny Fillets " Ends Deduct "received." Increase during work.	168666·70 7003·10 175669·80 24·26	7003·07
Bars for Threepences. Ounces. As by Mint Office Beam. 33643·00 As by Rolling Room Beam 33642·20 Showing a minus difference. 0·80	Threepenny Fillets. Threepenny Ends. Deduct "Received." Increase during work.	32661·25 987·55 33648·80 6·60	987·55 65029·43

IN CUTTING ROOM.			
RECEIVED.	RETURNED.		
		As by Cutting Room Beam.	As by Mint Office Beam.
Fillets for Florins. Ounces. As by Rolling Room Beam. 728342·56 As by Cutting Room Beam. 728339·29 Showing a minus difference. 3·27	Florin Blanks " Scissel Deduct "received." Increase during work.	Ounces. 433013·65 295327·34 728340·99 86·84	Ounces. 295327·98
Fillets for Shillings Ounces. As by Rolling Room Beam. 504599·40 As by Cutting Room Beam. 504598·71 Showing a minus difference. 0·69	Shilling Blanks " Scissel Deduct "received." Increase during work.	320518·84 184117·20 504636·04 37·33	184117·20
Fillets for Sixpences. Ounces. As by Rolling Room Beam. 168666·70 As by Cutting Room Beam 168666·25 Showing a minus difference. 0·45	Sixpenny Blanks. Sixpenny Scissel. Deduct "Received." Increase during work.	101407·09 67271·99 168679·08 12·83	67271·99
Fillets for Threepences. Ounces. As by Rolling Room Beam. 32661·25 As by Cutting Room Beam 32661·25 Showing no difference. 0·00	Threepenny Blanks. Threepenny Scissel. Deduct "Received."	19373·13 13293·73 32666·86	13293·73

IN CUTTING ROOM.			
RECEIVED.	RETURNED.		
		As by Cutting Room Beam.	As by Mint Office Beam.
	Increase during work.	5·61	560010·90

IN WEIGHING ROOM.			
RECEIVED.	RETURNED.		
		As by Weighing Room Beam.	As by Mint Office Beam.
Blanks for Florins. Ounces. As by Cutting Room Beam. 433013·65 As by Weighing Room Beam. 433010·55 Showing a minus difference. 3·10	Florin Blanks. " Rejected. Deduct "received." Loss during work.	Ounces. 423988·43 9020·63 433009·06 1·49	Ounces. 9020·63
Blanks for Shillings Ounces. As by Cutting Room Beam. 320518·84 As by Weighing Room Beam. 20516·07 Showing a minus difference. 2·77	Shilling Blanks.s Shilling Rejected Deduct "received." Loss during work.	315492·95 5021·79 320514·74 1·33	5021·79
Blanks for Sixpences. Ounces. As by Cutting Room Beam. 101407·09 As by Weighing Room Beam. 101406·65 Showing a minus difference. 0·44	Sixpenny Blanks. Sixpenny Rejected. Deduct "received." Increase during work.	99083·98 2223·63 101407·61 0·96	2223·63
Blanks for Threepences. Ounces. As by Cutting Room Beam. 19373·13 As by Weighing Room Beam 19372·96 Showing a minus difference. 0·17	Threepenny Blanks. Threepenny Rejected. Deduct "Received." Loss during work.	19326·87 46·03 319372·90 0·06	46·03 16312·08

IN ANNEALING ROOM.			
RECEIVED.	RETURNED.		
		As by Annealing Room Beam.	As by Mint Office Beam.
Blanks for Florins. Ounces. As by Weighing Room Beam. 423988·43	Florin Blanks	Ounces. 423606·37	Ounces.

IN ANNEALING ROOM.			
RECEIVED.	RETURNED.		
		As by Annealing Room Beam.	As by Mint Office Beam.
As by Annealing Room Beam.423984·25	Loss of Weight by Annealing and Blanching.	367·84	Supply. [123]
Showing a minus difference. 4·18	Deduct “received.”	423974·21	
Blanks for Shillings	Loss	10·04	
Ounces.			
As by Weighing Room Beam. 315492·95	Shilling Blanks	315190·47	
As by Annealing Room Beam.315490·42	Loss of Weight by Annealing and Blanching.	283·86	
Showing a minus difference. 2·53	Deduct “received.”	315474·33	
Blanks for Sixpences.	Loss	16·09	
Ounces.			
As by Weighing Room Beam. 99083·98	Sixpenny Blanks	98980·28	
As by Annealing Room Beam. 99083·67	Loss of Weight by Annealing and Blanching.	103·39	
Showing a minus difference. 0·31	Deduct “received.”	99083·67	
Blanks for Threepences.	Loss	0·00	
Ounces.			
As by Weighing Room Beam. 19326·87	Threepenny Blanks	19304·08	
As by Annealing Room Beam 19326·72	Loss of Weight by Annealing and Blanching.	22·64	47·67
Showing a minus difference. 0·15	Deduct “Received.”	19326·72	47·67
	Loss	0·00	

IN PRESS ROOM.			
RECEIVED.	RETURNED.		
		As by Press Room Beam.	As by Mint Office Beam.
Blanks for Florins.			
Ounces.		Ounces.	Ounces.
As by Annealing Room Beam.423606·37	Florin Coin	420367·28	420278·40
As by Press Room Beam. 423606·37	Florin Brockages.	2951·76	3069·61
Showing no difference. 0·00	Florin Pyx Pieces.	283·86	283·86
	Deduct “received.”	423602·90	

IN PRESS ROOM.			
RECEIVED.	RETURNED.		
		As by Press Room Beam.	As by Mint Office Beam.
Blanks for Shillings Ounces. As by Annealing Room Beam. 315190·47 As by Press Room Beam. 315190·47 Showing no difference. 0·00	Loss Shilling Coin Shilling Brockages. Shilling Pyx Pieces. Deduct “received.” Loss	3·47 313853·64 1227·88 107·87 315189·39 1·08	 313847·00 1228·17 107·87
Blanks for Sixpences. Ounces. As by Annealing Room Beam. 98980·28 As by Press Room Beam. 98980·28 Showing no difference. 0·54	Sixpenny Coin Sixpenny Brockages Sixpenny Pyx Pieces. Deduct “received.” Loss	98279·81 678·84 21·18 98979·83 0·45	98271·65 678·84 21·18
Blanks for Threepences. Ounces. As by Annealing Room Beam. 19304·08 As by Press Room Beam 19304·08 Showing no difference. 0·00	Threepenny Coin. Threepenny Brockages. Threepenny Pyx Pieces. Deduct “Received.” Loss	19145·45 154·85 3·02 19303·32 0·76	19159·20 154·85 3·02 857103·65

	Ounces.
Total received from Mint Office	1499104·00
” returned to ” ”	1498503·73
Loss	<u>600·27</u>

The proportional loss on this coinage was 0·199 ounces on each 500 ounces coined, instead of 1 in 500 as the Master had expected must be the case.

The mean rate of					
The Ends was	3·94	per cent.	on the	Rough Bars.	
” Scissel was	37·85	”	”	”	
” Rejected Blanks was	0·89	”	”	”	
” Brockages was	0·38	”	”	”	
” Coined Money was	56·87	”	”	”	
” Coined Money was	59·22	”	”	Clean Bars	

WAGES PAID TO WORKMEN BY PIECE-WORK SCALE.

	Florins.	Shillings.	Sixpences.	Threepences.
For Melting	13·215	9·176	3·079	0·584

	Florins.	Shillings.	Sixpences.	Threepences.
" Rolling	45·521	42·045	28·111	8·165
" Cutting	44·165	46·008	28·899	8·052
" Annealing	35·300	45·965	28·869	11·260
" Coining	29·192	39·231	24·570	9·572
Total wages paid	£167·393	£182·425	£113·528	£37·633
Rate paid for each 100 lbs. troy of Coined Money.	£0·475	£0·697	£1·386	£2·358

	Rate per cent. of Coined Money from Rough Bars.	Rate per cent. of Coined Money from Clean Bars.
Florins	55·26	57·76
Shillings	59·38	62·22
Sixpences	55·96	58·29
To the MASTER OF THE MINT. GEORGE F. ANSELL. May 6th 1858.		

It would appear necessary that I should now give^[124] a copy of one of the reports I made to the Master, showing for silver a similarly detailed history of its manipulation to that which was given for gold ([see page 92](#)); and as that was fully explained, it is felt unnecessary to say more of this report than to point out that the blanks are simply stamped in the press room, and that as this operation produces no alteration of weight, the losses recorded were due to abstractions of coined money, and were reported to the Master as such, while the loss exhibited in the annealing room was also due to the same cause. The loss which appears to have arisen in the weighing room arises from another cause. The blanks when operated upon lose little splinters of metal, which are preserved, and at the end of the coinage go to a fund of odd pieces, which are melted; and the ingot produced is assayed, its estimated amount of silver is determined, and this is called *supply*, and will be seen to be 47·67 ounces for the whole department, although it appears in the annealing room account, because it is usually cleaned in that room.

As regards the accounts for the rolling and cutting rooms, it would be a source of interest to the curious could they but examine the books and compare them for periods previously to and after 1857, when it would be demonstrated that nearly the whole of the losses above explained took place in those rooms. I stopped these *evaporations*, and removed the word *loss* which was habitually printed at the foot of each day's account, replacing it by the word GAIN, because that significant word expressed what should have happened, and that which under me did really take place in these rooms.

It is remarkable, but even in so simple a matter as the bronze coinage there also arises a large loss—that is, a great part of the profit is lost. I therefore propose to say a few words upon this subject, and see if reason will confirm the facts of experience. The average value of copper is about £100 a ton, and this may be assumed as the average price of bronze in ingots. It is perfectly well understood that bronze and copper coins are both accepted as token money; but some amongst us may remember when copper pennies were so coined as that each penny was intrinsically worth that sum, the intention being that any customer could check the honesty of the shopkeeper when he bought an ounce of tea or tobacco, by placing his “cart-wheel-penny” against

the commodity still in the scale. Experience, however, demonstrated that this was not a right footing for a subservient coinage. It was found that when copper in the market fluctuated to a price beyond 1*d.* per ounce, this coinage was collected and melted. Therefore it was determined to make the intrinsic far below the extrinsic value, and then to limit the legal tender to 2*s.* in nominal value in copper money.

In 1860 the copper token money was replaced by bronze token money, and the weight of this was still farther reduced, so that the intrinsic value of a penny became as nearly as possible one farthing—that is to say, a farthing's worth of copper or bronze had given to it in the act of coining an extrinsic value equal to a penny, by which a profit of 300 per cent. would appear to be realised. But since some of the bronze is coined into pence, halfpence, and farthings, and these vary in their proportional intrinsic value, it is felt to be fair to give the Government every possible latitude, because Lord Kinnaird evidently accepted those facts when he made his speech in the House of Lords.

Bronze costs £100 a ton, and a ton of bronze is, on an average, converted into £358 when coined—that is to say, £100 intrinsic value receives by the act of coining an extrinsic value equal to £358, and so yields a profit of £258. Yet the Chancellor of the Exchequer last year asked for, and obtained, £1,000 *to defray the loss on bronze*, the fact being that on the value coined—£10,010—there should have been a clear profit of £25,825. And it so happens that, by the return No. 157, 1869, a profit is admitted to the modest extent of £6,318 0*s.* 2*d.* Why was the £1,000 asked for, and what became of the £6,318 0*s.* 2*d.*? and further, why was that modest sum allowed to take the place of the proper profit, or £25,825?

The return above quoted, No. 157, 1869, shows that in the nine years, 1860-1868, the purchase value of copper bought for coinage was £489,071, and the same return shows that this intrinsic value was converted into money of the nominal value of £991,833; therefore, by deducting the cost from that sum, there appears a clear profit of £502,762 on bronze since 1860, which is at the moderate rate of £102 per cent. on the outlay. It will be well to investigate farther, and perhaps to take the contract given to Messrs. James Watt and Co. for 1,720 tons in 1860. The sum paid to that firm, added to the cost of copper supplied to them for the manufacture of bronze to produce 1,720 tons of coin, was £229,200; and the 1,720 tons of bronze money forwarded to the Mint by the terms of that contract was, in nominal value, when delivered, £696,293, so that the profit on that one transaction was £467,093, or at the rate of £203 per cent. on the outlay. The last coin (a penny) to complete this contract was struck by Messrs. Watt at 9 A.M. on Thursday, the 11th of June, 1863; so that the coinage, which has been conducted since that period, has been struck under the superintendence of the present Mint officials. The return No. 157, 1869, shows that, at the period when the Mint was receiving bronze coin from the contractors, and was at the same time coining bronze—viz., from 1860-1863, the profit was diluted from 258 to 108 per cent.; but when the Mint took the whole manufacture from 1864 to 1868, under Mr. John Graham alone, that profit was further lessened to 78·70 per cent., or rather less than £78 15*s.* was made where £258 should have been made.

The Mint officials may, and probably will, plead that this return, as well as that relating to silver, is “false;” still they must stand or fall by their own figures, or issue intelligible and trustworthy returns to the orders of the House of Commons; and these exhibit the fact that Messrs. Watt's contract gave a profit of 203 per cent.; the Mint and Messrs. Watt's contract jointly of 108 per cent.; the Mint solely, of only 78¾ per cent. It will probably be asserted that the purchase of the old copper money caused the apparent loss; but this argument will be disposed of if we take the case of 1863, up to which period 1,651 tons 8 cwts. 2 qrs. 27¾ lbs.; of worn copper coin had

been purchased at an absolute cost of £234·80 per ton. To assume then that this was the cause for diminished profit, let us see if it can be proved to demonstration. I have shown that a ton of bronze money is worth £358, and a ton of copper, when bought as old coin, £234·80; therefore, the copper so bought should yield profit of about £152·50 per centum. But it should do more than this; for I, of my own knowledge,—because I ordered it,—state that bronze was bought to a considerable amount at the then current rate of new copper. The following, however, is the worst that could happen:—If the whole of the old copper cost £234·80 per ton, when converted into bronze money, it would produce £358, or equal to £152·5 per cent. profit. Upon what ground then can the profit—when new copper at the lower price is bought—be reduced so low as to 108 per cent., and then fall to £78·70 per cent. on the copper purchased? Were any prosecutions made, or were men simply told their “services were no longer required?” If the accounts were investigated by a professional accountant, what amount of loss in weight—in addition to the loss by dirt which amounted to 0·41 per cent. on each ton melted—would be shown when they were balanced? Another curious inquiry might be made:—At what rate for labour was bronze money coined in the Royal Mint after the cessation of the contract? It might be found to be exorbitant.

Since, however, Lord Kinnaird directed attention to this subject, Mr. Fremantle has adjusted the return No. 157, as regards bronze, in a manner similar to, yet differing from, that relating to silver. I propose to insert copies of these returns, following the arrangement adopted for those relating to silver; and it will be seen that whereas Mr. Graham showed a profit of £502,761 11s. 11d. between the years 1860-68, *without an instance of loss*, Mr. Fremantle not only alters all the figures relating to the copper purchased, but reduces Mr. Graham’s profit to £403,376 18s. 6d. and shows *a loss* in 1864 to the extent of £31,896 13s. 4d.; which sum, deducted from £435,273 11s. 10d., yields the profit Mr. Fremantle is willing to admit. I am fully aware that “figures may be made to tell anything;” but until the year 1870 I was not aware that Parliament would permit its returns to be altered, amended, or adjusted without leave or a word of explanation. Surely such an occurrence, added to evidence previously adduced, will cause some independent member to “tackle” even Mr. Lowe in the House of Commons. Why Mr. Fremantle, in the case of silver, converts loss into profit, and in that of bronze reduces, apparently without reason, the amount of profit, should be explained. I have done my share in calling attention to the facts. It should be pointed out that the figures relating to 1869 in the following return are inserted, but not demonstrated, because I wish to mark emphatically the fact that it is Mr. Graham’s figures alone which are made to perform these operations, while those of Mr. Fremantle yield a profit of £12,476 4s. 6d., or at the rate of £137·35 per centum on the copper purchased.

**NO. 157.—RETURN TO AN ORDER OF THE HONOURABLE THE HOUSE
OF COMMONS, DATED 21ST APRIL, 1869, FOR ACCOUNT “OF ALL
GOLD, SILVER, AND COPPER MONEYS OF THE REALM COINED AT
THE MINT, FOR EACH YEAR FROM THE 1ST DAY OF JANUARY,
1859, TO THE 31ST DAY OF DECEMBER, 1868, &C.,” IN
CONTINUATION OF NO. 340, 1857-68.**

(Mr. AYRTON, [page 4.](#))

Date	Total Value of Copper Coined.	Purchase Value of Copper.	Profit on Coinage.	Loss on Coinage.
1860	£ s. d.	£ s. d.	£ s. d.	£ s. d.
Copper	134 8 0			
Bronze	37,856 0 0	9,185 2 11	28,805 5 1	
Bronze:				

Date	Total Value of Copper Coined.	Purchase Value of Copper.	Profit on Coinage.	Loss on Coinage.
1861	273,578 13 4	96,735 6 8	176,843 6 8	
1862	352,800 0 0	203,389 0 0	149,411 0 0	
1863	151,648 0 0	82,482 0 0	69,166 0 0	
1864	18,069 6 8	9,660 0 0	8,409 6 8	
1865	57,493 6 8	31,740 0 0	25,753 6 8	
1866	50,624 0 0	26,910 0 0	23,714 0 0	
1867	33,301 6 8	18,960 0 0	14,341 6 8	
1868	16,328 0 2	10,010 0 0	6,318 0 2	
	991,833 1 6	489,071 9 7	502,761 11 11	

THOS. GRAHAM, MASTER OF THE MINT.

ROYAL MINT, 19th April, 1869.

NO. 262.—RETURN TO AN ORDER OF THE HONOURABLE THE HOUSE OF COMMONS, DATED 30TH MAY, 1870, IN CONTINUATION OF PARLIAMENTARY PAPER, NO. 157, OF SESSION 1868-9.

(Mr. STANSFELD, [page 4.](#))

LEGEND:

(A) = Total Value of Copper Coined.

(B) = Total Amount Paid for Copper Bullion and Old Copper Coin.

Date	(A)	(B)	Profit on Coinage.	Loss on Coinage.
1860	£ s. d.	£ s. d.	£ s. d.	£ s. d.
Copper	134 8 0	16,317 10 0	21,672 18 0	
Bronze	37,856 0 0			
Bronze:				
1861	273,578 13 4	117,963 10 0	155,615 3 4	
1862	352,800 0 0	230,784 3 0	122,015 17 0	
1863	151,648 0 0	136,315 0 0	15,333 0 0	
1864	18,069 6 8	49,966 0 0		31,896 13 4
1865	57,493 6 8	16,518 0 0	40,975 6 8	
1866	50,624 0 0	9,273 0 0	41,351 0 0	
1867	33,301 6 8	6,260 0 0	27,041 6 8	
1868	16,328 0 2	5,059 0 0	11,269 0 2	
1869	20,832 0 0	8,355 15 6		
	1,012,665 1 6	596,811 18 6	435,273 11 10	31,896 13 4

C. W. FREMANTLE, DEPUTY MASTER OF THE MINT.

ROYAL MINT, 31st May, 1870.

The foregoing is demonstrative of the processes of coining as conducted in the Royal Mint, and of the system which prevails in that Department; I feel it right, however, to add an epitome of the trial of the pyx, a custom established by the Mint Indenture of 18th Edward III.

I will describe the trial which took place on the 17th of July, 1861, when the Lord Chancellor Westbury presided in the absence of the Queen, whom he represented as though she were

personally present. It should have commenced at nine o'clock in the morning, but some delay was caused by the inability to open the pyx chamber in Westminster Abbey, owing to the rusting of the locks. The Lord Chancellor was supported by the following members of the Most Honourable the Privy Council, Lord Granville, the Duke of Argyll, Sir Edward Ryan, and the Right Hon. Robert Lowe; Lord Monteagle attending officially with the trial plates in sealed packages. The jury of twelve men selected by the Goldsmiths' Company was then sworn in due form. Lord Monteagle next declared that the trial plates were standard according to law, and that they were in the same state, and that the seals were in the same state, as when deposited in the pyx chamber. The Lord Chancellor then charged the jury, telling them of the prerogative of the Crown, and that the Sovereign has a right to issue money bearing her image and superscription, and that the people take such money in commerce because it is guaranteed of certain purity, that this guarantee also insures the receiving of the money by foreign nations; and that these facts made the attesting of its purity a vital point, that all people might be assured by their verdict that the money issued is in accordance with the law, both as to weight and fineness. The Lord Chancellor then explained the laws relating to the coinage, and that the Master of the Mint, who was present, would, if their verdict were favourable, be discharged from further liabilities as to the coin he had made up to the 31st of December, 1860. After this the Lord Chancellor dwelt upon the antiquity of the trial of the pyx very briefly, and then explained that the word *pyx* is of Greek origin, and means "box"—the box in which is placed the money to be tested by the jury. He then explained the law as to the custody of the pyx and of its keys, and passed on to the latitude allowed for errors as to the standarding of the alloy, and for the remedy by weight for each pound of manufactured coin;^[125] stating that there had been one adverse verdict found by a jury at the trial of the pyx, but that the present verdict would doubtless be favourable to the Master of the Mint, who would then be discharged from further liabilities by Royal letters-patent. He then adjourned the jury to the Goldsmiths' Hall to make the necessary tests and trials of the coin, desiring that they should afterwards deliver their verdict to him, either at the House of Lords at six o'clock, or at his residence at eight o'clock. Lord Monteagle then handed the trial plates to the foreman of the jury. The trial plates are wide ribbons of gold and of silver, about as thick as a florin, and indented on the edges. The face of each plate is stamped with the obverse impression of a sovereign die; and the date of its manufacture, with the proportion of its alloy, is engraved on each plate.

The jury, having arrived at the Goldsmiths' Hall, proceeded to count and weigh the gold and silver moneys of Sir John Herschel, and of Thomas Graham, Esq.; they then selected a fixed number from the coins of each Mastership for trial by assay, following a process similar to that described at [pages 10-14](#). The jury also tested the coined money by weight, but did not test any individual coin (see new law, [pages 70-73](#)) as to its weight, because the old law only required the coined money to be within the limits of weight allowed, supposing a fixed number of any coins to be weighed against a standard pound. The limit, or remedy, was 12 grains on each pound weight troy of gold coin, or about 0·2568218 grain on each sovereign, and 24·00 grains on each pound weight troy of silver coin. After the assay the jury agreed upon a verdict, which was to the effect that the moneys of both Masterships were found by them to be within the remedy, both as regards weight and fineness. This verdict was delivered at eight o'clock P.M., at the Lord Chancellor's private residence. The Master of the Mint was present, with his Deputy Master and other officers, during the trial. The total amount in the pyx at this trial was £36,417 10s. in gold moneys, and £807 4s. 3d. in silver moneys, and from these the jury selected a total of 60·616 ounces of gold and 71·018 ounces of silver for the assays.

In the same manner the trial of the pyx was held at Westminster on the 19th January, 1866, the only difference consisting in the fact that at that time £34,927,008 8s. 0¼d. in coined gold, and

£1,556,100 11s. 10d. in coined silver were submitted to this test.

In an earlier page^[126] I have expressed my views as to the worthlessness of the farce called the *Trial of the Pyx*. While I studiously avoid altering that expression of opinion, I am sorry that the *last* Trial of the Pyx under the ancient law took place on Tuesday, the 15th day of February, 1870. I could have felt less regret at the removal of this ancient custom from one law to another, had there been provision for its better execution; but simply to do away with the sacred charm of antiquity, and to enact a troublesome deception, is worse than a blunder; it is, in my antiquarian view, a crime.

The *Trial of the Pyx*—now, without meaning, as well as a farce—is to be held, under the Act 33 Victoria, chap. 10, at least once in every year. Why not daily throughout the year—that its absurdity may be demonstrated more markedly? Had it been as it should have been, a real trial of the Pyx, instead of an examination with a foregone conclusion of certain coins; “at least once in every year,” would have been not only justifiable but right.

CONCLUDING REMARKS.

Throughout the foregoing pages I have endeavoured to point out the causes which lead to mismanagement in the Royal Mint, and I should be glad if I could see a reason to hope that this state of things was likely to terminate. However, there seems to exist an obstinate determination to maintain the present arrangements. If proof of this be required, it exists in the re-issue of Pistrucci's device on the sovereign of 1871, which meets with deserved condemnation from all classes of opinion. It is a wrong to Pistrucci's memory to append his initials to a coin so mutilated. While, however, Mr. Fremantle thinks it economy to continue—*vide* "European Mints"—the present system of the die department of the Royal Mint, the nation must bear the disgrace inherent to such determination. There is, however, reason to believe that this subject will be forced upon the Master of the Mint, for it is now attracting considerable attention. That St. George, seated on his horse, should be placed in a concavity, is perhaps intended to symbolise the necessity for washing from him the dirt in which he has of late been bathed; but that thus sinking the work on a coin is wise may be questioned, for our coins should represent medals, not saucers of tarnished copper. Instead, however, of altering the effigy of her Majesty, they at the Mint have made an unwise alteration of the eye alone, and which gives a very unpleasant expression. The necessity for the issue of this new coin arose from the fact of the alteration from base to standard coin ([see pages 54-56](#)), and its design is really a Hall mark to point out to the Mint authorities the real value of the coin. When purchased for melting, five hundred of these coins will be worth £500, but an equal number of coins of the previous issue is worth only £499; so that by the alteration the Bank of England now loses £2,000 which it before enjoyed from this source, quite independently of the profit demonstrated at [pages 126-127](#).

I have insisted on the necessity for a different system as applied to the chief engravership of the Royal Mint, and, consequently, for a new effigy of her Majesty on the coins of the realm. I had hoped that when the George and Dragon was issued, a new device would also appear, and that this did not take place is a subject of universal regret. There appears to be a want of

knowledge as to the cause for retaining the present obverse; I therefore mention it. The objection to change arose with his late Royal Highness the Prince Consort, who resisted any alteration of the design in the die of the face of her whom he loved so well. This fact probably prevents the issue of a coin from the new pattern die which is prepared in the Mint.

The want of knowledge exhibited in the Royal Mint is further demonstrated by the recommendations of Mr. Napier, who has not been at the pains to understand the cutting-out presses which he condemns. I concur, however, in that condemnation; for these machines should be small, powerful, and diaphanous, if I may so interpret that word; not heavy and light-obstructing as at present. Such a machine can readily replace the present machinery, which, with one coining press and its complete accompaniments, will find a fitting home in the South Kensington Museum. Had we in these days the inventive minds of past generations, we might hope to see the principle of the screw press of Mr. Boulton so modified and applied to the other presses in the Mint, that they would require but little more space than that occupied by themselves alone; indeed, such an instrument seems but a natural result of the existence of [Mr. Boulton's press](#), and coins, intended to represent medals, should be formed by a blow, not by gradual pressure. That the Chancellor of the Exchequer for the time being should be Master of the Mint is perhaps the strongest proof of the ignorance on the subject of the Parliament which passed the Act that can be produced. It keeps the Deputy-Master of the Mint, by his own admission, continually journeying between the Mint and the Treasury Chambers to instruct the Master, instead of, as he should be, attending to the arrangements of the Mint. Parliament, I hope, will yet see that the master of a working Department should be a practical, not merely a financial or political man.

When the new Coinage Act was passing through Parliament, a clause was introduced which has enabled any person to send gold to the Mint for coinage free from expense, except loss of interest. Colonel Tomline, M.P., has been the first to avail himself of this right, and I happened to be with that gentleman on the 4th February, 1871, when he received from the Mint £100 in coined gold, of which he gave me a specimen. This is the first time that a private person has attempted such an invasion of the prescriptive rights of the Bank of England, and by it he has made a clear gain of $1\frac{3}{4}d$.

(although in this instance he lost profit by assay and trade practice), for he sent an ingot which weighed 25·681 ounces—seller's weight at £3 17s. 9d. an ounce—and received back from the Mint 25·684 ounces of sovereigns, or in excess weight 0·003 ounce. The time occupied was twenty days. The principle thus established is a great one, and it is to be hoped that Colonel Tomline will carry his point against the Master of the Mint as regards a free coinage of silver; for while the Government claims the exclusive right to coin silver, its workshop—the Mint—refuses to issue coined silver to the public. That department sells threepenny pieces because the Bank of England will not take the trouble; but all other silver coin goes to that institution, to be issued by it to bankers alone in bulk.

Colonel Tomline will perhaps consider the wisdom of re-establishing an Exchange where those who have an excess of bronze or silver can exchange it for gold, and those who wish it can obtain the subservient coinages.

There are strong reasons why the Royal Mint, instead of the Bank of England, should purchase gold bullion and pay for it in Mint notes, but, leaving the full consideration of this part of the subject for a future opportunity, it seems that there can be little doubt of the fairness of my proposition to charge one pound for the coining of each thousand sovereigns, this sum being quite sufficient even under the present system to protect the Mint from loss, while it is certain that it would, under efficient management, yield a profit. If, then, beyond this degree of protection against pecuniary loss by the gold coinage the average amount of silver and bronze were coined, the Royal Mint should be—and under proper control would be—not only a self-supporting Department, but one of actual profit, just as the Post Office has become, instead of, as at present, a vast abyss into which a great amount of money is thrown, never again to appear. It must not be conceived that my wish or intention is to do violence to those who hold office; my desire has been to point out the irregularities which prevail, and at the same time, from experience gained under exceptional opportunities, and by perseverance under adverse circumstances, to develop the actual facts, so that when these are known the irregularities, to say the least of it, may be remedied.

I cannot assert that my descriptions explain the operations at the Mint as now conducted; but this I do feel, that if such suggestions as I have made for improvements in the operative department were alone followed in their

entirety, the result would be a clear saving of at least £4,000 a year; while, if the other propositions which I have endeavoured to demonstrate are carried into effect, it seems perfectly clear that the Chancellor of the Exchequer need not apply year after year to include in the estimates the large amount required to meet the losses of the Mint.

It is due to myself to state that the manuscript for my late edition was in the hands of the Publisher at the time of Mr. Graham's death, and on the occurrence of that event I, at great expense, altered many passages which I had written, with an especial view to Mr. Graham's replying to them.

I wish to state that my object is to point out the system and its results; and in case the feelings of any gentlemen should be hurt, I must apologise to them, pleading the necessity, when a discussion on the Mint is pending, that all who are acquainted with the subject should exhibit facts relating to it, that when the whole matter is fairly considered a true judgment should be arrived at; for, as Archbishop Whateley has truly said, "The deadliest of all falsehoods is the lie of suppression."

APPENDIX.

TO THE RIGHT HONOURABLE LORD KINNAIRD, K.T.

27, BERNARD STREET, RUSSELL SQUARE, W. C.

February 8th, 1871.

MY LORD,

I shall feel obliged if I may thank you for your Lordship's letter of the 7th inst., in which you request from me—in consequence of the remarks made on my character by Mr. Lowe in the House of Commons, and by the Duke of Argyll and Lord Lansdowne in the House of Lords—a history of my connection with the Mint.

In compliance with your Lordship's request, I proceed to put in writing an account of my association with and severance from the Royal Mint. That I may occupy as little of your Lordship's valuable time as possible, I have selected certain passages from the Treasury correspondence which relate to myself. I am not aware that I have at all unduly interpreted those letters, or that, by omitting parts, I have made them appear more favourable to myself than they were intended by their authors to be. If, however, the Government should so view these extracts, may I beg of your Lordship to move for the complete correspondence, the publication of which would, I believe, more fully demonstrate my "case." I possess copies of that correspondence, but extracts from it will convey all that concerns myself; to this there is one exception, and I beg to submit a considerable portion of one letter from the Master of the Mint to the Treasury, because its contents fully bear out my statement that great irregularities had existed and were checked. Indeed, I now invite your Lordship's attention to a passage in that letter because the Mint is at present conducted at extravagant expense, while its losses of bullion reach far towards the former average. Mr. Graham says, "The diminished waste of late years (1857-60) is further attended with increased economy in working, of which, indeed, a low waste return is one of the surest indications."

The correspondence which your Lordship has so generously placed at my disposal, I also include that my account may be complete, although necessarily brief. I should also state that, by the rules of the Civil Service, an extra clerk cannot approach the head of his Department without permission from his superior officer. Yet your Lordship will observe that the Master of the Mint not only received my reports, but addressed orders to, and treated me as, the chief on whom he relied, as is proved by his writings now copied as well as by those in my possession.

In the early part of the year 1856 the Master of the Mint found himself so beset with difficulties arising from irregularities committed by those who should have supported him, that he formed a determination to engage a person on whom he could rely in the Coining Department, and who at the same time had sufficient personal influence and knowledge of the work to check those irregularities which he knew were taking place, but which were beyond his personal supervision. These irregularities were known to the Government of that period, and had caused so vast an expense that that Government, which was presided over by Lord Palmerston, had intimated to the Master, that unless the Mint could be conducted more satisfactorily and economically, it would be broken up as an imperial establishment, and thus necessitate the placing of the coinage in the hands of contractors. With this view papers were printed and issued to various firms; but Mr. Graham, being very sincere and energetic in his desire to so conduct the Mint as that this necessity should not arise, sought the advice of engineers and others, with a view to finding such a man as combined within himself the qualities which he saw to be necessary. Finally, by the advice of Dr. A. W. Hofmann, he called upon me at my residence in October, 1856, and related to me in detail the facts above alluded to, and explained precisely the position of affairs at the Mint, giving me the names of the troublesome persons, and showing his own position to be so intolerable, that without some one on whom he could rely, it would be impossible for him to continue his Mastership. Mr. Graham then proceeded to tell me what had passed between Dr. Hofmann and himself as to my fitness for the work required, and explained clearly the impossibility of placing me in a secure position or in high office at first, but that if I would accept a supernumerary clerkship temporarily, he would, when I had effected the object of his desires, advise my promotion to the office then held by Mr. W. T. Brande, when that office should become vacant. With this understanding I agreed to accept the position proposed if

duly appointed; but that I might be able to carry out such reforms as were necessary, I insisted on sufficient power being given to me by himself or by the Government. In pursuance of this agreement, Mr. Graham recommended my appointment in a letter to the Treasury, dated 29th October, 1856, in the following terms:—

“I desired to introduce a young man, recommended by scientific and technical information available in coining, by energy of character, and by tried ability in the supervision of workmen—a faculty by no means common. After applying to Mr. William Fairbairn and to Mr. George Rennie, who both recommended candidates, and making inquiry in various other quarters, I have been led to propose the name of Mr. George Frederick Ansell as temporary clerk. Mr. Ansell was educated under Dr. Hofmann, and acted for some years as his laboratory assistant. He has since been Scientific Director in the Panopticon, Leicester Square, which was lately broken up. He appears to be a person of superior education and ability, and great activity and vigour of character, and, as I have been assured, has shown much discretion in the management of both pupils and workmen.”

Mr. Graham announced to me my appointment in the following note:—

“Royal Mint, November 6th, 1856.

“MY DEAR SIR,—

“I am happy to inform you that your nomination to office in the Mint has been approved by the Treasury.

“Would you favour me with a call to-day before three o’clock, if quite convenient; or, if not, to-morrow morning after eleven.

“Ever truly yours,

“GEORGE FREDERICK ANSELL, ESQ.”

“THOMAS GRAHAM.

I was thus appointed to a supernumerary clerkship, and took office in the Rolling Room of the Royal Mint; but before doing so I particularly inquired of Mr. Graham, in the presence of Mr. W. H. Barton, the then Deputy-Master, what authority I should have, and how far I should be supported if I attempted to enforce obedience, and whether I was at liberty to call for statements and examine original documents and books. He verbally authorised me to examine such books as I wished to examine, and desired Mr. Barton to give me such statements or information as I might ask for; and

as to my authority, he said, "If you order the men to dance a hornpipe on the table, they shall do it, and all orders shall pass through your hands." With such power, and a salary of £120 a year, or rather £10 a month, I took office on the 12th November, 1856.

An examination of the books in the Royal Mint will prove that I quickly corrected the irregularities in the Rolling Room, and increased the amount of finished work by 100 per cent. Loss of bullion became unknown in that room.

In the Adjusting and Cutting Room, however, the losses and rejected work continued to be astounding, the loss averaging seven ounces of gold per diem, and for days together running so high as twelve ounces. The Master had many times urged me to relinquish the direction of the Rolling Room, and to take charge of the Cutting Room. I, however, maintained that it was wiser "to begin at the beginning and clear up as I went." To this he reluctantly assented, but on the 25th of February, 1857, the officer who had charge of that department, as well as of the Weighing Room, exhibited so decided an opposition to the written order of the Master, that it became necessary for me to yield the point. On that day I changed duties with the officer alluded to. The Master wished me also to superintend and assist the officer so removed to the Rolling Room, and I made a note to that effect in the official book of that department, which doubtless still remains.

It had been the custom with officers in the Mint to consider it derogatory to understand practically the machines in use, and the officer whom I replaced did not know for years afterwards whether the cylinders used in the Drag Bench were fixed or movable, and described them as "going round like the rollers!!!" I had, however, a different opinion; and, notwithstanding the scorn and contempt of the "gentlemen" of the Mint, made myself minutely acquainted with all details, so that I could perform the manual labour of each workman; this had been observed, and when I took charge of the Adjusting and Cutting Rooms the foreman of those rooms barricaded himself in his recess with a determination to prevent me seeing his operations. I stepped from one copper trough to another, and thus placed myself at his side, greatly to his astonishment. His remarks were significant, "You mean to learn the duties?" "I do." To which he replied, "Man proposes, God disposes." I directed the removal of the barricades, and invited the foreman

into my private room, where a little conversation enabled him to understand his new officer.

I then delegated to Mr. Richard Pilcher the charge of the Weighing Room, and gave him the key of the stronghold in that room: while I remained in the Mint he retained that position, and still holds it. He served me faithfully and fearlessly.

Subsequently I weighed to the foreman of the Cutting and Adjusting Room, in the presence of himself and the workmen, a given weight of gold, which I watched in its progress, and weighed at each stop. As was to be expected, there was a final increase of weight: many explanations were asked, and given; but I firmly demanded all the gold so delivered, each day, and thus replaced *habitual loss* by habitual gain.

Without passing into too minute detail, it is sufficient to state, that while some men were removed, others were introduced. That these changes were made by me I can conclusively prove by the Master's written orders and letters to that effect. But the following letter to the Treasury seems to tell an indisputable tale.

EXTRACTS FROM LETTER OF MASTER OF THE MINT
TO THE LORDS OF THE TREASURY.

Royal Mint, August 6, 1857.

"The late foreman, who was dismissed.... The dismissal of J—— B——, late foreman in the Adjusting and Cutting Rooms, arose out of an investigation respecting the nature and extent of the waste of bullion in the operations of coining, which is a subject demanding the most watchful attention in the direction of a Mint. This loss is understood to have been represented by the moneyers as averaging 7 parts in 10,000 parts of gold coined, or £700 in one million sterling coined; and it has continued undiminished under the new arrangements. Such an amount of loss of metal appeared to me excessive, and certainly is so when compared with the corresponding loss in the Mint of Paris, which I am informed on good authority lies between 3 and 5 parts only in 10,000 parts of metal, and this with much hurry of execution, and with inferior machinery, and other circumstances against the Paris Mint.

"The appointment of the new officers, Mr. —— and Mr. Ansell, in the Coining Department gave me means of laying out the inquiry with effect. It is a subject of considerable difficulty from the circumstances that the losses to be detected, as they occur, are minute in themselves, although their cumulative effect is great; and such losses are often masked and concealed by legitimate change of weight in the bullion, due to oxidation of the alloy, adhering oil, &c. Without entering into details I may state that I have reasons

to believe the inquiry will result in a large reduction of our waste, and great profit to the Mint, which I trust will appear in the returns of the next gold coinage. A portion of the loss is traced to culpable proceedings on the part of the dismissed foreman, who, as tryer of fillets, appears to have occasioned much mischief by producing bad work (blanks out of remedy) seemingly with a purpose.

“By the vigorous action of two of these officers, Mr. — and Mr. Ansell, great economy has been enforced throughout the Coining Department. Mr. Ansell, since his appointment in November last, has been acting on the salary of a temporary clerk, £10 a month, with the addition of £2 10s. per month when engaged from 8 A.M. till 6 P.M. I now beg strongly to recommend that Mr. Ansell’s monthly salary shall be increased to £12 10s., in consideration of his valuable services, with the addition of £2 10s. per month when engaged from 8 A.M. to 6 P.M. as heretofore.

(The master then goes on to request my Lords to do this for Mr. Ansell.)

“THO. GRAHAM.

“*August 6, 1857* (No. 1114).”

In this letter the Master justly attributes the stoppage of the losses to the dismissal of J—— B——, at my recommendation, and in consequence of my investigation, as is proved by my Report to the Master, and by his letter to the Treasury, copies of which I can furnish; but that this was the case is demonstrated by the following order:—

“*Royal Mint, 10th July, 1857.*

“Ordered that J—— B—— be discharged from the service of the Mint, in consequence of frequent neglect and inaccuracy as Tryer in the Adjusting and Cutting Rooms of the Coining Department, in accordance with the Report of Mr. Ansell, of July 8th. Confirmed by Mr. Buckle.

(Signed)

“THO. GRAHAM.”

(EXTRACT.) LETTER FROM THE TREASURY TO THE MASTER OF THE MINT.

“*25th August, 1857.*

“My Lords sanction proposed arrangements of the 6th, but desire to receive a Report of the circumstances connected with the dismissal of J—— B——, in consequence of the investigation respecting waste of bullion.

(Signed)

“W. H. STEPHENSON.

“*Mint* (No. 1163), *27th August, 1857.*”

I, in consequence of this letter from the Treasury, went to Somerset House and to Doctors' Commons, and obtained many particulars relating to J—— B——; and the Master wrote a Report to the Treasury, of which I have, and can produce, a copy: it is dated 15th June, 1858.

(EXTRACT.) LETTER FROM THE LORDS OF THE TREASURY TO THE MASTER OF THE MINT.

"Treasury Chambers, 3rd September, 1858.

"SIR,

"In reply to your letter, dated 17th ultimo, recommending the grant of increased salaries to Messrs. —— and Ansell, I am commanded by the Lords Commissioners of Her Majesty's Treasury to state, that although they readily admit Messrs. —— and Ansell's claims upon their favourable consideration, they cannot authorise the full amount recommended for them: their Lordships are pleased to fix Mr. —— and Mr. Ansell's salary at two hundred and twenty pounds a year, including the allowance hitherto received by him for additional attendance—both these augmentations to take effect from the commencement of the present year.

(Signed)

"C. E. TREVELYAN.

"Treasury (No. 14409), 3rd September.

"THE MASTER OF THE MINT."

Notwithstanding these facts the Master of the Mint, towards the end of 1859, showed by his actions that he meant to introduce his bankrupt brother, to whose character your Lordship alluded in the House of Lords; and whose ignorance of the duties, coupled with great inequality of temper and intemperate habits, rendered him totally unfit for the office. Such being the unmistakable intention of Mr. Graham, who disregarded the advice tendered to him by influential persons, I addressed to him the letter which follows:—

TO THE MASTER OF THE MINT FROM GEORGE F. ANSELL, ESQ.

"Royal Mint, January 6th, 1860.

"SIR,

"I trust you will accord to me the regret with which I bring to your notice affairs relating to myself. Circumstances have placed me in a position which is very anomalous, while my salary is such that I find it quite impossible to

manage my payments so as to keep free from debt, and I now most unwillingly approach you upon the subject, hoping that you will permit me to place in your hands the grounds on which I rest my hope, that my application will receive your favourable and early consideration. On the 12th of November, 1856, I entered the service of the Mint as a Supernumerary Clerk, and subsequently passed the examination which was enforced by the Civil Service Commissioners, as that process was thought by you to be to my ultimate advantage.

“In my first interview with you I expressed surprise at the large losses which you said were occurring, notwithstanding the weight of oil which was added to the precious metals in the processes of coining. My conviction of the impossibility of this loss arising if proper supervision were exerted was so great that I at once stated to you, in the presence of Mr. Barton, that there should be no loss at all upon either gold or silver; but that if oil were used there should be an increase of weight.

“You, sir, gave me such authority in the Coining Department as would enable me to establish my proposition, which you then said was contrary to the facts as found in the Royal Mint, as well as in the Paris Mint.

“By your orders I commenced my duties in the Rolling Room, and was soon satisfied that the irregularities which existed would quite explain the losses which had taken place. I reported to you faithfully from day to day all irregularities, but upon none did I dwell so earnestly as upon the errors in, and differences of, weighing, attended by circumstances of which I feel it unnecessary to remind you, unless I say that they were reduced and overcome by no ordinary perseverance and at great personal risk.

“While the balances were in error it was impossible to determine how great the losses were, for they were stated to be more differences of weighing than anything else; but I felt it incumbent on me to experiment carefully on the gold which we were then coining, that I might thus satisfy myself and you of the truths of my proposition. The average of five preceding years gave a loss of 597·55 ounces on each million ounces coined; and the result of my first years experimenting was an average loss of 31·06 ounces on the million ounces coined. While the second year gave an absolute increase of weight of 9·22 ounces on the million ounces coined.

“Your mind will picture the interests which I disturbed, and give me credit for the moral courage which became necessary to support the violent enmity which my course drew down on me, for you are aware of the vigour with which I was met by adverse opinions from Messrs. —, —, —, and —; these gentlemen, as well as the Junior Clerks, who were supposed to be my colleagues in the Coining Department, openly maintaining that there were, and had been, no unnecessary losses. Threats of personal violence did not deter me from my duty; but the men being well plied with arguments by the Messrs. —, —, who were supported by Mr. —, rendered my course extremely difficult.

“Mr. — accused me of adding more oil than he used to permit, and so explained the—as he called it—“apparent stoppage of the losses;” but the fact of my using less oil than had ever been known was so notorious that he subsequently admitted it; nevertheless, the accusation induced me to

experiment, and these experiments resulted in the total abolition of the use of oil in the Rolling Room, and on the Cutters; while the minute amount which was necessary at the Draw-Bench was carefully wiped off before the blanks were cut out.

“Now that the use of oil is abolished gain is impossible, but the loss is invariably covered by the amount for which the sweep is sold, so that my proposition is now more satisfactorily proved than ever. Contemporaneously with these facts I had to encounter equally serious difficulties in the subject of rejected blanks, which reached 70 per cent. on the whole number of blanks cut; but the common average was from 30 to 35 per cent. By rigid experiments, conducted by my own hands, I reduced these abominations to 1·00 per cent. on silver, and to 4·00 per cent. on gold, without the aid of Mr. Pilcher’s excellent filing machine.

“In the year 1857-58, I coined by my own arrangements—although my plans were strongly combated at the time—the largest amount of gold coin ever yet produced in a given time, conducting the coinage without loss, and with a total of 4·50 per cent. of rejected blanks. A large quantity of this gold contained osmium-iridium, but I coined it without its being first refined.

“In 1859 I coined about £250,000 (64,790·224 ounces) of a peculiarly brittle gold, which was delivered to the Bank of England in June and July, 1859. I believe this to be the first time that brittle gold has been coined so as to produce perfectly good and tough coins.

“Permit me to remind you that all the elements of wrong still remain in the Mint, and would be re-exerted in the absence of your Mastership; for not one of those officers who permitted these things has been removed.

“As a consequence of the abolition of the use of oil in the Coining Department, the losses in the Melting House are greatly reduced.

“In addition to my works in the Coining Department, I beg to state that I have, by my own hands, reduced the losses in the Melting-House (this was done during the time of using oil in the Coining Department) 33 per cent.; and my figures were not only admitted by Mr. Mushet, but absolutely proved by him to be below the truth.

“It would be wrong in me to lead you to suppose that I have performed these works without active co-operation; I therefore freely acknowledge the services of Mr. — during these past two years, and I beg to be permitted to say that he is now rendering you most efficient services.

“If I have to acknowledge the services of Mr. — I feel myself bound in justice to say more of Mr. Richard Pilcher, who has, under peculiar circumstances, rendered you through me most valuable services; he has done many things greatly to the advantage of the Mint. In times past, Mr. Pilcher has borne a life best described as that of ‘a toad under a harrow,’ and his position remaining the same, although his present circumstances are altered, he would be again thrust back into the trouble he has just escaped from. I therefore, trusting that you will not think me presumptuous, beg of you to recommend to the Lords of the Treasury, as a temporary measure, that Mr. Pilcher be recognised as holding a position equal to that of a Junior Clerk, and that he may have sole charge of the Weighing Room, with a salary of

£200 a year, which would be an addition of only £8 18s. a year to his present pay. I assure you, most honestly, that Mr. Pilcher is thoroughly competent to conduct the duties of the Weighing Room, for he has conducted them without a single fault for three years.

“As regards myself, I feel that if you will give me your consideration, you will determine that my works are worthy a special application to the Lords of the Treasury; that my position and salary may be commensurate with my services, which have been rendered in profound confidence in your sense of justice towards those who do their duty to you.

“You are aware, sir, that of necessity, I have placed myself in direct antagonism with those by whom you are surrounded, and that I can therefore hope for reward at your hands only.

“With unmeasured hope and confidence,

“I am, Sir,

“Your most faithful servant,

“GEORGE F. ANSELL.

“TO THE MASTER OF THE MINT.”

After the “receipt of that letter” the Master told me he would at once establish me, but could not make a fresh appointment, because the Treasury looked with such distaste on new appointments; but he promised me “the first place:” “you shall be first—above them all.” That he admitted my claims is proved by the fact that he told a mutual friend—whom I had consulted, and by whose advice I wrote that letter—that “Mr. Ansell’s claims were stronger than he had represented;” but that this was his real opinion is clearly proved, not only by his former letters to the Treasury, but by the written orders and notes addressed to me, now in my possession, and by the letter which he wrote to the Treasury on the 7th June, 1860.

(EXTRACT.) LETTER FROM THE MASTER OF THE MINT TO THE
LORDS COMMISSIONERS OF HER MAJESTY’S TREASURY.

Royal Mint, 7th June, 1860.

“SIR,

“In continuation of the account of the waste or loss of bullion sustained in coining gold during the periods from 1851 to 1858 which was submitted to the Lords Commissioners of Her Majesty’s Treasury, in my letter of the 22nd May, 1858, I beg now to add a return of the waste upon the gold coinage of another year, 1858-9.

“The waste in the Melting and Coining Departments will be stated separately.

WASTE IN THE MELTING DEPARTMENT—GOLD COINAGE.

Date.	Weight of Gold Coinage.	Value of Gold Coinage.	Amount of Waste.	Value of Waste.
April 1, 1858, to March 31, 1859	Ounces. 736150·582	£ s. d. 2,866,386 6 6¾	Ounces. 248·804	£ s. d. 968 15 7¼

Date.	Value of Sweep recovered.	Value of Waste per Million.	Value of Sweep per Million.
April 1, 1858, to March 31, 1859	£ s. d. 231 2 1¼	£ s. d. 337 19 7	£ s. d. 80 12 6¼

“The net loss of the department will be obtained by deducting the value of the sweep recovered from the value of the waste, reported above, and is as follows:—

“On a gold coinage of £2,866,386 6s. 6¾d. the net loss by account in the Melting Department is £737 16s. 5¾d.; this gives on a million coined a net loss of £257 7s. 0¾d. Of the loss by account a portion is apparent only, and depends upon the dissipation during the melting of the oil and other impurities adhering to the gold scissel. The adventitious matter described was acquired by the scissel in the coining operations, and it has the effect of lowering the amount of loss reported in the Coining Department (as will be seen immediately) in the same proportion that it elevates the loss by account of the Melting Department. But after making allowances for the circumstance just stated, the loss of gold in melting remains higher in my opinion than it ought to be, and it will, I trust, be found to admit of some further reduction in future years.

WASTE IN COINING DEPARTMENT—GOLD COINAGE.

Date.	Weight of Coinage.	Value of Coinage.	Amount of Waste.	Value of Waste.
April 1, 1858, to March 31, 1859	Ounces. 736150·582	£ s. d. 2,866,386 6 6¾	Ounces. 16·603	£ s. d. 64 12 11¼

Date.	Value of Sweep recovered.	Value of Waste per Million.	Value of Sweep per Million.
April 1, 1858, to March 31, 1859	£ s. d. 209 16 7	£ s. d. 22 11 0¾	£ s. d. 73 4 0¾

“On a gold coinage of £2,866,386 6s. 6¾*d.* the account shows a waste amounting to £64 12s. 11½*d.*, with sweep recovered amounting to £209 16s. 7*d.* The Coining Department therefore returns a surplus by account of £145 3s. 7½*d.*; that is, £209 16s. 7*d.* less £64 12s. 11½*d.*

“A surplus by account instead of the usual loss was exhibited for the first time in the return of the preceding year. A true surplus or increase of metal is of course unattainable in any minting operation. The increase of weight recorded arises from adventitious matter acquired by the gold scissel in the Coining Department, to be again lost in the melting, as was just explained. The surplus by account of the Coining Department amounts to £50 13s. 0*d.* on £1,000,000 coined.

“A correct estimate of the whole waste in coining is obtained by combining the returns of the two departments:—

	£	s.	d.	
“Loss of Melting Department	257	7	0¾	per million.
Loss of Coining Department	50	13	0	”
Difference	206	14	0¾	”

“It thus appears that in 1858-9 the net loss in minting £1,000,000 gold was £206 14s. 0¾*d.*

“The annual loss since the reconstitution of the Mint in 1851 is as follows.

“Loss of bullion (waste, less sweep recovered) in minting £1,000,000 gold—

	£	s.	d.	
In 1851-52	479	5	8	
” 1852-53	921	10	7	
” 1853-54	795	11	7¾	
” 1854-55	661	2	4¼	
” 1855-56	724	1	3	
” 1856-57	691	14	1¾	
” 1857-58	140	4	7 ^[127]	
” 1858-59	206	14	0¾	
” 1859-60	187	0	0	partly by estimate.

“An additional year just completed, 1859-60, which appears in the preceding statement, is given subject to a subsequent correction of a few pounds more or less, as the sweep of that year (being still unsold) is taken by estimate.

“It appears by the table that the average loss on the gold coinage of the first six years is £784 0s. 0¾*d.*; and on the gold coinage of the last three years, £172 8s. 11½*d.* per million coined; showing an improvement of £611 11s. 1¼*d.* per million coined. On the twelve millions of gold coined, during the last three years, the saving exceeds twelve thousand pounds.

“The diminished waste of the late years is further attended with increased economy in working, of which, indeed, a low waste return is one of the surest indications. The proportion of sovereign blanks rejected in passing through the weighing machines, and which require to be remelted, causing thereby additional waste of metal and loss of labour, has been gradually diminished. This is shown by the following returns respecting the work performed in similar circumstances at various periods.

PROPORTION IN 100 SOVEREIGN BLANKS CUT.

“In 1855	21·49	per cent.			
” 1856	18·51	”			
” 1859	13·78	”	Under present officers.		
” 1860	4·11	”	”	”	

“The last small proportion of 4·11 per cent. of rejected blanks was further reduced to 2·07 per cent. by submitting the heavy portion of the blanks to the action of a new filing machine. I may be allowed to recall to your recollection that a Parliamentary grant of £1,100 was obtained by the Mint in 1856 for the purchase of two automaton filing and adjusting machines, which it was proposed to have constructed by Messrs. Napier and Sons, the eminent engineers. No part of this grant has been appropriated.

“The rapid amendment in the blanks which took place immediately afterwards led me to suspend the order for those machines, and ultimately to abandon the idea of any great expenditure for the object contemplated. In the meantime also a machine of a much more simple construction was contrived

by Mr. Richard Pilcher, of the Weighing Room, and was constructed in the Mint with no assistance from without. Pilcher's adjusting machine has proved sufficiently effective, costs nothing for labour, and has now been in constant operation for two years. The cost of making a pair of machines such as we now possess is estimated at £60, a sum which was saved to the public by the mode in which the work was executed. Trusting that the Lords Commissioners of her Majesty's Treasury will be disposed to consider favourably the merit and practical value of such an invention, I venture to solicit their Lordship's sanction to the application of £60, the sum just stated, to the benefit of the inventor, jointly with an ingenious mechanic on the establishment who gave material assistance in the construction of the machine. I would propose the following awards, if agreeable to their Lordships:—

“To Mr. R. Pilcher £40.

“To Meredith Jones £20.

“Their Lordships have afforded encouragement on previous occasions to voluntary efforts made within the Department in improving the machinery.

“On one additional point I would take the opportunity to offer information, and also submit a proposition for their Lordships' consideration. A correspondence between the Governor of the Bank of England and myself, on the property of the gold occasionally imported into the Mint to be coined, was formerly brought under your notice in my letter of the 21st of March, 1857. The evidence of the evil in question was not confined to the Royal Mint, but has, I believe, been felt at all other Mints, and also by goldsmiths generally, since the recent gold discoveries. It was, however, traced to the presence in the gold of a minute portion of antimony or arsenic (often not greater in quantity than one-tenth of a per cent.), which escapes the observation of the assayers. But no remedy in dealing with such gold presented itself, except the expensive one of having the brittle gold refined.

“An unusually large proportion of the gold received for coinage last year was of this defective character; but the whole of it was successfully coined, notwithstanding, and no part returned to the Bank to be refined, as on former occasions.

“This improvement in the practice of the Mint is the result of a laborious investigation made in the coining department, of which I am happy to assign the chief merit to Mr. Ansell. It now appears that the antimony or arsenic acts injuriously when the gold blanks are allowed to cool gradually, but not when cooled suddenly after annealing; the gold appearing to have time to crystallise and become granular under the influence of the antimony particles in the one case, but not in the other. The improvement may be justly represented as one of considerable value. It saves entirely much extra labour hitherto applied to brittle gold without any beneficial result. It will also lead to the diminution of waste, of which indeed brittle gold was always a fruitful source.

“The services of Mr. Ansell in effecting this improvement appear to me well deserving of recognition; and I would strongly recommend the award to that gentleman of a moderate grant of such amount as their Lordships may be pleased to accord, on account of these services. Mr. Ansell is a valuable

officer, and no one has contributed more to the reduction in other ways of the expenditure of the establishment. Both he and Mr. —, another junior officer of the Coining Department, are also engaged at salaries (...) which are certainly very moderate, considering the responsible duties these gentlemen are called upon to perform.

(Signed)

“THO. GRAHAM.

“To GEORGE A. HAMILTON, ESQ., Treasury.”

I quote thus much of this letter to prove to your Lordship that THE MASTER ADOPTED MY RECOMMENDATIONS; indeed, he specifically stated to me that the recommendations for Mr. Pilcher and myself were made with a view to support a future application. I would also observe that the actual money saving effected by the coining of the brittle gold here alluded to was £1,562 7s. 11¾d., and I desire to be very clear in stating that I copied this letter from Mr. Graham’s own writing. That a copy of it was sent to the Treasury would seem to be certain, for the following reply was received and handed to me by the Master:—

“FROM THE TREASURY TO THE MASTER OF THE MINT.

Treasury Chambers, 22nd September, 1860.

“SIR,

“I am directed by the Lords Commissioners of Her Majesty’s Treasury to acquaint you that the general result of your Report of the 7th June last on the loss of bullion in coining gold in the year 1858-1859 is, in the opinion of my Lords, very creditable to the Department under your management. My Lords willingly accede to your recommendation that £40 should be granted to Mr. R. Pilcher, and £20 to Meredith Jones for their services in the invention and construction of an adjusting machine under the circumstances represented in your Report. My Lords will also be prepared to grant a suitable payment to Mr. Ansell, in recognition of his services in obviating the difficulty in coining gold containing minute portions of antimony or arsenic, and they request that you will report to them the amount that you recommend to be granted to Mr. Ansell.

(Signed)

“GEORGE A. HAMILTON.

“THE MASTER OF THE MINT.

“Mint Number, 2988, 24th September, 1860.”

In answer to this the Master wrote to the Treasury, recommending a payment of £100 to Mr. Ansell, and the Treasury reply was as follows:—

FROM THE TREASURY TO THE MASTER OF THE MINT.

“Treasury Chambers, 19th October, 1860.

“SIR,

“With reference to your letter of the 13th instant, I am directed by the Lords Commissioners of Her Majesty’s Treasury to acquaint you that my Lords are pleased to authorise, in accordance with your recommendation, the grant of £100 to Mr. Ansell in recognition of his services in obviating the difficulty of coining gold containing a minute portion of antimony or arsenic.

(Signed)

“GEORGE A. HAMILTON.

“THE MASTER OF THE MINT.

“Mint Number, 3024, 20th October, 1860.”

Under my advice, the Master addressed a letter to Mr. Pilcher, conveying to that gentleman the £40, having previously written to me the following letter:—

THE MASTER OF THE MINT TO G. F. ANSELL, ESQ.

“Royal Mint, 20th October, 1860.

“DEAR SIR,

“Your success in obviating the difficulty of coining gold containing a minute portion of antimony or arsenic is a benefit to this department which calls for my most grateful thanks. In recognition of your services, I have the pleasure, with the approbation of the Lords Commissioners of Her Majesty’s Treasury, to present to you the sum of one hundred pounds, for which cheque is enclosed.

“I have the honour to remain,

“Dear Sir,

“Most faithfully yours,

(Signed)

“THO. GRAHAM.

“GEORGE F. ANSELL, ESQ.

“Mint Number, 3026, 1860.”

I was not at the time aware that the Master wrote to the Treasury on the 8th of June, 1860. That he did so, would seem to be proved by the following letter in reply; but that he should write on the next day after his letter of the 7th June, 1860, renders it possible that he made some alteration after that letter left my hand. I have not, my Lord, seen the letter written on the 8th June and alluded to in the next.

(EXTRACT.) LETTER FROM THE TREASURY TO THE MASTER OF THE MINT.

“Treasury Chambers, 25th September, 1861.

“SIR,

“I am commanded by the Lords Commissioners of Her Majesty’s Treasury to inform you that, having in consideration the very favourable opinion expressed in your Report of the 8th June, 1860, as to the ability and vigilance displayed by Messrs. — and Ansell, and the saving which has been effected in the processes at the Mint under your direction, their Lordships are pleased to sanction the increase of £30 a year to the salaries of those gentlemen, to take effect from the commencement of the present quarter, as recommended in your letter of the 6th instant.

(Signed)

“GEORGE A. HAMILTON.

“THE MASTER OF THE MINT.

“Mint Number, 454, 26th September, 1861.”

On the introduction to the Mint of Mr. John Graham all was changed, and it became clear that I was to be sacrificed. Mr. Barton, then Deputy-Master, therefore advised me to bring the matter to an issue by applying for the first vacant office. This I did, but my note was not even acknowledged; the Master, however, obtained a “Minute” from the Lords of the Treasury to the effect that *extra clerks were not at liberty to apply for promotion*, and this was conveyed to me by a messenger. To the office for which I had thus applied, Mr. John Graham who was then an extra clerk, was subsequently appointed. Upon the occurrence of the death of Mr. W. T. Brande, whose office had been specifically promised to me, I spoke to the Master of the Mint; but he assured me “The Treasury would fill up the vacancy, for they were dissatisfied with the whole of the Mint appointments.” He, however, appointed his brother, notwithstanding his acknowledged incompetence and unfitness. It was on the occurrence of that event that Mr. Barton strongly “advised me, on the event of the next vacancy to apply for it over the Master’s head, if you can get an influential person to back you.”

Finding the ground thus removed from my feet, I sought counsel with my friends, and determined to renew my studies in chemistry—having, in consequence of the Master’s promises, relinquished that profession when I entered the Mint.

It now becomes necessary to show by what occurrences my name was brought to your Lordship’s notice, and this I will do briefly.

When I had completed my laboratory in Bernard Street, I called on Mr. Robert Hunt, F. R. S., and asked him what subject would be likely to do me

most service. He at once kindly suggested “a means to find fire-damp in coal mines—one that will act unerringly, and require no attention from man.” I at once adopted the subject for study, and Mr. Hunt gave me many kind letters of introduction that I might “see my enemy at home,” especially a note to Mr. Oakes of Riddings House, Alfreton, whose generosity and hospitality were very prominent. I told that gentleman the object of my visit, and talked with him at considerable length, for I found in him a man singularly desirous to help his fellow-men. Mr. Oakes invited his brother to take me to a pit at a distance of about four miles, where I should be almost sure to find some fire-damp, and to this pit Mr. Oakes, junior, most kindly took me. The coal pit was itself a very remarkable one, and contained singular evidences in the history of the past. It will ever be associated with the invention of my “Fire-damp Indicator,” for it was in this pit that the idea for its formation occurred to me. If, therefore, this instrument should fortunately be the means of saving life, that benefit will have arisen from the kindness of Mr. Oakes. When I had made my instruments, I took them to the Mint, and showed them to Mr. Graham. Diffusion being fitly styled GRAHAM’S LAW, I took it for granted he would be pleased to find that force applied practically. I said to him, “I have proposed, Mr. Graham, to indicate the existence of fire-damp in coal mines.” He replied, “Ah, yes; that subject comes up every seven years.” I placed an instrument on his table, and asked if he would look at it, saying, “I propose to use your law of diffusion.” Before I had finished the word *diffusion* he cut in upon me with an exclamation of great force, “Good God! what would I have given for that thought!” I said, “God gave it to me.” To which he said, “I was on the Coal Mines Commission, and I did all I could to find a means of discovering the presence of that dreadful gas.” He refused to examine my apparatus, on the ground that he objected to the smell of gas, but said, “the honour of being associated with you in it would be so great that I shall be glad if you will allow me to pay all expenses, and you can make your experiments in my laboratory; the apparatus shall be left there.” I, however, declined his offer.

I exhibited the apparatus, as was but natural, in the first instance to Mr. Robert Hunt, at his residence, and afterwards to many other scientific men. It was subsequently noticed in the *Times*, in very high terms. A few days after that event, I received a note from Dr. Angus Smith, F.R.S., asking if I could “exhibit my experiments to Lord Kinnaird at 2 P.M. to-morrow.” I gladly accepted the proposed appointment, and suggested whether “his

Lordship could obligingly come to me at 27, Bernard Street," which he courteously did, accompanied by Mr. P. H. Holland and Dr. Angus R. Smith. I then found that Lord Kinnaird was Chairman, and Mr. Holland a Commissioner, of the Royal Commission of Mines. Your Lordship, as you may remember, honoured me by remaining in company with Mr. Holland and Dr. A. Smith for more than two hours, examining minutely every detail. From that day you have spared neither time nor labour to induce coal owners to adopt and use my "Indicator,"—which your Lordship also brought under the notice of Her Majesty, who deputed H.R.H. the Duke of Edinburgh to examine it on her behalf. This His Royal Highness graciously did at my residence. It has also been distinguished by mention in the Preface to the last edition of Ure's Dictionary, in which work it is fully described. In practical operation it has proved to be entirely successful even in the hands of working colliers, who have, on more than one occasion, said to me, "When men's lives are as valuable as those of horses, your instruments will be used; but, you see, they buy horses, and men they get for nothing."

I took occasion to speak with your Lordship about my troubled position at the Mint, and asked permission to make known to you all the circumstances. You having thus heard from me a statement of my views, advised me to await the arrival of the proper time, and stated that then you would assist me. I have already said that Mr. Barton advised me to apply over Mr. Graham's head for promotion when a vacancy should occur, and I therefore waited that event. Unfortunately for me, Mr. Barton was the first to be removed by death, he having died on the 25th August, 1868. I had, therefore, no powerful friend left in the Mint.

It was amidst such circumstances that I sought the assistance of your Lordship; which commenced with the following correspondence:—

**FROM THE RIGHT HON. LORD KINNAIRD, K.T.,
TO THE RIGHT HON. B. DISRAELI.**

*"Rossie Priory, Inchtute, N.B.,
"August 28th, 1868.*

"DEAR SIR,

"May I ask you to read the enclosed—I cannot, of course, ask you as from myself to attend to the application therein—but having become acquainted with Mr. Ansell during the time I was Chairman of the Mines Commission, I was so struck with his intelligence and ingenuity, combined

with a great knowledge of chemistry, that I felt he was a very valuable public servant. Mr. Ansell has invented a machine for the discovery of fire-damp in mines, which, if generally adopted, would be the means of preventing great loss of life from explosions in coal mines, and I am convinced that if you could find time you would be much interested in witnessing his experiments.

“Mr. Ansell has no friends in high quarters to bring his claims under your notice; I venture therefore to do so, however unwillingly, because I feel that he is deserving, and especially well fitted for the appointment. With many apologies for troubling you,

“I remain,
“Yours faithfully,
(Signed) “KINNAIRD.

“THE RIGHT HON. B. DISRAELI.”

ENCLOSURE.

“Royal Mint, August 28th, 1868.

“SIR,

“I most respectfully request your attention to the following statement of my claim for promotion consequent on the death of the Deputy-Master of the Mint, Mr. Barton. By a letter dated 29th October, 1856, addressed to the Treasury, I was recommended by Mr. Thomas Graham, F.R.S., the Master of the Mint, as fit to be appointed to a Supernumerary Clerkship in the Royal Mint.

“On the 12th of November, 1856, I was appointed to that office at a salary of £120 per annum, having previously passed the necessary examination before the Civil Service Commissioners. The internal management of the Mint then was, and had for some time previously, been in great confusion; the workmen were disorderly and dishonest, peculation was of daily occurrence, and it was universally believed that the processes of coining could not be carried on without considerable loss. Moreover, in consequence of the various modes by which bullion was then weighed, it was almost impossible to detect dishonesty. Nor was this all, for the blank coins were so imperfectly executed that it was always necessary to reject and return to the melting-pot from 15 to 35 per cent.

“From 1851 to 1856 the average ‘loss by coining,’ as it was then called, was

In gold, £581·42 per million pounds sterling.
In silver, £271·00 per million ounces coined.

“During the same years the average number of blank coins rejected was

In gold, 17·92 per cent.
In silver, 25·00 ”

“At this period, owing to the use of oil on the metal in the processes of coining, an additional loss was incurred in the melting. The average loss by melting during the same period being,

In gold, £356·33 per million pounds sterling.
In silver, £210·75 per million ounces coined.

“Soon after I entered the Mint I was taken into the confidence of the Master.

“I satisfied him that by proper management and vigilance no loss ought to be incurred by the processes of coining; that if due care were taken, scarcely any of the blank coins, when completed, would be rejected, and that a further saving to the Government would be effected by abolishing the use of oil in the processes of coining.

“Mr. Graham entrusted to me the sole task of carrying out these reformatations, at the same time promising that if I were successful I should be amply rewarded by promotion.

“You, Sir, will be able to judge of my success by the following results:—

“Instead of any ‘loss by coining’ since 1856 to the present date, there has been, on the average, a positive gain:—

In gold, £20 per million pounds sterling coined;

While for silver, £55·75 per million ounces coined is the total average loss.

“During the same time the average number of blank coins rejected has been

In gold, 2·07 per cent.
In silver, 1·00 ”

“During the same period the average loss by melting has been reduced to,

In gold, £271·90 per million pounds sterling.
In silver, £57·90 per million ounces coined.

“In 1857, as you, Sir, will doubtless perfectly remember, there arose a very great pressure on the Bank of England for coined gold; any endeavour to supply this unusual demand necessitated great exertion on the part of the Mint, and the Master of the Mint placed the matter entirely in my hands. I produced, without a single instance of loss by peculation, and with an average of only 6 per cent. rejected blank coins, 1,100 journey weights of sovereigns per week, being more by 200 journey weights than had ever before been obtained. I thus was the humble means of saving the country from experiencing the disastrous effects consequent on the Bank of England being unable to meet the demand for gold—a state of things almost universally expected at the time as the unavoidable result of the then monetary crisis.

“In 1858, during the temporary absence on leave of the Master Melter, the Master of the Mint, by a written order, directed me to conduct the melting

department. This I did with such success as to reduce the then ordinary loss by melting by no less than 33 per cent.

“By a letter to the Treasury, dated 17th August, 1858, Mr. Graham recommended my salary to be increased to £220 per annum. This was done soon afterwards.

“In 1859 I endeavoured to coin a large quantity of brittle gold, which had up to that time, not only in the Royal Mint, but also in the mints of other countries, been invariably rejected, under the impression that it was totally unfit for coining purposes in consequence of its extreme brittleness. I coined and sent to the Bank of England upwards of a quarter of a million pounds sterling of this brittle gold, the coins having proved to be so tough that they could not be broken by ordinary means. For this service I received from Mr. Graham a letter, dated 20th October, 1860, conveying to me ‘his most grateful thanks,’ and from the Lords of the Treasury a small gratuity of £100, awarded to me in consequence of a letter to the Treasury from the Master of the Mint, dated 13th October, 1860, recommending that such a donation should be made.

“By a subsequent letter, dated September 6th, 1861, Mr. Graham recommended the Lords of the Treasury to raise my salary to £250 per annum. His recommendation was at once acceded to on the part of the Treasury.

“In 1862, at the request of the Master of the Mint, I undertook the destruction of the old copper money, and destroyed from 10 to 16 tons per diem, at the comparatively small cost of 8s. 4d. per ton. Before this work was put under my charge 2 tons per diem had never been destroyed; and in the Paris Mint, where a similar process of demonetising had been in operation, the cost had never been less than £10 per ton.

“On several occasions offers were made to me of situations (unconnected with the Royal Mint) in which I should have earned a much greater salary than that which I now receive, but I have always been induced to refuse them by the express promises made to me from time to time by the Master of the Mint, Mr. Graham, that he would ensure my being placed in a safe position in the Mint.

“In the month of November, 1863, the office of Assistant Coiner became vacant. No one, so far as I am aware, had performed so many or so important services as to give him such a claim to that appointment as I possessed; but, much to my surprise, Mr. Graham ignored my claims, and appointed to the vacant office his brother, Mr. John Graham, a gentleman considerably junior to me in the service of the Mint. Smarting under what I conceived the injustice I had experienced from the Master of the Mint, I wrote perhaps too strongly, but for this I have apologised, and he has accepted my apology, but I feel that it is probable that the Master of the Mint will, consequently on the estrangement between us, not recommend me for the appointment, and I can only rely on my services (into which I hope you will kindly cause inquiry to be made), as giving me a claim, and therefore venture to address you by the advice of Lord Kinnaird.

“The unwillingness which I have cause to fear on the part of the Master of the Mint to recommend me will not, I am sure, arise from a belief of my

unfitness—as in 1864 he offered to secure to me the appointment of Chief Coiner and Melter to the Hong Kong Mint, at a salary of £1,000 a year—an appointment which, being the father of a family, I was compelled to decline.

“I have to apologise for thus venturing to address you, which I could only think of doing under the belief that you will take into your favourable consideration the application of one who has endeavoured faithfully and zealously to fulfil his duty, and not without some success, as I have endeavoured to point out.

“I have the honour to be, Sir,
“Your most obedient servant,
(Signed) “GEORGE P. ANSELL.

“THE RIGHT HONOURABLE B. DISRAELI,
“Hughendon Manor, Buckinghamshire.”

This application, I would remind your Lordship, was made while Mr. Disraeli was First Minister of the Crown, and was acknowledged as follows:

“10, Downing Street, Whitehall, August 31st, 1868.

“MY LORD,

“I am directed by Mr. Disraeli to acknowledge the receipt of your Lordship’s letter of the 28th inst., and to acquaint you that he has made a note of its contents.

“I have the honour to be, my Lord,
“Your obedient servant,
(Signed) “MONTAGU CORRY.

“THE LORD KINNAIRD.”

On the 3rd September, 1868, Mr. Montagu Corry called on the Master of the Mint and remained with him some considerable time.

On the 28th September I received by a messenger, without any further intimation, the following note:—

THE MASTER OF THE MINT TO G. F. ANSELL, ESQ.

“Royal Mint, 28th September, 1868.

“DEAR SIR,

“I regret to have to inform you that it is not in my power to continue your present engagement as an extra officer in the coining department while the Mint is only so partially occupied as it has been for some time past.

“Your assistance will not, therefore, be required beyond the 31st December next.

“I remain, faithfully yours,
(Signed) “THO. GRAHAM.

“G. F. ANSELL, ESQ.”

Which I immediately forwarded to your Lordship, asking advice, when you were so good as to write as follows:—

**THE RT. HON. LORD KINNAIRD, K.T.,
TO THE RT. HON. B. DISRAELI.**

“Rossie Priory, Inchtute, N.B., September 30th, 1868.

“DEAR SIR,

“I venture to trouble you on behalf of Mr. Ansell, of the Royal Mint, who I had advised to apply to you for the appointment of Deputy Master of the Mint, as from his long experience in the Department, and great ability as a chemist, I felt he deserved promotion, and was in every way fitted for the office. It seems however, that his application has drawn down the ire of the Master of the Mint, who, I suppose, wished to have some friend of his own, consequently Mr. Ansell has received a very curt dismissal (copy enclosed).

“It is very hard on a public servant who has done his duty for so many years to be thus summarily dismissed, and I therefore venture again to trouble you on the subject, feeling confident that your sense of justice will dispose you to view the matter favourably.

“Yours faithfully,
(Signed) “KINNAIRD.

“THE RIGHT HON. B. DISRAELI, GROSVENOR GATE.”

Your Lordship also wrote letters to the noblemen and gentlemen who had supported my application.

On the 2nd December, 1868, Mr. Disraeli, on the eve of his resignation, appointed Mr. C. W. Fremantle—his private secretary—to the vacant office of Deputy-Master of the Mint.

**THE RT. HON. LORD KINNAIRD, K.T.,
TO THE RT. HON. W. E. GLADSTONE.**

“*Rossie Priory, Inchtute, N.B., December 25th, 1868.*”

“DEAR MR. GLADSTONE,

“I should not have troubled you at present, as you still must have much to occupy your time and attention; but the case I have to bring before you is *pressing*, as, if anything is to be done in the matter, initiatory steps must be taken before the 31st.

“During the time I was engaged on the Mines Commission, I became acquainted with Mr. G. F. Ansell, of the Mint, who has invented a most ingenious instrument for detecting fire-damp in mines, which, if generally adopted in working coal pits, would, I am perfectly satisfied, be the means of saving life to a very great extent. I have, therefore, taken great interest in Mr. Ansell, and on the occasion of the death of the Deputy-Master of the Mint, I and other friends of his applied to the late Government to appoint him to the vacant office, and we believed that he would get the appointment for which he is so well qualified. Some delay, however, took place, which, we were led to believe, was owing to the Master of the Mint, Mr. Graham, being opposed to it, and he at the same time, on the plea that Mr. Ansell was a supernumerary clerk, though *he had been twelve years* in the service, gave him notice to quit on the 31st of this month, and the office of Deputy has been filled up.

“Now Mr. Ansell, who is a very able chemist, and has got the very highest testimonials, gave up his profession under a promise that he was to get a permanent appointment in order to undertake a most difficult task—that of detecting peculations which were taking place in the Mint. In this he succeeded, and was highly complimented by the Master of the Mint in letters to the Treasury which can be seen.

“The fact is, he saved the country, as the tables will show, upwards of £27,000, to say nothing of the advantage of putting a stop to peculations, and introducing order and economy in the working of the Mint.

“Mr. Graham, though having borne, and still bearing, witness to Mr. Ansell’s high qualifications to the office, has conceived a strong dislike to Mr. Ansell, because, as I understand, Mr. Ansell remonstrated with him, and has spoken to the Master in strong terms, on the appointment of his brother, Mr. John Graham, to an office in the Mint.

“In consequence of this ill-will, and of Mr. Ansell having applied to the late Government for the appointment of Deputy-Master, Mr. Ansell has been dismissed, *after twelve years’* service, on the *plea* that there is no work for him. Although a fresh coinage is about to take place immediately, and I think there is every probability of these peculations being again carried on if inefficient men are appointed, and there is no one to check them as Mr. Ansell did, I think I am justified, for the sake of the public service, independent of the hardship of the case, in asking you to *cause an inquiry* to be made into the reason of a public servant being dismissed after twelve years of service, and, in the mean time, to order the dismissal to be suspended till the inquiry is made.

“If you would get one of the Junior Lords of the Treasury to hear Mr. Ansell’s statement, he will, I am sure, be able to satisfy himself as to the truth

of what I have stated, and be able to enlighten you as to what has taken place in the office of the Mint. With many apologies for troubling you,

“I remain, yours faithfully,
(Signed) “KINNAIRD.

“THE RT. HON. W. E. GLADSTONE, &c., &c.”

On the 30th December your Lordship wrote me a letter, in which you said:—

“Mr. Gladstone writes me, ‘I have at once transmitted your letter to the Chancellor of the Exchequer, to whose cognisance it belongs.’ I have written to Mr. Lowe.”

Your Lordship again wrote on the 5th January, 1869, enclosing a letter

**FROM THE RT. HON. ROBERT LOWE
TO THE RT. HON. LORD KINNAIRD, K.T.**

“11, Downing St., Whitehall, 4th January, 1869.

“MY DEAR LORD,

“Mr. Ansell, in whom you have interested yourself, is, I have ascertained, a temporary clerk in the Mint, and the duration of his employment must accordingly depend on the necessities of the Department, of which the Master of the Mint is the only judge. The Treasury could not, therefore, interfere with Mr. Graham’s discretion in such a case.

“I have, however, felt it my duty to make a particular inquiry as to the nature of the expectations held out to Mr. Ansell when his engagement commenced, and which you have been informed amounted to a promise that he was to get a permanent employment; but I am informed by the Master of the Mint—the only person who could, I imagine, have made such a promise—in the most positive terms, that he never gave Mr. Ansell, either before or after he went to the Mint, any encouragement to look for a permanent engagement. Under these circumstances, whatever the merits of Mr. Ansell may be, I am afraid I cannot undertake so serious a proceeding as to interfere with the course which the Master of the Mint may consider it proper to pursue in this matter.

“Believe me,
“Yours truly,
(Signed) “ROBERT LOWE.

“LORD KINNAIRD.”

On the 6th January, 1869, I, by the advice of your Lordship, took the opinion of eminent counsel, before whom I laid proofs of my statements. Counsel advised me to draw up a Memorial, and ask your Lordship to present it to the Treasury. This I did, with the assistance of a well-known solicitor, who used infinite pains. The Memorial was as follows:—

“TO THE RIGHT HONOURABLE THE LORDS COMMISSIONERS
OF HER MAJESTY’S TREASURY.

“THE HUMBLE MEMORIAL OF GEORGE FREDERICK ANSELL,
of No. 27, Bernard Street, Russell Square,
Analytical Chemist.

“Sheweth as follows:—

“In the month of October, 1856, the Master of the Mint called upon your Memorialist, and offered to nominate him to a position in Her Majesty’s Mint, and promised that he should be promoted if he succeeded in effecting the object the Master of the Mint had in view, which was to reform certain abuses which prevailed extensively in the coining department, in which there was then, and had been for some time previously, great waste, owing to the carelessness and peculations of those employed in conducting some of the processes in that department. There were also great irregularities and disobedience of orders and disregard of regulations. All those circumstances combined, caused considerable loss in the operation of coining, and the Master of the Mint informed your Memorialist that he understood it to be the intention of the Government to put the coinage out to contract, unless it could be conducted with greater economy in the Mint. Upon the understanding and in the belief that he was to be promoted when any vacancy in a higher office should occur, your Memorialist accepted the offer made to him, and relinquished the professional pursuits he was engaged in prosecuting, and agreed to devote the whole of his time and energies to the service required of him. Consequently upon this, the Master of the Mint, in a communication dated the 29th of October, 1856, submitted a proposition for the consideration of your Lordships’ predecessors in the following terms:—

““Under this designation I desire to introduce a young man recommended by scientific or technical information available in coining, by energy of character, and by tried ability in the supervision of workmen—a faculty by no means common.... I have been led to propose the name of Mr. George Frederick Ansell as temporary clerk. Mr. Ansell was educated under Dr. Hofmann, and acted for some years as his laboratory assistant. He has since been scientific director in the Royal Panopticon, in Leicester Square, which was lately broken up. He appears to be a person of superior education and ability, and great activity and vigour of character, and, as I have been assured, has shown much discretion in the management of both pupils and workmen. Mr. Ansell, if appointed, would be placed in the rolling room, and have charge of putting the bars into work.’

“On the 4th of November, 1856, your Lordships’ predecessors approved of the recommendation of the Master of the Mint, and on the 12th of November, 1856, your Memorialist entered upon his service. Before taking active duty, he inquired of the Master what latitude was allowed in case of any difference of weight between the bullion given to the men and the return. The Master replied that he did not know, but that there had always been considerable loss in the room your Memorialist was to enter. Your Memorialist assured the Master that there ought to be no loss, and stated that if he were supported he would demonstrate it. The Master of the Mint promised to give your Memorialist full authority, and also assured him that he might depend upon promotion if he were able to carry out what he had stated.

“In a very short time after your Memorialist had entered upon his duties the losses were stopped, the men became obedient to orders, and all irregularities ceased; but it was not without exciting considerable ill-will amongst some of the men and officers employed that your Memorialist succeeded in accomplishing the ends in view.

“Previously to your Memorialist entering upon his duties, speculation had been of almost daily occurrence. In consequence of the various modes by which the bullion was then weighed, it was almost impossible to detect dishonesty. In addition to this, the blank coins were so imperfectly made, that it had always been necessary to reject and return to the melting-pot quantities ranging from 15 to 35 per cent.

“During the six years, 1851 to 1856, the average loss by coining, as it was termed, was as follows:—

In gold, £581·420 per million pounds sterling.
In silver, £271·000 per million ounces coined.

In addition to this, a loss was incurred in the melting, owing to the use of oil in the processes of rolling and cutting out, and the average of this loss during the same period was:—

In gold, £356·330 per million pounds sterling.
In silver, £210·750 per million ounces coined.

“During the same period, the average number of blank coins rejected was:

—

In gold, 17·92 per cent.
In silver, 25·00 ”

“The Master of the Mint entrusted solely to your Memorialist the task of carrying out such reforms, and taking such steps as your Memorialist considered necessary to put a stop to the speculations and waste above mentioned.

“The result of such superintendence by your Memorialist has been that, during the years from 1857 to 1866, instead of any loss by coining, there has been, on the average, a positive gain, inasmuch as the gain on gold amounted to £20 per million sterling, whilst the loss in silver was reduced from £271 per million ounces, to £55 per million ounces. During the same period, the average number of blanks rejected has been, in gold, only 2·07 per cent., and

in silver, only 1·00 per cent. The average loss by melting was also reduced to the following:—

In gold, £271·900 per million pounds sterling.

In silver, £57·900 per million ounces coined.

“The following tabular statement, compiled from the official books of the Royal Mint, shews the average loss by peculation during the six years 1851-1857, and the saving effected under your Memorialist’s management during the years 1857-1866:—

Coining Department.					
Date.	Money coined.	Absolute loss by peculations.	Averages per £1,000,000 coined.		Absolute saving effected by stopping peculation.
			Loss.	Gain.	
	£	£	£	£	£
1851-57	40,345,185	20,030	491	—	—
1857-66	50,894,385	Nil.	Nil.	26	27,736

Melting Department.		
Date.	Average loss by melting per £1,000,000 coined.	Absolute average saving effected by the disuse of oil on the scissel.
	£	£
1851-57	261	—
1857-66	233	28

“The result from the above table is that in the period 1851-1857 £40,345,185 in gold was coined at a loss to the Government by peculations and waste in coining of upwards of £20,000, whilst by the gains and savings effected by your Memorialist, amounting to £545 in each £1,000,000 coined, the gain to the Government during the years 1857-1866 amounted to £27,736, and this in the gold alone, and quite independently of the saving effected in the coining of silver.

“In the year 1857, shortly after the appointment of your Memorialist, there was a great demand and pressure on the Bank of England for gold

coin, and the endeavour to supply this unusual demand necessitated great exertion on the part of the Mint. The Master of the Mint placed the matter entirely under the direction of your Memorialist, who was the means of producing, without a single instance of loss by peculation and with an average of only 6 per cent. of rejected blanks, 1,100 journey weights of sovereigns per week, being more by 200 journey weights than had ever before been obtained. This extra production undoubtedly contributed largely to meet the demand for gold and relieve the pressure on the Bank during the monetary crisis of that year.

“In the year 1858, during the temporary absence on leave of the Master Melter, the Master of the Mint directed your Memorialist to conduct the melting department, and during such superintendence of it your Memorialist reduced the then ordinary loss in melting by no less than 33 per cent.

“In the month of August of that year the Master of the Mint recommended your Memorialist’s salary to be increased from £120 per annum to £220.

“In the year 1859 a large quantity of brittle gold had been brought to the Mint, and up to that time that kind of gold had been invariably rejected by the Royal Mint, as well as by the Mints of other countries, under the impression that it was totally unfit for coining in consequence of its extreme brittleness. Your Memorialist endeavoured to turn this gold to account, and succeeded in coining it; and upwards of a quarter of a million sterling of this coined gold was sent to the Bank of England, and the coins proved to be so tough that they could not be broken by the ordinary means. For this service your Memorialist received from the Master of the Mint a letter expressing his ‘most grateful thanks,’ and upon his recommendation, in the month of October, 1860, a gratuity of £100 was awarded to your Memorialist by your Lordships’ predecessors, and in the year 1861 your Memorialist’s salary was raised from £220 to £250 per annum. Your Memorialist also originated and developed the plans by which the new bronze money was coined previous to its issue in 1860. Those plans are still followed, and in consequence of your Memorialist’s investigations of the subject, Messrs. Heaton and Messrs. Grenfell were referred to him for information and assistance by the Master of the Mint.

“In the year 1862 your Memorialist, at the request of the Master of the Mint, undertook the destruction of all the old copper coinage, and destroyed

from 10 to 16 tons per diem, at the comparatively small cost of 8s. 4d. per ton. Before this work was put under your Memorialist's charge, scarcely so much as 2 tons per diem had ever been destroyed; and in the Paris Mint, where a similar process had been in operation, the cost had never been less than £10 per ton.

"Down to this period, offers had been made to your Memorialist on several occasions of situations unconnected with the Royal Mint, at a much higher salary than he was in receipt of there, but your Memorialist was induced to decline them in consequence of the assurances of the Master of the Mint that his position there should be greatly improved. In the month of November, 1863, the office of Assistant Coiner became vacant. No one, so far as your Memorialist is aware, had performed so many or such important services in the establishment as he had done; but his claims were ignored, and the Master of the Mint, passing over him, appointed his own brother, Mr. John Graham (who was considerably junior in the service), to the vacant post.

"Upon the death of the late Mr. Barton, Deputy-Master of the Mint, your Memorialist applied to the Right Honourable the then First Lord for promotion, but he appointed Mr. Fremantle, his private secretary, to Mr. Barton's office, and no other change or promotion was made.

"On the 28th of September, 1868, your Memorialist received his dismissal from the service, in the following letter from the Master of the Mint—

'Royal Mint, 28th September, 1868.'

'DEAR SIR,

'I regret to have to inform you that it is not in my power to continue your present engagement as an extra officer in the coining department while the Mint is only so partially occupied as it has been for some time past. Your assistance will not therefore be required beyond the 31st of December next.

'I remain,

'Faithfully yours,

'THO. GRAHAM.

'G. F. ANSELL, ESQ.'

"Notwithstanding the intimation conveyed by the above letter that your Memorialist's services were no longer required in consequence of the

coining department being only partially occupied, it is the fact that another gentleman, who was an extra clerk in the bronze store department, and considerably your Memorialist's junior (but who is private secretary to Mr. John Graham, the brother of the Master), has been appointed to the same post as your Memorialist held; and it is also the fact that the Mint is now executing a new coinage of about £5,000,000 of gold, and your Memorialist believes that it will be found, on investigation, that losses have again occurred, upon this and the last preceding coinage, which was not superintended by your Memorialist.

“Upon a reference to the communications to the Treasury from the Master of the Mint, under the dates of

29th October, 1856,
6th August, 1857,
17th August, 1858,
7th and 8th July, 1860,

it will be seen that your Memorialist's services have been referred to by the Master of the Mint in terms of high approbation, and your Memorialist can confidently assert that his dismissal has not been owing to any want of efficiency or attention to the public service on his part; and, on the other hand, others who have less effectually served the public interest, and have been junior to your Memorialist in the service, have been promoted over his head. The Master of the Mint at one time offered to recommend your Memorialist to an appointment in the Mint at Hong Kong; but the climate would have been unsuitable to your Memorialist and his family, and he was compelled to decline it.

“Your Memorialist has spent twelve of the best years of his life in the public service at an insignificant salary, and has effected savings to the public to the amount of £35,000 and upwards in the gold coinage alone, besides many other savings in other branches of his department, and has now been dismissed without, as he humbly submits, any adequate reason at three months' notice (which your Memorialist is advised is insufficient according to the usual course of law) and without compensation of any kind.

“Your Memorialist therefore humbly prays your Lordships to cause inquiry to be made into the circumstances stated by your Memorialist, and his special services in connection with the Royal

Mint, and that your Lordships will be pleased to continue his services in some other department in which they may be made available, or to award some compensation to him in consideration of the considerable sums he has been the means of saving to the public during the course of his employment at the Royal Mint, and of his abrupt dismissal by the Master without any just cause.

“GEORGE F. ANSELL.

“10th February, 1869.”

On the 19th February, your Lordship enclosed to me the following letter:

**FROM C. RIVERS WILSON, ESQ.,
TO THE HON. A. KINNAIRD, M.P.**

“11, Downing Street, Whitehall, 15th February, 1869.

“DEAR SIR,

“The Chancellor of the Exchequer has received the Memorial from Mr. Ansell, forwarded in your letter of Saturday, and has caused it at once to be laid before the Board of Treasury, in compliance with the wish expressed by Lord Kinnaird.

“I am, dear Sir,

“Yours faithfully,

(Signed)

“C. RIVERS WILSON.

“HON. ARTHUR KINNAIRD, M.P.”

On the 22nd February, 1869, Mr. John Graham died, thus leaving the office of Chief Coiner vacant. I immediately informed your Lordship of this unexpected event, and you, on the 23rd, wrote to the Hon. A. Kinnaird, asking that gentleman to go immediately and see Mr. Lowe, and to take him a copy of my book of testimonials; and on the 24th your Lordship wrote to the Hon. A. Kinnaird, asking him to see Mr. Gladstone, in whose gift the office of Chief Coiner is. On the 27th February, 1869, a friend suggested that I should “ask Lord Kinnaird to mediate between Mr. Graham and myself.” Your Lordship permitted me to make that suggestion, and wrote as follows:—

**THE RT. HON. LORD KINNAIRD, K.T., TO
THOMAS GRAHAM, ESQ., F.R.S., D.C.L.**

“Rossie Priory, Inchtute, N.B., March 1st, 1869.

“SIR,

“I do not know whether you are aware that Mr. G. F. Ansell, by my advice, has presented a Memorial to the Lords of the Treasury, praying for compensation for his dismissal from the Mint.

“I considered the case so hard that a gentleman who had served the country for about twelve years, ably and efficiently, as shown by letters from the Treasury and from yourself, should be dismissed on three months’ notice, that I resolved to bring this case before Parliament, but before doing so I thought it best that Mr. Ansell should memorialise the Treasury, stating his case and past services.

“Since the Memorial was sent in, I have heard of the loss you have sustained, and consequently have applied for the appointment for Mr. Ansell to the office lately filled by your brother.

“I think it right you should know this, and I am not without hopes that my recommendation may meet with your concurrence, as though I am aware that you had at one time some difference with Mr. Ansell, yet he has at the same time told me of many kindnesses he has received from you—shown also on a late occasion, when you gave him a very good testimonial on his being a candidate for a Gas Inspectorship, which proves the high opinion you have of Mr. Ansell’s qualifications.

“He has assured me that, should he get the appointment, he will act most cordially with you, and be most anxious to forget all that is past.

“I am prepared, on hearing from you that you concur in Mr. Ansell’s appointment, to write to Mr. R. Lowe to tell him that I wish the Memorial withdrawn.

“Your obedient servant,
(Signed) “KINNAIRD.

“T. GRAHAM, ESQ., F.R.S., D.C.L.,
“Master of the Royal Mint, London.”

Envelope marked “Private,”

**FROM T. GRAHAM, ESQ., F.R.S., D.C.L., TO
THE RT. HON. LORD KINNAIRD, K.T.**

“4, Gordon Square, 3rd March, 1869.

“MY LORD,

“On returning from Scotland I find your letter of the 1st instant, the receipt of which I beg to acknowledge.

“I beg to remain, my Lord,
“Your Lordship’s most obedient servant,
(Signed) “THO. GRAHAM.

“THE RIGHT HON. LORD KINNAIRD.”

On the 6th March, 1869, your Lordship wrote me to the effect that Mr. Lowe had stated that he “still entertained a belief that Mr. Ansell was never on the staff of the Mint, and he cannot force a supernumerary on Mr. Graham, but he HAS SENT THE MEMORIAL TO MR. GRAHAM TO ANSWER.” I feel that the Right Hon. Robert Lowe thus committed a singular act of injustice to me. Those who will read the Memorial cannot but be struck with the fact that my sole complaint is against Mr. Thomas Graham, yet Mr. Lowe remits the case to the accused for *his* decision. Such an act is a remarkable comment on the promise of the Government *that those who would faithfully do their duty, irrespective of their nominal position, should be rewarded.*

This injustice induced your Lordship to take further steps, and you sent me the following letter:—

**FROM THE RIGHT HON. W. E. GLADSTONE
TO HON. A. KINNAIRD, M.P.**

“March 5th, 1869.

“MY DEAR KINNAIRD,

“I am not sure whether the arrangements to be made at the Mint will place any vacancy at my disposal, but if they do I will carefully consider the subject of your letter respecting Mr. Ansell.

“Ever yours sincerely,
(Signed) “W. E. GLADSTONE.

“THE HON. A. KINNAIRD.”

On the receipt of this letter your Lordship directed me to send a copy of my “Treatise on Coining” to Mr. A. West for Mr. Gladstone. Up to this date the case had been urged upon Mr. Gladstone, not only by your Lordship, but in addition by five Cabinet Ministers and five members of the House of Commons; yet there was, as in the case of Mr. Disraeli, some mysterious cause why I was defeated, and on the 27th of March, 1869, I received the following:—

FROM THE LORDS OF THE TREASURY TO G. F. ANSELL, ESQ.

(No. 3727, 24th March).

“Treasury Chambers, 27th March, 1869.

“SIR,

“In reply to your Memorial of the 10th of February last, I am directed by the Lords Commissioners of her Majesty’s Treasury to acquaint you that my Lords have made inquiry into the circumstances which led to the cessation of your employment as a temporary clerk at the Royal Mint, and they do not find that they are such as in their opinion would justify them in awarding you any compensation in consequence thereof.

“I am further desired to state that your removal from the service was entirely within the discretion of the Master of the Mint, who distinctly denies ever having held out to you any expectation of promotion or of permanent employment; and that your statements as to the value of your services at the Mint are not corroborated by Mr. Graham.

“In conclusion, I am desired to add, with regard to your application that your services may be continued in some other department, that my Lords have no appointment at their disposal to which they can nominate you.

“I am, Sir,

“Your obedient servant,

(Signed) “WILLIAM LAW,”

George F. Ansell, Esq.,

“27, Bernard Street, Russell Square, W.C.”

“EX^{ad}.

Whereupon your Lordship took steps, which I will explain presently; but I beg first to append a few extracts from my diary, written day by day at a time when, had my confidence in Mr. Graham been shaken, I could have induced him to make a written agreement. I instead placed in him so profound a confidence that I placed my future, *as he knew*, entirely in his keeping.

“9th March, 1858.

“I expressed an opinion that in the event of the Master being ill, or seeing fit to resign, I should be summarily dismissed the Mint, and that the —— and —— would come into power. I also related the grounds for my opinion, and the Master said that ‘the Treasury were too much awake to all affairs to admit of my dismissal or removal, or of such a change at any time under any circumstances; and that, beside, Mr. Sterry was now just

preparing his last Report of the waste per quarter, and which would effectually close —— and ——'s chance.'”

“9th July, 1858.

“I showed the Master, and he read Mr. Halder Wood's letter. I sought his advice, and he advised me to remain at the Mint, promising that, so soon as Mr. Brande or Mr. Buckle died, I should be placed in their position, for the Treasury would not make a place for me. I wrote this to Mr. Wood, who soon replied, offering me a place which was next to himself, and advising me ‘not to trust Graham's promises.’”

“September 2nd, 1857.

“A note from Mr. F. Versmann, declining a position which I had placed at his disposal under these circumstances. Mr. Corcoran offered to introduce me to Mr. Payne, who subsequently offered to engage me on the following terms: I was to attend three days a week at Farnham, where I was to have a residence and £250 a year. The other three days I was at perfect liberty to use and to spend in London, in no way tied to Mr. Payne. I consulted Mr. Graham. He promised most faithfully that I should be ‘above all,’ and advised me to decline Mr. Payne's offer. I did decline it, and to prove to Mr. Graham that the offer was *bona fide*, I handed to him Mr. Payne's letter, with a request that he would recommend a fit person, and I would introduce him. Mr. Graham gave the letter to Mr. Versmann, who in his note says he declines because ‘I am sorry indeed that Mr. Payne finds it necessary for the chemist to reside on the spot; but, after all, I am sure I am right in not giving up my chances in London.’ Mr. Corcoran has at this day a perfect recollection of the whole matter, and writes me ‘I believe you would have accepted, but you had a prospect of a certain position in the Royal Mint.’”

“October 5th, 1859.

“In a conversation about my position, which had arisen from the opposition offered to me, the Master said, ‘Yes, that is the difficulty, you see the Treasury will not give you a position while we have so much old lumber in the way of officers who do nothing.... And in truth I will tell you I had hoped that the Act of last session would have rendered retirement at sixty-five compulsory, and then I would have placed you in a far higher position in the coining department; but that clause was unwisely withdrawn, and here I am, still encumbered by people utterly useless.’ I said, ‘Is it probable that I

shall have to wait for my promotion till the death or retirement of one of my so-called 'superior officers,' or shall I now stand a chance of a more immediate position of safety?' He replied with great earnestness, 'It is a year since anything was done for you, so I will do something for you before the end of early winter, and you stand a far better chance now that Sir Charles Trevelyan is gone from the Treasury, for he always stood in your way, because his own nominations had turned out so very disastrously to this place; but I will promise you I will set you firmly directly the men return to London—directly the Ministers are in town.'”

22nd April, 1869.

Being determined to find the reason why I failed to be appointed, you, as stated in your Lordship's letter, called on Mr. Graham, at the Mint. Mr. Graham spoke of his desire to serve Mr. Ansell, and offered “to appoint him to the position from which Mr. C. W. Goodwin was to be retired with an increased salary.”

I had in writing begged your Lordship to allow me to decline this offer, because I saw through the malevolent malice which induced it, and enclosed a statement with extracts from my diary.

Four days later, it appears that your Lordship called again on Mr. Thomas Graham, in the Mint, where he was in consultation with Dr. Lyon Playfair, M.P., and Mr. C. W. Fremantle, when you, as stated in your letter, read and heard read Mr. Graham's letter to the Treasury, and your Lordship wrote to Mr. Lowe in relation to this interview with Mr. Graham, and sent me Mr. Lowe's reply.

“Downing Street, Whitehall, April 29th, 1869.

“DEAR LORD KINNAIRD,

“Had Mr. Graham consulted me, I should have advised him not to produce any documents to you and Dr. Lyon Playfair, for the simple reason that I have already decided the case, and that it ought not to be opened without my consent. That consent I feel myself bound to withhold. I cannot permit any one in this office to act as a Court of Appeal on what I have decided. I have already given to your Lordship such explanations as the case seemed to require, and I have nothing to add to it.

“I am, my lord,

“Faithfully yours,

(Signed) “ROBERT LOWE.

“THE RIGHT HON. LORD KINNAIRD.”

**THE RIGHT HON. LORD KINNAIRD, K.T.,
TO THE RIGHT HON. ROBERT LOWE.**

“Rossie Priory, Inchtute, N.B., May 4th, 1869.

“DEAR MR. LOWE,

“The only explanation in re Ansell which I received from you, was that Mr. Ansell’s Memorial, which complained of Mr. Graham’s conduct, had been remitted to that gentleman for an answer, and that, after receiving it, you considered further inquiry was unnecessary.

“I then asked to see the answer, and was refused, though, at the same time you *assured me* that it did not reflect in the least on Mr. Ansell’s character. When I did see Mr. Graham’s answer, I was struck with the animus which pervaded its tone, and can hardly understand how such a difference ‘between us’ can exist as to what *does* and *does not* reflect on a man’s character. That the impression conveyed to my mind by the document is the natural one, is corroborated by the declaration of Dr. Lyon Playfair on reading it; for he declared, ‘most strongly, that if published, it would prevent Mr. Ansell obtaining, not only any official, but any other employment.’ And certainly, if the statements therein be true, such a result would be deserved; however, the document bears on the face of it so manifest a perversion of facts, and exhibits such a malevolent feeling towards Mr. Ansell on the part of Mr. Graham, on *whose word alone* you have ‘decided the case,’ that I am satisfied that the publication of this document would rather tell in Mr. Ansell’s favour than otherwise. I pointed out to you that the appointment by Mr. Graham of his brother, whose unfortunate habits were well known, to a position of great responsibility in the Mint, and the peculations and losses which, by Mr. Graham’s own admission, had taken place, demanded a special inquiry into the management, and however much I might wish to screen a man of Mr. Graham’s scientific acquirements, I do not feel inclined to follow your advice and ‘ask Mr. Gladstone to give Mr. Ansell another appointment,’ instead of exposing the mismanagement of the Mint, and I cannot understand how, if you believe Mr. Graham’s answer to be reliable, you could recommend me to take such a course.

“Yours faithfully,
(Signed) “KINNAIRD.

“THE RIGHT HON. ROBERT LOWE, Downing Street.”

**THE RIGHT HON. LORD KINNAIRD, K.T., TO
THE RIGHT HON. W. E. GLADSTONE.**

“Rossie Priory, Inchtute, N.B., May 4th, 1869.

“DEAR MR. GLADSTONE,

“Since I saw you on the subject of Mr. Ansell’s appointment to the vacant office at the Mint, I have seen Mr. Graham’s answer to Mr. Ansell’s Memorial, and I earnestly hope that you will not think of filling up the vacant

office without a full inquiry into the management of that Department, which I consider is imperatively called for, and I do not think that in making that appointment you would be justified in being guided by the advice of the Master of the Mint, who formerly appointed to that office his brother, whose drunken habits were so notorious as to render him quite unfit to be continued in such a charge.

“I see no reason to doubt Mr. Ansell’s statements that great peculations did take place at the Mint, as well as waste in coining. This is admitted by the Master himself, and clearly brought out in the Memorial of Mr. Ansell, who was employed for twelve years as supernumerary, and received an acknowledgment more than once from the Treasury in consequence of the saving he had effected. Mr. Graham, it is true, denies the facts, and endeavours to blast Mr. Ansell’s character, and thus to throw discredit on the evidence Mr. Ansell could give.

“Mr. Graham further states in his answer that Mr. Ansell was dismissed on the complaint of his brother, while the letter of dismissal following immediately on Mr. Ansell’s application to Mr. Disraeli to be put on the permanent staff bears, that it was in consequence of the ‘coining department being so partially occupied,’ though at the same time a new coinage had been ordered.

“Mr. Graham’s answer bears on the face of it such a perversion of facts that it appears to me an inquiry is absolutely necessary, and I hope you will not oppose a Motion for a Committee of Inquiry into the management of the Mint, unless some other mode of inquiry is adopted.

“Believe me,
“Yours faithfully,
(Signed) “KINNAIRD.

“THE RIGHT HON. W. E. GLADSTONE.”

Your Lordship will perhaps allow me to bring to your remembrance that Mr. Thomas Graham died on the 16th of September, 1869, many months after you had made great exertions in my favour. This is the more necessary because some have thought that Mr. Graham was not aware of my charges against him, whereas he had a full knowledge of all that was passing, and his removal made it more difficult for me to prove my statements or to obtain justice. It had been, I believe, your Lordship’s intention to allow “my case,” as it has been called, to go into abeyance till, on the occasion of your motion in the House of Lords for a Committee to inquire into the management of the Mint, others gratuitously attacked me, instead of replying to your Lordship exact statements made on the 18th July, 1870, when you once more brought the subject of the Mint before the House of Lords. As that

debate became of personal interest to myself, I beg to quote your Lordship's concluding speech.

“LORD KINNAIRD: I am quite ready, as far as Professor Graham is concerned, to acquit him individually and personally from anything improper; all I have to say with regard to Professor Graham is that he was not a fit person for the office, as he continued the jobbery which he found in existence, and which had prevailed there ever since the Mint was a Government office. He appointed his own brother, a calico printer, to an office for which he was entirely unfit, not only on the ground of general incompetence, but from his well-known habits of intemperance. I am quite ready to admit that the bulk of the information I possess upon the subject is obtained from Mr. Ansell, who was for many years employed by the Master of the Mint, and who, as the returns show, succeeded in making even the gold coinage a paying department, as it should be, instead of a losing department, as it is; but my statements are also founded on, and confirmed by, returns presented to both Houses of Parliament. Mr. Ansell also put a stop to speculation, and made himself very unpopular among the officials in consequence. Mr. Ansell gave up very valuable appointments in order to fulfil his duties at the Mint, under the repeated promise of promotion when a vacancy should occur; and I am afraid I am responsible for what is called his dismissal, for, when I applied to the late Prime Minister for his promotion, Mr. Graham, who, in common with the other officials, feared an exposure from Mr. Ansell's promotion, wrote him a letter of dismissal as follows:—

‘Royal Mint, September 28, 1868.

‘DEAR SIR,

‘I regret to have to inform you that it is not in my power to continue your present engagement as an extra officer in the coining department, while the Mint is only so partially occupied as it has been for some time past. Your assistance will not, therefore, be required beyond the 31st of December next.

‘I remain, faithfully yours,

‘THO. GRAHAM.

‘G. F. ANSELL, ESQ.’

“After this dismissal the profit ceased and the loss recurred. I accordingly applied to the Prime Minister and the Chancellor of the Exchequer for an inquiry, but I was treated rather summarily, and the

Chancellor of the Exchequer said he deprecated any inquiry. Only two or three months, however, after giving him that notice, and while he was still engaged in the Mint, Mr. Graham warmly recommended him for the office of Gas Inspector to the Corporation of London, stating that he was a well-informed practical chemist, and ‘a man of tried integrity’—‘a man of *tried integrity!*’ Now, within three months after the Master of the Mint dismissed this officer—who, as the returns will show, had saved the country some thousands—he writes this testimonial:—

‘I beg to express the very favourable opinion which I entertain of Mr. G. F. Ansell’s qualifications for the office of Inspector of Gas to the Corporation of London. Mr. Ansell is a well-informed practical chemist, and has given much attention for several years to gas. He is the inventor of an ingenious instrument for indicating the presence of fire-damp in the atmosphere of mines, which is much admired by scientific and practical men. Mr. Ansell is also a man of tried integrity and great energy. From what I have seen of him at the Mint I would say that he would devote himself entirely and conscientiously to the work of the new office, if appointed.’

“THE DUKE OF ARGYLL: Who signed that letter?

“LORD KINNAIRD: Thomas Graham. And, notwithstanding this high testimony to the value of his services, this man was dismissed because he knew too much of what was going on in the Mint. I challenge the Mint authorities to prove a single statement in his book to be false. If they can, why do they not contradict him? They dare not. Mr. Ansell’s book has been some months before the public, and if the Mint could have challenged his statements, which are very damaging, they would certainly have done so. I challenge contradiction. I am very glad the noble Lord has given me the opportunity of making this statement with regard to Mr. Graham, a talented man, but quite unable to cope with the clever men about him, unaided by Mr. Ansell. Now that Mr. Ansell has been dismissed, you see what is the result. The noble Lord has told your Lordships of reforms which have been made at the Mint; but I can assure him I expect little good to result from those reforms, for the very men who then had charge of these departments, and who permitted the peculations and mismanagement which Mr. Ansell stopped, are the men who are now promoted to the chief management of the operative departments of the Mint. I hope, therefore, the matter will not be allowed to drop, for I am sure there is no department in this or any other country more corrupt than the Mint.”

“Motion (by leave of the House) *withdrawn*.”—
(*House of Lords, Monday 18th July, 1870. Hansard’s Report*)

I beg to express my great obligation to your Lordship for so complete a determination to obtain justice for me as you have exhibited, and which has required not only labour given ungrudgingly, but time, which I am sure would have been of inestimable value to others. My power to thank is weak, but I am entirely grateful. I am also obliged to your Lordship for sending me a copy of Hansard’s Report, which I, in accordance with your Lordship’s permission, print elsewhere, but reserve its conclusion as a fitting finish to this statement, which I submit as a true epitome of my unfortunate—for myself—connection with the Royal Mint.

I have the honour to remain,

My Lord,

Most gratefully your Lordship’s obedient and obliged servant,

GEORGE F. ANSELL.

THE RIGHT HON. LORD KINNAIRD, K.T., F.R.G.S.

Lord-Lieutenant of Perthshire, &c., &c., &c.

INDEX.

- Accounts of the Royal Mint, [14-15](#), [36](#), [83](#), [84](#), [85](#), [94](#), [102](#), [103](#),
[104](#), [106](#), [107](#), [115](#), [136](#), [142](#), [149](#), [155](#), [173](#).
- Acts of Parliament relating to the Coinage, [3](#), [36](#), [56](#), [71](#), [72](#), [73](#),
[109](#), [124](#), [150](#), [157](#), [159](#), [166](#).
- Adjusting of Fillets, [9](#), [19](#), [20](#), [22](#), [23](#), [25](#), [27](#), [29](#), [31](#), [32](#), [42](#), [112-113](#).
- Alloy, Copper is used as, [4](#), [6](#), [7](#), [8](#), [13](#), [21](#), [31](#), [46](#), [48](#), [49](#), [53-56](#), [58](#),
[82](#), [86](#), [87](#), [138](#), [141](#), [142](#), [143](#), [157-159](#).
- ” Extra, [49](#), [136](#), [137](#), [138](#), [141](#).
- ” Impurities contained in, [6](#), [13-14](#), [81](#).
- ” The best, for Coinage purposes, [49](#), [81](#), [157](#), [158](#).
- ” ” Coining Dies, [81](#), [157](#).
- Analysis of Bullion, Advantages attending, [13-14](#), [81](#), [157-158](#).
- Annealing and Blanching, Debases Standard of Coinage,
[31](#), [48](#), [49](#), [50](#), [53-56](#), [57](#), [58](#).
- ” Furnace, [21](#), [48](#), [49](#), [57](#), [62](#), [74](#).
- ” Loss occasioned by, [9](#), [21](#), [22](#), [31](#), [48](#), [49](#), [53-56](#), [57](#),
[86](#), [92](#), [138](#), [161](#), [162](#).
- Ansell, George F., Diary of, [194-195](#).
- ” Fire-damp Indicator, [182-183](#), [187](#), [193](#), [198](#).
- Ansell’s Standard Gauge, [23-25](#), [35](#).
- Antimony in Standard Gold attacks Iron, [19](#).
- Argyll, Duke of, [ix](#), [166](#), [171](#), [198](#).
- Artists as regards Dies for Coining, [64](#), [67](#), [158](#), [168](#).
- Arts, Journal of the Society of, [66](#).
- Assay Pieces taken, [4](#), [7](#), [9](#), [10](#), [53-56](#).
- ” ” Weight of, [10](#), [56](#).
- ” Proofs, [12](#), [13](#).
- ” Reports, preservation of, [14](#), [54](#), [146](#).
- ” Unreported Fractions, [127](#), [130](#), [142](#), [143](#).
- Assayers, Two Resident, suggested, [14](#), [81](#).
- Assaying Apparatus, Messrs Matthey, Johnson, and Co.’s, [12-13](#).
- ” Cost of, [3](#), [13](#), [120-121](#), [122](#), [149](#).
- ” Explanation of Process of, [10-14](#), [56](#).
- ” Loss by, [53](#), [56](#), [102](#), [103](#), [149](#).
- Ayrton, Right Hon. A. S., [148](#), [152](#), [154](#), [165](#).
- Balance, Cotton’s Automaton, described, [16](#), [37-42](#), [48](#), [124](#).
- ” ” Bradshaw’s Improvements in, [40-42](#).
- ” ” Pilcher’s Improvements in, [37](#), [39](#), [40](#), [41](#), [42](#).
- ” Napier’s, for Mint Office, [15](#), [16](#).
- ” Short’s Improved, for Bullion, described, [16-18](#).
- ” Testing the Accuracy of a, [16](#).
- Bank of England, [v](#), [3](#), [36](#), [37](#), [51](#), [54](#), [55](#), [64](#), [65](#), [66](#), [70](#), [71](#), [101](#),
[102](#), [103](#), [109](#), [110](#), [122](#), [123](#), [126](#), [150](#), [155](#), [156](#), [176](#), [185](#).
- ” ” ” Imports Gold for Coining and makes profit thereby,
[v](#), [3](#), [4](#), [126-127](#), [170](#).
- Bars, by Assay below standard fineness, [49](#), [53-56](#), [168](#).
- ” Ends sheared from, [20](#), [92](#), [112](#), [113](#), [116](#), [136](#), [160](#), [161](#).
- ” Proportion of Coined Money obtained from, [84](#), [92](#), [111](#), [112](#), [113](#), [114](#), [115](#), [117](#), [160](#), [161](#).

- ” Rolling of, [18-25](#).
- ” Used for Coining, Dimensions of, [9](#), [31](#), [74](#), [75](#).
- ” ” ” Uniformity of, desirable, [9](#), [31](#), [49](#), [53](#).

Barton, Sir John, [29](#), [31](#), [32](#).

Barton’s Draw-Bench, [27-33](#).

Barton, Mr. W. H., xvi., [29](#), [30](#), [56](#), [63](#), [86-87](#), [106](#), [172](#),
[175](#), [182](#), [183](#), [184](#), [191](#).

Becker, Mr. C., [25](#).

Blanching and Annealing Blanks debases Gold Coinage, [31](#), [48](#), [49](#), [50](#), [53-56](#), [57](#), [58](#).

Blanching, Effects of, [2](#), [48-49](#), [53-56-57](#), [58](#), [74](#), [75](#), [86](#), [92](#), [138](#), [143](#), [161](#).

Blanks, Annealing and Blanching, debases standard of Coinage, [31](#), [48](#), [49](#), [50](#), [51](#), [53-56](#), [57](#), [58](#).

- ” Drying of, by sawdust, [57-58](#).
- ” Dumb, [46](#), [61-62](#).
- ” Examination of, [35](#), [41](#), [42](#), [46](#), [50](#), [61](#), [70](#).
- ” Filing the heavy, [42-46](#).
- ” Weighing of Individual, [23](#), [31](#), [35-36](#), [37-42](#), [43](#), [108-110](#).
- ” Weighing of Gold, [37](#), [42-44](#), [46](#).

Boulton, Mr. Matthew, [61](#), [69](#), [169](#).

Boulton’s Coining Press described, [59-61](#), [68](#), [69](#), [80](#), [169](#).

Bradshaw, Mr. William, [40](#), [41](#).

Brande, Mr. W. T., [xv](#), [83](#), [120](#), [172](#), [176](#), [182](#), [195](#).

Britt.—Brit., [75-76](#).

Brittle Gold, Coining of, [v](#), [6](#), [49-53](#), [98](#), [112](#), [176](#), [180](#), [181](#), [185](#), [190](#).

- ” ” Coins from, Distinctively Marked, [50](#).

Brockages Imperfect Coins, [68-69](#), [70](#), [92](#), [161](#).

Bronze Coinage, [9](#), [33](#), [64](#), [68](#), [73](#), [74-76](#), [93](#), [94](#), [118](#), [121](#), [129](#), [147](#), [162-165](#).

- ” ” Loss of Profit on, [129](#), [147](#), [162-165](#).
- ” ” Weight and Value of, [163](#), [164](#), [165](#).

Bullion, Analysis of, [13-14](#), [81](#), [157-158](#).

- ” necessity for Analyses of, [13-14](#), [81](#).

Bumping, Prevention of, [12](#).

Chancellor of the Exchequer, as regards the Coinage, [3](#), [104](#), [120-121](#), [122-123-125](#),
[126](#), [127](#), [145](#), [146](#), [150-151](#), [163](#), [164](#), [170](#), [188](#), [193](#), [196](#).

Charcoal, its use when Melting Gold and Silver, [6](#), [8](#), [138-139](#).

Chimney Chambers to Condense volatilized
Gold and Silver, [83](#), [88](#), [103](#), [104-105](#), [128](#), [145](#), [147](#), [162](#), [166-167](#).

Clipping Coins (the edges of), [59](#), [62](#), [64](#), [125](#), [141](#).

Coinage, Integrity and Accuracy of the, [14](#), [42-44](#), [53-56](#), [70](#), [71-72](#), [123-126](#), [127](#), [157-159](#).

Coinage, a Free, [3](#), [169](#).

- ” Debasement of the, [31](#), [48](#), [49](#), [50](#),
[53-56](#), [57](#), [58](#), [120-121](#), [122-123](#), [157-159](#).
- ” Debates on the, [52](#), [72](#), [105](#), [106](#), [146-148](#), [149](#).
- ” International, Commission on, [36](#), [117](#), [120](#).

Coined Money, Annealing of, [50](#), [51](#), [52](#).

- ” ” Rate per cent. obtained from Bars, [20](#), [30](#), [44](#), [84](#),
[89](#), [92](#), [107-115](#), [117](#), [140](#), [161](#), [176](#), [179](#).
- ” ” Weight and Value of Bronze, [68](#), [73](#), [162-163](#), [165](#).
- ” ” Weight and Value of Gold, [3](#), [36](#), [56](#), [68](#), [70](#), [73](#), [90](#), [91](#), [92](#),
[96](#), [97](#), [111](#), [112](#), [114](#), [117](#), [119](#), [121](#), [123](#), [124](#).
- ” ” Weight and Value of Silver, [3](#), [56](#), [65](#), [68](#), [71](#), [73](#), [132](#), [133](#),
[134](#), [135](#), [146](#), [147](#), [149](#), [150](#), [154](#), [155](#), [156](#), [161](#).

Coining, Article on, [vii](#), [4](#), [71](#).

- ” by Contract, [v](#), [127](#), [128](#).

- ” Cost of, [42](#), [43](#), [44](#), [51](#), [70](#), [92](#), [115](#), [117](#), [118-120](#), [121-123](#), [130](#), [141](#), [149-150](#), [155](#), [161](#).
- ” Early Methods of, [1-2](#).
- ” Gold, Gain in weight by, [86](#), [87](#), [88](#), [89](#), [90](#), [91](#), [92](#), [93](#), [94](#), [95](#), [98](#), [99](#), [100](#), [101-104](#), [145](#), [168](#), [169](#), [173](#), [175](#), [178](#), [184](#), [189](#), [190](#).
- ” Gold, Loss of Weight by, [9](#), [16](#), [21](#), [22](#), [31](#), [49](#), [51](#), [53](#), [56-57](#), [82-83](#), [84-86](#), [87-88](#), [89](#), [90](#), [91](#), [92](#), [94](#), [95](#), [98](#), [99-104](#), [117](#), [119-120](#), [122](#), [125](#), [126](#), [129](#), [145](#), [173](#), [174](#), [175](#), [176](#), [177-179](#), [180](#), [184-189](#), [190](#).
- ” Press, Description of Boulton’s, [58-61](#), [68](#), [69](#), [80](#), [169](#).
- ” Press requires improvements, [58](#), [61](#), [69-70](#), [169](#).
- ” Proposal to fix the rate of Cost of, [115-123](#).
- ” Royal Prerogative, [149](#), [150](#), [166](#).
- ” Silver, Gain in Weight by, [iv](#), [3](#), [95](#), [130](#), [131](#), [146-148](#), [151](#), [160](#).
- ” ” ” Loss of Weight by, [v](#), [16](#), [21](#), [22](#), [74](#), [82](#), [95](#), [105](#), [106](#), [126](#), [130](#), [131-134](#), [139-140](#), [141](#), [142](#), [143](#), [145-148](#), [149](#), [151](#), [155](#), [161](#), [162](#), [175](#), [184](#), [189](#).

Coins, Amount of Legal Tender in, [68](#), [76](#), [151](#), [162](#), [166](#).

- ” Cause of Cracked, [46](#), [61-62](#).
- ” Diameter of, [63](#), [68](#).
- ” Edges of Crenated, Milled, Grained, [2](#), [48](#), [59](#), [61-62](#), [64](#), [80](#).
- ” ” Lettered, [2](#), [62-63](#).
- ” Examination for Imperfect, [46](#), [62](#), [70](#).
- ” Imperfect, Brockages, [68](#), [69](#), [70](#), [92](#), [161](#).
- ” Incused, how produced, [69](#).
- ” Pounding of, [35](#), [49](#), [53](#), [71-72](#), [73](#), [125](#), [167](#).
- ” Protecting Edge is necessary to, [2](#), [46](#), [48](#), [62](#), [64](#), [67](#), [158](#), [159](#).
- ” Ringing of, [46](#), [62](#), [70](#).
- ” Sweating of, [59](#), [125](#), [141](#).
- ” Weighing of, after Coining, [39](#), [43](#), [45](#), [70](#), [108-110](#).
- ” Weight of Current and New, [36](#), [39](#), [66](#), [68](#), [70](#), [71](#), [72](#), [73](#), [123](#), [124](#), [167](#).

Collar for Coining Milled, Lettered, [48](#), [59](#), [60](#), [61-62](#), [63](#), [64](#), [67-68](#).

Comber, Mr. P. F., [74](#).

Copper Moneys, [68](#), [76](#), [157](#), [162](#), [164](#).

Copper on Gold, Influence of Oxide of, [6](#), [49](#).

- ” is used as an Alloy, [4](#), [6](#), [7-8](#), [21](#), [46](#), [48](#), [49](#), [138](#).

Corcoran, Mr. Bryan, [195](#).

Corry, Mr. Montagu, [186](#).

Cotton, Mr. William, [16](#), [37](#), [39](#), [41](#), [42](#), [124](#).

Cotton’s Automaton Balance, [16](#), [37-42](#), [48](#), [124](#).

Crenated or Milled Edge on Coined Money, [2](#), [48](#), [59](#), [61-62](#), [64](#), [80](#).

Cutting-out Press described, [2](#), [33-35](#), [169](#).

Dies for Coining, [1](#), [50](#), [57](#), [59](#), [60](#), [64](#), [68](#), [76-81](#), [157](#), [158](#), [167](#).

Differences in Weighing Bullion, [14-16](#), [89](#), [92](#), [104](#), [127](#), [137](#), [138](#), [143](#), [160-161](#), [176](#), [184](#).

Difficulties in introducing Improvements, [9](#), [15](#), [40](#), [44](#), [46](#), [51](#).

Disraeli, Right Hon. B., [183-186-187](#), [191](#).

Draw-bench, Description of the, [27-33](#).

- ” ” Opposition to use of, [29-30](#), [31-32](#).
- ” ” Proper mode of using, [32-33](#).

Edge-Compressor, [46-48](#).

Edge, Protecting, to Coins, necessity for, [2](#), [46](#), [48](#), [62](#), [64](#), [67](#), [158](#), [159](#).

Edinburgh, H.R.H. the Duke of, [183](#).

Ends sheared from Bars, [20](#), [92](#), [112](#), [113](#), [116](#), [136](#), [160](#), [161](#).
 Engravings, Initial Letters to Illustrate, [5](#).
 European Mints, Reports upon, [9](#), [30](#), [34](#), [42](#), [44](#), [55](#), [61](#), [74](#), [83](#), [99-100](#),
[105](#), [106](#), [125](#), [147](#), [148](#), [168](#).
 Evelyn, John, [62](#).
 Exchequer, Chancellor of the,
 as regards the Coinage, [3](#), [104](#), [120-121](#), [122-125](#), [126](#), [127](#), [145](#), [146](#),
[150-151](#), [163](#), [164](#), [170](#), [188](#), [193](#), [196](#).
 Fairbairn, Mr. W., [xvi.](#), [172](#).
 Fenton, Mr. William, [33](#).
 File, Pilcher's, for reducing Heavy Blanks, [42-46](#), [176](#), [179](#), [180](#).
 Fillets, Adjusting of, [9](#), [19](#), [20](#), [22](#), [23](#), [25](#), [27](#), [29](#), [31](#), [32](#), [42](#), [112-113](#).
 " of uniform thickness, [25](#), [30-31](#).
 Fillets, Dumb, [46](#), [61-62](#), [89](#), [92](#), [112](#), [116](#).
 " Flatted, [25](#), [26](#), [27](#).
 " Rate per cent. of, obtained from Bars, [20](#), [92](#), [108](#), [112-113](#), [116](#), [160-161](#).
 " Testing accuracy of, [22](#), [23-24](#).
 Fire-damp Indicator, [182](#), [183](#), [187](#), [193](#), [198](#).
 Flat end, explanation of term, [27](#).
 Flatting Mill described, [25](#), [26](#).
 Forbes, Mr. George, [64](#), [155](#).
 Fourpenny-pieces, Coinage of, required, [10](#).
 Fremantle, Mr. C. W., [iii.](#), [iv.](#), [v.](#), [vii.](#), [9](#), [14](#), [15](#), [30](#), [31](#), [55](#), [56](#), [59](#), [72](#), [91](#),
[97](#), [99](#), [100](#), [102](#), [106](#), [118](#), [125](#), [127](#), [128](#), [129](#), [133](#), [135](#),
[144](#), [147](#), [152](#), [153](#), [154](#), [155](#), [164-165](#), [168](#), [187](#), [191](#), [195](#).
 French War Indemnity, Weight of the, [71](#).
 Furnace, Annealing, [21](#), [48](#), [49](#), [57](#), [62](#), [74](#).
 Gauges used in Coining, [22](#), [23-25](#), [35](#).
 Gauging Mill, Description of, [22-24](#), [26-27](#), [30-31](#).
 Gausby, Mr. Robert, [93](#).
 Gibson, Mr. E. S., [37](#), [62](#).
 Gladstone, Right Hon. W. E., [iii.](#), [viii.](#), [76](#), [101](#), [115](#), [187](#), [188](#), [192](#), [194](#).
 Goodwin, Mr. C. W., [iii.](#), [195](#).
 Gold, Annealing of, [22](#), [31](#), [48](#), [49](#), [50](#), [53-56](#), [57-58](#), [74](#).
 " Assay of Standard, described, [10-14](#), [53-56](#), [81](#), [122](#).
 " Brittle, Coining of, [6](#), [49-53](#), [98](#), [105](#), [112](#), [176](#), [180](#), [181](#), [185](#), [190](#).
 " Coin, when worn, invariably below standard, [48](#), [49](#), [53-56](#), [168](#).
 " Coinage, Statement showing details of, [89](#), [92](#).
 " Coining, Authority for, [3](#), [158](#), [166](#).
 " Coins, Weight and Value of Worn and Light, [53-56](#), [66](#), [67](#).
 " Copper is used to alloy, [4](#), [6](#), [7-8](#), [21](#), [46](#), [48-49](#), [53-56](#).
 " Gain by Coining, [86](#), [87](#), [88-89](#), [90-91](#), [92](#), [93](#), [94](#), [95](#), [98](#), [99](#), [100](#),
[101-104](#), [145](#), [168](#), [169](#), [173](#), [175](#), [178](#), [184](#), [189](#), [190](#).
 " Loss by Annealing, [21](#), [22](#), [31](#), [48](#), [49](#), [53-56](#), [57](#), [86](#), [92](#).
 " " Coining, [9](#), [16](#), [21](#), [22](#), [31](#), [49](#), [51](#), [53](#), [56-57](#), [82-83](#), [84-86](#),
[87-88](#), [89](#), [90](#), [91](#), [92](#), [94](#), [95](#), [98](#), [99-104](#), [117](#), [119-120](#), [122](#), [125](#),
[126](#), [129](#), [145](#), [173](#), [174](#), [175](#), [176](#), [177-179](#), [180](#), [184-189](#), [190](#).
 " " Melting, [86](#), [87](#), [96](#), [97](#), [98](#), [99-104](#), [105](#), [106](#), [145](#), [178](#), [184](#), [190](#).
 " Melting of, described, [5-7](#), [46](#), [53](#), [56](#), [129](#), [144](#).
 " Price of Standard, [3](#), [55](#), [66](#), [67](#), [70](#), [73](#), [84](#), [120](#), [122](#), [130](#).
 " Refining of, [14](#), [49](#), [50](#), [52](#), [53](#), [81-82](#), [104](#), [108](#).
 Gold Standard, [3](#), [4](#), [36](#), [53-66](#), [70](#), [73](#), [84](#), [120](#).

" " affected by Antimony, [14](#), [19](#), [49](#), [50](#),
[51](#), [176](#), [180](#), [181](#), [185](#), [190](#).
 " " affected by Arsenic, [14](#), [49](#), [50](#), [51](#), [176](#), [180](#), [181](#), [185](#), [190](#).
 " " affected by Lead, [6](#), [14](#), [49](#), [51](#), [81](#).
 " " " Mercury, [14](#), [81](#).
 " " " Osmium-Iridium, [81](#).
 " " " Oxide of Copper, [6](#), [49](#).
 " " " Palladium, [81](#).
 " " " Platinum, [81](#).
 " " " Tin, [14](#), [81](#).
 " " " Zinc, [6](#), [14](#).
 " Sweep recovered after Coinage of, [82](#), [83](#), [86](#), [88](#), [90](#), [91](#), [93](#), [98](#), [106-107](#).
 " " " Melting of, [96](#), [97](#), [98](#), [106-107](#).
 " Weight and Value of Coined, [56](#), [84](#), [90](#), [91](#), [92](#), [96](#), [97](#), [106](#), [111](#), [114](#), [121](#), [167](#), [178](#), [190](#).
 " and Silver Refining of, [81-82](#), [104](#), [138](#), [160](#).
 " while at the Mint bears no interest, [126](#).
 Graham, Mr. John, [15](#), [44](#), [71](#), [94](#), [98](#), [108](#), [112](#), [163](#), [182](#), [185](#), [187](#), [191](#), [196](#), [197](#).
 " " Thomas, [iii](#), [iv](#), [vii](#), [viii](#), [ix](#), [xi](#), [xv](#), [xvi](#), [50](#), [51](#), [54](#), [71](#), [83](#), [93](#),
[118](#), [119](#), [144](#), [153](#), [154](#), [155](#), [164-165](#), [170](#), [172](#), [174](#), [180](#),
[182](#), [183](#), [187](#), [193](#), [194](#), [195](#), [196](#), [197](#), [198](#).
 Graphite Pots for Melting, [5](#), [129](#).
 Granville, Right Hon. Earl, [166](#).
 Hatchett, Experiments of, [49](#), [157](#).
 Hatchett, Mr. C., [49](#), [157](#).
 Hawkins, Mr. Edward, [62](#).
 Herschel, Sir J. F. W., [54](#), [167](#).
 Hofmann, Dr. A. W., [xv](#), [xvi](#), [172](#), [189](#).
 Holland, Mr. P. H., [183](#).
 Hubbard, Mr. J. G., [55](#), [121](#).
 Hunt, Mr. Robert, [182](#), [183](#).
 Imperfect Coins, Brockages, [68](#), [69](#), [70](#), [92](#), [161](#).
 Importation of Ingots for Coining, [3](#), [4](#), [127](#), [170](#).
 Improvements, Difficulties attending introduction of, [9](#), [15](#), [40](#), [44](#), [46](#), [51](#).
 Indemnity, Weight of the French War, [71](#).
 Indenture, Weight and Fineness of each Coin was fixed by Mint, [36](#), [71](#), [108](#), [166](#).
 Indicator, Fire-damp, [182](#), [183](#), [187](#), [193](#), [198](#).
 Ingots, "Potting" of, [4](#).
 Integrity and Accuracy of the Coinage, [14](#), [42-44](#), [53-56](#), [70](#), [71-72](#), [123-125](#), [127](#), [157-159](#), [166-167](#).
 International Coinage, Report of Commissioners on, [36](#), [119](#), [120](#).
 Italy, Bronze Coinage of, [64](#).
 Jones, Mr. Meredith, [47](#), [179](#), [180](#).
 Journey, Weight a, [36](#), [70](#).
 Keys of Strongholds, [173](#).
 Kinnaird, Right Hon. Lord, [iii-v](#), [vii](#), [ix](#), [xiii](#), [51](#), [52](#), [54](#), [56](#), [91](#), [97](#),
[100](#), [101](#), [102](#), [105](#), [106](#), [108](#), [126](#), [133](#), [135](#), [145-147](#), [148](#), [149](#), [151](#), [152](#), [153](#),
[163](#), [164](#), [171](#), [183](#), [186](#), [188](#), [192-199](#).
 " Hon. A., M.P., [192](#), [194](#).
 Lansdowne, Marquess of, [iv](#), [ix](#), [52](#), [99](#), [100](#), [104](#), [105](#), [106](#),
[146](#), [147-148](#), [149](#), [152](#), [171](#).

Legal Tender, Amount of Coins in, [68](#), [76](#), [151](#), [162](#), [166](#).
 Loss by Annealing, [9](#), [21](#), [22](#), [31](#), [48](#), [49](#), [53-56](#), [86](#), [92](#), [138](#), [161](#), [162](#).
 " by Coining Bronze, [129](#), [147](#), [162-165](#).
 " " Gold, [9](#), [16](#), [21](#), [22](#), [31](#), [49](#), [51](#), [53](#), [56-57](#), [82](#), [83](#), [84](#), [86](#), [87-88](#),
 [90](#), [91](#), [92](#), [94](#), [95](#), [98](#), [99-104](#), [117](#), [119](#), [120](#), [122](#), [125](#), [126](#),
 [129](#), [145](#), [173](#), [174](#), [175](#), [176](#), [177-179](#), [180](#), [184-189](#), [190](#).
 " " Silver, [v](#), [16](#), [21](#), [22](#), [74](#), [82](#), [95](#), [105](#), [106](#), [126](#), [130](#), [131-133](#),
 [139-140](#), [141](#), [142](#), [143](#), [145-148](#), [149](#).
 " by Melting Gold, [86-87](#), [96](#), [97](#), [98](#), [99-104](#), [105](#), [106](#), [145](#), [178](#), [184](#), [190](#).
 " by Melting Silver, [131](#), [134-135](#), [136-142](#), [143](#), [144](#), [184](#), [189](#).
 " of Profit on Silver Coinage, [v](#), [129](#), [130](#), [146](#), [147](#), [148-151-156](#).
 Lowe, Right Hon. Robert, [iii](#), [v](#), [ix](#), [104](#), [120](#), [127](#), [150](#), [151](#),
 [164](#), [166](#), [171](#), [188](#), [192](#), [193](#), [196](#).
 Lowndes, Mr. W., [2](#), [159](#).
 Lowry, Mr. J. W., [37](#).

 Marking Machine, Jones's, [46-48](#).
 Martin, Mr. John, [30](#).
 Mathison, Mr. Gilbert F., [99](#).
 Melting Gold, Loss by, [53](#), [86](#), [87](#), [96](#), [97](#), [98](#), [99-104](#), [105](#), [106](#), [145](#),
 [178](#), [184](#), [190](#).
 Melting Gold, Process of, described, [5-7](#), [46](#), [53](#), [56](#), [129](#), [144](#).
 Melting Silver, Process of, described, [7-9](#), [46](#), [129](#), [138-145](#).
 Melting Silver, Gain by, [134-135](#), [136](#), [137](#), [142](#), [143](#).
 " " Loss by, [131](#), [134-135](#), [136-142](#), [143](#), [144](#), [184](#), [189](#).
 " " Value of Sweep recovered from Gold, [96](#), [97](#), [98](#), [106-107](#).
 " " Value of Sweep recovered from Silver, [106](#), [107](#), [131](#), [134](#), [135](#), [138](#), [139](#), [140](#), [144](#).
 Metals, Annealing of, *see* Annealing.
 Miller, Mr. John, [66](#).
 " " Wm., [64](#), [65](#), [67](#), [155](#), [156](#).
 Mint Indenture formerly fixed Weight and Fineness of Coins, [36](#), [71](#), [108](#), [166](#).
 Mint Notes, Issue of, for Bullion, [170](#).
 Mint premises, building of New, [v](#), [127](#), [128](#).
 Money, Hammered and Milled, [1](#), [2](#), [159](#).
 Moneyers, The Company of, [82](#), [87](#), [99](#), [112](#).
 Monteagle, Lord, [166](#).
 Murray, Mr. John, [31](#).

 Napier, Mr. James M., [15](#), [16](#), [34](#), [37](#), [40](#), [42](#), [44](#), [59](#), [61](#),
 [74](#), [108](#), [147](#), [148](#), [169](#).

 Oakes, Mr., [182](#).
 Oil on Scissel, [29](#), [32](#), [33](#), [35](#),
 [86](#), [87](#), [98](#), [99](#),
 [131](#), [136](#), [139](#),
 [140](#), [174](#), [175](#), [176](#).
 " to Lubricate Fillets, [32](#), [33](#), [176](#).
 Orders, "Circulating," [94](#).
 Office, System of Promotion to, [v](#), [viii](#), [15](#), [52](#),
 [56](#), [94](#), [108](#), [115](#), [128](#),
 [153](#), [169](#), [171](#), [173](#),
 [175](#), [176](#), [177](#), [182](#).

 Parliament, Acts of, relating to the Coinage, [3](#), [36](#), [56](#), [71](#),

[72](#), [73](#), [109](#), [124](#),
[150](#), [157](#), [159](#), [166](#).
 Palmerston, Lord, [xv](#), [140](#), [171](#).
 Pepys, Samuel, Diary of, [158](#), [159](#).
 Percy, Dr., [6](#), [52](#).
 Pilcher, Mr. Richard, [37](#), [39](#), [42](#), [44](#),
[47](#), [177](#), [179](#), [180](#).
 Pilcher's File, [42-46](#), [176](#), [179](#), [180](#).
 Playfair, Dr. Lyon, [iii](#), [iv](#), [195](#), [196](#).
 Pounding of Coins, its use and abuse, [35](#), [49](#), [53](#), [71-72](#),
[73](#), [125](#), [167](#).
 Proclamations relating to the Coinage, [68](#), [76](#).
 Promotion to Office, System of, [v](#), [viii](#), [15](#), [52](#),
[56](#), [94](#), [108](#), [115](#),
[128](#), [153](#), [169](#), [171](#),
[173](#), [175](#), [176](#), [177](#), [182](#).
 Pyx Pieces, Weight of, [56](#), [92](#), [112](#), [113](#), [114](#), [161](#).
 " Suggestions as to the Trial of the, 53-[54](#), [56](#).
 " The Trial of the, [49](#), [56](#), [70](#), [123-124](#), [166-167](#).

 Queen, Her Majesty the, [166](#), [168](#), [183](#).

 Rejected Blanks, [30](#), [32](#), [35](#), [42-44](#),
[46](#), [92](#), [107-110](#), [161](#),
[170](#), [179](#), [184](#), [189](#).
 Rejected Coins, [42-44](#), [46](#), [107-110](#).
 " explanation of term, [44](#).
 Remedy allowed, [32](#), [39](#), [43](#), [108](#).
 " explanation of term, [39](#), [167](#).
 " practically used, [32](#), [71-72](#), [108-110](#), [112](#).
 Rennie, Mr. G., [xvi](#), [172](#).
 Reports on Mint, [14](#), [118-119](#), [127](#), [147](#).
 Roberts, Mr. W. C., [v](#), [52](#), [53](#), [55](#),
[56](#), [57](#), [74](#), [99](#), [100](#),
[105](#), [129](#), [144](#), [147](#),
[153](#), [154](#).
 Rolling Mill, [18-20](#), [22-23](#), [26-27](#), [30-31](#).
 Rolling Mills, Accuracy to be obtained by, [23](#), [25](#), [27](#), [30-31](#).
 Rollers, Steel, [26](#), [27](#), [30](#), [31](#).
 Rossie, Lord, *see* Kinnaird, Lord.
 Ryan, Sir Edward, [166](#).

 Scissel, Meaning of the word, [2](#), [35](#).
 " Melting of, [8](#), [136-137](#).
 " Oil upon, [29](#), [32](#), [33](#), [35](#),
[86](#), [87](#), [98](#), [99](#),
[131](#), [139](#), [140](#), [174](#),
[175](#), [176](#).
 Seyd, Mr. Ernest, [54-55](#), [130](#).
 Shears, Description of, [19](#), [20](#), [21](#),
[25](#), [30](#), [31](#).
 Short, Mr. S. R., [16](#).
 Silver copper used to alloy, [7-8](#), [46](#), [49](#).
 Silver, Annealing of, [2](#), [31](#), [72-74](#), [143](#).

[161](#), [162](#).
 " as an Alloy for Gold, [13](#), [53](#), [81](#).
 " Assaying of Standard, [13](#).
 " Authority for Coining, [3](#), [166](#), [167](#).
 " Coinage, Loss of Profit or Seigniorage on, [v](#), [129-130](#), [146](#),
 [147](#), [148-151-156](#).
 " " Profit or Seigniorage derived from, [iv](#), [3](#),
 [146-148](#), [151](#), [152](#),
 [153](#), [154-156](#), [157-159](#).
 " " Tabular Statement showing details of, [159](#), [160](#),
 [161](#), [162](#).
 " " Weight and Value of, [65](#), [132-135](#), [146](#),
 [149](#), [150](#), [151](#), [152](#),
 [154](#), [156](#), [160](#),
 [161](#), [167](#).
 " Coining of, [7](#), [67](#), [72](#), [74](#),
 [116](#), [118](#), [121](#), [129](#),
 [143](#), [159-161](#).
 " Coins, Weight and Value of Worn and Light, [64-65](#), [66](#), [67](#),
 [130](#), [146](#), [152](#), [155](#), [156](#).
 " Gain by Coining, [95](#), [130](#), [131](#), [160](#).
 " " Melting, [134-135](#), [136](#), [137](#).
 " Loss by Annealing, [31](#), [72-74](#), [143](#), [161](#), [162](#).
 " " Coining, [v](#), [16](#), [21](#), [22](#), [74](#),
 [82](#), [95](#), [105](#), [106](#), [126](#),
 [130](#), [131-133](#), [139-140](#),
 [141](#), [142](#), [143](#), [145-148](#), [149](#).
 " " Melting, [131](#), [134-135](#), [136-142](#),
 [143](#), [144](#), [184](#), [189](#).
 " Melting of, [7-9](#), [46](#), [129](#), [138-145](#).
 " Standard, [3](#), [8](#), [73](#), [129](#), [130](#), [142](#), [157](#).
 " Sweep recovered from Coinage, [82](#), [83](#), [86](#), [106-107](#),
 [131](#), [132](#), [133](#), [144](#).
 " Sweep recovered from Melting, [106-107](#), [131](#), [134](#), [135](#),
 [138](#), [139](#), [140](#), [144](#).
 Simon, Thomas, Petition Crown of, [62-64](#).
 Smith, Colonel J. T., [140-141](#), [142-143](#).
 " Dr. Angus R., [183](#).
 " Mr. Richard, [6](#).
 Soap for Lubricating Fillets, [32](#), [33](#).
 Sovereign, Legal and Current Weight of, [39](#), [66](#), [68](#), [70](#),
 [73](#), [120](#), [122-123](#), [124](#).
 " Sweating of, [59](#), [125](#), [141](#).
 Steel Rollers, [26-27](#), [30-31](#).
 " Used for Dies, [76-81](#), [157-158](#).
 Sterry, Mr. Charles, [194](#).
 Strongholds, the Keys of the, [173](#).
 Supply, or Scrap Bullion, [86](#), [98](#), [161](#), [162](#).
 Sweep, explanation of term, [86](#), [106](#), [140](#).
 " recovered from Gold Coinage, [52](#), [82](#), [83](#), [86](#),
 [88](#), [90](#), [91](#), [93](#),
 [98](#), [106-107](#).
 " " " Silver Coinage, [82](#), [83](#), [86](#), [106-107](#),
 [131](#), [132](#), [133](#), [144](#).

” ” ” Gold Melting, [96](#), [97](#), [98](#), [106-107](#).
” ” ” Silver Melting, [106](#), [107](#), [131](#), [134](#),
[135](#), [138](#), [139](#), [140](#), [144](#).

Times newspaper, Letters in the, [37](#), [52](#), [54-55](#), [72](#).

Tomlino, Colonel, [169](#).

Tomlinson's Cyclopædia, article on coining, [viii](#), [4](#), [71](#).

Trevelyan, Sir Charles, [195](#).

Trier, Duties of the, [26](#), [31](#), [33](#), [35-36](#).

Value of Gain, *see* Gold, Silver, Gain by Coining.

” Loss sustained, [31](#), [65](#), [67](#), [83](#),
[84](#), [86](#), [89-91](#), [96-97](#),
[98-99-100](#), [101](#), [102](#),
[104](#), [105](#), [125](#), [128](#),
[129](#), [131-135](#), [137](#), [143](#),
[144](#), [145-149](#), [150](#), [151](#),
[153](#), [154](#), [155-156](#), [161](#),
[163](#), [164](#), [165](#), [174](#), [178](#),
[179](#), [190](#).
” Money Coined in each Coinage, *see* Coined Money.
” Standard Gold, *see* Gold, price of Standard.
” Sweep recovered, *see* Sweep recovered.

Versmann, Mr. F., [195](#).

Volatilization of Gold and Silver in Coinage operations, [83](#), [88](#), [103](#), [104-105](#),
[128](#), [145](#), [147](#), [162](#).

Wages System, Suggested new one, [115-119](#), [121-123](#).

” ” under which paid for Coining, [46](#), [115-119](#).

” ” under which paid for Melting, [46](#), [116-117](#).

Warrington, Mr. Robert, [81](#).

Waste, *see* Coining, Loss by.

Weighing Bullion, Differences made in, [14-16](#), [89](#), [92](#),

[104](#), [127](#), [137](#),
[138](#), [143](#), [160-161](#), [176](#), [184](#).

” ” System of, [14-16](#), [136](#), [137](#), [138](#).

” Machines, Cotton's Automaton, [16](#), [37-42](#), [48](#), [124](#).

” ” Bradshaw's Improvements in, [40-42](#).

” ” Pilcher's Improvements in, [37](#), [39-40](#), [42](#).

” ” Napier's, for Mint Office, [15](#), [16](#).

” ” Short's, for Bullion, [16-18](#).

Wellington, Duke of, [54](#).

Westbury, Lord, [166](#), [167](#).

Wilson, Mr. C. Rivers, [14](#), [127](#), [147](#).

Wood, Mr. Halder, [195](#).

Worn Coins below Standard by Assay, [48-49](#), [53-56](#), [57](#).

” ” Weight and Value of Gold, [53-56](#), [66](#), [67](#).

” ” ” Silver, [64-65](#), [66](#), [67](#),
[130](#), [146](#), [152](#), [155-156](#).

Worn-out Melting-Pots, [139](#), [140](#).

Zinc causes difficulty in Coining Bronze, [75](#).

” ” ” ” Gold, [6](#), [14](#).

“THE ROYAL MINT.”—THIRD EDITION

OPINIONS OF THE PRESS.

Especially worthy of official attention, and is accompanied by suggestions for an amended system.—*Times*.

According to Mr. Ansell, who is every way entitled to credit, great loss has frequently arisen through the promotion of incompetent men who cannot handle properly the fine instruments employed.... The whole subject is of special interest at the present time, and Mr. Ansell's contribution to the facts of the discussion will be frequently appealed to.—*Economist*.

From an extensive knowledge of the practical working of the Royal Mint, Mr. Ansell has given the educated world a very clear and comprehensive account of the operations of that institution, and has endeavoured to show—and, we are convinced, has succeeded in showing—the nature of the defects through which it has for a long period proved no inconsiderable burden to the country. The writer was elected to his post through his known scientific and technical attainments; and the work now before us is worthy of his reputation.... The practical aim of the book is to prove that the Royal Mint should no longer be a burden to the country, and that the losses now entailed could, by a better system of management and supervision, be reduced practically to *nil*.—*Money Market Review*.

Mr. Ansell enters fully into the very interesting details of the various processes involved in coining for the Bank; from the reception of the ingots to their conversion into the circulating medium of the realm. The text is much assisted by numerous well-executed woodcuts and steel engravings, which serve to explain the internal economy of the Mint, and the various pieces of elaborate machinery required to produce the enormous quantity of money which must be kept in use in a country having the extended commercial interests of England.... Mr. Ansell's book is extremely opportune, and deserves careful perusal.—*Examiner and London Review*.

The author of this book is, unquestionably, thoroughly versed in the metallurgical, chemical, and mechanical details of the subjects treated therein.—*Morning Advertiser*.

Mr. Ansell having been employed in the Mint, is well qualified to speak on the subject of which he treats. At this juncture, when discussions with regard to that establishment and the coinage are before the Legislature, the book will be well worthy of the perusal of those taking part in the controversy.—*Standard*.

An interesting work relating to the operations of that establishment.—*Daily News*.

Those interested in the subject may investigate the alleged shortcomings of the administration by its light, and ascertain if the Bill of Mr. Lowe will effect any of the needed reforms, and to what extent. The contents of the book are of an important character, and merit careful consideration.—*Bullionist*.

Mr. Ansell has at command a thoroughly practical, official, and scientific knowledge of his subject; and from the position which he occupied, is entitled to speak with authority. His opinions, therefore, on the economical bearings of the coinage discussion, both social and political, are of real interest.... The chief value of the work is that we are here shown, by one who for a long time held responsible office, what he considers to be the faults of the existing system.... Now that the Mastership of the Mint has become vacant, and the question is before the country how it shall be filled, the interest of the book, great in itself, is enhanced.—*Financier*.

We wish to direct attention to a very valuable work entitled "The Royal Mint," by Mr. G. F. Ansell. The whole question of coinage is under the consideration of the Chancellor of the Exchequer, who is expected to make several changes in the organisation and management of the Mint, therefore Mr. Ansell's work will no doubt be carefully perused by Members of the House of Commons, as it throws much light on every branch of working up the gold and silver into coins.—*London Mercantile Journal*.

This work will be read with much interest as public attention has so recently been drawn to the management of the Mint. The author writes from his own personal experience, having been engaged as an authoritative member of the staff under the late master, Mr. Thomas Graham. The author asserts that each of his statements is the result of his actual experience, and capable of proof. The work is amply illustrated with some very fine engravings, which are fully explained in the text. Mr. Ansell's suggestions for the better scientific and official management of the Mint should command serious attention.—*Public Opinion*.

This is a most important work, and it has been published opportunely.... We cannot even indicate the many points of importance to which Mr. Ansell draws our attention, but while giving the work general commendation, we heartily advise its perusal by all who are in any way interested in the subject.—*Civil Service Gazette*.

The present position of affairs at the Royal Mint, and the interesting service which has been rendered by Lord Kinnaid in the discussion in the House of Lords on the Coinage Bill, cannot fail to secure for Mr. Ansell's skilfully prepared book more than an ordinary amount of attention.... The management of the Mint is a matter of national importance, and the practical experience and thorough knowledge of mechanics possessed by Mr. Ansell have enabled him to produce a comprehensive and valuable book of reference.—*Dundee Advertiser*.

We receive with much satisfaction so lucid an account of the working of the Mint, and the highly important suggestions for its better management, as set forth in Mr. G. F. Ansell's excellent treatise.... The book now under notice, which is dedicated to Lord Kinnaid, will tend very much to enlighten all who may take an interest in this very important subject.—*Court Journal*.

Mr. Ansell frequently pauses to point out how, according to the conclusions formed from his personal observations and mechanical knowledge, the processes of manufacture might be simplified or cheapened. He insists that if the mechanical improvements suggested by him were followed out “the result would be a clear saving of £4,000 a year.”—*Echo*.

No one needs any longer to complain of want of information as to the manner in which the coinage of the country is prepared, or what are the duties of those who are entrusted with the important duty of supervising its mintage. In a comparatively small compass this book tells all that can be said on these heads, besides giving other details, the purpose of which is as practical as the manner of description is precise and positive.—*Bell's Weekly Messenger*.

In addition to the information to be found in the book regarding the coining of money, those interested in mechanical details will doubtless read with pleasure the minute description of the whole of the machinery in use in the Mint. The book has done and will do good service.—*South London Press*.

Mr. Ansell's book is cleverly written, and will probably lead to enquiry.—*City Press*.

This is in many respects a remarkable work, for it displays both scientific and technical knowledge on the subject of the coinage to an extent that at the present time can scarcely be surpassed.... The book is full of interest, for the facts are carefully marshalled, and the illustrations with which the volume is enriched are apposite and excellent. We can, therefore, unhesitatingly recommend the work to such of our readers as take an interest in such subjects.—*Scotsman*.

We will close with two quotations from this really valuable book, containing, as it does, so much that is interesting, and showing so intimate an acquaintance on the author's part with the subject of which it treats.—*Edinburgh Evening Courant*.

It is clear, however, that the evils proverbially connected with money are in full force at the Royal Mint: anxiety weighs down the higher authorities, heart-burnings prevail amongst the lower, irregularities exist amongst the subordinates. It is probable that some change has taken place in the management of matters since the author was in the position of an eye-witness; but the general value of his work is not likely to have been much diminished thereby.—*Illustrated London News*.

LONDON: EFFINGHAM WILSON, ROYAL
EXCHANGE.

PRINTED BY VIRTUE AND CO., CITY ROAD, LONDON.

Transcriber's Notes:

Deprecated spellings or archaic words were not corrected.

Footnotes and illustrations have been moved so that they do not break up paragraphs and so that they are next to the text they illustrate.

Typographical and punctuation errors have been silently corrected.

Footnotes:

[1] See page [196](#).

[2] See page [186](#).

[3] See page [198](#).

[4] See page [197](#).

[5] See page [146-148](#).

[6] See page [148-149](#).

[7] See page [169](#).

[8] See pages [126](#), [144-156](#).

[9] See pages [126-127](#), [168](#).

[10] See pages [50-53](#).

[11] See pages [129-138](#).

[12] See pages [3](#), [73](#).

[13] This word is explained at page [35](#).

[14] See pages [9](#), [53](#).

[15] That is, made by Payen, of Paris.

[16] See pages [81-82](#).

[17] See p. [39](#).

[18] See page [112](#).

[19] See page [23](#).

[20] As the cylinder wears, the screws of C may be loosened to permit the shifting of the abraded part, so that the whole circumference of the cylinder may be used.—[See also page 31](#).

[21] See page [48](#).

[22] See page [39](#).

[23] See p. [169](#).

[24] See pages [42-44](#).

[25] See pages [71-73](#).

[26] See pages [73](#), [123](#), [124](#).

[27] That is, coins which, although otherwise perfect in manufacture, are not within the limited variations of weight permitted by the Mint regulations.

[28] See pages [108](#), [109](#).

[29] See page [53](#).

[30] See page [6](#).

[31] These coins, struck between marked dies, may be recognised by a small line which I placed in the centre of the ribbon at the back of the head representing her Majesty on the obverse.

[32] Besides many verbal congratulations, the Master wrote me the following letter:—

Registered No. 3026, 1860.

Royal Mint, 20th October, 1860.

DEAR SIR,

Your success in obviating the difficulty of coining gold containing a minute portion of antimony or arsenic is a benefit to this department, which calls for my most grateful thanks. In recognition of your services I have the pleasure, with the approbation of the Lords Commissioners of Her Majesty's Treasury, to present to you the sum of one hundred pounds, for which cheque is enclosed.

I have the honour to be,

Dear Sir,

Most faithfully yours,

Tho. Graham

George F. Ansell, Esq.

[33] See pages [48](#), [49](#).

[34] See pages [48](#), [49](#).

[35] See pages [166](#), [167](#).

[36] See pages [123-125](#).

[37] See page [80](#).

[38] In Evelyn's "Discourse of Medals," 1697, p. 224, occurs the following passage:—"Mr. *Slingsby*, to whom I suggested the *Decus & Tutamen* out of a

Viniet in *Cardinel de Richlieu's Greek Testament*, printed at the *Louvre*, hindering his intended Addition (*in armis*), which neither would have become the *impress*, nor stood gracefully in the circle.”

[39] *Dei Gratia* was also omitted on some of the coins of George I.

[40] 33rd Vict., c. 10, cl. 4.

[41] See pages [42-45](#), [108](#), [109](#).

[42] See pages [73](#), [123](#), [124](#).

[43] See pages [39](#), [73](#), [124](#).

[44] See pages [48](#), [49](#).

[45] See page [9](#).

[46] See pages [86-88](#), [140](#).

[47] See pages [29](#), [32-33](#).

[48] See pages [98-99](#).

[49] See page [178](#).

[50] See pages [107](#), [117](#).

[51] See page [89](#).

[52] See page [27](#).

[53] See pages [99-106](#).

[54] See pages [35](#), [87](#).

[55] The moneyers were those contractors who conducted the coinages previously to the Mint being formed into a governmental department in 1851.—*Vide* Parliamentary Reports, 1837, 1849, 1852.

[56] See page [178](#).

[57] Mr. John Graham, who had been engaged “gratuitously” for many months “to get his hand in,” was appointed, on the 23rd April, 1861, to an extra clerkship, at £20 a month, as Inspector of Bronze Coins,—“as a warming-pan.”

[58] See page [90](#).

[59] See page [95](#).

[60] Specially referred to on page [94](#).

[61] Specially referred to on page [94](#).

[62] Specially referred to on page [94](#).

[63] Specially referred to on page [94](#).

[64] See page [90](#).

[65] See page [96](#).

[66] See page [96](#).

[67] See pages [97](#), [101](#).

[68] See page [102](#).

[69] A coinage having been in progress on the 31st of March, 1869, no waste account was made up at the close of the financial year 1868-69.

[70] See page [94](#).

[71] See page [83](#).

[72] See page [90](#).

[73] See also page [91](#).

[74] See page [136](#).

[75] See page [102](#).

[76] See page [102](#).

[77] See pages [91](#), [97](#), [133](#), [135](#).

[78] See Master's Letter to Treasury, page [179](#).

[79] See my Report to the Master of the Mint, March 8, 1860.

[80] See Remedy explained at page [39](#).

[81] See page [44](#).

[82] See pages [44-45](#).

[83] See pages [46](#), [92](#).

[84] See pages [46](#), [92](#).

[85] See page [111](#).

[86] See page [110](#).

[87] Explained on page [115](#).

[88] Explained on page [115](#).

[89] Explained on page [115](#).

[90] Explained on page [115](#).

[91] See page [114](#).

[92] See page [117](#).

[93] See page [92](#).

[94] See page [111](#).

[95] See page [111](#).

[96] By reference to the table at page 84, it will be seen that the amount of coin obtained in 1866 was below the worst produced at any other period since 1851.

[97] See Report by Commissioners on International Coinage, p. 93.

[98] See the Mint Books.

[99] See page [117](#).

[100] See page [125](#).

[101] See page [129](#).

- [102] See pages [44](#), [179](#).
- [103] See pages [3](#), [127](#).
- [104] See pages [39](#), [73](#).
- [105] See page [71](#).
- [106] See page [178](#).
- [107] See page [136](#).
- [108] See page [162](#).
- [109] See page [152](#).
- [110] See pages [146](#), [152](#).
- [111] See pages [65-67](#), [152](#), [155](#), [156](#).
- [112] See pages [127](#), [137-138](#).
- [113] See page [88](#).
- [114] See page [134](#).
- [115] See pages [127](#), [130](#).
- [116] See page [134](#).
- [117] See page [134](#).
- [118] See pages [131](#), [136](#).
- [119] See Report by Colonel J. T. Smith on Indian Mints from 1841-2 to 1852-3. Printed by authority at Madras, 1855.
- [120] See pages [137-138](#).
- [121] See pages [132-134](#).
- [122] I am prepared to give the names of the parties dismissed and the dates of dismissal.—G. F. A.
- [123] See page [162](#).
- [124] See pages [89](#), [160-161](#).
- [125] Specified by the Mint Indenture, which was legalised by the Act of Parliament 56 Geo. III. cap. 68, sec. ii.
- [126] See pages [53-56](#).
- [127] Understated at £99 14s. 5½d. in previous letter of May 22, 1858, in consequence of the sweep recovered per million having been taken at £136 13s. 9d. instead of £93 3s. 7d., the correct amount.

*** END OF THE PROJECT GUTENBERG EBOOK THE ROYAL MINT

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.